

Waste
Isolation
Pilot
Plant

CH-TRAMPAC



Revision 7
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LIST OF ABBREVIATIONS AND ACRONYMS

°C	degrees Celsius
°F	degrees Fahrenheit
AFGGR	(maximum) allowable flammable gas generation rate
ATGGR	allowable total gas generation rate
ALARA	“as low as reasonably achievable”
ANS	American Nuclear Society
ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
Be	beryllium
BeO	beryllium oxide
Be ₂ C	beryllium carbide
CBFO	Carlsbad Field Office
CCC	Criticality Control Container
CCO	Criticality Control Overpack
CFR	Code of Federal Regulations
CH	contact-handled
CH-TRAMPAC	<u>C</u> ontact- <u>H</u> andled <u>T</u> ransuranic Waste <u>A</u> uthorized <u>M</u> ethods for <u>P</u> ayload <u>C</u> ontrol
CH-TRUCON	<u>CH-TRU</u> Waste <u>C</u> ontent Codes
Ci	curie(s)
DOE	U.S. Department of Energy
DQO	data quality objective
EPA	U.S. Environmental Protection Agency
eV	electron volt
FGE	fissile gram equivalent
FGGR	flammable gas generation rate
FI	flammability index
GCF	group contribution factor
ICV	inner containment vessel
ID	identification
in.	inch(es)

lbs.	pound(s)
LTO	(Unified Flammable Gas Test Procedure) Long-Term Objective
MLEL	mixture lower explosive limit
mrem/hr	millirem/hour
MS	mass spectrometry
NRC	U.S. Nuclear Regulatory Commission
OCA	outer confinement assembly
OPCTCD	Overpack Payload Container Transportation Certification Document
PATCD	Payload Assembly Transportation Certification Document
PCTCD	Payload Container Transportation Certification Document
ppm	parts per million
PTGGR	payload total gas generation rate
PTGGRL	payload total gas generation rate limit
Pu	plutonium
QA	quality assurance
QAPP	Quality Assurance Program Plan (for the TRUPACT-II Gas Generation Test Program)
RSS	root sum of squares
SAR	safety analysis report
SWB	standard waste box
TDOP	ten-drum overpack
TFGC	total flammable gas concentration
TFGVC	total flammable gas/VOC concentration
TFVC	total flammable VOC concentration
TGGR	total gas generation rate
TRAMPAC	<u>TRU</u> Waste <u>A</u> uthorized <u>M</u> ethods for <u>P</u> ayload <u>C</u> ontrol (programmatic or waste-specific)
TRU	transuranic
TRUPACT-II	<u>Trans</u> uranic <u>P</u> ackage <u>T</u> ransporter-II
UFGTP	Unified Flammable Gas Test Procedure
VOC	volatile organic compound
WIPP	Waste Isolation Pilot Plant

LIST OF CH-TRU PAYLOAD APPENDICES

The [CH-TRU Payload Appendices](#) contain supporting documentation and other reference materials for the limits and compliance methods defined in the CH-TRAMPAC. All appendices referenced in the CH-TRAMPAC are found in the [CH-TRU Payload Appendices](#). The table of contents for the [CH-TRU Payload Appendices](#) is shown below.

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 - 6.13 Shipment of CH-TRU Waste Packaging Configurations with Unvented Heat-Sealed Bag Layers

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1.0 INTRODUCTION

The Contact-Handled Transuranic Waste Authorized Methods for Payload Control (CH-TRAMPAC) is the governing document for shipments in the Transuranic Package Transporter-II (TRUPACT-II) and HalfPACT packagings. All users of the TRUPACT-II and the HalfPACT shall comply with all payload requirements outlined in this document, using one or more of the methods described. Supporting information for the limits and compliance methods defined in this document is contained in the [CH-TRU Payload Appendices](#).¹

1.1 Scope

The TRUPACT-II and the HalfPACT have been developed as Type B packagings to provide a safe means of transporting contact-handled (CH) transuranic (TRU) wastes and other authorized payloads such as tritium-contaminated materials. The shipment of tritium-contaminated waste shall be as described in [Appendix 6.11](#) of the [CH-TRU Payload Appendices](#).¹ Both packagings consist of an outer confinement assembly (OCA), an inner containment vessel (ICV), and two aluminum honeycomb spacer assemblies, with the HalfPACT being a shorter version of the TRUPACT-II. The CH-TRAMPAC defines the authorized contents for both packagings. The defined payload is, for the most part, independent of packaging. Information that is specific to the TRUPACT-II or the HalfPACT packaging is identified as such within the CH-TRAMPAC.

¹ U.S. Department of Energy (DOE), [CH-TRU Payload Appendices](#), U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico.

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1.2 Purpose

The purposes of the CH-TRAMPAC are to:

- Define the applicable requirements for a payload to be transported in the TRUPACT-II or HalfPACT packaging
- Describe the acceptable methods of compliance that shall be used to prepare and characterize the CH-TRU materials or other payload materials prior to transport in a TRUPACT-II or HalfPACT packaging
- Identify the quality assurance (QA) program that shall be applied to these methods.

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1.3 Requirements

Requirements are established to ensure compliance of the payload with the transportation parameters of the TRUPACT-II and HalfPACT packagings. The CH-TRAMPAC defines payload requirements under the following categories:

- Container and Physical Properties ([Chapter 2.0](#))
- Nuclear Properties ([Chapter 3.0](#))
- Chemical Properties ([Chapter 4.0](#))
- Gas Generation ([Chapter 5.0](#))
- Payload Assembly ([Chapter 6.0](#))
- Quality Assurance ([Chapter 7.0](#)).

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1.4 Methods of Compliance

This document describes allowable methods to be used for determining compliance with each payload requirement and the controls imposed on the use of each method. Each generator or storage site shall select and implement a single method, or a combination of methods, to ensure that the payload is compliant with each requirement and is qualified for shipment. These methods shall be delineated in a programmatic or waste-specific data TRU Waste Authorized Methods for Payload Control (TRAMPAC).

Each shipper shall document and demonstrate compliance with the CH-TRAMPAC by one of the following methods:

- A programmatic TRAMPAC, which defines the process in which payload compliance is met, will be prepared by the shipper and approved by the U.S. Department of Energy (DOE) Carlsbad Field Office (CBFO). Implementing procedures of the TRAMPAC will be reviewed by the DOE-CBFO for completeness and compliance as part of the audit process.

Compliance with transportation requirements applies to the following two categories of waste:

1. Waste Generated Under a Formal Certification Program. Payload containers in this category (designated “100 Series”) are characterized individually based on process knowledge and visual examination at the time of waste generation. For sites using a set of site/equipment-specific procedures for payload control for compliance, an independent verification of compliance must be performed prior to transport for no less than 10 percent of the 100 Series payload containers transported from each site per year. This independent verification may consist of a second operator verifying the container contents or waste records during the waste generation process or radiography.
 2. Waste Generated Prior to Site Implementation of a Formal Certification Program. Payload containers in this category (designated “200 Series”) are characterized based on process knowledge. An independent verification of compliance shall be performed prior to transport for 200 Series waste using a statistical sampling program comprised of the methods identified in this section (visual examination, radiography, measurement, etc.).
- For small quantity shipments, a waste-specific data TRAMPAC will be prepared by the shipper and approved by the DOE-CBFO. The waste data are evaluated against the requirements in this document. A small quantity shipment may be made by any waste generator who does not have a DOE-CBFO approved programmatic TRAMPAC or a waste generator with a limited number of containers not addressed within their programmatic TRAMPAC.

A summary of the methods of compliance that shall be used for TRUPACT-II and HalfPACT payload control is provided in the following sections.

1.4.1 Visual Examination

Visual examination at the time of waste generation may be used to qualify waste for transport (e.g., 100 Series waste). The operator(s) of a waste generating area shall visually examine the physical form of the waste according to site/equipment-specific procedures and remove all prohibited waste forms prior to placement in the payload container. Observation of the waste generation process by an independent operator may be used as an independent verification of compliance prior to closure of the payload container. Visual examination under a sampling program may be used as independent verification (e.g., to verify the absence of prohibited items in 200 Series waste).

1.4.2 Visual Inspection

Visual inspection may be used to evaluate compliance with specific restrictions (e.g., visual inspection of payload container type, marking, etc.). Visual inspection by a second operator may be considered independent verification.

1.4.3 Radiography

Radiography, or equivalent nondestructive examination techniques, may be used as an independent verification to qualify waste for transport after the payload container is closed (e.g., to nondestructively examine the physical form of the waste and to verify the absence of prohibited waste forms). A radiography system normally consists of an X-ray-producing device, an imaging system, an enclosure for radiation protection, a waste container handling system, an audio/video recording system, and an operator control and data acquisition station. Some variation within a given nondestructive examination system will exist between sites. Site/equipment-specific QA and quality control procedures ensure that radiography system operator(s) are properly trained and qualified. [Appendix 5.1](#) of the [CH-TRU Payload Appendices](#)¹ describes typical real-time radiography procedures.

Other radiographic methods must meet the same performance objectives for real-time radiography (i.e., to nondestructively examine the physical form of the waste and to verify the absence of prohibited items in a payload container) and must be controlled by procedures similar to those described in [Appendix 5.1](#) of the [CH-TRU Payload Appendices](#)¹ for real-time radiography.

1.4.4 Records and Database Information

Information obtained from existing site records and/or databases or knowledge of process may be used to qualify waste for transport (e.g., as a basis for reporting the absence of prohibited waste forms within waste containers). This information may be verified using radiography ([Section 1.4.3](#)) and/or a waste sampling program ([Section 1.4.6](#)).

¹ U.S. Department of Energy (DOE), [CH-TRU Payload Appendices](#), U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico.

1.4.5 Administrative and Procurement Controls

Site-specific administrative and procurement controls may be used to show that the payload container contents are monitored and controlled and to demonstrate the absence of prohibited items.

1.4.6 Sampling Programs

Sampling programs comprised of the statistical application of other methods identified in this section may be used as an independent verification of compliance (e.g., for 200 Series waste). A site-specific statistical sampling program designed to address all payload requirements needing verification is recommended. [Appendix 5.3](#) of the [CH-TRU Payload Appendices](#)¹ describes previous sampling programs at the DOE sites.

1.4.7 Measurement

Direct measurement or evaluation based on analysis using the direct measurement may be used to qualify waste (e.g., direct measurement of the weight or analysis of assay data to determine decay heat).

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1.5 CH-TRUCON

The CH-TRU Waste Content Codes (CH-TRUCON) is a catalog of TRUPACT-II and HalfPACT authorized contents and a description of the methods utilized to demonstrate compliance with the CH-TRAMPAC.

1.5.1 Required Elements

Each content code within the CH-TRUCON must contain the following elements:

CONTENT CODE: Identifies the two-letter site abbreviation that designates the physical location of the waste and the three-digit code that designates waste generation relative to implementation of a formal certification program and the physical and chemical form of the waste. Content code identifiers are defined in the CH-TRUCON.

CONTENT DESCRIPTION: Identifies the physical form of the waste (e.g., describing whether it is inorganic or organic, solidified or solid). This is similar to the waste material type titles in [Section 5.1](#).

STORAGE SITE: Provides the location of the waste, if the location is different than the generating site. If the generating site and storage site are the same, this section is not required to be included in the content code.

GENERATING SITE: Provides the location of waste generation.

WASTE DESCRIPTION: Provides basic information regarding the nature and/or main components of the waste.

GENERATING SOURCE(S): Lists processes and/or buildings at each site that generate the waste in each content code.

WASTE FORM: Provides more detailed information on the waste contents, how the waste is processed, and/or specific information about the chemistry of constituents.

WASTE PACKAGING: Describes, in detail, techniques necessary for waste packaging in a given content code. This includes a description of the waste confinement layers, the number of layers of confinement used in packaging waste, and the mechanism for bag, can, or container closure.

ASSAY: Describes the types of radioactive materials measurement techniques or other methods utilized to obtain fissile material content and decay heat values for a particular content code.

RESIDUAL LIQUIDS: Describes the procedures used by the sites to ensure that the limits imposed on liquids (<1% by volume) are met for each content code.

EXPLOSIVES/COMPRESSED GASES: Identifies the methods used to preclude the presence of explosives or compressed gases.

PYROPHORICS: Describes the controls in place at each site to ensure that nonradioactive pyrophoric materials in TRU waste are excluded, reacted to render nonreactive, or are immobilized prior to placement in waste.

CORROSIVES: Describes the controls in place to ensure that corrosive materials in TRU waste either are not present or are neutralized or immobilized prior to placement in a payload container.

CHEMICAL COMPATIBILITY: Describes the controls in place to ensure chemical compatibility for the waste contents and the TRUPACT-II or HalfPACT packaging. All chemicals/materials in the waste for a specific content code are restricted to the allowable chemical lists ([Table 4.3-1](#) through [Table 4.3-8](#)) and the 5% limit on total materials not listed as specified in [Section 4.3](#).

PAYLOAD CONTAINER VENTING AND ASPIRATION: Details how payload containers that have been stored in an unvented condition (i.e., no filter and/or unpunctured liner) will be aspirated to ensure equilibration of any gases that may have accumulated in the closed container. This procedure is required only for unvented waste.

ADDITIONAL CRITERIA: Provides details on how the waste qualifies for shipment by meeting additional transport requirements (e.g., venting payload containers and liners).

SHIPPING CATEGORY: Shipping categories based on the above parameters for each content code are summarized in the CH-TRUCON.

MAXIMUM ALLOWABLE FLAMMABLE (GAS/VOC) LIMITS: The maximum allowable flammable (gas/VOC) limit for each payload container assigned to the shipping category is determined in accordance with [Section 5.2](#).

1.5.2 Use and Approval

All containers must have a content code approved by the Waste Isolation Pilot Plant (WIPP) CH-TRU Payload Engineer to be eligible for shipment. Any site requiring the transportation of TRU waste in the TRUPACT-II or HalfPACT that is not described in an approved content code must request the revision or addition of a content code by submitting a request to the WIPP CH-TRU Payload Engineer.

The WIPP CH-TRU Payload Engineer has the authority to approve a content code request only if compliance with the transportation requirements of the CH-TRAMPAC document can be demonstrated. Any submittal not meeting the requirements of the CH-TRAMPAC shall not be approved for inclusion in the CH-TRUCON or be used as the basis for a shipment in the TRUPACT-II or HalfPACT. The WIPP CH-TRU Payload Engineer does not have the authority to change the transportation requirements for the TRUPACT-II or the HalfPACT as specified in the CH-TRAMPAC document without approval from the U.S. Nuclear Regulatory Commission (NRC).

Requests for revisions to content codes shall be submitted to the WIPP CH-TRU Payload Engineer and may include the following:

- Minor revisions to the content code descriptions (e.g., changes to buildings or sources generating the waste)
- Changes to method(s) used to characterize the waste (e.g., the use of an alternate approved assay method)
- Changes to the packaging descriptions (e.g., the use of a different number of bag layers)
- Addition of a new authorized waste form from a given site (e.g., shipment of filter waste)

- Minor revisions to the chemical list for a specific content code.

The process for requesting a content code addition or revision is as follows:

1. The site prepares a request containing sufficient information to satisfy all of the necessary elements identified in [Section 1.5.1](#). If the request is for a content code revision, only the revised elements require preparation and documentation. The site shall ensure that the information submitted in the form of a content code addition or revision accurately describes the waste and waste generating processes based on site knowledge.
2. The site submits the request (e.g., draft content code or revised content code elements) to the WIPP CH-TRU Payload Engineer for review.
3. The WIPP CH-TRU Payload Engineer shall review the submittal for completeness and satisfactory demonstration of compliance with all the transportation requirements of the CH-TRAMPAC. As part of this review, the WIPP CH-TRU Payload Engineer's responsibilities may include a review to ensure that each of the previously identified elements is complete, the calculation or verification of new payload shipping categories to accommodate changes in packaging configurations using the Payload Shipping Category Worksheet ([Table 2.2-1](#) through [Table 2.2-4](#) in [Appendix 2.2](#) of the [CH-TRU Payload Appendices](#)¹), and the analysis of compliance with the list of allowable materials (for new waste forms or for changes in chemical composition) pursuant to [Section 4.3](#). The WIPP CH-TRU Payload Engineer shall not approve any submittal that does not demonstrate compliance with every transportation requirement for the TRUPACT-II and HalfPACT.
4. Upon completion of the review, the WIPP CH-TRU Payload Engineer shall send notification to the site indicating the status of the request. If the request is denied, the WIPP CH-TRU Payload Engineer shall indicate in the notification the reason why the request was not accepted and shall identify which elements of the submittal are incomplete or out of compliance.
5. If the request is approved, a site may begin using the new or revised content code once notification is received from the WIPP CH-TRU Payload Engineer. Sites may not use proposed content code additions or revisions to make shipments in the TRUPACT-II or HalfPACT prior to receipt of notification from the WIPP CH-TRU Payload Engineer.
6. The WIPP CH-TRU Payload Engineer shall record all approved content code additions or revisions in the CH-TRU CON. The current revision of the CH-TRU CON shall be available to sites.

¹ U.S. Department of Energy (DOE), [CH-TRU Payload Appendices](#), U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico.

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1.6 Compliance Program

1.6.1 Transportation Certification Official

The site Transportation Certification Official is responsible for administratively verifying the compliance of payload containers and the payload assembly with all transportation requirements. The site Transportation Certification Official shall approve by signature on the transportation certification documents every payload for transport.

1.6.2 DOE-CBFO

The DOE-CBFO is responsible for the performance of compliance verification audits, which are conducted prior to the first shipment and periodically thereafter to evaluate TRUPACT-II and HalfPACT payload compliance. Audit activities include document review and interview of site operators on a job-function basis relative to meeting the applicable criteria. Where specific technical ability is required (e.g., chemical compatibility, isotopic inventory, and assay), technical experts are included on the audit team. DOE-CBFO will grant or deny waste transportation authorization based on objective evidence of the audit and the recommendation of the audit team's report. Compliance verification audits are not required at sites that document compliance by preparing waste-specific data TRAMPACs that are reviewed and approved by the DOE-CBFO.

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1.7 Quality Assurance

The QA requirements applicable to the use of the TRUPACT-II and HalfPACT packagings are defined by Title 10, Code of Federal Regulations (CFR), Part 71 (10 CFR 71), Subpart H. The use and maintenance of the TRUPACT-II and the HalfPACT by the user are conducted under a QA program approved by the appropriate DOE field office. The compliance of a payload to be transported in the TRUPACT-II or the HalfPACT is determined by the user under a QA program approved by the DOE-CBFO (see [Chapter 7.0](#)).

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2.0 CONTAINER AND PHYSICAL PROPERTIES REQUIREMENTS

2.1 Authorized Payload Containers

2.1.1 Requirements

Payload containers transported within the TRUPACT-II or HalfPACT shall comply with the specifications in [Section 2.9](#).

The list of authorized payload containers and the associated number of containers in a TRUPACT-II and HalfPACT payload are summarized in [Table 2.1-1](#).

Table 2.1-1 – Authorized Payload Containers

Payload Container Type	Number in Payload ^①	
	TRUPACT-II	HalfPACT
55-Gallon Drum	14	7
Standard Pipe Overpack ^②	14	7
S100 Pipe Overpack	14	7
S200 Pipe Overpack	14	7
S300 Pipe Overpack	14	7
85-Gallon Drum ^③	8	4
100-Gallon Drum	6	3
Standard Waste Box (SWB)	2	1
Ten-Drum Overpack (TDOP)	1	NA ^④
SC-30G1 Shielded Container	NA ^④	3
SC-30G2 Shielded Container	NA ^④	2
SC-30G3 Shielded Container	NA ^④	1
SC-55G1 Shielded Container	NA ^④	2
SC-55G2 Shielded Container	NA ^④	1
Criticality Control Overpack (CCO)	14	7

Notes:

- ① Payloads shall be comprised of a single payload container type. Exception: As specified in [Sections 2.9.2, 3.1, and 6.2.4](#), if standard pipe overpacks are assembled in a mixed payload with 55-gallon drums, the standard pipe overpacks shall be considered 55-gallon drums and subject to the applicable limits for 55-gallon drums.
- ② Standard pipe overpacks must be assembled into seven-packs of 6-inch standard pipe overpacks or 12-inch standard pipe overpacks only. Exception: Does not apply when assembled in a mixed payload with 55-gallon drums.
- ③ The term “85-gallon drum” in this document includes 75- to 88-gallon drums.
- ④ NA = Not applicable.

2.1.2 Methods of Compliance and Verification

Compliance shall be by one, or a combination, of the following methods:

- Visual inspection to the specifications of [Section 2.9](#).
- Administrative and procurement controls demonstrating that payload containers have been procured to the specifications of [Section 2.9](#).

In addition to meeting the specifications of [Section 2.9](#) at the time of procurement, the integrity of the payload container shall be visually inspected prior to transport to ensure that payload container is in good and unimpaired condition (e.g., no significant rusting and is of sound structural integrity). For shielded container shipments, associated dunnage assemblies shall also be visually inspected prior to transport to ensure they are in a good and unimpaired condition and properly installed. Compliance shall be documented in accordance with site-specific procedures prior to shipment.

2.2 Dunnage

2.2.1 Requirements

A shipper shall use empty 55-, 85-, or 100-gallon drums, an SWB, or shielded containers as dunnage to complete a payload configuration if too few loaded payload containers are available that meet transportation requirements. The dunnage container(s) must meet the specifications of [Section 2.9](#) with the exception that dunnage containers shall have at least one open vent port (i.e., not filtered or plugged). Note that empty 55-gallon drum(s) can be used as dunnage container(s) to complete a payload of pipe overpacks or CCOs.

Dunnage containers shall be marked in accordance with [Section 2.4](#).

2.2.2 Methods of Compliance and Verification

Compliance shall be verified through visual inspection and documented prior to shipment.

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2.3 Container/Assembly Weight and Center of Gravity

2.3.1 Requirements

2.3.1.1 Requirements for Container/Assembly Weight

Each payload container shall comply with the maximum gross weight limits summarized in [Table 2.3-1](#).

Table 2.3-1 – Payload Container Maximum Gross Weight Limits

Payload Container	Maximum Gross Weight (pounds)
55-Gallon Drum	1,000
Standard Pipe Overpack (6-inch diameter) ^①	328
Standard Pipe Overpack (12-inch diameter) ^①	547
S100 Pipe Overpack ^①	550
S200 Pipe Overpack ^①	547
S300 Pipe Overpack ^①	547
85-Gallon Drum ^②	1,000
100-Gallon Drum	1,000
SWB	4,000
TDOP	6,700
SC-30G1 Shielded Container	2,260
SC-30G2 Shielded Container	3,160
SC-30G3 Shielded Container	6,300
SC-55G1 Shielded Container	3,410
SC-55G2 Shielded Container	6,500
CCO	350

Note:

- ① Compliance with the maximum contents and maximum pipe component weight limits specified in [Table 2.9-7](#), [Table 2.9-12](#), [Table 2.9-16](#), or [Table 2.9-20](#), as applicable, is also required.
- ② The term “85-gallon drum” in this document includes 75- to 88-gallon drums.

Each loaded TRUPACT-II and associated payload assembly shall comply with the maximum gross weight limits provided in [Table 2.3-2](#).

Table 2.3-2 – TRUPACT-II and Payload Assembly Maximum Gross Weight Limits

Assembly	Maximum Gross Weight (pounds)
Payload (Contents)	7,265
Loaded TRUPACT-II (Package)	19,250

Each loaded HalfPACT and associated payload assembly shall comply with the maximum gross weight limits provided in [Table 2.3-3](#).

Table 2.3-3 – HalfPACT and Payload Assembly Maximum Gross Weight Limits

Payload	Maximum Gross Weight (pounds)
Payload (Contents)	7,600
Loaded HalfPACT (Package)	18,100

Maximum gross weight limits for payload assemblies of either the TRUPACT-II or HalfPACT include the weight of the loaded payload container(s) and any additional payload assembly items (e.g., dunnage containers, pallets, spacers, guide tubes, slip sheets, reinforcing plates, banding material, slings, axial and radial dunnage, etc., as specified in [Section 2.9](#)). Actual payload assembly weights are limited by “as-built” TRUPACT-II and HalfPACT weights and U.S. Department of Transportation requirements for a loaded transport vehicle.

Payload containers, payload assemblies, and loaded TRUPACT-II or HalfPACT packages shall be acceptable for transport only if the weights plus associated errors are less than or equal to the maximum gross weights specified in [Table 2.3-1](#) and [Table 2.3-2](#) or [Table 2.3-3](#).

2.3.1.2 Requirements for Center of Gravity

For TRUPACT-II payloads, the total weight of payload container(s) in the top layer of the payload assembly shall be less than or equal to the total weight of payload container(s) in the bottom layer of payload assembly. In the case of drums overpacked in a TDOP, the total weight of the top layer of drums shall be less than or equal to the total weight of the bottom layer of drums.

No center of gravity requirements exist for HalfPACT payloads.

2.3.2 Methods of Compliance and Verification

2.3.2.1 Methods of Compliance and Verification for Container/Assembly Weight

Compliance shall be by measurement. Reference weights of well-defined components such as those making up the payload handling materials (e.g., dunnage containers, pallets, spacers, guide tubes, slip sheets, reinforcing plates, banding material, slings, axial and radial dunnage, etc.) may

be analytically established as a bounding value or statistically established by measurement plus associated error to the 95th Upper Tolerance Limit. Measured weights shall be determined using a calibrated scale or load cell. The scale or load cell calibrations shall be in accordance with the National Institute of Standards and Technology Handbook 44¹, or an equivalent standard. The error associated with measured weight is determined from the scale calibration tolerance. If multiple scales are used, a bounding value based on the highest scale calibration tolerance may be used to determine the measurement error. Reference weights have an associated error of zero due to the error being incorporated into the reference value.

One of two methods shall be used to calculate the total gross weight of a payload container, payload assembly, or a loaded package:

- If the total gross weight is obtained by summing the weights of individually established sub-components, it shall include the square root of the sum of the squares of the individual errors.
- If the total gross weight is obtained by weighing all components as an assembly, it shall include the error.

The weight and associated error of each payload container shall be recorded in the Payload Container Transportation Certification Document (PCTCD) or the Overpack Payload Container Transportation Certification Document (OPCTCD), as appropriate (see [Section 6.2.1](#)). The weight and the associated error of the loaded package shall be recorded on the Payload Assembly Transportation Certification Document (PATCD) (see [Section 6.2.2](#)).

2.3.2.2 Methods of Compliance and Verification for Center of Gravity

The weight and associated error of each payload container and of the total payload shall be calculated and reported on the PATCD (see [Section 6.2.2](#)), which requires the documentation of verification of compliance with the center of gravity requirements for TRUPACT-II payloads.

No center of gravity requirements exist for HalfPACT payloads.

¹ National Institute of Standards and Technology, *Specifications, Tolerances, and Other Technical Requirements for Weighing and Measuring Devices*, NIST Handbook 44, National Institute of Standards and Technology, Office of Weights and Measures, Washington, D.C.

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2.4 Container Marking

2.4.1 Requirements

Each payload container shall be labeled with a unique container identification number.

If an empty 55-, 85-, or 100-gallon drum or shielded container is used as dunnage to complete a payload configuration (see [Section 2.2](#)), the dunnage container shall be labeled with the following information:

- Unique container identification number
- “EMPTY” or “DUNNAGE.”

If a seven-pack of only dunnage 55-gallon drums, a four-pack of only dunnage 85-gallon drums, a three-pack of only dunnage 100-gallon drums, or a dunnage SWB is used in the TRUPACT-II, the container(s) shall be labeled only “EMPTY” or “DUNNAGE,” and the unique container identification number label is not required for these containers.

2.4.2 Methods of Compliance and Verification

Compliance shall be through visual inspection of each payload container and dunnage and the unique container identification number/label shall be recorded on the PCTCD or the OPCTCD (see [Section 6.2.1](#)).

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2.5 Filter Vents

2.5.1 Requirements

Each payload container to be transported in the TRUPACT-II or HalfPACT, including all payload containers that are overpacked in other payload containers, shall have one or more filter vents or equivalent venting mechanisms. Filter vents and equivalent venting mechanisms shall meet the specifications of [Table 2.5-1](#).

The use of a heat-sealed filtered bag as the innermost layer of confinement to package CH-TRU waste is limited to Waste Material Types I.3, II.1, III.1, and III.3 provided that there is no potential for contact of the filters with water. Waste Material Types II.3 and III.2, which by definition include a metal can as the innermost layer of confinement, may use heat-sealed filtered bags as confinement layers outside of the innermost metal can. Because Waste Material Type II.2 (inorganic solids packaged in metal cans) does not generate flammable gas, heat-sealed filtered or unfiltered bags may be used as confinement layers outside of the innermost metal can. For other waste material types, heat-sealed filtered bags are not allowed as the innermost layer of confinement.

The use of filtered bags in waste packaging configurations must be specified in approved content codes. [Appendix 3.11](#) of the [CH-TRU Payload Appendices](#)¹ describes the use of filtered bags as confinement layers.

Filter vents for the outer payload container and any overpacked payload container shall be legibly marked to ensure both (1) identification of the supplier and (2) date of manufacture, lot number, or unique serial number.

The filter vent shall have an operating temperature range from –40 degrees Celsius (°C) to +80°C (-40 degrees Fahrenheit [°F] to +176°F) and a minimum melting temperature of 100°C (+212°F). The filter vent threads shall be compatible with the bung in the container or shall be self-tapping.

¹ U.S. Department of Energy (DOE), [CH-TRU Payload Appendices](#), U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico.

Table 2.5-1 – Minimum Filter Vent Specifications

Container/Confinement Layer Type	Minimum Filter Vent Specification		
	Flow Rate (ml/min of air, STP ^① , at 1 inch of water) ^②	Efficiency (percent)	Total Hydrogen Diffusivity (mol/s/mol fraction ^③ at 25°C) ^④
Payload Containers			
55-Gallon Drum	25	NA ^⑤	1.90E-6
85-Gallon Drum	25	NA ^⑤	3.70E-6
100-Gallon Drum	25	NA ^⑤	3.70E-6
SWB	25	NA ^⑤	7.40E-6
TDOP	25	NA ^⑤	3.33E-5 ^⑥
Shielded Container	25	>99.9	3.70E-6
Components of Fixed Payload Container Designs			
Pipe Component	25	>99.9	1.90E-6
Criticality Control Container	25	>99.9	1.90E-6
Inner Confinement Layers			
Filtered Bag	25	NA ^⑤	1.075E-5
Filtered Confinement Layer (e.g., Metal Can, 30-gallon drum, etc.)	25	NA ^⑤	1.90E-6 ^⑦

Notes:

- ① ml/min of air, STP = Milliliter(s) per minute of air, standard temperature and pressure
- ② Filters tested at a different pressure gradient shall have a proportional flow rate (e.g., 25 ml/min at 1 inch of water = 693 ml/min at 1 pound per square inch).
- ③ mol/s/mol fraction = Moles per second per mole fraction
- ④ Total hydrogen diffusivity may be achieved through the use of multiple filter vents or equivalent venting mechanisms.
- ⑤ NA = Not applicable
- ⑥ If used to overpack other containers, the TDOP is required to meet the minimum total filter vent diffusivity requirements detailed in [Appendix 6.10](#) of the [CH-TRU Payload Appendices](#).
- ⑦ For filtered confinement layers that do not meet the minimum total filter vent diffusivity, a minimum hydrogen release rate (mol/s) resistance must be determined by testing or analysis as described in [Appendix 3.8](#) of the [CH-TRU Payload Appendices](#).

2.5.2 Methods of Compliance and Verification

Compliance shall be by one, or a combination, of the following methods:

- Administrative and procurement controls demonstrating that filter vents have been procured to the specifications of [Section 2.5.1](#).
- Visual inspection to the specifications of [Section 2.5.1](#).
- Sampling by measurement of filter characteristics to the specifications of [Section 2.5.1](#).

If sampling by measurement is selected as the compliance method, the test methods used to determine the compliance of filter vents or equivalent venting mechanisms with the minimum performance-based requirements specified in [Table 2.5-1](#) and this section shall be directed by procedures under a QA program. For equivalent venting mechanisms, compliance with the minimum total hydrogen diffusivity per container type specified in [Table 2.5-1](#) shall be demonstrated through documentation for the equivalent venting mechanism used.

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2.6 Residual Liquids

2.6.1 Requirements

Liquid waste is prohibited in payload containers, except for residual amounts in well-drained containers. The total volume of residual liquid in a payload container shall be less than 1 percent (volume) of the payload container.

2.6.2 Methods of Compliance and Verification

Compliance shall be by one, or a combination, of the following methods:

- Review of records and database information, which may include knowledge of process
- Radiography
- Visual examination
- Sampling program.

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2.7 Sharp or Heavy Objects

2.7.1 Requirements

Sharp or heavy objects in the waste shall be blocked, braced, or suitably packaged as necessary to provide puncture protection for the payload containers packaging these objects.

2.7.2 Methods of Compliance and Verification

Compliance shall be by one, or a combination, of the following methods:

- Review of records and database information, which may include knowledge of process
- Radiography
- Visual examination
- Sampling program.

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2.8 Sealed Containers

A sealed container is defined as a closed container that has no quantified hydrogen release.

2.8.1 Requirements

Sealed containers that are greater than 4 liters (nominal) are prohibited except for Waste Material Type II.2 packaged in a metal container; Waste Material Type II.2 in metal cans does not generate any flammable gas (see [Appendix 3.2](#) of the [CH-TRU Payload Appendices](#)¹).

A closed container that is greater than 4 liters (nominal) must be evaluated as a confinement layer and its hydrogen gas release must be established as described in [Appendix 3.8](#) of the [CH-TRU Payload Appendices](#). Closed containers that are less than or equal to 4 liters (nominal) shall be evaluated for the capability to hold pressure. Per [Section 4.2](#), pressurized containers are prohibited.

2.8.2 Methods of Compliance and Verification

Compliance shall be by one, or a combination, of the following methods:

- Review of records and database information, which may include knowledge of process
- Radiography
- Visual examination
- Sampling program.

¹ U.S. Department of Energy (DOE), [CH-TRU Payload Appendices](#), U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico.

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2.9 Payload Container/Assembly Configuration Specifications

2.9.1 55-Gallon Drum

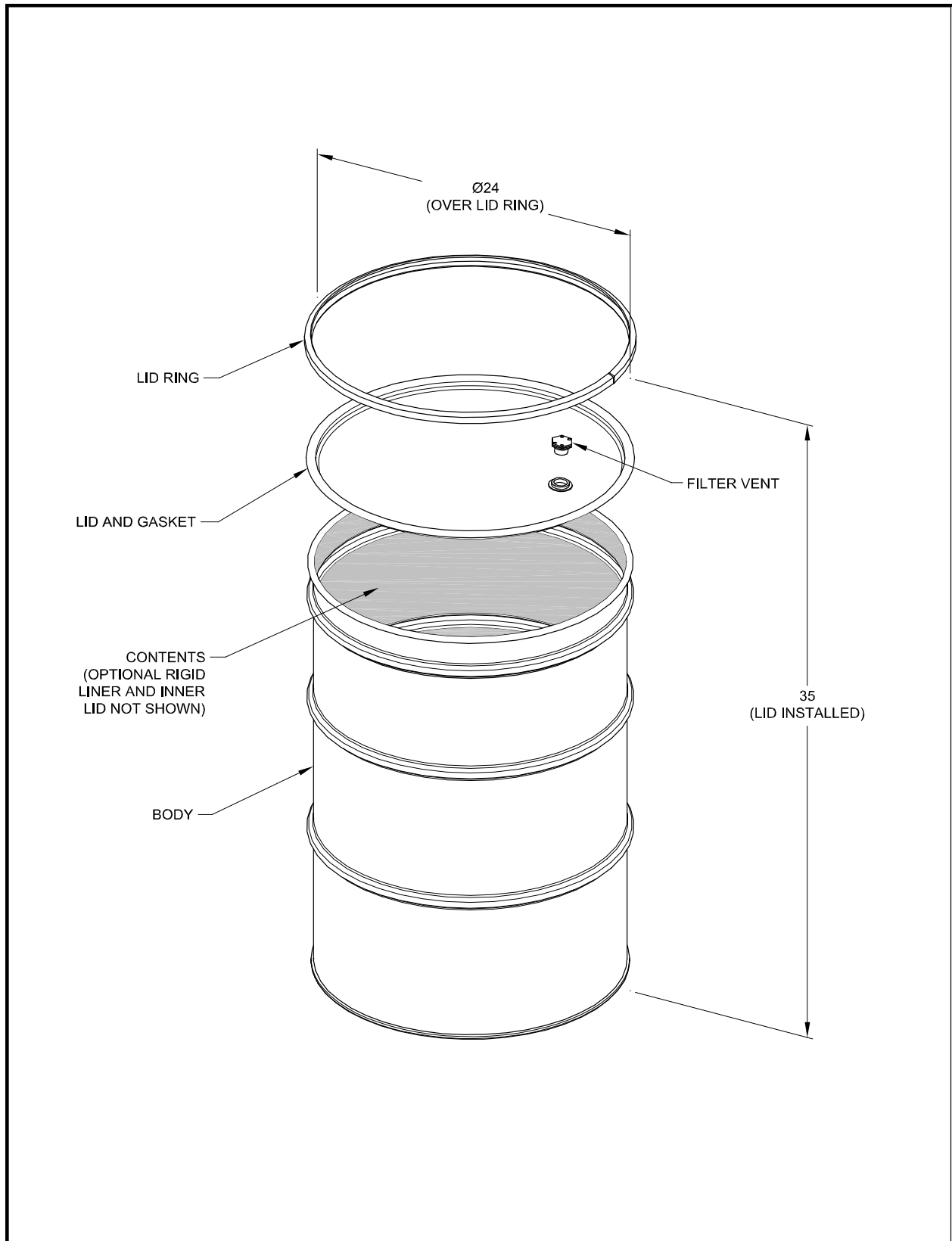
The 55-gallon drum is authorized for transport within the TRUPACT-II and the HalfPACT (Figure 2.9-1). The TRUPACT-II will accommodate 14 55-gallon drums (Figure 2.9-2), and the HalfPACT will accommodate seven 55-gallon drums (Figure 2.9-3). The approximate dimensions of the 55-gallon drum are given in Table 2.9-1.

Table 2.9-1 – 55-Gallon Drum Dimensions

Dimension	Approximate Measurement (inches)	
	Inside Dimension	Outside Dimension
Height	33¼	35
Diameter	22½	24

The 55-gallon-drum body, lid, and lid ring are constructed of steel. A gasket, of tubular or foam styrene-butadiene, neoprene rubber, or equivalent, is required for outer drum lid closure. The drum must be vented as specified in Section 2.5. An optional, rigid, polyethylene liner and lid may be used inside the drum. If a lid is used with the liner, the liner lid must contain a 0.3-inch (in.) minimum diameter hole, or a filter with a hydrogen release rate equivalent to or greater than a 0.3-in. minimum diameter hole. Otherwise, the liner must be treated as any other confinement layer with the associated resistance of the liner hole diameter or filter calculated in accordance with Appendix 2.2 of the CH-TRU Payload Appendices.¹ An optional steel inner lid with an optional gasket may be used inside the drum (i.e., a double-lid drum). A double-lid drum with a filtered inner lid will be considered the same as a drum with a filtered inner confinement layer. Table 2.9-2 presents the 55-gallon drum construction materials. Table 2.9-3 specifies the weights associated with the 55-gallon drum that are applicable to shipment within the TRUPACT-II and the HalfPACT.

¹ U.S. Department of Energy (DOE), CH-TRU Payload Appendices, U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico.

**Figure 2.9-1 – 55-Gallon Drum**

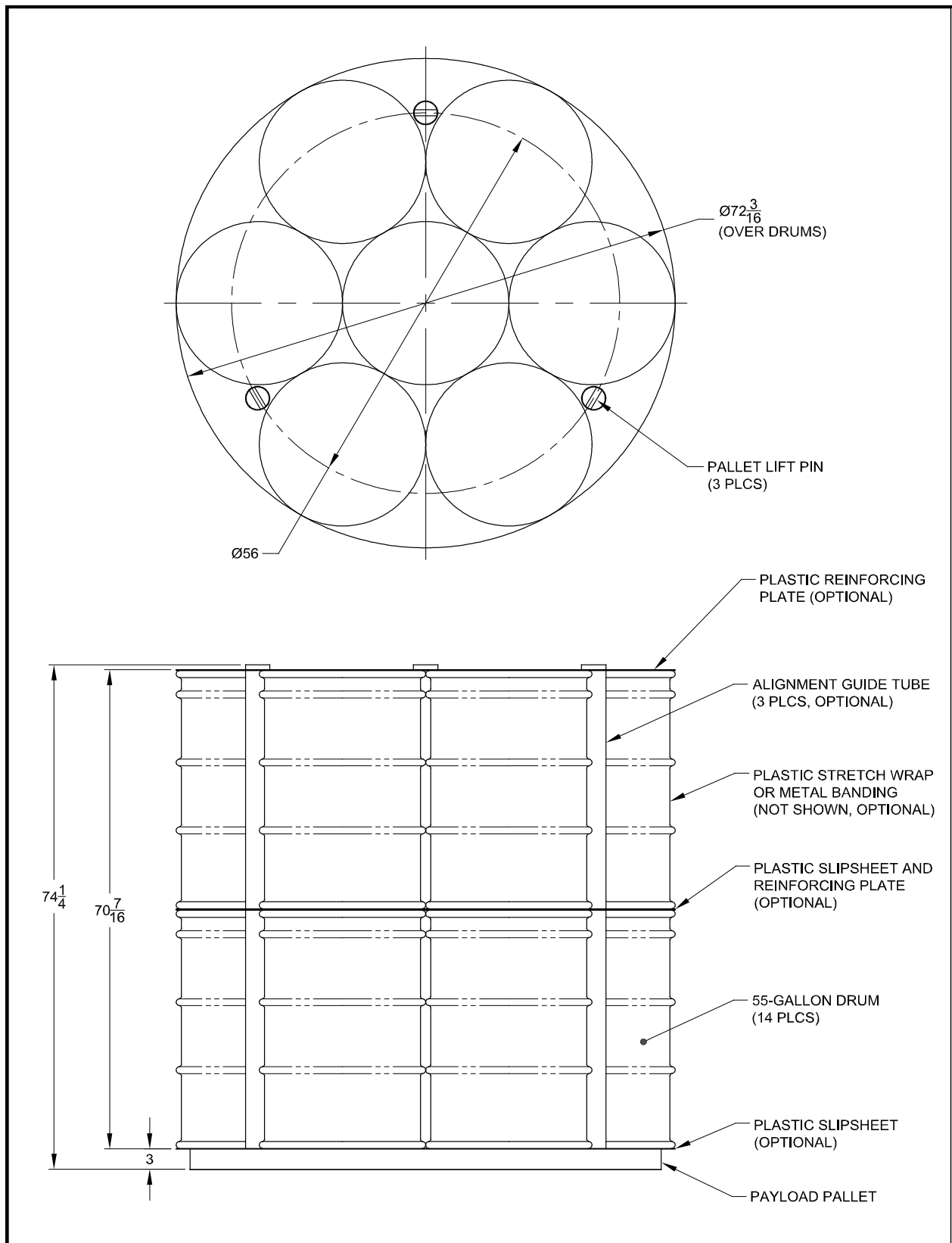


Figure 2.9-2 – TRUPACT-II: Fourteen 55-Gallon Drum Payload

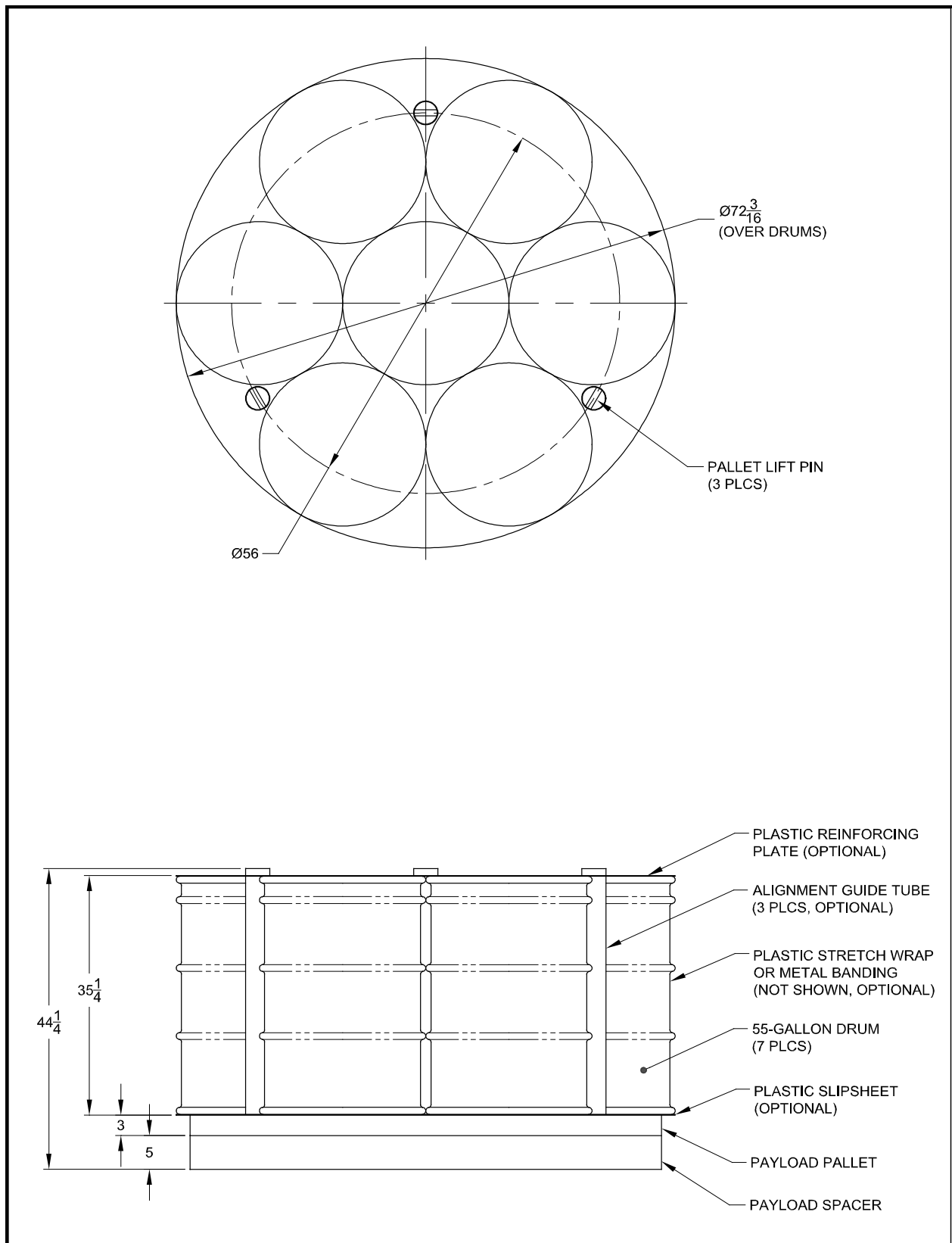
**Figure 2.9-3 – HalfPACT: Seven 55-Gallon Drum Payload**

Table 2.9-2 – 55-Gallon Drum Materials of Construction

55-Gallon Drum Component	Material
Body, lid, and lid ring	Steel
Rigid liner and liner lid (optional)	Polyethylene
Inner lid (optional)	Steel
Gasket	<u>Type I</u> - Tubular styrene-butadiene, or equivalent <u>Type II</u> - Foam styrene-butadiene, or equivalent <u>Type III</u> - Neoprene rubber, or equivalent

Table 2.9-3 – 55-Gallon Drum Weights

Component	Weight (pounds)	
	Approximate Empty	Maximum Gross
55-gallon drum	60	1,000

CH-TRU waste may be directly loaded into a 55-gallon drum. Empty 55-gallon drums used as dunnage to complete a payload must also meet the specifications presented in this section with the exception that dunnage containers shall have at least one open vent port (i.e., not filtered or plugged). [Table 2.9-4](#) identifies material content forms authorized for transport within a 55-gallon drum.

Table 2.9-4 – 55-Gallon Drum: Material Content Forms Authorized for Transport

Form Number	Description
1	Direct load: Solids, any particle size (e.g., fine powder or inorganic particulates)
2	Direct load: Solids, large particle size (e.g., sand, concrete, or debris)
3	Direct load: Solids, large objects (e.g., metal cans containing waste)

2.9.2 Standard Pipe Overpack

The standard pipe overpack consists of a pipe component positioned by fiberboard/plywood dunnage within a 55-gallon drum with a high-density polyethylene rigid liner and lid ([Figure 2.9-4](#)). [Appendix 1.3.1](#) of the [TRUPACT-II Shipping Package Safety Analysis Report \(SAR\)](#)², [Appendix 1.3.1](#) of the [HalfPACT SAR](#)³, and [Appendix 4.1](#) of the [CH-TRU Payload Appendices](#)¹ provide detailed descriptions of this payload container. The TRUPACT-II will accommodate 14 standard pipe overpacks ([Figure 2.9-2](#)), and the HalfPACT will accommodate 7 standard pipe overpacks ([Figure 2.9-3](#)). Standard pipe overpacks may be assembled in a TRUPACT-II or HalfPACT payload with 55-gallon drums. Standard pipe overpacks in this payload assembly are considered 55-gallon drums and are subject to the specifications of [Section 2.9.1](#). The pipe component is available in two sizes as specified in [Table 2.9-5](#).

Table 2.9-5 – Pipe Component Dimensions

Pipe Component Size	Dimension	Maximum Measurement (inches)	Bolt Size (inches)	Number of Bolts	Minimum Bolted Flange Diameter (inches)
6-inch	Diameter	6.7 Outside Diameter	3/4	8	11
	Height	27.5 Overall			
12-inch	Diameter	12.8 Outside Diameter	7/8	12	16.3
	Height	27.5 Overall			

The pipe component body, lid, and bolt flange are constructed of stainless steel. A butyl rubber or ethylene propylene O-ring is required for pipe component closure. One or more bolts may have tamper-resistant heads and/or may have a thread-locking compound applied to the threads. The pipe component and 55-gallon drum must be vented as specified in [Section 2.5](#). [Table 2.9-6](#) presents the pipe component construction materials.

² U.S. Department of Energy (DOE), [TRUPACT-II Shipping Package Safety Analysis Report](#), USNRC Certificate of Compliance 71-9218, U.S. Department of Energy, Carlsbad Area Office, Carlsbad, New Mexico.

³ U.S. Department of Energy (DOE), [HalfPACT Shipping Package Safety Analysis Report](#), USNRC Certificate of Compliance 71-9279, U.S. Department of Energy, Carlsbad Area Office, Carlsbad, New Mexico.

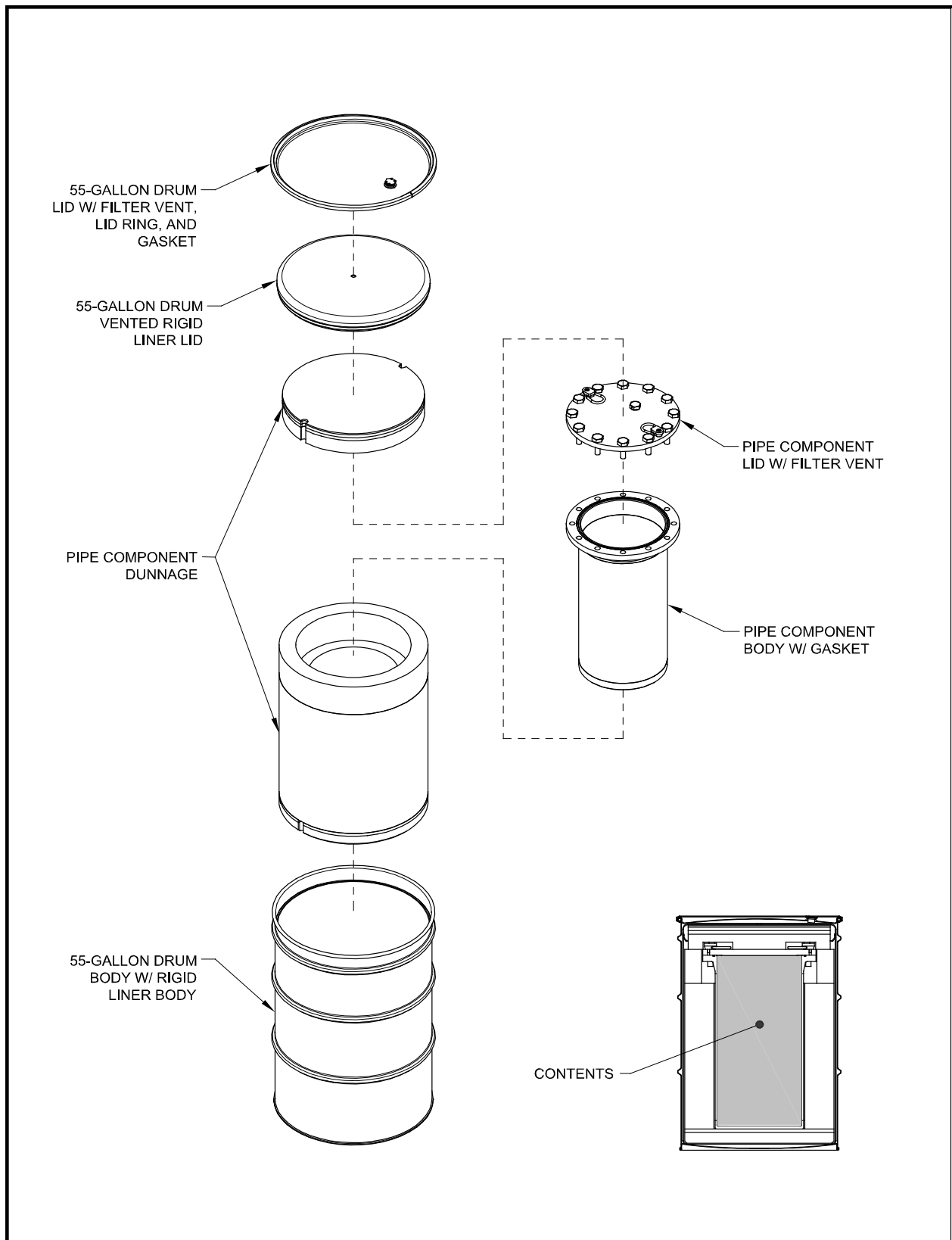
**Figure 2.9-4 – Standard Pipe Overpack**

Table 2.9-6 – Pipe Component Materials of Construction

Component	Material
Body, lid, and bolt flange	Stainless steel
Gasket	Butyl rubber or ethylene propylene O-ring

Table 2.9-7 specifies the weights associated with the pipe component that are applicable to shipment within the TRUPACT-II and the HalfPACT.

Table 2.9-7 – Pipe Component Weights

Pipe Component Size	Pipe Component Weight (pounds)	
	Maximum Contents	Maximum Gross
6-in.-diameter pipe component only	66	153
12-in.-diameter pipe component only	225	407

The maximum allowable weights of the pipe overpacks are bounded by the as-tested weights. Compliance with the maximum gross weight limits for both the pipe component and the pipe overpack is required. Table 2.9-8 specifies the weights associated with the standard pipe overpack that are applicable to shipment within the TRUPACT-II and the HalfPACT.

Table 2.9-8 – Standard Pipe Overpack Weights

Size of Pipe Component Assembled	Maximum Gross Weight (pounds)
6-in.-diameter pipe component assembled in a 55-gallon drum	328
12-in.-diameter pipe component assembled in a 55-gallon drum	547

Table 2.9-9 identifies material content forms authorized for transport within a pipe overpack.

Table 2.9-9 – Standard Pipe Overpack: Material Content Forms
Authorized for Transport

Form Number	Description
1	Direct load: Solids, any particle size (e.g., fine powder or inorganic particulates)
2	Direct load: Solids, large particle size (e.g., sand, concrete, or debris)
3	Direct load: Solids, large objects (e.g., metal cans containing waste)

2.9.3 S100 Pipe Overpack

The S100 pipe overpack is a shielded version of the standard pipe overpack described in [Section 2.9.2](#). It consists of a 6-in. diameter pipe component positioned within a 55-gallon drum by means of fiberboard/plywood dunnage and neutron shielding materials, as shown schematically in [Figure 2.9-5](#). The 6-in. diameter pipe component used in the S100 pipe overpack is identical to the 6-in. diameter pipe component for the standard pipe overpack described in [Section 2.9.2](#). The 12-in. diameter pipe component is not used in the S100 pipe overpack. [Appendix 1.3.1](#) of the [TRUPACT-II SAR](#)², [Appendix 1.3.1](#) of the [HalfPACT SAR](#)³, and [Appendix 4.2](#) of the [CH-TRU Payload Appendices](#)¹ provide detailed descriptions of this payload container. The TRUPACT-II will accommodate 14 S100 pipe overpacks ([Figure 2.9-2](#)), and the HalfPACT will accommodate 7 S100 pipe overpacks ([Figure 2.9-3](#)). The 6-in. diameter pipe component dimensions are specified in [Table 2.9-5](#). The pipe component materials of construction are specified in [Table 2.9-6](#). The pipe component and 55-gallon drum must be vented as specified in [Section 2.5](#).

The neutron shield around the side of the pipe component may be in the form of a casting (such as a commercial neutron shielding casting compound), a solid monolith (such as a molded or machined unit of solid plastic), or fabricated component (such as a tightly wound roll of plastic film or other built-up fabrication). The end plugs and shield sleeve within the pipe component are made of a rigid plastic. Nominal dimensions are specified in [Table 2.9-10](#).

Table 2.9-10 – Shield Nominal Dimensions

Item	Inside Diameter (inches)	Outside Diameter (inches)	Length (inches)
Side neutron shielding ^①	7.0	21.5	26.7
Shield sleeve	3.25	6.0	12.0
End shield plugs	Not applicable	6.0	6.5

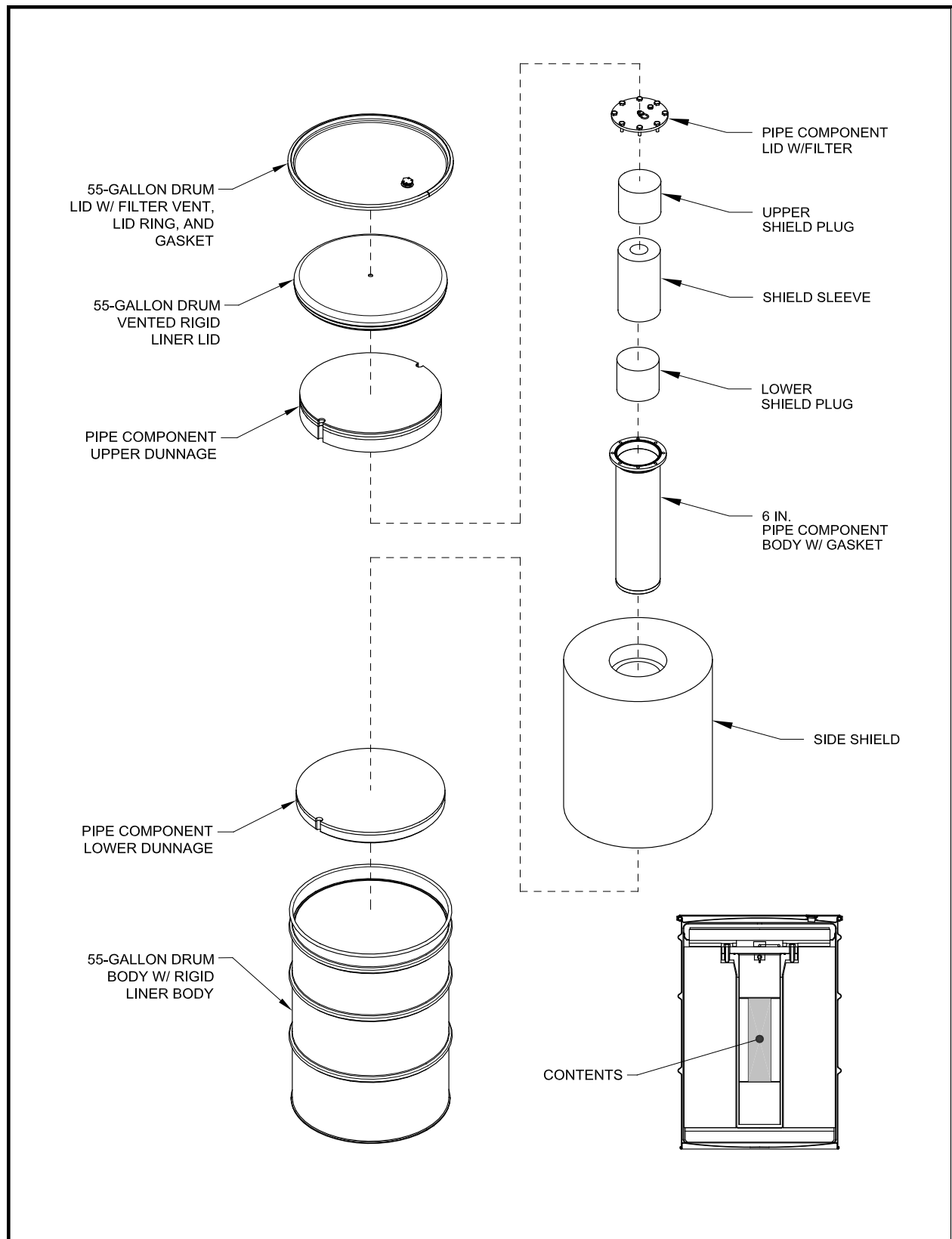
Note:

① Dimensions do not include the outer high-density polyethylene rigid drum liner.

The shielding materials of construction are specified in [Table 2.9-11](#).

Table 2.9-11 – Shield Materials of Construction

Item	Material
Side neutron shielding	High-density polyethylene or equivalent neutron shielding material (see Appendix 4.2 of the CH-TRU Payload Appendices ¹)
End shield plugs and shield sleeve	High-density polyethylene

**Figure 2.9-5 – S100 Pipe Overpack**

The maximum allowable contents weight of the pipe component is 66 pounds (lbs.) and the maximum allowable gross weight of the loaded pipe component is 153 lbs., which is consistent with the 6-in. diameter pipe component weights specified in [Table 2.9-7](#). The maximum allowable gross weight of the S100 pipe overpack is 550 lbs., which accommodates the weight of the side neutron shielding, as described in [Appendix 4.2](#) of the [CH-TRU Payload Appendices](#).¹ [Table 2.9-12](#) summarizes the nominal individual and maximum total weights associated with the shielded 6-in. diameter pipe component that are applicable to shipment within the TRUPACT-II and the HalfPACT.

Table 2.9-12 – 6-inch Diameter Pipe Component Weights

Item	Type	Weight (pounds)
End shield plugs and shield sleeve	Nominal	22
Payload	Nominal	44
Total (Contents)	Maximum	66
Total (Pipe Component)	Maximum	153

The authorized material content form for the S100 pipe overpack is specified in [Table 2.9-13](#).

Table 2.9-13 – S100 Pipe Overpack: Material Content Form Authorized for Transport

Form Number	Description
1	Sealed neutron sources

2.9.4 S200 Pipe Overpack

The S200 pipe overpack is a shielded version of the standard pipe overpack described in [Section 2.9.2](#). It consists of a gamma-shield insert located by rigid polyurethane foam dunnage inside a 12-in. diameter pipe component positioned within a 55-gallon drum by means of fiberboard/plywood dunnage. A schematic of the S200 pipe overpack is shown in [Figure 2.9-6](#), with detailed descriptions provided in [Appendix 1.3.1](#) of the [TRUPACT-II SAR](#)², [Appendix 1.3.1](#) of the [HalfPACT SAR](#)³, and [Appendix 4.3](#) of the [CH-TRU Payload Appendices](#).¹ The 12-in. diameter pipe component used in the S200 pipe overpack is identical to the 12-in. diameter pipe component described for the standard pipe overpack in [Section 2.9.2](#). The 6-in. diameter pipe component is not used in the S200 pipe overpack. The TRUPACT-II will accommodate 14 S200 pipe overpacks ([Figure 2.9-2](#)), and the HalfPACT will accommodate 7 S200 pipe overpacks ([Figure 2.9-3](#)). The 12-in. diameter pipe component dimensions and materials of construction are specified in [Table 2.9-5](#) and [Table 2.9-6](#), respectively. The pipe component and 55-gallon drum must be vented as specified in [Section 2.5](#). The gamma-shield insert is a lead two-component assembly consisting of a cylindrical body with an integral bottom cap and a detachable lid. The shield insert is available in two sizes as specified in [Table 2.9-14](#). A mixture of S200-A and S200-B pipe overpacks may be assembled into a single payload.

Table 2.9-14 – Shield Insert Nominal Dimensions

Size	Thickness (inches)	Inside Diameter (inches)	Inside Height (inches)	Outside Diameter (inches)	Outside Height (inches)
S200-A	1.000	8.125	8.125	10.125	10.625
S200-B	0.600	8.125	16.125	9.325	17.825

The shield insert body, lid, and dunnage materials of construction are specified in [Table 2.9-15](#).

Table 2.9-15 – Shield Insert Materials of Construction

Item	Material
Body, lid	Lead
Dunnage	Rigid polyurethane foam

The maximum allowable weight of 12-in. diameter pipe component contents (shield insert assembly plus payload) is 225 lbs., and the maximum gross weight of the loaded 12-in. diameter pipe component is 407 lbs., which are consistent with the 12-in. diameter pipe component weights specified in [Table 2.9-7](#). The maximum allowable gross weight of the loaded S200 pipe overpack is 547 lbs., which is consistent with the weights specified in [Table 2.9-8](#). [Table 2.9-16](#) summarizes the nominal individual and maximum total weights associated with the shield insert assembly components and payload that are applicable to shipment within the TRUPACT-II and the HalfPACT.

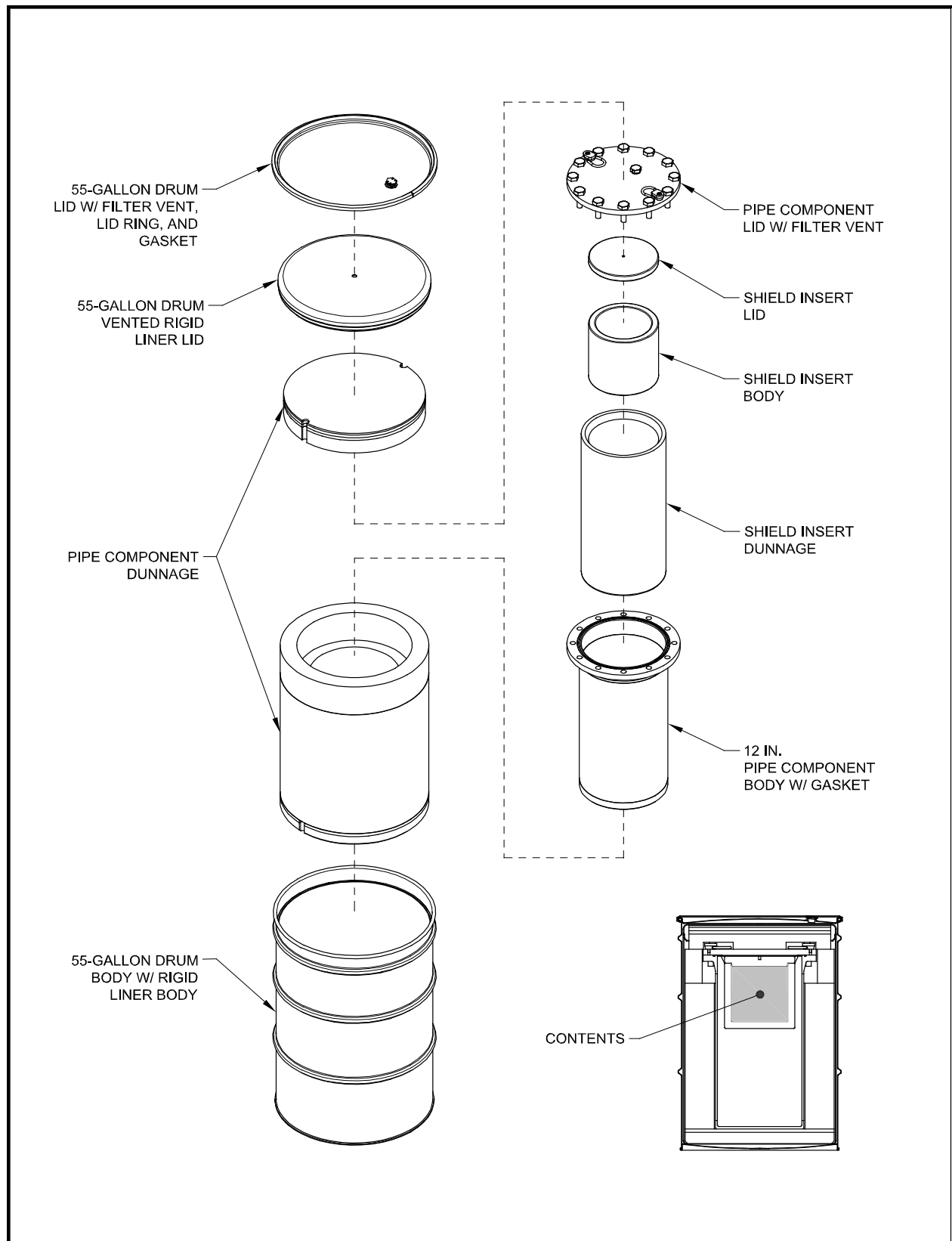
**Figure 2.9-6 – S200 Pipe Overpack**

Table 2.9-16 – 12-inch Pipe Component Weights

Item	Type	Weight	
		S200-A (pounds)	S200-B (pounds)
Shield Insert Body	Nominal	134	129
Shield Insert Lid	Nominal	43	27
Shield Insert Dunnage	Nominal	18	15
Payload	Nominal	25	50
Total (Contents)	Maximum	225	225
Total (Pipe Component)	Maximum	407	407

Table 2.9-17 identifies material content forms authorized for transport within an S200 pipe overpack.

Table 2.9-17 – S200 Pipe Overpack: Material Content Forms Authorized for Transport

Form Number	Description
1	Direct load: Solids, any particle size (e.g., fine powder or inorganic particulates)
2	Direct load: Solids, large particle size (e.g., sand, concrete, or debris)
3	Direct load: Solids, large objects (e.g., metal cans containing waste)

2.9.5 S300 Pipe Overpack

The S300 pipe overpack is a shielded version of the standard pipe overpack described in [Section 2.9.2](#). It consists of a neutron-shield insert located inside a 12-in. diameter pipe component positioned within a 55-gallon drum by means of fiberboard/plywood dunnage. A schematic of the S300 pipe overpack is shown in [Figure 2.9-7](#), with detailed descriptions provided in [Appendix 1.3.1](#) of the [TRUPACT-II SAR](#)², [Appendix 1.3.1](#) of the [HalfPACT SAR](#)³, and [Appendix 4.4](#) of the [CH-TRU Payload Appendices](#).¹ The 12-in. diameter pipe component used in the S300 pipe overpack is identical to the 12-in. diameter pipe component described for the standard pipe overpack in [Section 2.9.2](#). The 6-in. diameter pipe component is not used in the S300 pipe overpack. The TRUPACT-II will accommodate 14 S300 pipe overpacks ([Figure 2.9-2](#)), and the HalfPACT will accommodate 7 S300 pipe overpacks ([Figure 2.9-3](#)). The 12-in. diameter pipe component dimensions and materials of construction are specified in [Table 2.9-5](#) and [Table 2.9-6](#), respectively. The pipe component and 55-gallon drum must be vented as specified in [Section 2.5](#). The neutron shield insert is a two-part assembly consisting of a cylindrical body and stepped lid made from solid, high-density polyethylene with dimensions given in [Table 2.9-18](#).

Table 2.9-18 – Shield Insert Nominal Dimensions

Inside Diameter (inches)	Inside Height (inches)	Outside Diameter (inches)	Outside Height (inches)
3.5	17.0	11.75	24.75

The shield insert body and lid materials of construction are specified in [Table 2.9-19](#).

Table 2.9-19 – Shield Insert Materials of Construction

Item	Material
Body, lid	High-Density Polyethylene

The maximum allowable weight of 12-in. diameter pipe component contents (shield insert plus payload) is 225 lbs., and the maximum gross weight of the loaded 12-in. diameter pipe component is 407 lbs., which are consistent with the 12-in. diameter pipe component weights specified in [Table 2.9-7](#). The maximum allowable gross weight of the loaded S300 pipe overpack is 547 lbs., which is consistent with the data given in [Table 2.9-8](#). [Table 2.9-20](#) summarizes the nominal individual and maximum total weights associated with the shield insert and payload that are applicable to shipment within the TRUPACT-II and the HalfPACT.

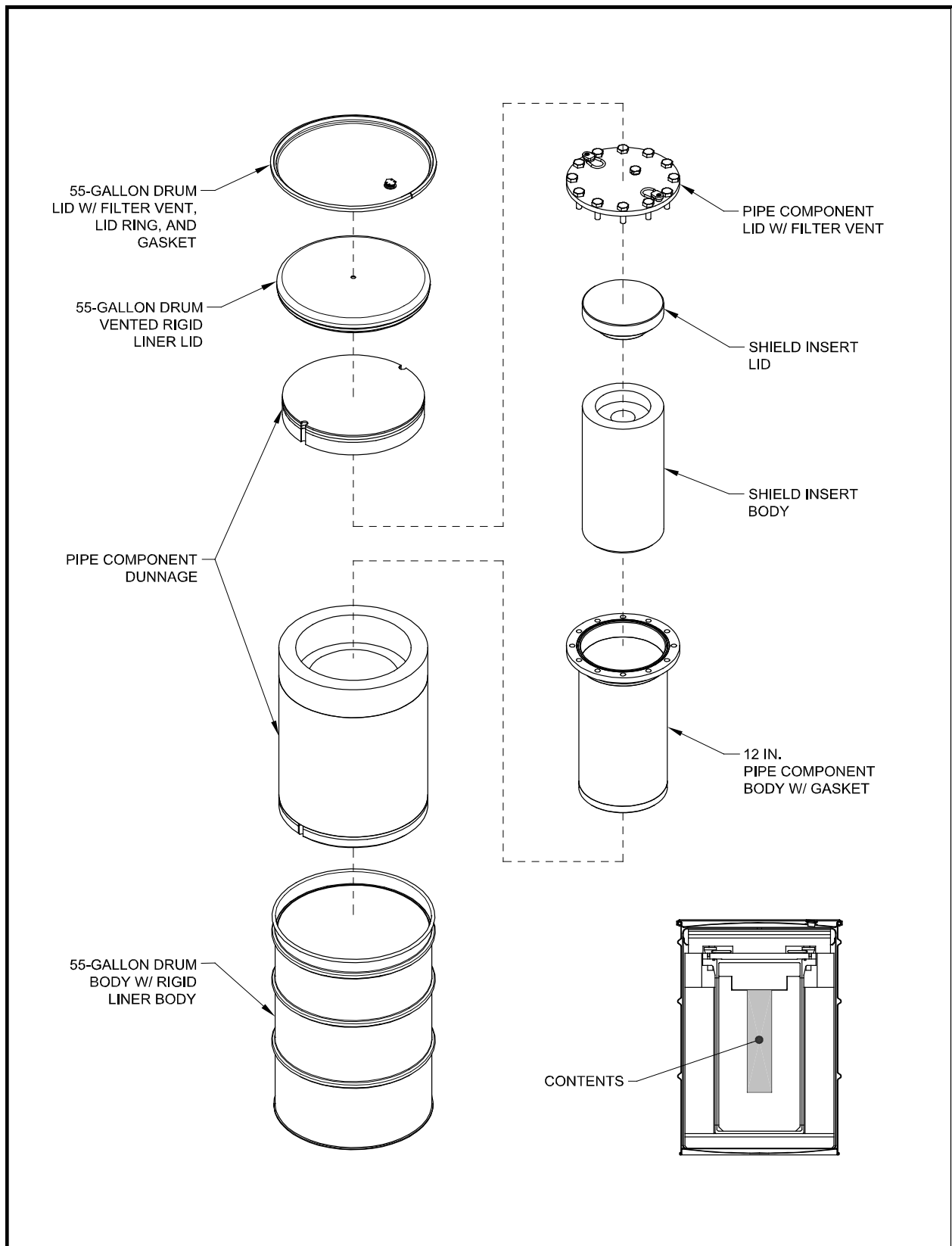
**Figure 2.9-7 – S300 Pipe Overpack**

Table 2.9-20 – 12-inch Pipe Component Weights

Item	Type	Weight (pounds)
Shield Insert Body and Lid	Nominal	87
Payload	Nominal	138
Total (Contents)	Maximum	225
Total (Pipe Component)	Maximum	407

Table 2.9-21 identifies material content forms authorized for transport within an S300 pipe overpack.

Table 2.9-21 – S300 Pipe Overpack: Material Content Form Authorized for Transport

Form Number	Description
1	Sealed neutron sources

2.9.6 85-Gallon Drum

The 85-gallon drum is authorized for transport within the TRUPACT-II and the HalfPACT. The term “85-gallon drum” refers to drums with a range of dimensions yielding 75 to 88 gallons. The short 85-gallon drum is authorized for transport within the TRUPACT-II, and the short and tall 85-gallon drums are authorized for transport within the HalfPACT (Figure 2.9-8 and Figure 2.9-9). The TRUPACT-II will accommodate eight short 85-gallon drums (Figure 2.9-10), and the HalfPACT will accommodate four short 85-gallon drums (Figure 2.9-11) or four tall 85-gallon drums (Figure 2.9-12). The approximate dimensions of the 85-gallon drum are given in Table 2.9-22.

Table 2.9-22 – 85-Gallon Drum Dimensions

Dimension	Approximate Measurement (inches)			
	Short Drum		Tall Drum	
	Inside Dimension	Outside Dimension	Inside Dimension	Outside Dimension
Height	33¼	35	38¼	40¼
Diameter	27⅛	29¾	26	28⅝

The 85-gallon drum body, lid, and lid ring are constructed of steel. A gasket, of tubular or foam styrene-butadiene, neoprene rubber, or equivalent, is required for outer drum lid closure. The drum must be vented as specified in Section 2.5. An optional, rigid, polyethylene liner and lid may be used inside the drum. If a lid is used with the liner, the liner lid must contain a 0.3-in. minimum diameter hole, or a filter with a hydrogen release rate equivalent to or greater than a 0.3-in. minimum diameter hole. Otherwise, the liner must be treated as any other confinement layer with the associated resistance of the liner hole diameter or filter calculated in accordance with Appendix 2.2 of the CH-TRU Payload Appendices.¹ An optional steel inner lid with an optional gasket may be used inside the drum (i.e., a double-lid drum). A double-lid drum with a filtered inner lid will be considered the same as a drum with a filtered inner confinement layer. Table 2.9-23 presents the 85-gallon drum construction materials. Table 2.9-24 specifies the weights associated with the 85-gallon drum that are applicable to shipment within the TRUPACT-II and the HalfPACT.

CH-TRU waste may be directly loaded into an 85-gallon drum or may be loaded into a 55-gallon drum, which is then overpacked in an 85-gallon drum (referred to as an 85-gallon drum overpack). If an empty 85-gallon drum is used as dunnage to complete a payload of four or eight 85-gallon drums, it must also meet the specifications presented in this section with the exception that dunnage containers shall have at least one open vent port (i.e., not filtered or plugged). Table 2.9-25 identifies material content forms authorized for transport within an 85-gallon drum.

Table 2.9-23 – 85-Gallon Drum Materials of Construction

85-Gallon Drum Component	Material
Body, lid, and lid ring	Steel
Rigid liner and liner lid (optional)	Polyethylene
Inner lid (optional)	Steel
Gasket	<u>Type I</u> - Tubular styrene-butadiene, or equivalent <u>Type II</u> - Foam styrene-butadiene, or equivalent <u>Type III</u> - Neoprene rubber, or equivalent

Table 2.9-24 – 85-Gallon Drum Weights

Component	Weight (pounds)	
	Approximate Empty	Maximum Gross
85-gallon drum	81	1,000

Table 2.9-25 – 85-Gallon Drum: Material Content Forms Authorized for Transport

Form Number	Description
1	Direct load: Solids, any particle size (e.g., fine powder or inorganic particulates)
2	Direct load: Solids, large particle size (e.g., sand, concrete, or debris)
3	Direct load: Solids, large objects (e.g., metal cans containing waste)

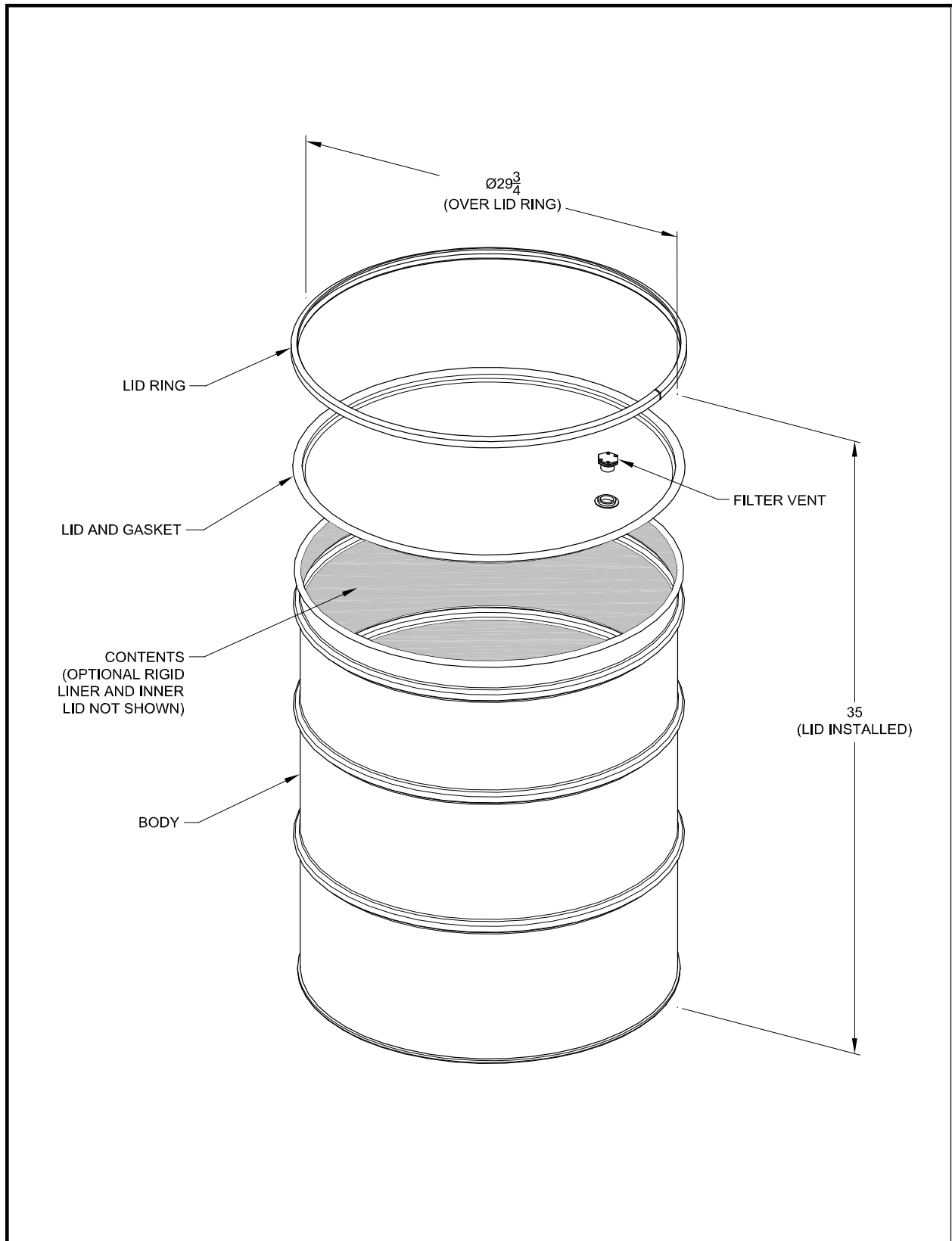
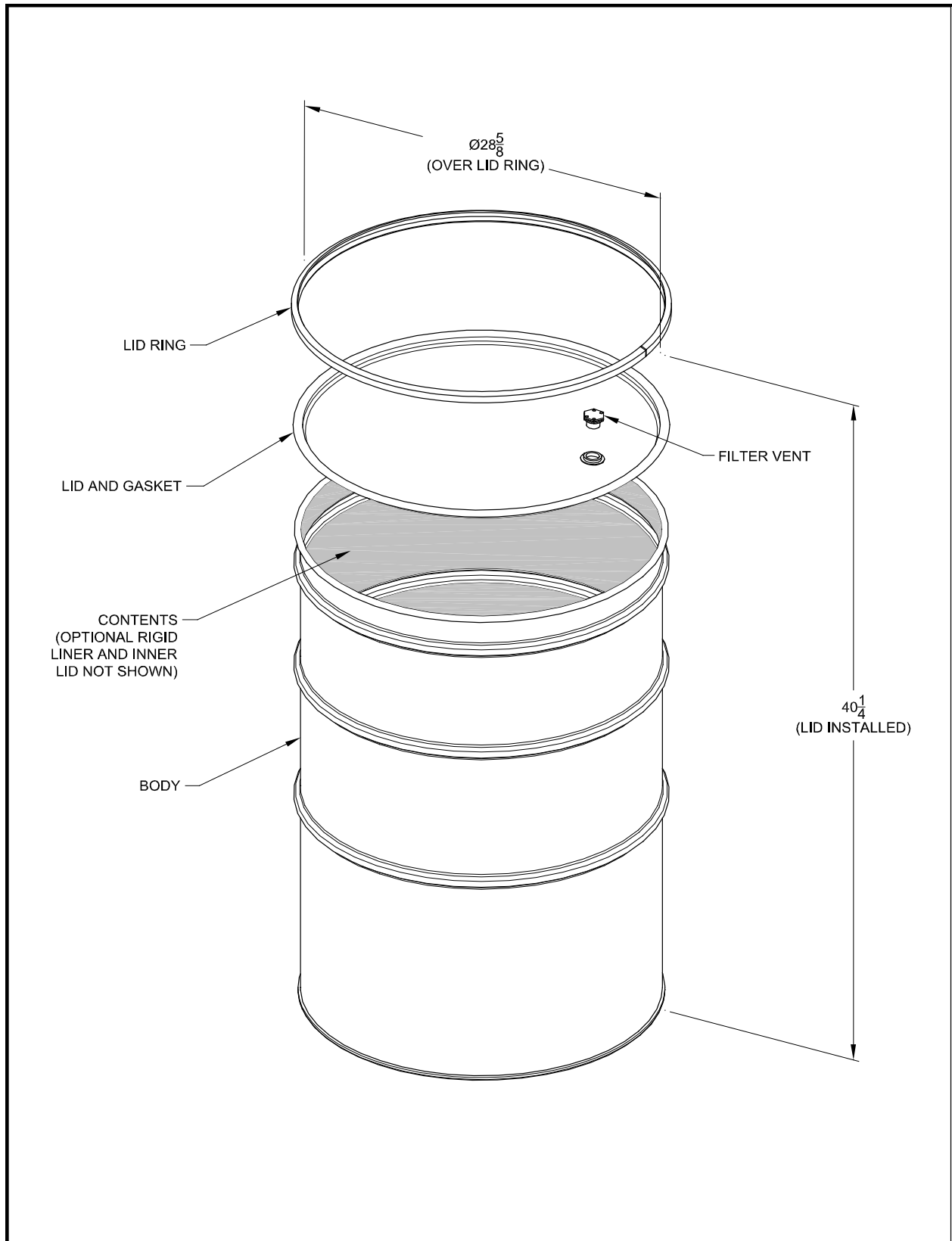


Figure 2.9-8 – 85-Gallon Drum (Short)

**Figure 2.9-9 – 85-Gallon Drum (Tall)**

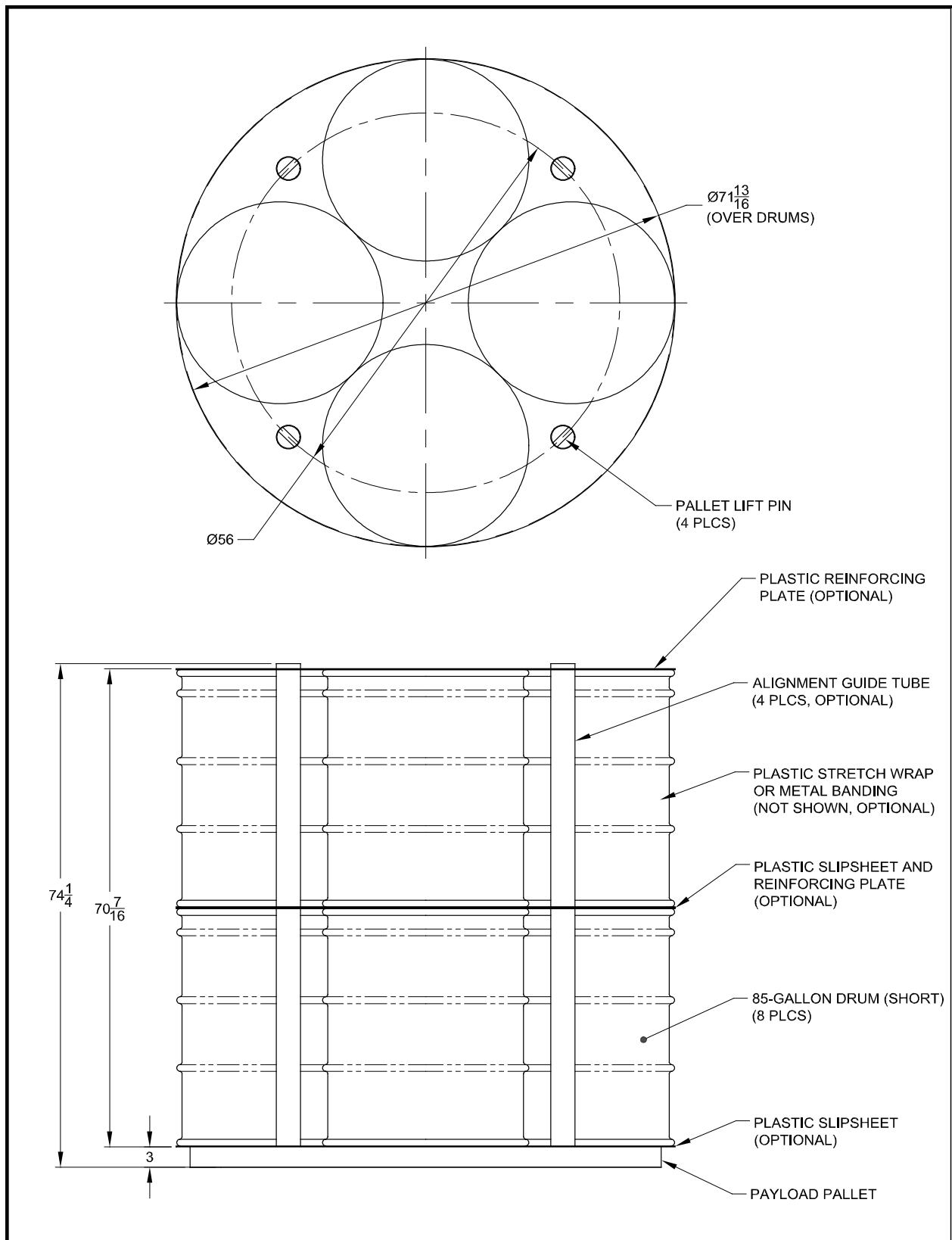


Figure 2.9-10 – TRUPACT-II: Eight 85-Gallon Drum (Short) Payload

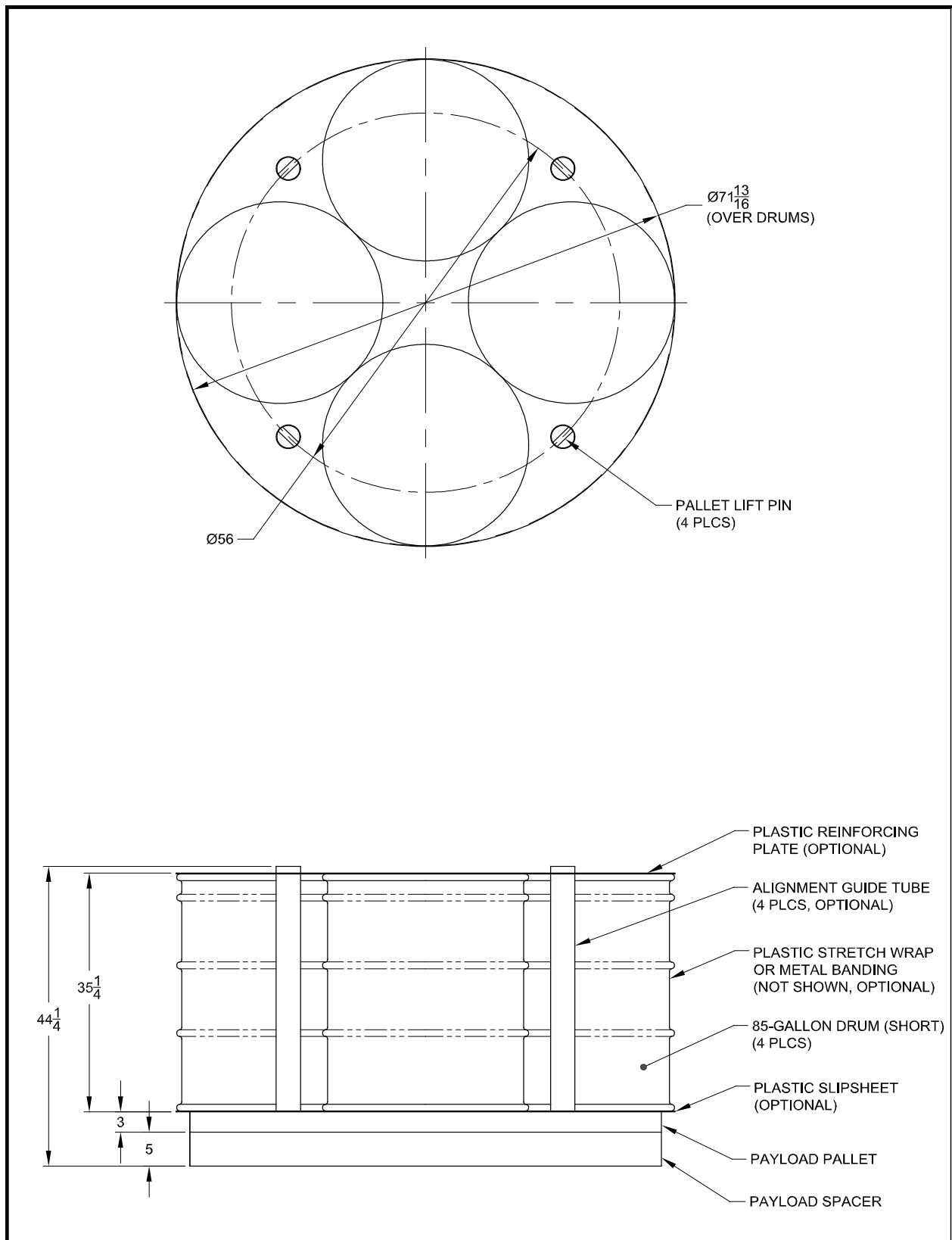


Figure 2.9-11 – HalfPACT: Four 85-Gallon Drum (Short) Payload

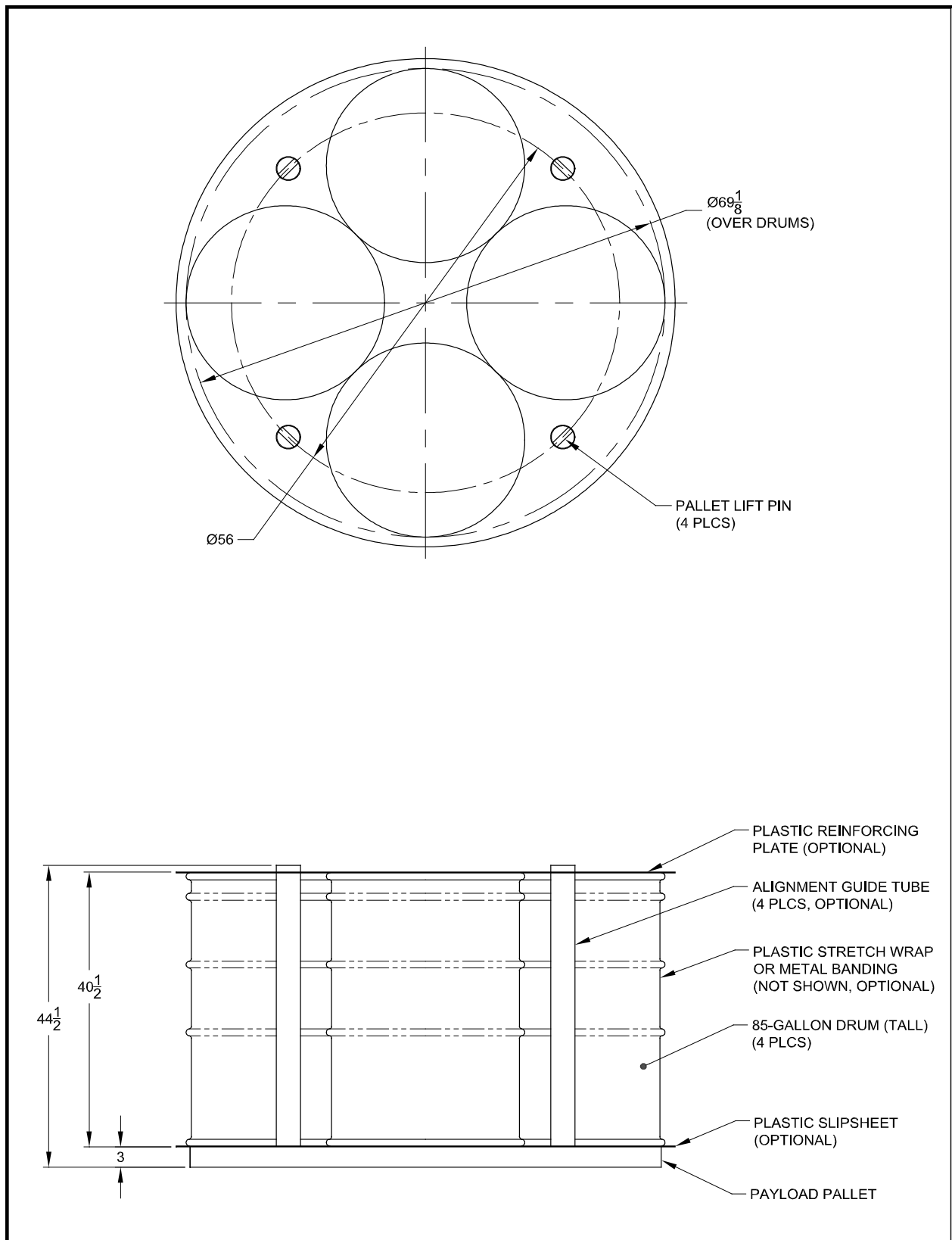


Figure 2.9-12 – HalfPACT: Four 85-Gallon Drum (Tall) Payload

2.9.7 100-Gallon Drum

The 100-gallon drum is authorized for transport within the TRUPACT-II and the HalfPACT (Figure 2.9-13). The TRUPACT-II will accommodate six 100-gallon drums (Figure 2.9-14), and the HalfPACT will accommodate three 100-gallon drums (Figure 2.9-15). The approximate dimensions of the 100-gallon drum are given in Table 2.9-26.

Table 2.9-26 – 100-Gallon Drum Dimensions

Dimension	Approximate Measurement (inches)	
	Inside Dimension	Outside Dimension
Height	33	35
Diameter	30	32

The 100-gallon drum body, lid, and lid ring are constructed of steel. A gasket, of tubular or foam styrene-butadiene, neoprene rubber, or equivalent, is required for outer drum lid closure. The drum must be vented as specified in Section 2.5. An optional, rigid, polyethylene liner and lid may be used inside the drum. If a lid is used with the liner, the liner lid must contain a 0.3-in. minimum diameter hole, or a filter with a hydrogen release rate equivalent to or greater than a 0.3-in. minimum diameter hole. Otherwise, the liner must be treated as any other confinement layer with the associated resistance of the liner hole diameter or filter calculated in accordance with Appendix 2.2 of the CH-TRU Payload Appendices.¹ An optional steel inner lid with an optional gasket may be used inside the drum (i.e., a double-lid drum). A double-lid drum with a filtered inner lid will be considered the same as a drum with a filtered inner confinement layer. Table 2.9-27 presents the 100-gallon drum construction materials. Table 2.9-28 specifies the weights associated with the 100-gallon drum that are applicable to shipment within the TRUPACT-II and HalfPACT.

Table 2.9-27 – 100-Gallon Drum Materials of Construction

100-Gallon Drum Component	Material
Body, lid, and lid ring	Steel
Rigid liner and liner lid (optional)	Polyethylene
Inner lid (optional)	Steel
Gasket	<u>Type I</u> - Tubular styrene-butadiene, or equivalent <u>Type II</u> - Foam styrene-butadiene, or equivalent <u>Type III</u> - Neoprene rubber, or equivalent

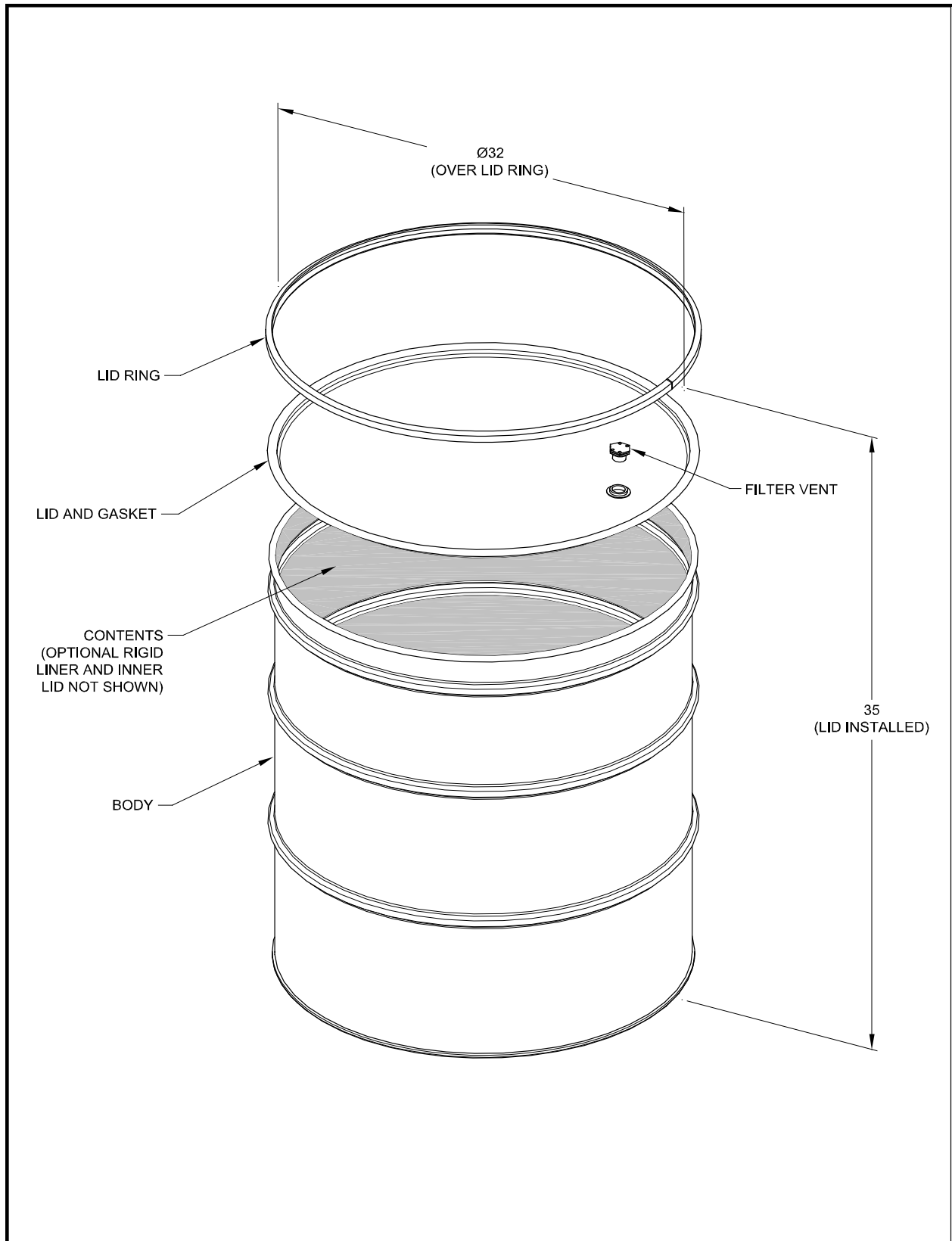


Figure 2.9-13 – 100-Gallon Drum

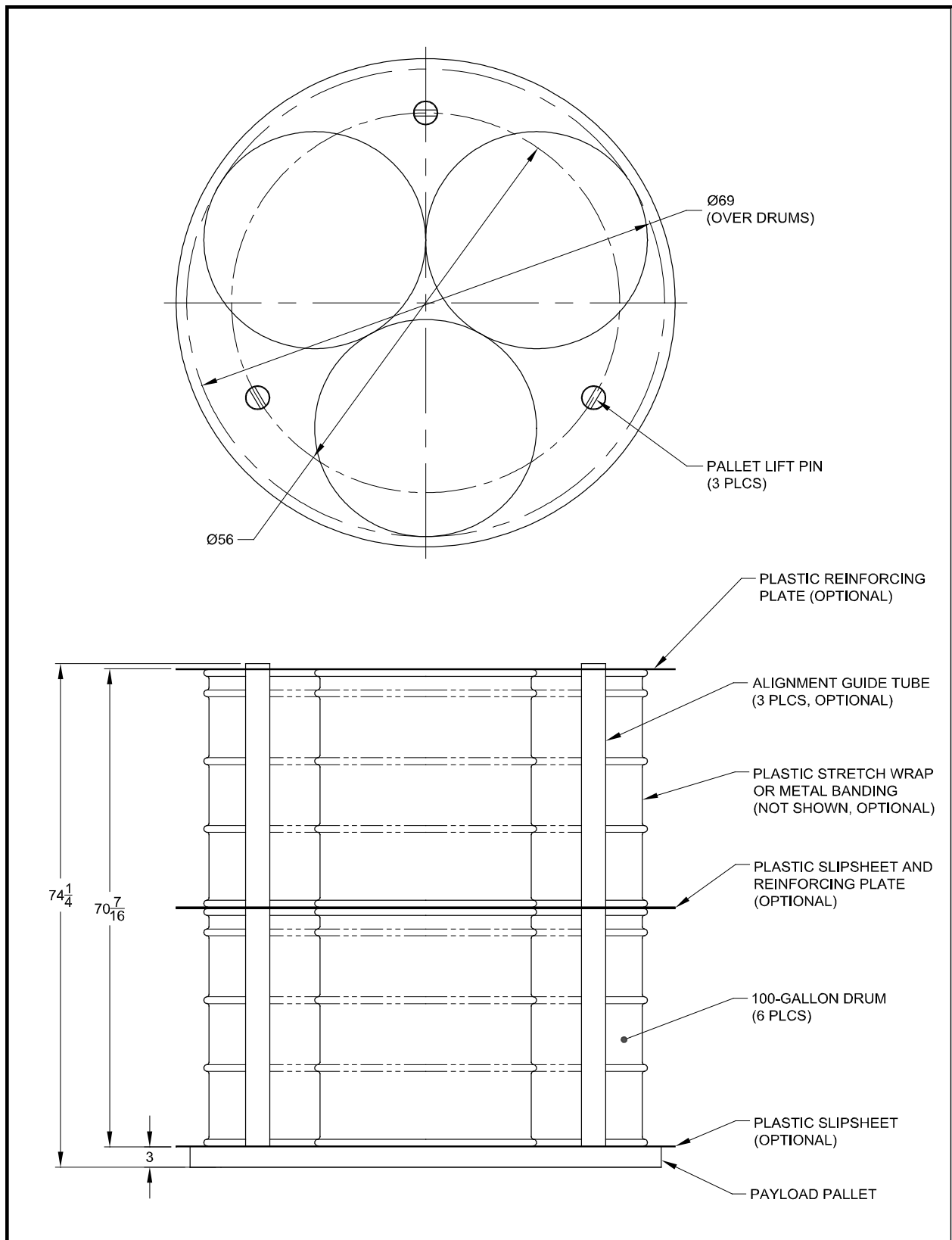


Figure 2.9-14 – TRUPACT-II: Six 100-Gallon Drum Payload

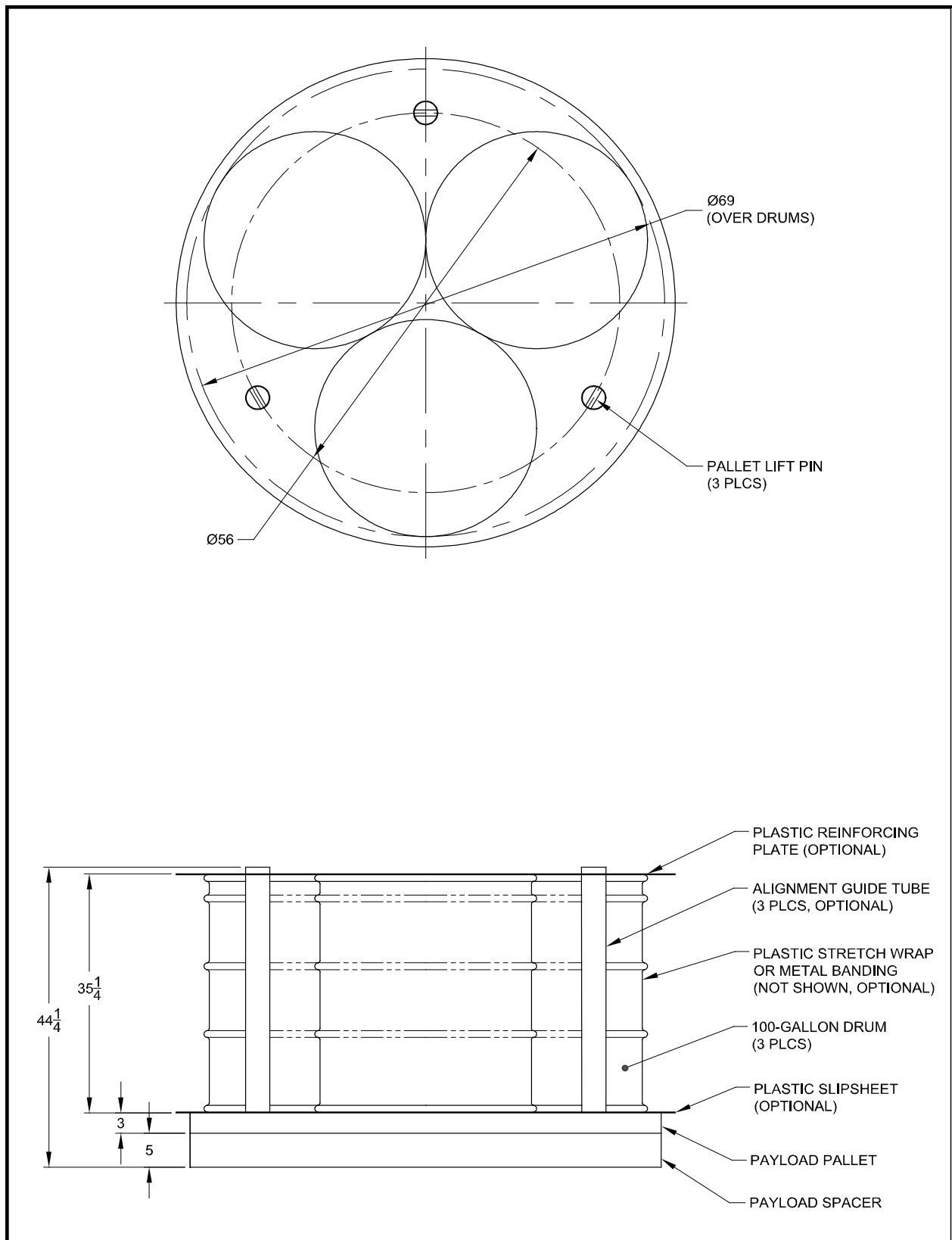


Figure 2.9-15 – HalfPACT: Three 100-Gallon Drum Payload

Table 2.9-28 – 100-Gallon Drum Weights

Component	Weight (pounds)	
	Approximate Empty	Maximum Gross
100-gallon drum	95	1,000

CH-TRU waste may be directly loaded into a 100-gallon drum. Empty 100-gallon drums used as dunnage to complete a payload must also meet the specifications presented in this section with the exception that dunnage containers shall have at least one open vent port (i.e., not filtered or plugged). [Table 2.9-29](#) identifies material content forms authorized for transport within a 100-gallon drum. For all material content forms, if the waste consists of compacted 55-gallon drums, the drums shall allow for hydrogen gas release through puncture holes in the sides of the 55-gallon drum corresponding to the drum headspace and/or through obvious failures in drum integrity due to the compaction process (breach of drum). For material content Form Number 4 (see [Table 2.9-29](#)), if grout is used to encapsulate the waste, the grout formulation (e.g., use of gas-permeable grout) and/or application (e.g., grouting around a pipe used as a gas release pathway) shall allow for hydrogen gas release through or around the grout to prevent accumulation of hydrogen inside the encapsulated waste.

Table 2.9-29 – 100-Gallon Drum: Material Content Forms Authorized for Transport

Form Number	Description
1	Direct load: Solids, any particle size (e.g., fine powder or inorganic particulates)
2	Direct load: Solids, large particle size (e.g., sand, concrete, or debris)
3	Direct load: Solids, large objects (e.g., metal cans containing waste)
4	Direct load: Encapsulated solids (e.g., solid particles or objects surrounded by a layer of cementitious grout or similar encapsulant)

2.9.8 Standard Waste Box

The SWB is designed for efficient loading of the ICV in the TRUPACT-II and the HalfPACT (Figure 2.9-16). The TRUPACT-II will accommodate two SWBs (Figure 2.9-17), and the HalfPACT will accommodate one SWB (Figure 2.9-18). The approximate dimensions of the SWB are given in Table 2.9-30.

Table 2.9-30 – SWB Dimensions

Dimension	Approximate Measurement (inches)	
	Inside	Outside
Height	36 $\frac{5}{16}$	36 $\frac{7}{8}$
Length	68 $\frac{3}{4}$	71
Width	52	54 $\frac{1}{2}$

The SWB is constructed of steel. The SWB is closed by a neoprene or equivalent gasket and steel screws. The SWB contains four filter ports. The SWB must be vented as specified in Section 2.5. Ports without filters must be plugged during transport. Table 2.9-31 presents the SWB construction materials. Table 2.9-32 specifies the weights associated with the SWB that are applicable to shipment within the TRUPACT-II and the HalfPACT.

Table 2.9-31 – SWB Materials of Construction

SWB Component	Material
Body panels (lid, bottom, sides, ends)	Steel
Lift attachments	Steel plate
Upper perimeter tube assembly	Steel tube
Offset tubes	Steel tube
Lid perimeter reinforcement	Steel plate
Gasket	Neoprene or equivalent
Screws	Steel
Filter ports	Steel

CH-TRU waste may be directly loaded into an SWB, or may be loaded into up to four 55-gallon drums, up to two 85-gallon drums, or up to two 100-gallon drums, and overpacked in an SWB (i.e., “SWB/55 Overpack”, “SWB/85 Overpack”, or “SWB/100 Overpack”, respectively). An empty SWB used as dunnage to complete a TRUPACT-II payload of two SWBs must also meet the specifications presented in this section with the exception that dunnage containers shall have at least one open vent port (i.e., not filtered or plugged). Table 2.9-33 identifies material content forms authorized for transport within an SWB.

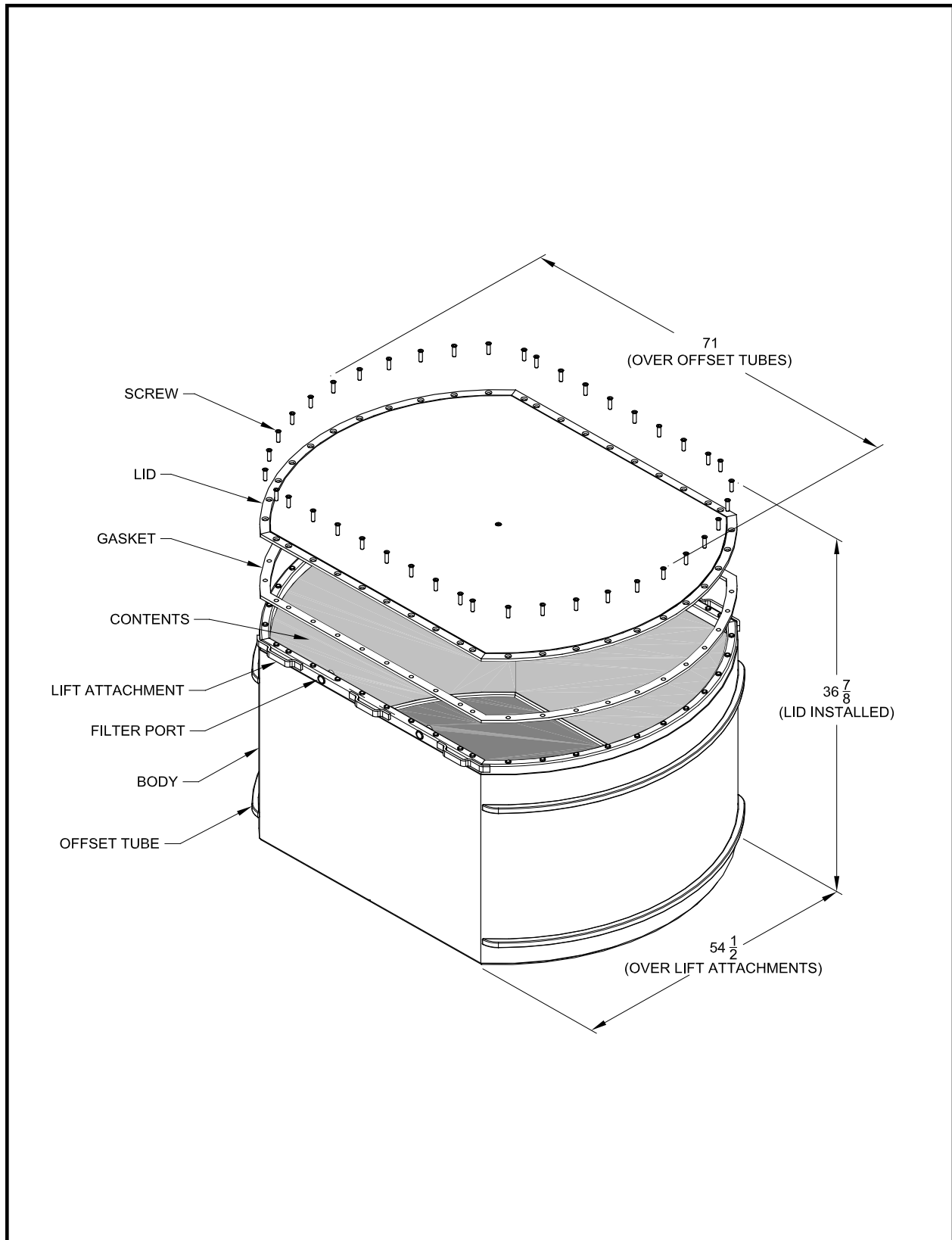


Figure 2.9-16 – Standard Waste Box

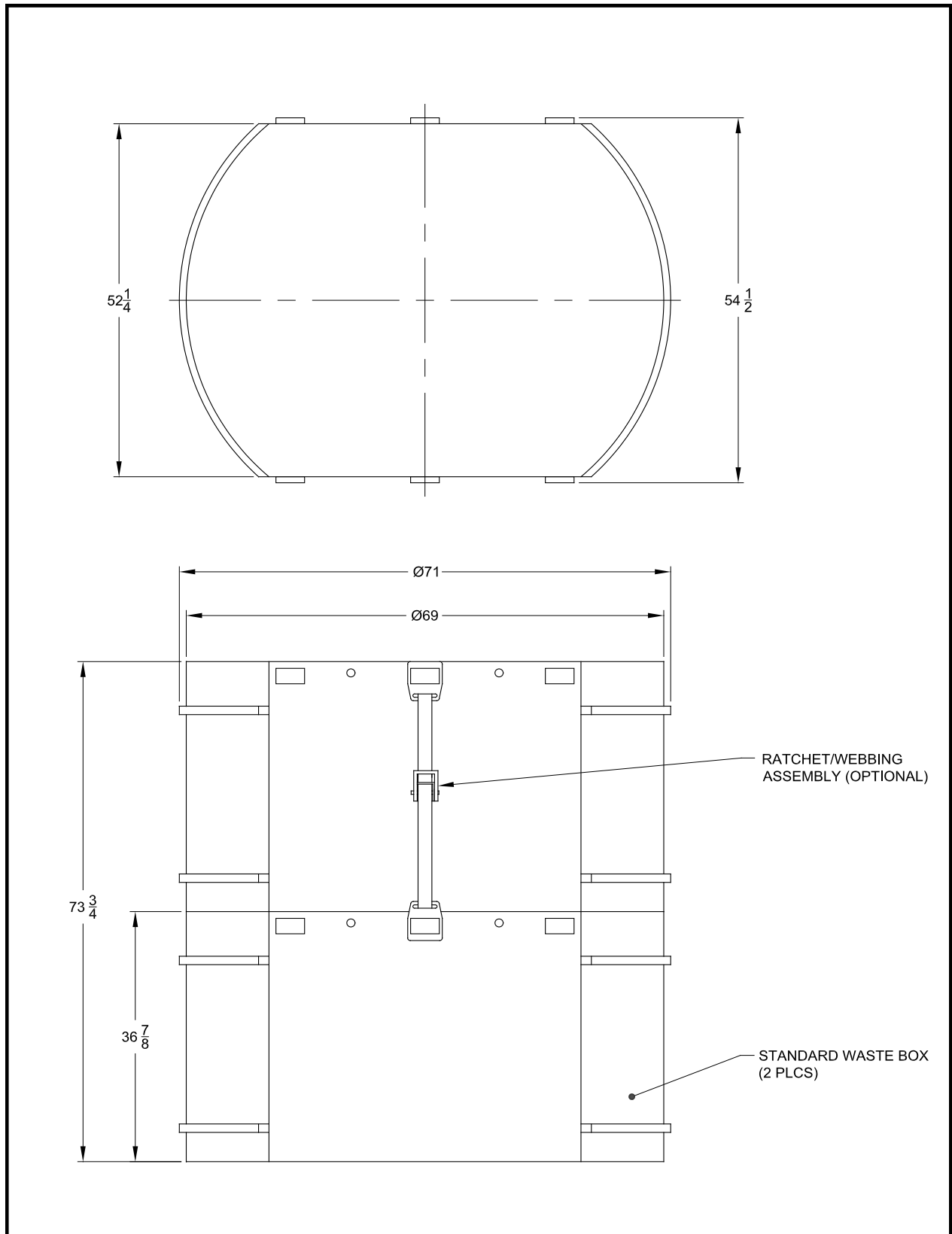


Figure 2.9-17 – TRUPACT-II: Two Standard Waste Box Payload

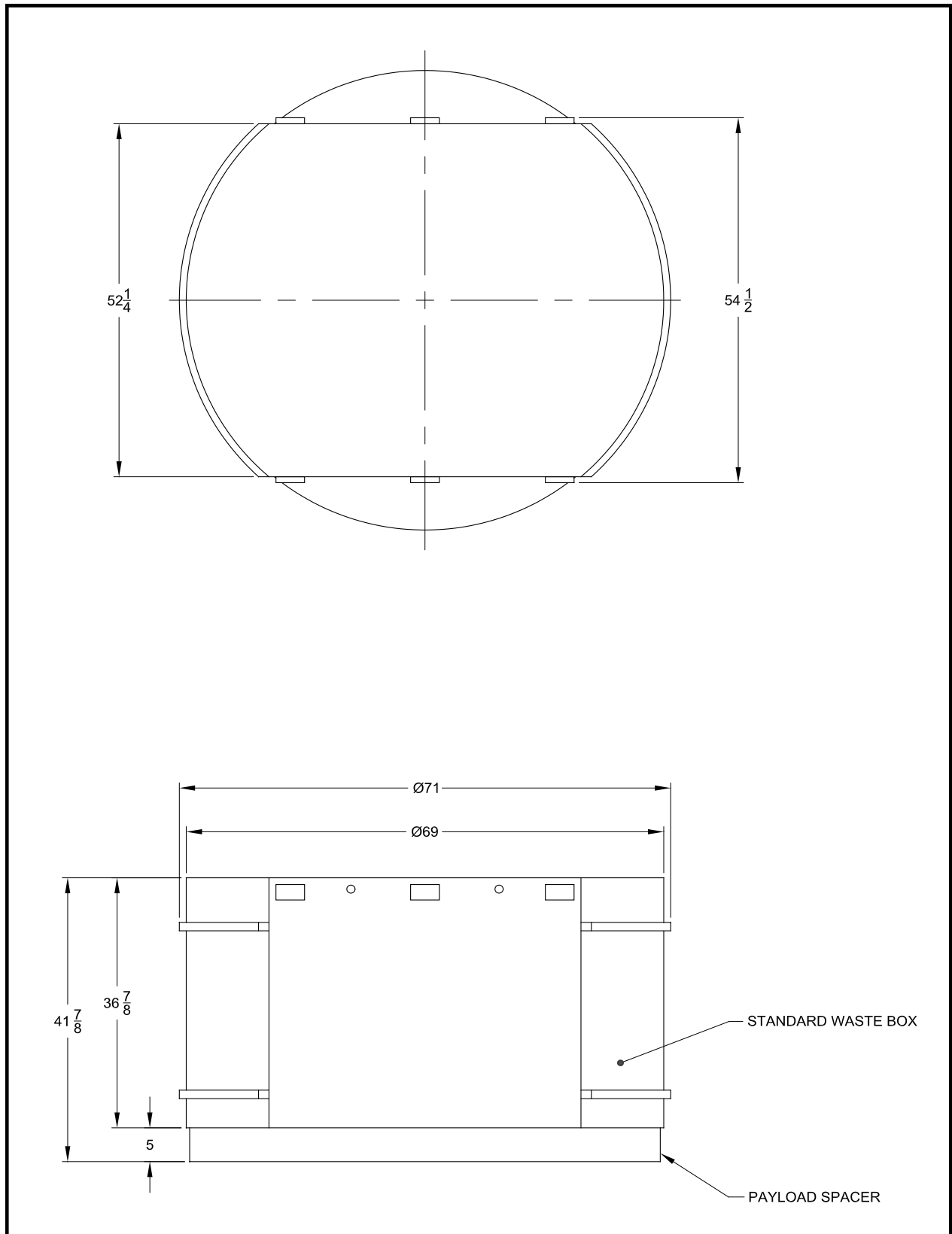


Figure 2.9-18 – HalfPACT: One Standard Waste Box Payload

Table 2.9-32 – SWB Weights

Component	Weight (pounds)	
	Approximate Empty	Maximum Gross
SWB	640	4,000

Table 2.9-33 – SWB: Material Content Forms Authorized for Transport

Form Number	Description
1	Direct load: Solids, any particle size (e.g., fine powder or inorganic particulates)
2	Direct load: Solids, large particle size only (e.g., sand, concrete, or debris)
3	Direct load: Solids, large objects (e.g., metal cans containing waste)
4	Direct load: Large, bulky dense objects with sharp and obtrusive members or components with dispersible Form 1 or Form 2 (e.g., steel plate, electric motors, steel pipe, or concrete blocks) ^①

Note:

① Blocked, braced, or suitably packaged as necessary to provide puncture protection for the SWB.

2.9.9 Ten-Drum Overpack

The TDOP is designed for efficient loading in the ICV of the TRUPACT-II (Figure 2.9-19). The TRUPACT-II will accommodate one TDOP (Figure 2.9-20). The approximate dimensions of the TDOP are given in Table 2.9-34.

Table 2.9-34 – TDOP Dimensions

Dimension	Approximate Measurement (inches)	
	Inside	Outside
Height	72 ⁵ / ₈	73 ¹ / ₈
Diameter	68 ³ / ₄	71 ¹ / ₄

The TDOP is constructed of steel. The TDOP is closed by a neoprene or equivalent gasket and steel screws. The TDOP contains ten filter ports. The TDOP must be vented as specified in Section 2.5. Ports without filters must be plugged during transport. Table 2.9-35 presents the TDOP construction materials. Table 2.9-36 specifies the weights associated with the TDOP that are applicable to shipment within the TRUPACT-II.

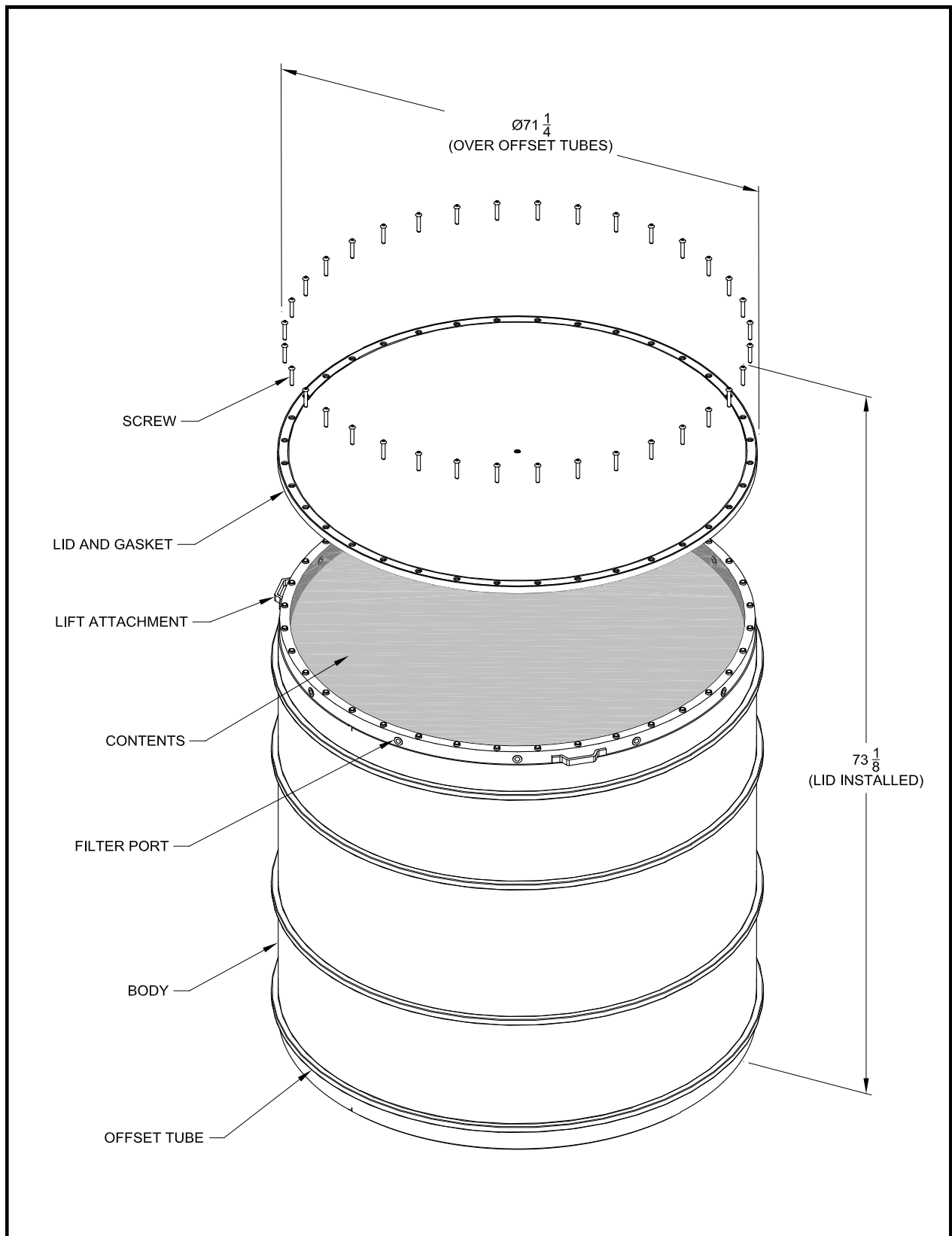
Table 2.9-35 – TDOP Materials of Construction

TDOP Component	Material
Body, lid, and bottom	Steel
Lift attachments	Steel
Offset tubes	Steel tubing
Gasket	Neoprene or equivalent
Screws	Steel
Filter ports	Steel

Table 2.9-36 – TDOP Weights

Component	Weight (pounds)	
	Approximate Empty	Maximum Gross
TDOP	1,700	6,700

CH-TRU waste may be directly loaded into a TDOP, or may be loaded into up to ten 55-gallon drums, up to six 85-gallon drums, up to four 100-gallon drums, or an SWB, and overpacked in a TDOP (i.e., “TDOP/55 Overpack”, “TDOP/85 Overpack”, “TDOP/100 Overpack” or “TDOP/SWB Overpack”, respectively). Table 2.9-37 identifies material content forms authorized for transport within a TDOP.

**Figure 2.9-19 – Ten-Drum Overpack**

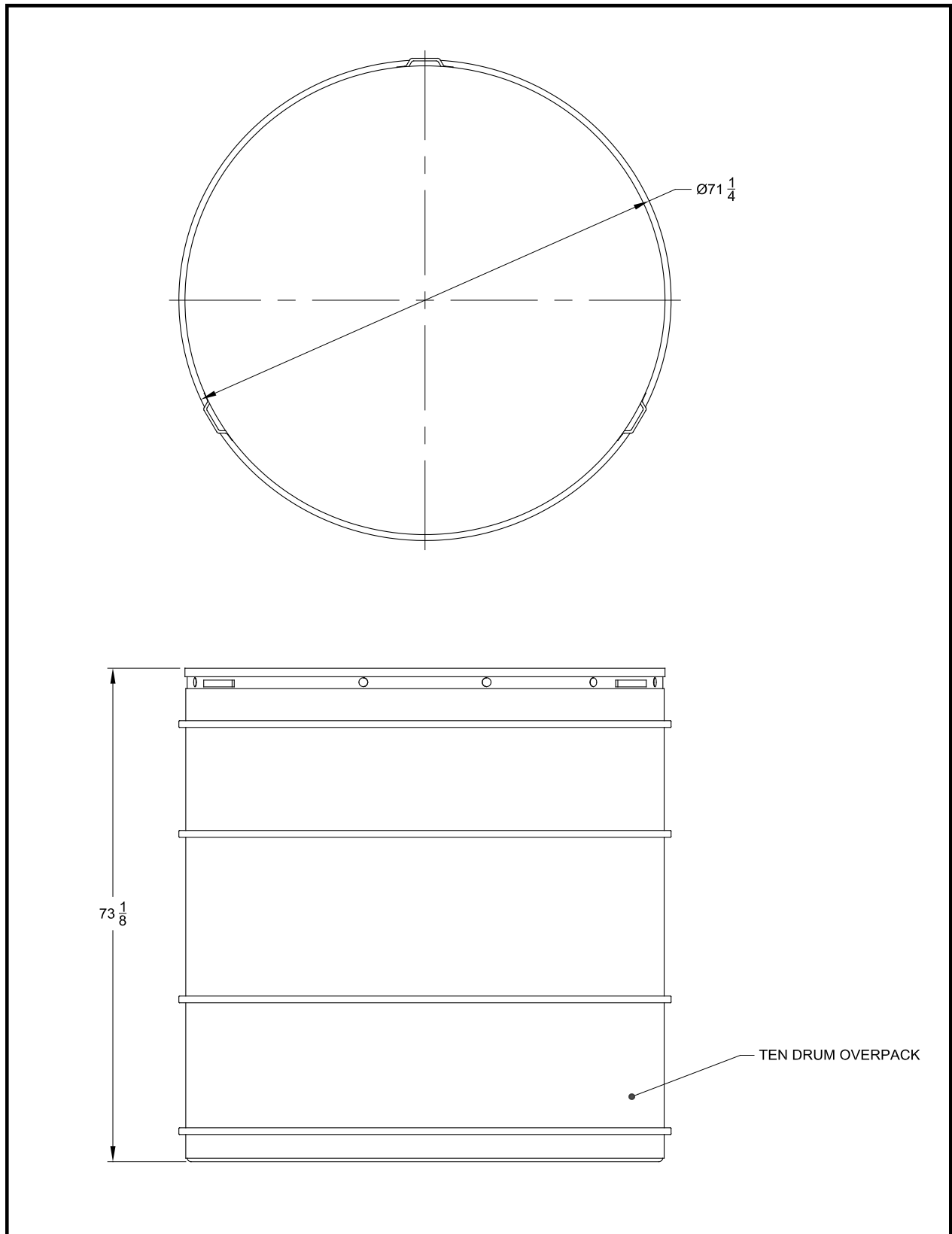


Figure 2.9-20 – TRUPACT-II: One Ten-Drum Overpack Payload

Table 2.9-37 – TDOP: Material Content Forms Authorized for Transport

Form Number	Description
1	Direct load: Solids, any particle size (e.g., fine powder or inorganic particulates)
2	Direct load: Solids, large particle size only (e.g., sand, concrete, or debris)
3	Direct load: Solids, large objects (e.g., metal cans containing waste)
4	Direct load: Large, bulky dense objects with sharp and obtrusive members or components with dispersible Form 1 or Form 2 (e.g., steel plate, electric motors, steel pipe, or concrete blocks) ^①

Note:

① Blocked, braced, or suitably packaged as necessary to provide puncture protection for the TDOP.

2.9.10 SC-30G1 Shielded Container

The SC-30G1 shielded container is authorized for transport within the HalfPACT ([Figure 2.9-21](#)). [Appendix 1.3.1](#) of the [HalfPACT SAR](#)³ and [Appendix 4.5](#) of the [CH-TRU Payload Appendices](#)¹ provide detailed descriptions of this payload container and associated dunnage and pallet (Drawing 163-008). The HalfPACT will accommodate three SC-30G1s ([Figure 2.9-22](#)). When assembled in the HalfPACT, the SC-30G1 is surrounded by energy absorbing dunnage assemblies (both axial and radial). The approximate dimensions of the SC-30G1 are given in [Table 2.9-38](#).

Table 2.9-38 – SC-30G1 Shielded Container Dimensions

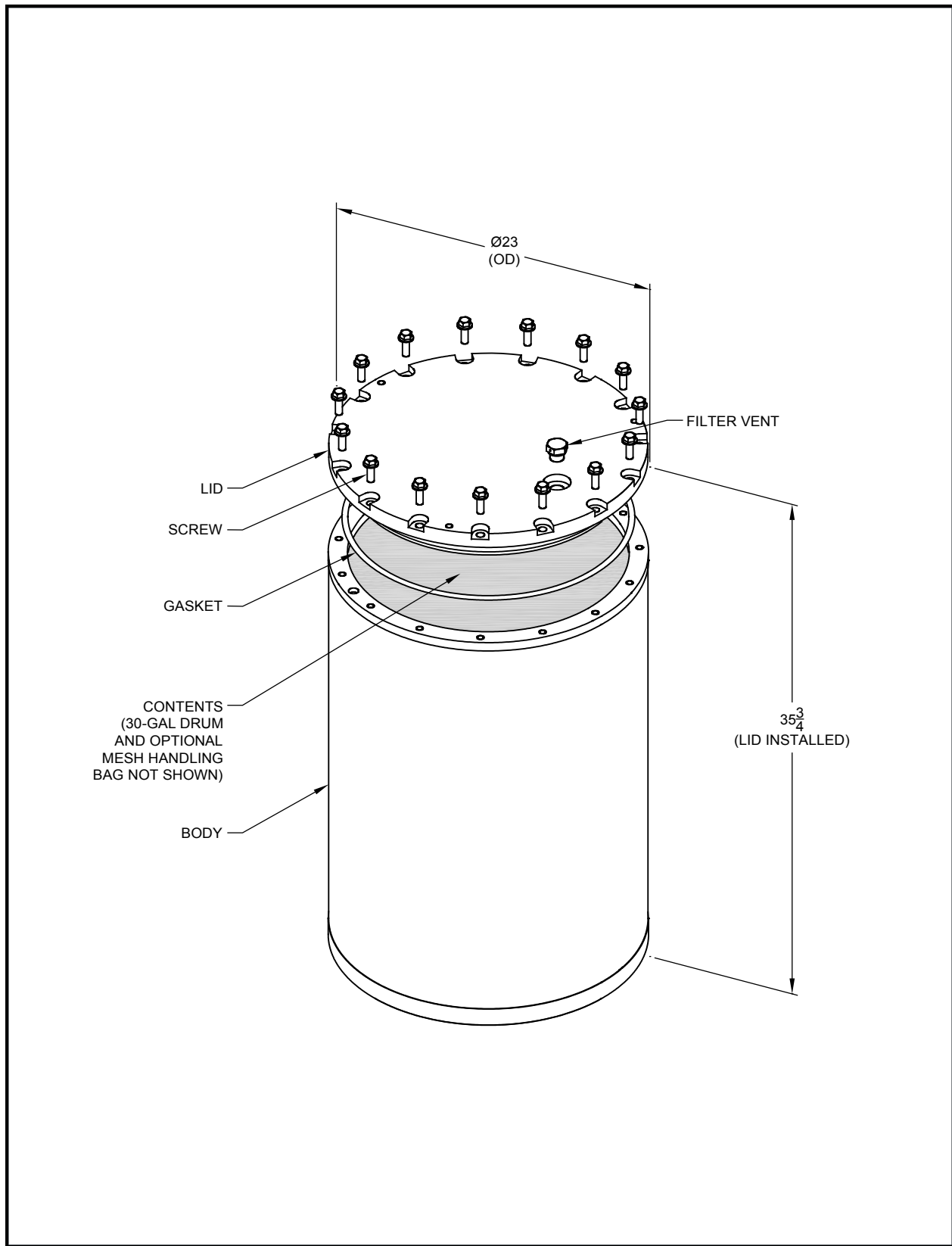
Dimension	Approximate Measurement (inches)	
	Inside Dimension	Outside Dimension
Height	29¾	35¾
Diameter	20⅜	23

The SC-30G1 body side wall is constructed of 1-in. nominal lead with ⅝-in. nominal steel. The SC-30G1 lid and base are constructed of 3-in. steel. The lid is bolted on with 15, ½-in. Grade 8 closure bolts. A gasket of silicone rubber is required for lid closure. The SC-30G1 and its 30-gallon steel drum contents must be vented as specified in [Section 2.5](#). The SC-30G1 vent port includes a lead shield plug. [Table 2.9-39](#) presents the SC-30G1 materials of construction.

Table 2.9-39 – SC-30G1 Shielded Container Materials of Construction

SC-30G1 Component	Material
Lid, body base, and body flange	Carbon steel
Body side wall	Lead and carbon steel
Filter shield plug	Lead
Gasket	Silicone rubber

[Table 2.9-40](#) specifies the weights associated with the SC-30G1 that are applicable to shipment within the HalfPACT.

**Figure 2.9-21 – SC-30G1 Shielded Container**

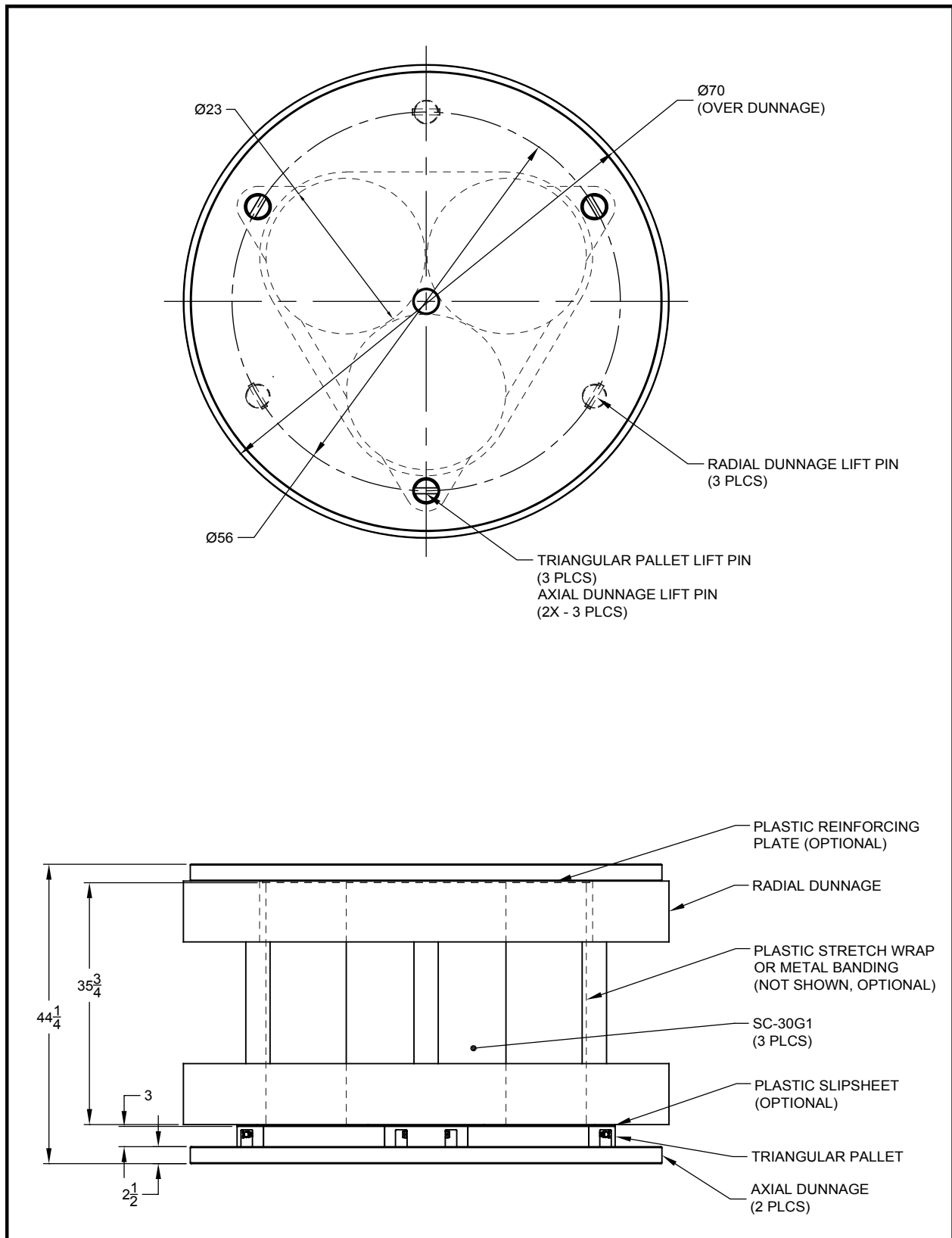


Figure 2.9-22 – HalfPACT: Three SC-30G1 Shielded Container Payload

Table 2.9-40 – SC-30G1 Shielded Container Weights

Component	Weight (pounds)	
	Approximate Empty	Maximum Gross
SC-30G1	1,726	2,260

CH-TRU waste must be loaded into a vented 30-gallon steel drum with an inside lever lock, which is then loaded into the SC-30G1 with the aid of an optional mesh handling bag. Empty SC-30G1s used as dunnage to complete a payload must also meet the specifications presented in this section with the exception that dunnage containers shall not include a 30-gallon drum and shall have an open vent port (i.e., not filtered or plugged).

[Table 2.9-41](#) identifies material content forms authorized for transport within an SC-30G1.

Table 2.9-41 – SC-30G1 Shielded Container: Material Content Forms Authorized for Transport

Form Number	Description
1	Direct load: Solids, any particle size (e.g., fine powder or inorganic particulates)
2	Direct load: Solids, large particle size (e.g., sand, concrete, or debris)
3	Direct load: Solids, large objects (e.g., metal cans containing waste)

2.9.11 Criticality Control Overpack

The CCO consists of a criticality control container (CCC) positioned by plywood upper and lower dunnage assemblies within a 55-gallon drum (Figure 2.9-23). Appendix 1.3.1 of the TRUPACT-II SAR², Appendix 1.3.1 of the HalfPACT SAR³ and Appendix 4.6 of the CH-TRU Payload Appendices¹ provide detailed descriptions of this payload container (Drawing 163-009). The TRUPACT-II will accommodate 14 CCOs (Figure 2.9-2), and the HalfPACT will accommodate 7 CCOs (Figure 2.9-3). The approximate dimensions of the CCO are given in Table 2.9-42.

Table 2.9-42 – CCO Dimensions

Dimension	Approximate Measurement (inches)	
	Inside Dimension (CCC)	Outside Dimension (drum)
Height	26 $\frac{7}{8}$	35
Diameter	6	24

The CCC body is constructed of steel 6-inch Class 150 standard blind and slip flanges and Schedule 40 pipe. The lid is bolted on with 8, $\frac{3}{4}$ -in. steel heavy hex head bolts. A gasket of aramid-inorganic/nbr is required for lid closure. The 55-gallon drum and CCC must be vented as specified in Section 2.5. Table 2.9-43 presents the CCO materials of construction.

Table 2.9-43 – CCO Materials of Construction

CCO Component	Material
Drum	Steel
Body, lid, base, and flange	Steel
Dunnage	Plywood
Gasket	Aramid-inorganic/nbr

Table 2.9-44 specifies the weights associated with the CCO that are applicable to shipment within the TRUPACT-II and HalfPACT.

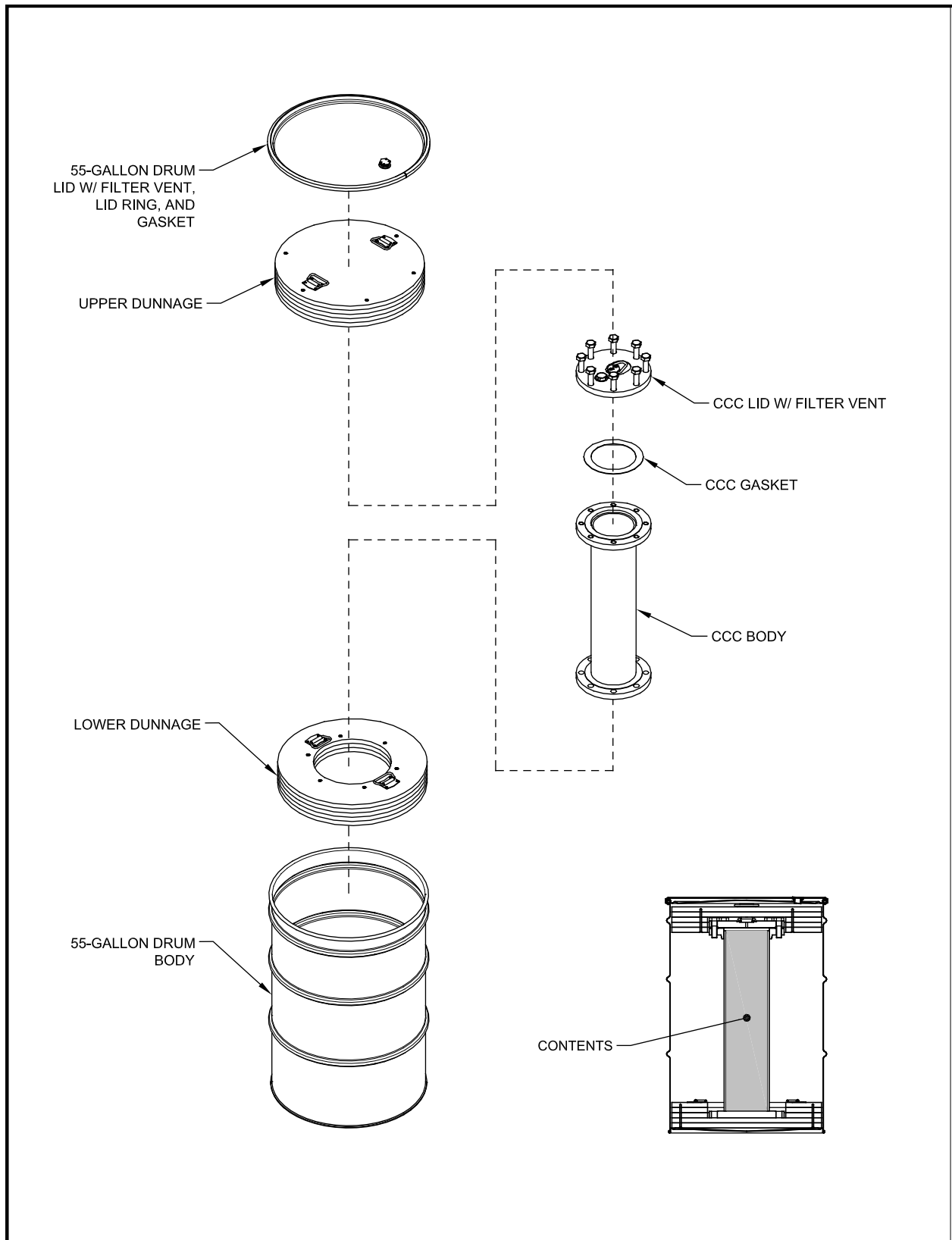


Figure 2.9-23 – Criticality Control Overpack

Table 2.9-44 – CCO Weights

Component	Weight (pounds)	
	Approximate Empty	Maximum Gross
CCO	230	350

Table 2.9-45 identifies material content forms authorized for transport within a CCO.

Table 2.9-45 – CCO: Material Content Forms Authorized for Transport

Form Number	Description
1	Direct load: Solids, any particle size (e.g., fine powder or inorganic particulates)
2	Direct load: Solids, large particle size (e.g., sand, concrete, or debris)
3	Direct load: Solids, large objects (e.g., metal cans containing waste)

2.9.12 SC-30G2 Shielded Container

The SC-30G2 shielded container is authorized for transport within the HalfPACT ([Figure 2.9-24](#)). [Appendix 1.3.1](#) of the [HalfPACT SAR](#)³ and [Appendix 4.7](#) of the [CH-TRU Payload Appendices](#)¹ provide detailed descriptions of this payload container and associated dunnage and pallet (Drawing 163-010). The HalfPACT will accommodate two SC-30G2s ([Figure 2.9-25](#)). When assembled in the HalfPACT, the SC-30G2 is partially surrounded by energy absorbing dunnage assemblies (both top axial and radial). The approximate dimensions of the SC-30G2 are given in [Table 2.9-46](#).

Table 2.9-46 – SC-30G2 Shielded Container Dimensions

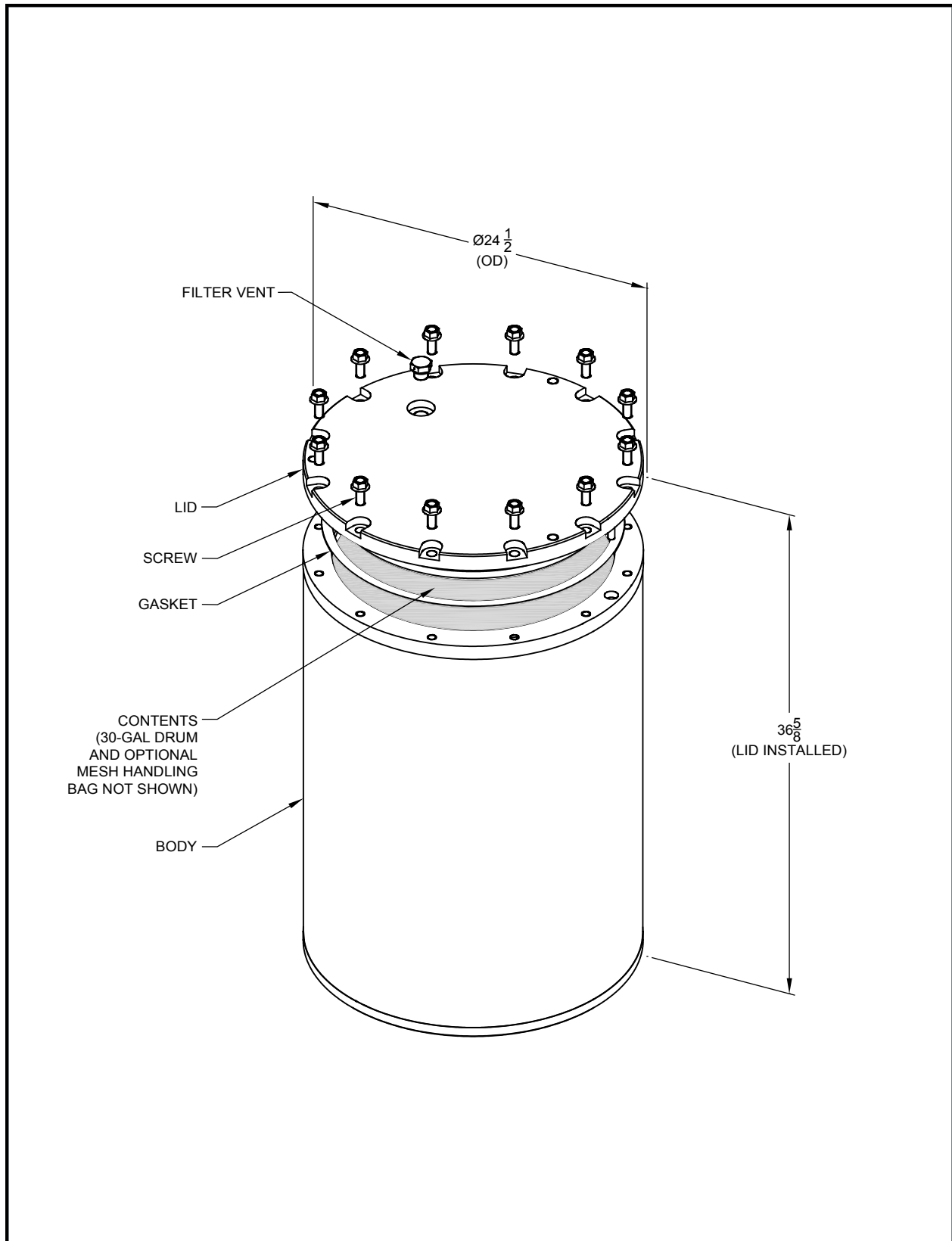
Dimension	Approximate Measurement (inches)	
	Inside Dimension	Outside Dimension
Height	29 ³ / ₄	36 ⁵ / ₈
Diameter	20 ³ / ₈	24 ¹ / ₂

The SC-30G2 body side wall is constructed of 1½-in. nominal lead with ⅝-in. nominal steel. The SC-30G2 lid is constructed of ¾-in. nominal lead with 3⅛-in. nominal steel. The SC-30G2 base is constructed of 1¼-in. nominal lead with 1¾-in. nominal steel. The lid is bolted on with 12, ⅝-in. Grade 8 closure bolts. A gasket of silicone rubber is required for lid closure. The SC-30G2 and its 30-gallon steel drum contents must be vented as specified in [Section 2.5](#). The SC-30G2 vent port includes a lead shield plug. [Table 2.9-47](#) presents the SC-30G2 materials of construction.

Table 2.9-47 – SC-30G2 Shielded Container Materials of Construction

SC-30G2 Component	Material
Body flange	Carbon steel
Lid, body base, and body side wall	Lead and carbon steel
Filter shield plug	Lead
Gasket	Silicone rubber

[Table 2.9-48](#) specifies the weights associated with the SC-30G2 that are applicable to shipment within the HalfPACT.

**Figure 2.9-24 – SC-30G2 Shielded Container**

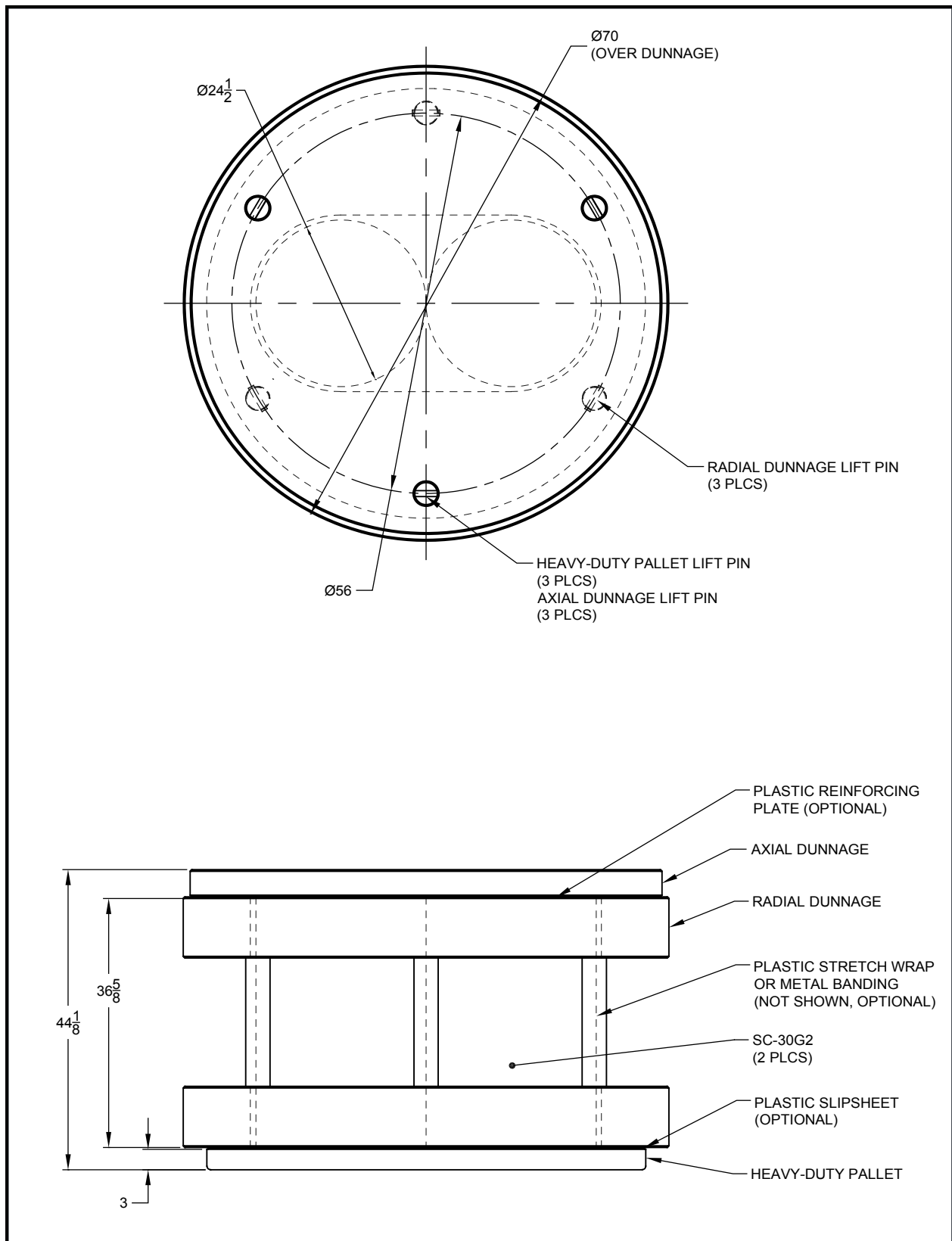


Figure 2.9-25 – HalfPACT: Two SC-30G2 Shielded Container Payload

Table 2.9-48 – SC-30G2 Shielded Container Weights

Component	Weight (pounds)	
	Approximate Empty	Maximum Gross
SC-30G2	2,610	3,160

CH-TRU waste must be loaded into a vented 30-gallon steel drum with an inside lever lock, which is then loaded into the SC-30G2 with the aid of an optional mesh handling bag. Empty SC-30G2s used as dunnage to complete a payload must also meet the specifications presented in this section with the exception that dunnage containers shall not include a 30-gallon drum and shall have an open vent port (i.e., not filtered or plugged).

[Table 2.9-49](#) identifies material content forms authorized for transport within a SC-30G2.

Table 2.9-49 – SC-30G2 Shielded Container: Material Content Forms Authorized for Transport

Form Number	Description
1	Direct load: Solids, any particle size (e.g., fine powder or inorganic particulates)
2	Direct load: Solids, large particle size (e.g., sand, concrete, or debris)
3	Direct load: Solids, large objects (e.g., metal cans containing waste)

2.9.13 SC-30G3 Shielded Container

The SC-30G3 shielded container is authorized for transport within the HalfPACT ([Figure 2.9-26](#)). [Appendix 1.3.1](#) of the [HalfPACT SAR](#)³ and [Appendix 4.8](#) of the [CH-TRU Payload Appendices](#)¹ provide detailed descriptions of this payload container and associated dunnage and pallet (Drawing 163-011). The HalfPACT will accommodate one SC-30G3 ([Figure 2.9-27](#)). When assembled in the HalfPACT, the SC-30G3 is fitted with clearance-limiting end caps and surrounded by an energy absorbing dunnage assembly (lateral). The approximate dimensions of the SC-30G3 are given in [Table 2.9-50](#).

Table 2.9-50 – SC-30G3 Shielded Container Dimensions

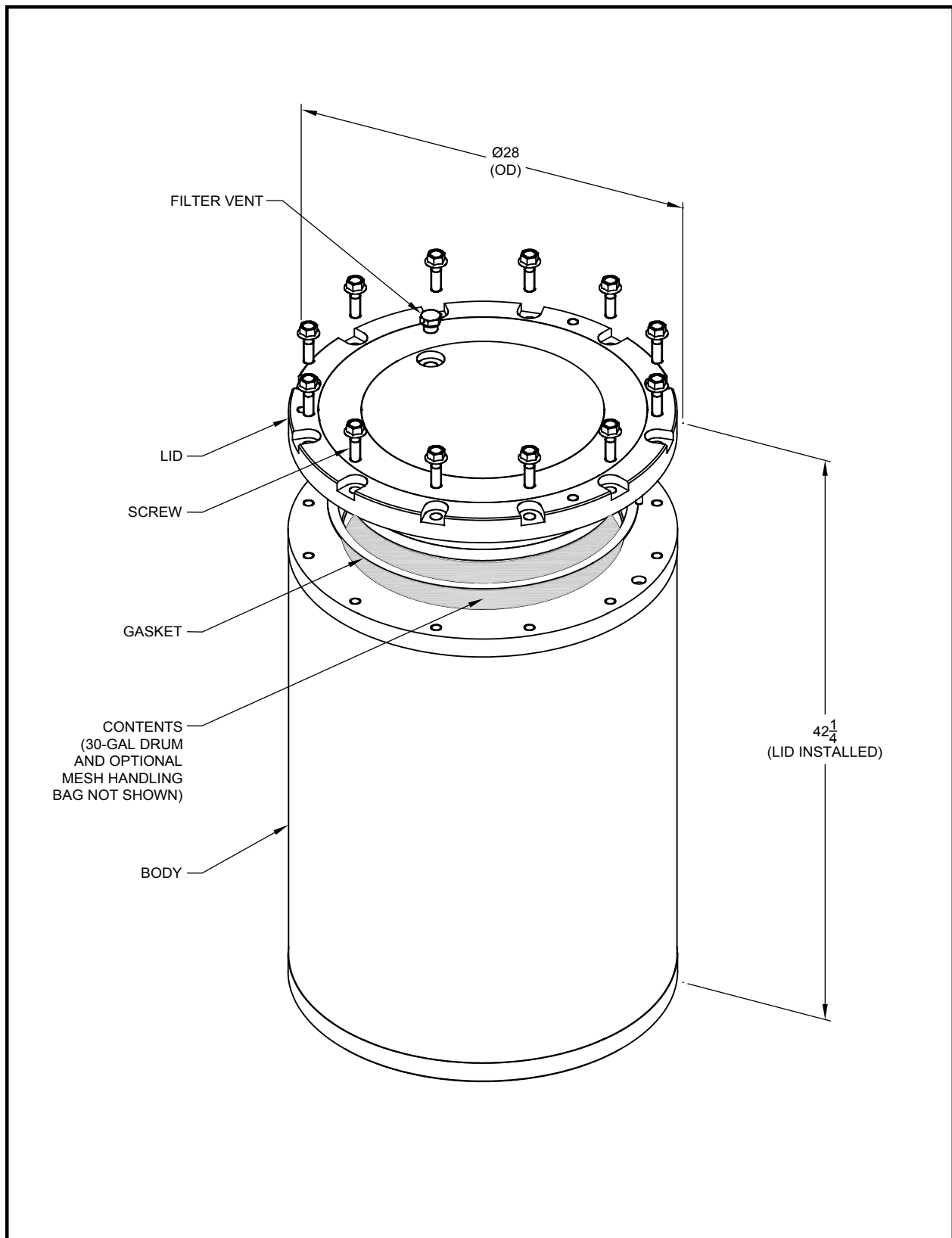
Dimension	Approximate Measurement (inches)	
	Inside Dimension	Outside Dimension
Height	29 $\frac{3}{4}$	42 $\frac{1}{4}$
Diameter	20 $\frac{3}{8}$	28

The SC-30G3 body side wall is constructed of 2 $\frac{3}{4}$ -in. nominal lead with 1-in. nominal steel. The SC-30G3 lid is constructed of 2 $\frac{1}{4}$ -in. nominal lead with 4 $\frac{1}{2}$ -in. nominal steel. The SC-30G3 base is constructed of 2 $\frac{1}{2}$ -in. nominal lead with 3 $\frac{1}{4}$ -in. nominal steel. The lid is bolted on with 12, $\frac{3}{4}$ -in. Grade 8 closure bolts. A gasket of silicone rubber is required for lid closure. The SC-30G3 and its 30-gallon steel drum contents must be vented as specified in [Section 2.5](#). The SC-30G3 lid also incorporates a $\frac{3}{4}$ -in. thick lead shield ring. [Table 2.9-51](#) presents the SC-30G3 materials of construction.

Table 2.9-51 – SC-30G3 Shielded Container Materials of Construction

SC-30G3 Component	Material
Body flange	Carbon steel
Lid, body base, and body side wall	Lead and carbon steel
Lid shield ring	Lead
Gasket	Silicone rubber

[Table 2.9-52](#) specifies the weights associated with the SC-30G3 that are applicable to shipment within the HalfPACT.

**Figure 2.9-26 – SC-30G3 Shielded Container**

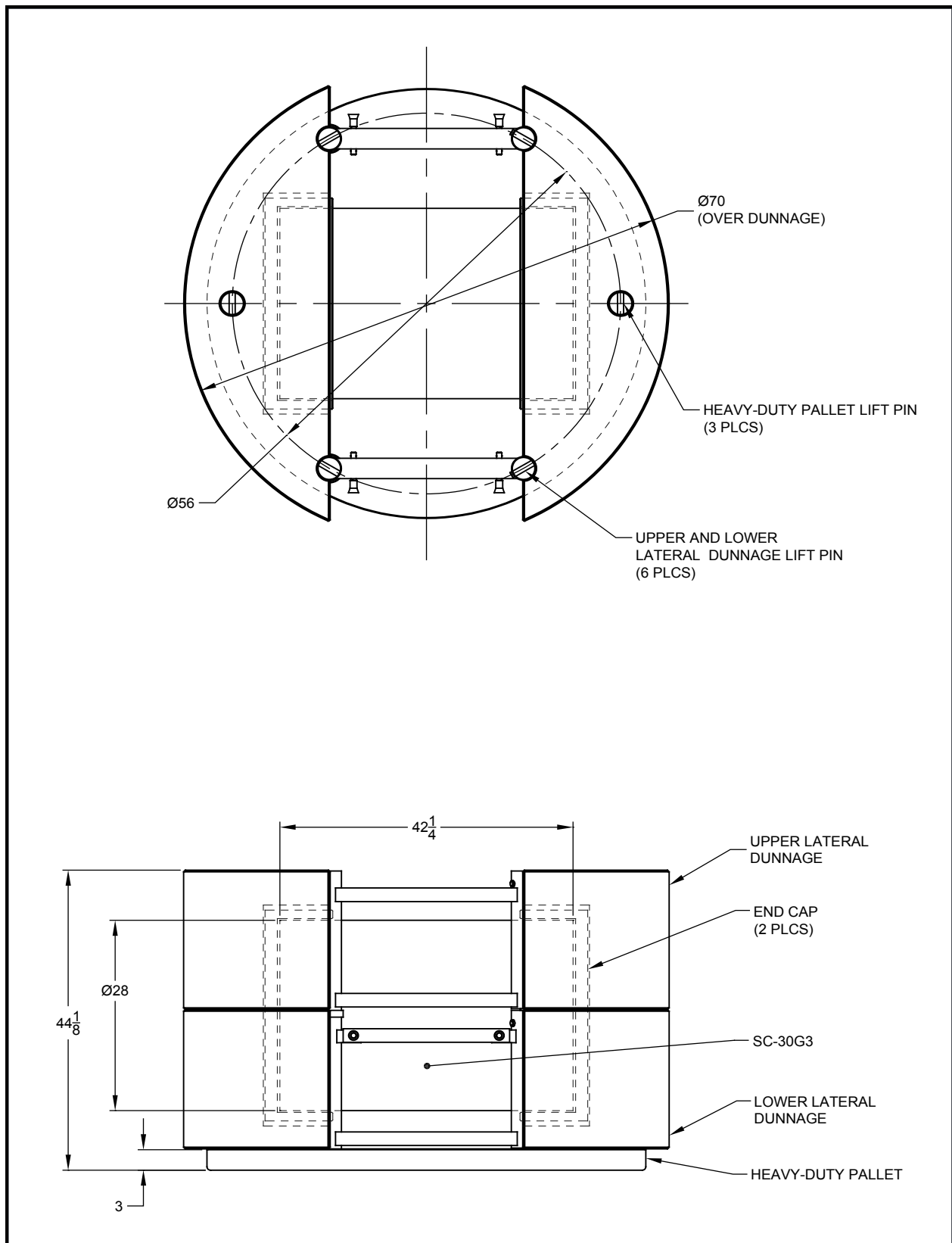


Figure 2.9-27 – HalfPACT: One SC-30G3 Shielded Container Payload

Table 2.9-52 – SC-30G3 Shielded Container Weights

Component	Weight (pounds)	
	Approximate Empty	Maximum Gross
SC-30G3	5,750	6,300

CH-TRU waste must be loaded into a vented 30-gallon steel drum with an inside lever lock, which is then loaded into the SC-30G3 with the aid of an optional mesh handling bag.

[Table 2.9-53](#) identifies material content forms authorized for transport within a SC-30G3.

Table 2.9-53 – SC-30G3 Shielded Container: Material Content Forms Authorized for Transport

Form Number	Description
1	Direct load: Solids, any particle size (e.g., fine powder or inorganic particulates)
2	Direct load: Solids, large particle size (e.g., sand, concrete, or debris)
3	Direct load: Solids, large objects (e.g., metal cans containing waste)

2.9.14 SC-55G1 Shielded Container

The SC-55G1 shielded container is authorized for transport within the HalfPACT ([Figure 2.9-28](#)). [Appendix 1.3.1](#) of the [HalfPACT SAR](#)³ and [Appendix 4.9](#) of the [CH-TRU Payload Appendices](#)¹ provide detailed descriptions of this payload container and associated dunnage and pallet (Drawing 163-012). The HalfPACT will accommodate two SC-55G1s ([Figure 2.9-29](#)). When assembled in the HalfPACT, the SC-55G1 is partially surrounded by an energy absorbing dunnage assembly (radial). The approximate dimensions of the SC-55G1 are given in [Table 2.9-54](#).

Table 2.9-54 – SC-55G1 Shielded Container Dimensions

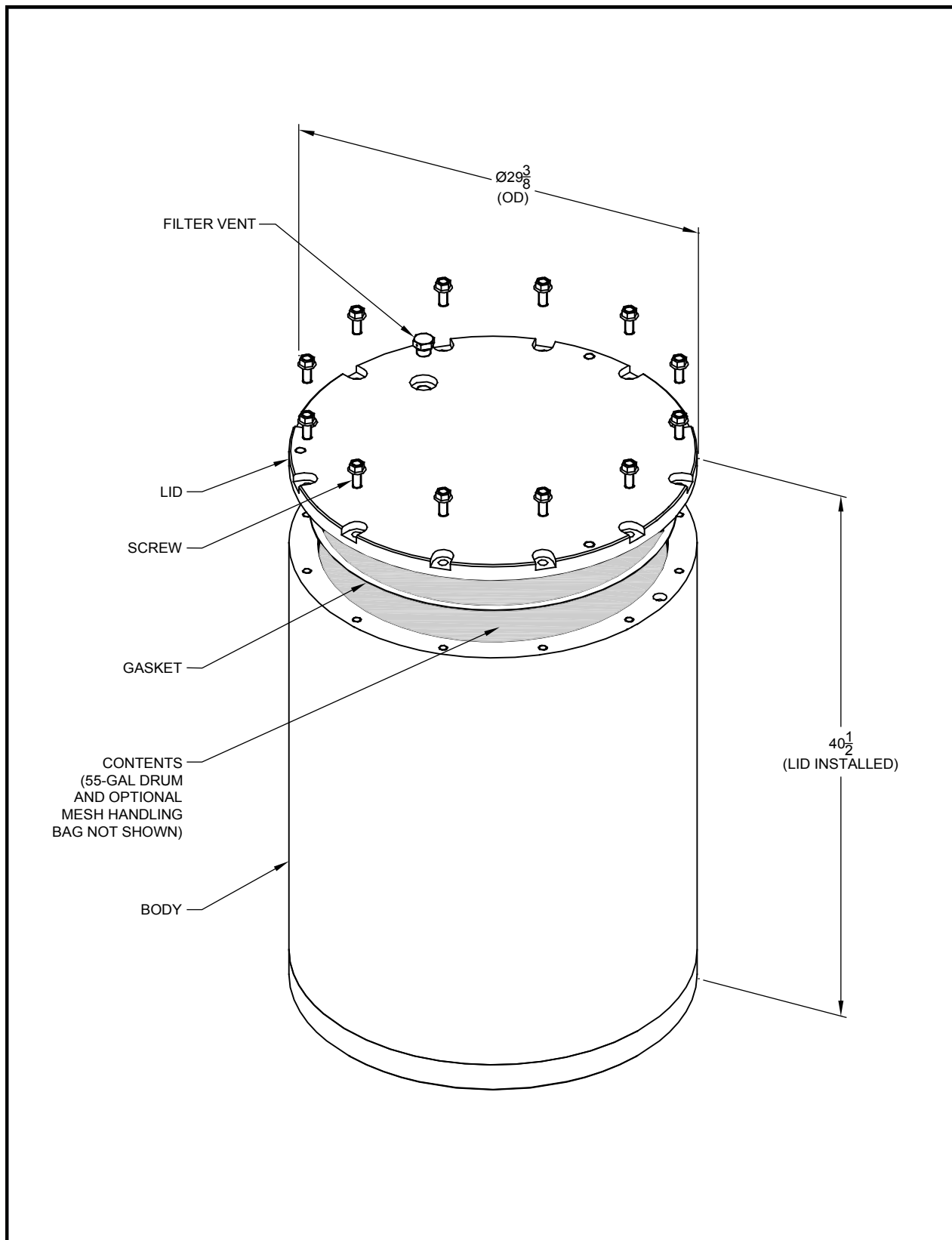
Dimension	Approximate Measurement (inches)	
	Inside Dimension	Outside Dimension
Height	35 ³ / ₄	40 ¹ / ₂
Diameter	25	29 ³ / ₈

The SC-55G1 body side wall is constructed of 2¹/₄-in. nominal steel. The SC-55G1 lid is constructed of 2³/₈-in. nominal steel. The SC-55G1 base is constructed of 2³/₈-in. nominal steel. The lid is bolted on with 12, ⁵/₈-in. Grade 8 closure bolts. A gasket of silicone rubber is required for lid closure. The SC-55G1 and its 55-gallon steel drum contents must be vented as specified in [Section 2.5](#). The SC-55G1 vent port includes a lead shield plug. [Table 2.9-55](#) presents the SC-55G1 materials of construction.

Table 2.9-55 – SC-55G1 Shielded Container Materials of Construction

SC-55G1 Component	Material
Lid and body	Carbon steel
Filter shield plug	Lead
Gasket	Silicone rubber

[Table 2.9-56](#) specifies the weights associated with the SC-55G1 that are applicable to shipment within the HalfPACT.

**Figure 2.9-28 – SC-55G1 Shielded Container**

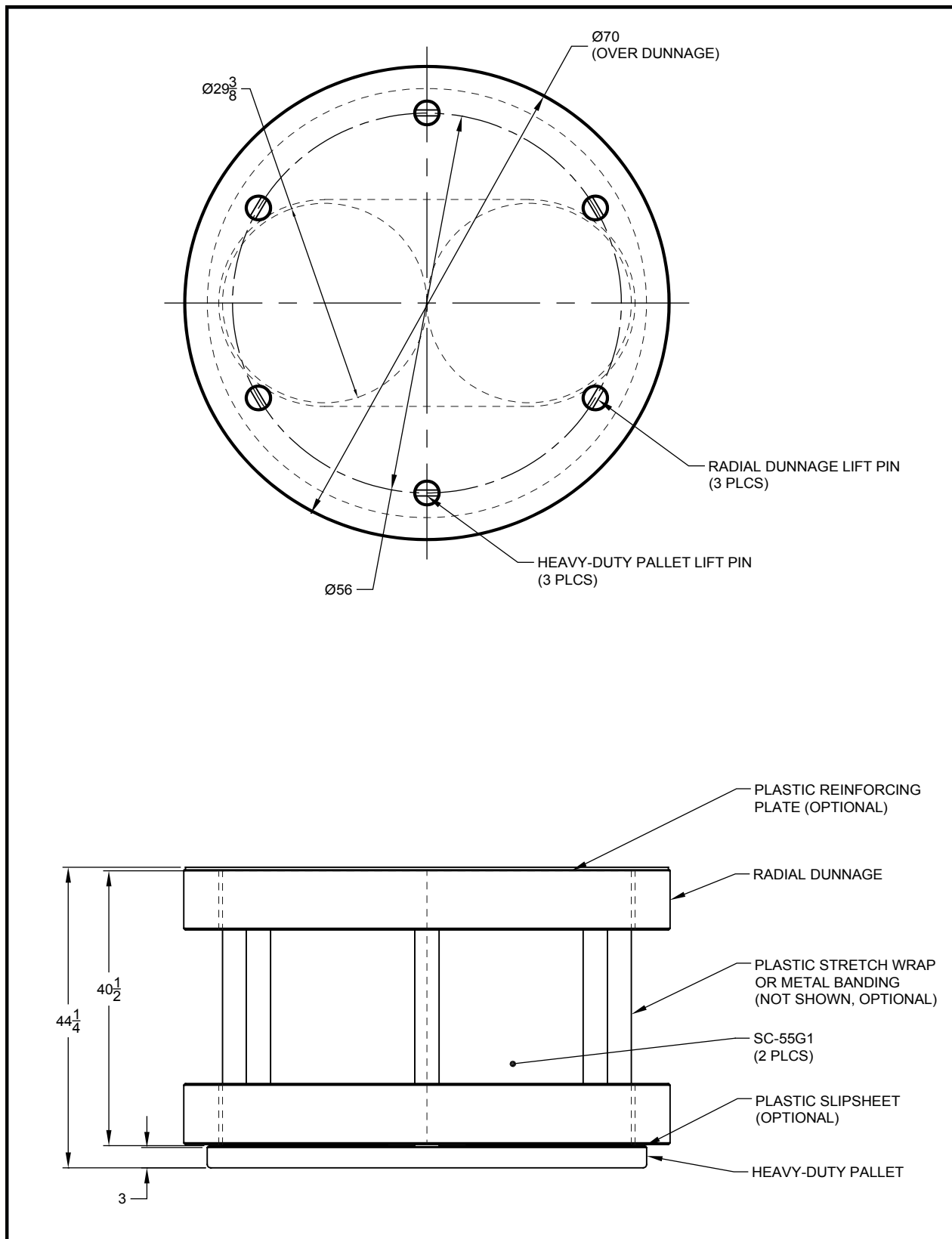


Figure 2.9-29 – HalfPACT: Two SC-55G1 Shielded Container Payload

Table 2.9-56 – SC-55G1 Shielded Container Weights

Component	Weight (pounds)	
	Approximate Empty	Maximum Gross
SC-55G1	2,810	3,410

CH-TRU waste must be loaded into a vented 55-gallon steel drum with an inside lever lock, which is then loaded into the SC-55G1 with the aid of an optional mesh handling bag. Empty SC-55G1s used as dunnage to complete a payload must also meet the specifications presented in this section with the exception that dunnage containers shall not include a 55-gallon drum and shall have an open vent port (i.e., not filtered or plugged).

[Table 2.9-57](#) identifies material content forms authorized for transport within a SC-55G1.

Table 2.9-57 – SC-55G1 Shielded Container: Material Content Forms Authorized for Transport

Form Number	Description
1	Direct load: Solids, any particle size (e.g., fine powder or inorganic particulates)
2	Direct load: Solids, large particle size (e.g., sand, concrete, or debris)
3	Direct load: Solids, large objects (e.g., metal cans containing waste)

2.9.15 SC-55G2 Shielded Container

The SC-55G2 shielded container is authorized for transport within the HalfPACT ([Figure 2.9-30](#)). [Appendix 1.3.1](#) of the [HalfPACT SAR](#)³ and [Appendix 4.10](#) of the [CH-TRU Payload Appendices](#)¹ provide detailed descriptions of this payload container and associated dunnage and pallet (Drawing 163-013). The HalfPACT will accommodate one SC-55G2 [Figure 2.9-31](#)). When assembled in the HalfPACT, the SC-55G2 is surrounded by an energy absorbing dunnage assembly (lateral). The approximate dimensions of the SC-55G2 are given in [Table 2.9-58](#).

Table 2.9-58 – SC-55G2 Shielded Container Dimensions

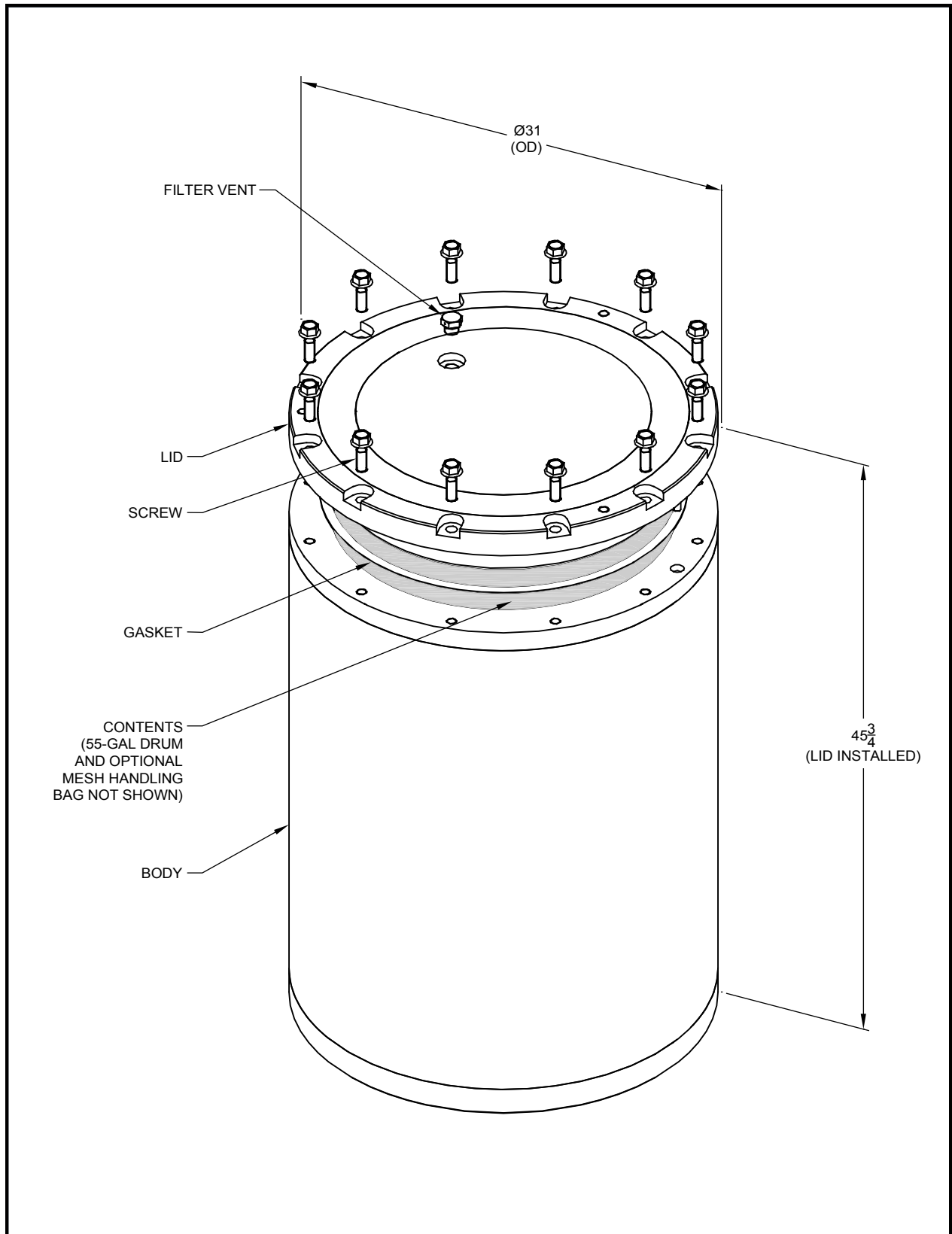
Dimension	Approximate Measurement (inches)	
	Inside Dimension	Outside Dimension
Height	35 ³ / ₄	45 ³ / ₄
Diameter	25	31

The SC-55G2 body side wall is constructed of 2-in. nominal lead and 1-in. nominal steel. The SC-55G2 lid is constructed of 1½-in. nominal lead and 4¼-in. nominal steel. The SC-55G2 base is constructed of 1¾-in. nominal lead and 2½-in. nominal steel. The lid is bolted on with 12, ¾-in. Grade 8 closure bolts. A gasket of silicone rubber is required for lid closure. The SC-55G2 and its 55-gallon steel drum contents must be vented as specified in [Section 2.5](#). The SC-55G2 lid also incorporates a ½-in. thick lead shield ring. [Table 2.9-59](#) presents the SC-55G2 materials of construction.

Table 2.9-59 – SC-55G2 Shielded Container Materials of Construction

SC-55G2 Component	Material
Body flange	Carbon steel
Lid, body base, and body side wall	Lead and carbon steel
Lid shield ring	Lead
Gasket	Silicone rubber

[Table 2.9-60](#) specifies the weights associated with the SC-55G2 that are applicable to shipment within the HalfPACT.

**Figure 2.9-30 – SC-55G2 Shielded Container**

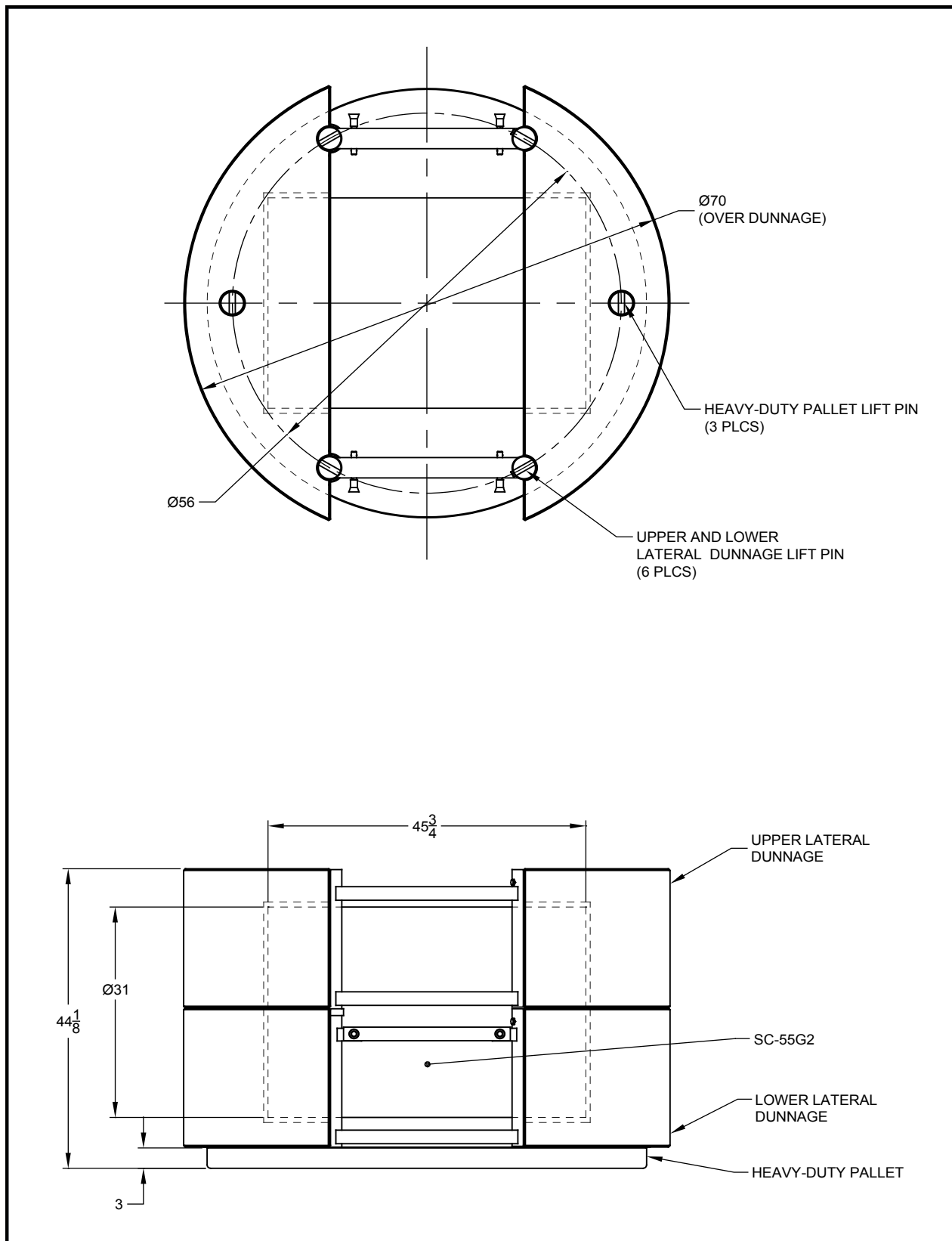


Figure 2.9-31 – HalfPACT: One SC-55G2 Shielded Container Payload

Table 2.9-60 – SC-55G2 Shielded Container Weights

Component	Weight (pounds)	
	Approximate Empty	Maximum Gross
SC-55G2	5,900	6,500

CH-TRU waste must be loaded into a vented 55-gallon steel drum with an inside lever lock, which is then loaded into the SC-55G2 with the aid of an optional mesh handling bag.

[Table 2.9-61](#) identifies material content forms authorized for transport within a SC-55G2.

Table 2.9-61 – SC-55G2 Shielded Container: Material Content Forms Authorized for Transport

Form Number	Description
1	Direct load: Solids, any particle size (e.g., fine powder or inorganic particulates)
2	Direct load: Solids, large particle size (e.g., sand, concrete, or debris)
3	Direct load: Solids, large objects (e.g., metal cans containing waste)

3.0 NUCLEAR PROPERTIES REQUIREMENTS

3.1 Nuclear Criticality

3.1.1 Requirements

Each payload container and associated package shall be acceptable for transport only if the Pu-239 fissile gram equivalent (FGE) plus two times the measurement error (i.e., two standard deviations) is less than or equal to the limits specified in [Table 3.1-1](#). For payloads taking credit for Pu-240 poisoning, the minimum Pu-240 content for the payload shall be determined after the subtraction of two times the measurement error (i.e., two standard deviations).

A payload composed of both 55-gallon drums and standard pipe overpack payload containers shall be acceptable for transport only if the Pu-239 FGE plus two times the measurement error (i.e., two standard deviations) is less than or equal to the applicable 55-gallon drum payload limit.

If payload containers with different properties (e.g., machine-compacted and non-machine-compacted waste) are assembled in the same payload, the most-restrictive payload container and payload assembly limits and requirements shall apply to all payload containers.

Table 3.1-1 – Summary of FGE Limits

FGE Limits with No Credit for Pu-240 Poisoning				
Contents[®]	Payload Container	Fissile Limit per Payload Container (Pu-239 FGE)	Fissile Limit per TRUPACT-II Package (Pu-239 FGE)	Fissile Limit per HalfPACT Package (Pu-239 FGE)
Not machine compacted with "negative" special reflector status	Drum	325	325	325
	Pipe Overpack	200	2,800	1,400
	SWB	325	325	325
	TDOP	325	325	NA [®]
	Shielded Container	325	NA [®]	325
	CCO	380	5,320	2,660
Not machine compacted with "positive" special reflector status	Drum	100	100	100
	Pipe Overpack	140	1,960	980
	SWB	100	100	100
	TDOP	100	100	NA [®]
	Shielded Container	Unauthorized	NA [®]	Unauthorized
	CCO	Unauthorized	Unauthorized	Unauthorized
Machine compacted with "negative" special reflector status	Drum	250	250	250
	Pipe Overpack	200	2,800	1,400
	SWB	250	250	250
	TDOP	250	250	NA [®]
	Shielded Container	245	NA [®]	245
	CCO	Unauthorized	Unauthorized	Unauthorized
Machine compacted with controls and "negative" special reflector status	Drum [®]	200	325	325
	Pipe Overpack	Unauthorized	Unauthorized	Unauthorized
	SWB	Unauthorized	Unauthorized	Unauthorized
	TDOP	Unauthorized	Unauthorized	NA [®]
	Shielded Container	Unauthorized	NA [®]	Unauthorized
	CCO [®]	380	5,320	2,660
Machine compacted with "positive" special reflector status	Drum	Unauthorized	Unauthorized	Unauthorized
	Pipe Overpack	Unauthorized	Unauthorized	Unauthorized
	SWB	Unauthorized	Unauthorized	Unauthorized
	TDOP	Unauthorized	Unauthorized	NA [®]
	Shielded Container	Unauthorized	NA [®]	Unauthorized
	CCO	Unauthorized	Unauthorized	Unauthorized
FGE Limits with Credit for Pu-240 Poisoning				
Contents[®]	Minimum Pu-240 Content in Payload (grams)	Fissile Material Limit per Payload Container/Package (Pu-239 FGE)		
Not machine compacted with "negative" special reflector status in drums, SWB(s), or TDOP	5	340		
	15	360		
	25	380		

Notes:

- ① Special reflector materials are identified in [Table 6.2-1](#) of the [TRUPACT-II](#) and [HalfPACT SARs](#) and the methodology for establishing special reflector status is defined in [Section 6.2.8](#) and [Section 6.2.10](#) of the [TRUPACT-II](#) and [HalfPACT SARs](#), respectively.
- ② Not applicable.
- ③ The contents shall be machine-compacted waste in the form of “puck” drums and meeting the following conditions:
 - The packing fraction of puck drum contents shall not be greater than 70%, the minimum combined thickness of steel separating the fissile masses in two axially adjacent overpack drums shall be 0.12 inch, and, for TRUPACT-II payloads, the payload assembly shall utilize a polyethylene slipsheet and reinforcing plate, each with a nominal thickness of 0.15 inch, that axially separates the top and bottom tier of payload containers, or
 - One compacted puck drum spacer shall be used in the bottom of each drum packaging the puck drums as per the specifications in [Appendix 1.3.1](#) of the [TRUPACT-II SAR](#)¹ and [Appendix 1.3.1](#) of the [HalfPACT SAR](#)² and, for TRUPACT-II payloads, the payload assembly shall utilize a polyethylene slipsheet and reinforcing plate, each with a nominal thickness of 0.15 inch, that axially separates the top and bottom tier of payload containers.
 - Compliance with these conditions shall be documented in site-specific documents.
- ④ Limited to a maximum of 2,000 grams of plastic per CCO.

3.1.2 Methods of Compliance and Verification

Compliance with the FGE requirements involves the following steps:

- Determination of the isotopic composition
- Determination of the quantity of radionuclides
- Calculation of the FGE and compliance evaluation.

Each of these steps is discussed in detail below.

Isotopic Composition

The isotopic composition of the waste may be determined from direct measurements taken on the product material during the processing or post-process certification at each site, analysis of the waste, or from existing records. The isotopic composition of the waste need not be determined by direct analysis or measurement of the waste unless process information is not available.

Pu-239 FGE for other fissile or fissionable isotopes, including special actinide elements, shall be obtained using the American National Standards Institute (ANSI)/American Nuclear Society (ANS) method ANSI/ANS-8.1³, ANSI/ANS-8.15⁴, or an equivalent method.⁵ Decay heat and

¹ U.S. Department of Energy (DOE), [TRUPACT-II Shipping Package Safety Analysis Report](#), USNRC Certificate of Compliance 71-9218, U.S. Department of Energy, Carlsbad Area Office, Carlsbad, New Mexico.

² U.S. Department of Energy (DOE), [HalfPACT Shipping Package Safety Analysis Report](#), USNRC Certificate of Compliance 71-9279, U.S. Department of Energy, Carlsbad Area Office, Carlsbad, New Mexico.

³ American National Standards Institute/American Nuclear Society (ANSI/ANS), *Nuclear Criticality Safety in Operations with Fissionable Materials Outside Reactors*, ANSI/ANS-8.1-2014, American National Standards Institute/American Nuclear Society, Washington, D.C.

specific activity shall be obtained using Radcalc⁶, or an equivalent nuclear properties database. [Table 3.1-2](#) lists the Pu-239 FGE, as well as the decay heat and specific activity, of many radionuclides.

The following are two examples of analytical methods for determining isotopic composition:

- Mass spectrometry (MS)
- Gamma ray pulse height analysis.

Depending on the mixture of radionuclides present in the waste, one or more methods may be required. These assay methods are described in [Appendix 5.2](#) of the [CH-TRU Payload Appendices](#).⁷

MS is a primary method for determining the radioisotopic composition in product material (e.g., plutonium isotopic composition). The isotope analyses shall be performed in accordance with the following American Society for Testing and Materials (ASTM) MS methods: ASTM C 696, ASTM C 697, and ASTM C 759^{8,9,10}, or equivalent methods.

Gamma ray pulse height analysis or MS is used to determine the isotopic composition for gamma-emitting radionuclides. Gamma ray pulse height analysis shall be performed in accordance with ASTM C 1030¹¹, or an equivalent method.

Quantity of Radionuclides

The quantity of the radionuclides in each payload container shall be estimated by either a direct measurement or records of the individual payload container, a summation of assay results from individual packages in a payload container, or a direct measurement on a representative sample of a waste stream (such as solidified inorganics). An assay refers to one of several radiation measurement techniques that determine the quantity of nuclear material in TRU wastes. Assay instruments detect and quantify the primary radiation (alpha, gamma, and/or neutron) emanating

⁴ American National Standards Institute/American Nuclear Society (ANSI/ANS), *Nuclear Criticality Control of Special Actinide Elements*, ANSI/ANS-8.15-2014, American National Standards Institute/American Nuclear Society, Washington, D.C.

⁵ Larson, S., *Fissile Gram Equivalent (FGE) Conversion Factors for use at WIPP*, Atkins-NS-DAC-NWP-17-01 Rev. 0, Atkins Nuclear Solutions, July 2017.

⁶ EnergySolutions Federal Services, Inc., *Radcalc 4.1 User Manual*, FSWO-SQA-026, Revision 0, May 2009, EnergySolutions Federal Services, Inc., Richland, Washington.

⁷ U.S. Department of Energy (DOE), [CH-TRU Payload Appendices](#), U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico.

⁸ American Society for Testing and Materials (ASTM), *Standard Test Methods for Chemical, Mass Spectrometric, and Spectrochemical Analysis of Nuclear-Grade Uranium Dioxide Powders and Pellets*, ASTM C 696, American Society for Testing and Materials, Washington, D.C.

⁹ American Society for Testing and Materials (ASTM), *Standard Test Methods for Chemical, Mass Spectrometric, and Spectrochemical Analysis of Nuclear-Grade Plutonium Dioxide Powders and Pellets*, ASTM C 697, American Society for Testing and Materials, Washington, D.C.

¹⁰ American Society for Testing and Materials (ASTM), *Standard Test Methods for Chemical Mass Spectrometric, and Spectrochemical, Nuclear, and Radiochemical Analysis of Nuclear-Grade Plutonium Nitrate Solutions*, ASTM C 759, American Society for Testing and Materials, Washington, D.C.

¹¹ American Society for Testing and Materials (ASTM), *Standard Test Method for Determination of Plutonium Isotopic Composition by Gamma-Ray Spectrometry*, ASTM C 1030, American Society for Testing and Materials, Washington, D.C.

from specific radionuclides, or a secondary radiation emitted from neutron interrogation techniques. The measured quantity of radiation is then used to calculate the quantity of other radionuclides and the total quantity of Pu-239 FGE. That calculation requires knowledge of the isotopic composition of the waste. Combinations of gamma spectroscopy and neutron measurements are often needed to calculate the quantity of nonfissile radionuclides.

The following are five examples of assay methods for quantifying radionuclides in TRU waste:

- Passive gamma (hyper-pure germanium, germanium, lithium-drifted germanium, sodium iodide: transmission-corrected and noncorrected)
- Radiochemical assay (alpha and gamma spectroscopy)
- Passive neutron coincidence counting assay
- Passive-active neutron assay
- Calorimetry.

These assay methods are described in [Appendix 5.2](#) of the [CH-TRU Payload Appendices](#).⁷

General assay requirements that apply to all sites are as follows:

- Each site shall select and use the assay method(s) of its choice, provided the method(s) is/are approved by the DOE-CBFO under the programmatic or waste-specific data TRAMPAC (see [Section 1.4](#)) and the prescribed controls are implemented.
- The site's waste content code descriptions shall list the specific assay method(s) and its/their application(s).
- Site/equipment-specific operating and QA procedures or the waste-specific data package shall describe the assay method(s) and the controls imposed on the assay operations. The controls include performing calibration and background measurements. The calibration and background measurements shall fall within the stated acceptable ranges before assays are performed.
- Site/equipment-specific QA plans and procedures or the QA associated with the waste-specific data package shall include oversight of assay methods and controls.
- Each site shall provide a specialized training program for assay operators.

Calculation of the FGE and Compliance Evaluation

The FGE of each payload container shall be calculated from the isotopic composition and quantity of radionuclides. The FGE value plus two times the measurement error (i.e., two standard deviations) shall be less than or equal to the applicable limit for each payload container. The Pu-239 FGE of each payload container shall be recorded in the PCTCD and OPCTCD, as applicable (see [Section 6.2.2](#)).

The total Pu-239 FGE for a payload shall be calculated and recorded in the PATCD ([Section 6.2.2](#)). The total Pu-239 FGE error is the square root of the sum of the squares of twice the individual Pu-239 FGE errors. This error shall be added to the FGE of the payload and compared to the limit.

Table 3.1-2 – Pu-239 Fissile Gram Equivalent, Decay Heat, and Specific Activity of Many Radionuclides

Nuclide		Atomic Number	Pu-239 FGE [®]	Decay Heat [®] (W/g)	Specific [®] Activity (Ci/g)
H	3	1	0.00E+00	3.25E-01	9.61E+03
Be	10	4	0.00E+00	3.34E-05	2.23E-02
C	14	6	0.00E+00	1.31E-03	4.48E+00
Na	22	11	0.00E+00	8.84E+01	6.25E+03
Al	26	13	0.00E+00	3.50E-04	1.92E-02
P	32	15	0.00E+00	1.18E+03	2.86E+05
P	33	15	0.00E+00	7.07E+01	1.56E+05
S	35	16	0.00E+00	1.24E+01	4.28E+04
Cl	36	17	0.00E+00	5.35E-05	3.30E-02
K	40	19	0.00E+00	2.55E-08	7.07E-06
Ca	45	20	0.00E+00	8.15E+00	1.78E+04
Sc	46	21	0.00E+00	4.26E+02	3.39E+04
V	49	23	0.00E+00	2.17E-01	8.08E+03
Cr	51	24	0.00E+00	2.00E+01	9.25E+04
Mn	54	25	0.00E+00	3.86E+01	7.76E+03
Fe	55	26	0.00E+00	7.99E-02	2.38E+03
Fe	59	26	0.00E+00	3.86E+02	4.98E+04
Co	57	27	0.00E+00	7.18E+00	8.44E+03
Co	58	27	0.00E+00	1.91E+02	3.18E+04
Co	60	27	0.00E+00	1.74E+01	1.13E+03
Ni	59	28	0.00E+00	3.39E-06	7.98E-02
Ni	63	28	0.00E+00	5.83E-03	5.65E+01
Cu	64	29	0.00E+00	7.23E+03	3.86E+06
Zn	65	30	0.00E+00	2.88E+01	8.24E+03
As	73	33	0.00E+00	5.43E+00	2.23E+04
Se	75	34	0.00E+00	3.50E+01	1.46E+04
Se	79	34	0.00E+00	1.36E-06	4.12E-03
Kr	85	36	0.00E+00	5.35E-01	3.92E+02
Rb	86	37	0.00E+00	3.63E+02	8.15E+04
Rb	87	37	0.00E+00	4.19E-11	8.66E-08
Sr	85	38	0.00E+00	7.41E+01	2.37E+04
Sr	89	38	0.00E+00	1.01E+02	2.90E+04
Sr	90	38	0.00E+00	1.42E-01	1.38E+02

Nuclide		Atomic Number	Pu-239 FGE ^②	Decay Heat [®] (W/g)	Specific [®] Activity (Ci/g)
Y	88	39	0.00E+00	2.23E+02	1.39E+04
Y	90	39	0.00E+00	3.01E+03	5.44E+05
Y	90m	39	0.00E+00	4.41E+04	1.09E+07
Y	91	39	0.00E+00	8.86E+01	2.45E+04
Zr	88	40	0.00E+00	4.31E+01	1.79E+04
Zr	90m	40	0.00E+00	2.13E+09	1.55E+11
Zr	93	40	0.00E+00	2.87E-07	2.52E-03
Zr	95	40	0.00E+00	1.09E+02	2.15E+04
Nb	93m	41	0.00E+00	4.37E-02	2.39E+02
Nb	94	41	0.00E+00	1.96E-03	1.90E-01
Nb	95	41	0.00E+00	1.89E+02	3.93E+04
Nb	95m	41	0.00E+00	5.64E+02	3.81E+05
Tc	99	43	0.00E+00	8.55E-06	1.69E-02
Tc	99m	43	0.00E+00	4.44E+03	5.27E+06
Ru	103	44	0.00E+00	1.08E+02	3.23E+04
Ru	106	44	0.00E+00	1.98E-01	3.31E+03
Rh	103m	45	0.00E+00	7.24E+03	3.26E+07
Rh	106	45	0.00E+00	3.40E+07	3.55E+09
Pd	107	46	0.00E+00	2.84E-08	5.14E-04
Ag	108	47	0.00E+00	2.70E+06	7.26E+08
Ag	108m	47	0.00E+00	7.73E-02	7.93E+00
Ag	109m	47	0.00E+00	1.35E+06	2.62E+09
Ag	110	47	0.00E+00	3.00E+07	4.18E+09
Ag	110m	47	0.00E+00	7.99E+01	4.76E+03
Cd	109	48	0.00E+00	3.20E-01	2.59E+03
Cd	113m	48	0.00E+00	2.47E-01	2.25E+02
Cd	115m	48	0.00E+00	9.63E+01	2.55E+04
In	114	49	0.00E+00	6.32E+06	1.38E+09
In	114m	49	0.00E+00	3.12E+01	2.29E+04
In	115m	49	0.00E+00	1.21E+04	6.08E+06
Sn	113	50	0.00E+00	1.76E+00	1.00E+04
Sn	119m	50	0.00E+00	1.99E+00	3.75E+03
Sn	121m	50	0.00E+00	1.29E-02	5.38E+01
Sn	123	50	0.00E+00	2.58E+01	8.22E+03
Sn	126	50	0.00E+00	1.31E-05	1.23E-02
Sb	124	51	0.00E+00	2.32E+02	1.75E+04

Nuclide		Atomic Number	Pu-239 FGE [®]	Decay Heat [®] (W/g)	Specific [®] Activity (Ci/g)
Sb	125	51	0.00E+00	3.30E+00	1.04E+03
Sb	126	51	0.00E+00	1.54E+03	8.36E+04
Sb	126m	51	0.00E+00	1.10E+06	8.15E+09
Te	123	52	0.00E+00	4.24E-19	3.16E-14
Te	123m	52	0.00E+00	1.28E+01	8.89E+03
Te	125m	52	0.00E+00	1.52E+01	1.82E+04
Te	127	52	0.00E+00	3.59E+03	2.64E+06
Te	127m	52	0.00E+00	5.24E+00	9.44E+03
Te	129	52	0.00E+00	7.52E+04	2.10E+07
Te	129m	52	0.00E+00	5.37E+01	3.01E+04
I	125	53	0.00E+00	6.13E+00	1.76E+04
I	129	53	0.00E+00	8.87E-08	1.72E-04
I	131	53	0.00E+00	4.23E+02	1.24E+05
Cs	134	55	0.00E+00	1.32E+01	1.29E+03
Cs	135	55	0.00E+00	6.11E-07	1.15E-03
Cs	137	55	0.00E+00	9.68E-02	8.69E+01
Ba	133	56	0.00E+00	6.89E-01	2.55E+02
Ba	137m	56	0.00E+00	2.10E+06	5.38E+08
Ce	141	58	0.00E+00	4.14E+01	2.85E+04
Ce	142	58	0.00E+00	4.23E-16	5.04E-14
Ce	144	58	0.00E+00	2.08E+00	3.18E+03
Pr	143	59	0.00E+00	1.26E+02	6.74E+04
Pr	144	59	0.00E+00	5.53E+05	7.56E+07
Pr	144m	59	0.00E+00	6.83E+04	1.89E+08
Pm	146	61	0.00E+00	2.23E+00	4.43E+02
Pm	147	61	0.00E+00	3.41E-01	9.28E+02
Pm	148	61	0.00E+00	1.27E+03	1.64E+05
Pm	148m	61	0.00E+00	2.75E+02	2.15E+04
Sm	146	62	0.00E+00	3.73E-07	2.45E-05
Sm	147	62	0.00E+00	3.14E-10	2.30E-08
Sm	151	62	0.00E+00	3.10E-03	2.63E+01
Eu	150	63	0.00E+00	6.05E-01	6.56E+01
Eu	152	63	0.00E+00	1.33E+00	1.74E+02
Eu	154	63	0.00E+00	2.44E+00	2.70E+02
Eu	155	63	0.00E+00	3.58E-01	4.86E+02
Gd	152	64	0.00E+00	2.85E-13	2.18E-11

Nuclide		Atomic Number	Pu-239 FGE ^②	Decay Heat [®] (W/g)	Specific [®] Activity (Ci/g)
Gd	153	64	0.00E+00	3.06E+00	3.55E+03
Tb	160	65	0.00E+00	9.26E+01	1.13E+04
Ho	166m	67	0.00E+00	1.88E-02	1.80E+00
Tm	168	69	0.00E+00	5.54E+01	8.35E+03
Ta	182	73	0.00E+00	5.56E+01	6.26E+03
Au	198	79	0.00E+00	1.06E+03	2.45E+05
Hg	203	80	0.00E+00	2.75E+01	1.38E+04
Tl	204	81	0.00E+00	6.53E-01	4.63E+02
Tl	207	81	0.00E+00	5.59E+05	1.90E+08
Tl	208	81	0.00E+00	6.97E+06	2.96E+08
Tl	209	81	0.00E+00	6.81E+06	4.09E+08
Pb	209	82	0.00E+00	5.39E+03	4.61E+06
Pb	210	82	0.00E+00	2.07E-02	7.68E+01
Pb	211	82	0.00E+00	7.57E+04	2.47E+07
Pb	212	82	0.00E+00	2.58E+03	1.39E+06
Pb	214	82	0.00E+00	1.03E+05	3.28E+07
Bi	207	83	0.00E+00	5.35E-01	5.44E+01
Bi	210	83	0.00E+00	2.86E+02	1.24E+05
Bi	211	83	0.00E+00	1.64E+07	4.11E+08
Bi	212	83	0.00E+00	2.46E+05	1.47E+07
Bi	213	83	0.00E+00	8.06E+04	1.94E+07
Bi	214	83	0.00E+00	5.68E+05	4.42E+07
Po	209	84	0.00E+00	4.94E-01	1.68E+01
Po	210	84	0.00E+00	1.44E+02	4.49E+03
Po	211	84	0.00E+00	4.67E+09	1.04E+11
Po	212	84	0.00E+00	9.48E+15	1.79E+17
Po	213	84	0.00E+00	6.38E+14	1.26E+16
Po	214	84	0.00E+00	1.50E+13	3.22E+14
Po	215	84	0.00E+00	1.32E+12	2.95E+13
Po	216	84	0.00E+00	1.43E+10	3.48E+11
Po	218	84	0.00E+00	1.01E+07	2.78E+08
At	211	85	0.00E+00	3.10E+04	2.06E+06
At	217	85	0.00E+00	6.87E+10	1.61E+12
Rn	219	86	0.00E+00	5.36E+08	1.30E+10
Rn	220	86	0.00E+00	3.49E+07	9.19E+08
Rn	222	86	0.00E+00	5.10E+03	1.54E+05

Nuclide		Atomic Number	Pu-239 FGE [®]	Decay Heat [®] (W/g)	Specific [®] Activity (Ci/g)
Fr	221	87	0.00E+00	6.70E+06	1.74E+08
Fr	223	87	0.00E+00	1.01E+05	3.87E+07
Ra	223	88	0.00E+00	1.82E+03	5.12E+04
Ra	224	88	0.00E+00	5.50E+03	1.60E+05
Ra	225	88	0.00E+00	2.83E+01	3.92E+04
Ra	226	88	0.00E+00	2.85E-02	9.89E-01
Ra	228	88	0.00E+00	3.79E-02	2.73E+02
Ac	225	89	0.00E+00	2.03E+03	5.80E+04
Ac	227	89	0.00E+00	3.63E-02	7.23E+01
Ac	228	89	0.00E+00	1.86E+04	2.23E+06
Th	227	90	0.00E+00	1.12E+03	3.07E+04
Th	228	90	0.00E+00	2.68E+01	8.20E+02
Th	229	90	0.00E+00	6.50E-03	2.13E-01
Th	230	90	0.00E+00	5.82E-04	2.06E-02
Th	231	90	0.00E+00	6.01E+02	5.32E+05
Th	232	90	0.00E+00	2.66E-09	1.10E-07
Th	234	90	0.00E+00	9.52E+00	2.32E+04
Pa	231	91	0.00E+00	1.44E-03	4.72E-02
Pa	233	91	0.00E+00	5.08E+01	2.08E+04
Pa	234	91	0.00E+00	2.13E+04	1.97E+06
Pa	234m	91	0.00E+00	3.40E+06	6.87E+08
U	232	92	4.50E-01	7.09E-01	2.21E+01
U	233	92	9.00E-01	2.81E-04	9.63E-03
U	234	92	7.63E-03	1.79E-04	6.22E-03
U	235	92	6.43E-01	5.99E-08	2.16E-06
U	236	92	0.00E+00	1.74E-06	6.39E-05
U	237	92	0.00E+00	1.66E+02	8.16E+04
U	238	92	0.00E+00	8.53E-09	3.36E-07
U	239	92	0.00E+00	9.17E+04	3.35E+07
U	240	92	0.00E+00	8.49E+02	9.26E+05
Np	237	93	1.29E-02	2.06E-05	7.05E-04
Np	238	93	0.00E+00	1.35E+03	2.59E+05
Np	239	93	0.00E+00	6.12E+02	2.32E+05
Np	240	93	0.00E+00	1.22E+05	1.21E+07
Np	240m	93	0.00E+00	6.40E+05	1.06E+08
Pu	236	94	7.50E-01	1.84E+01	5.30E+02

Nuclide		Atomic Number	Pu-239 FGE ^②	Decay Heat ^③ (W/g)	Specific ^③ Activity (Ci/g)
Pu	238	94	8.82E-02	5.68E-01	1.71E+01
Pu	239	94	1.00E+00	1.93E-03	6.20E-02
Pu	240	94	2.25E-02	7.07E-03	2.27E-01
Pu	241	94	2.43E+00	3.29E-03	1.03E+02
Pu	242	94	8.18E-03	1.17E-04	3.95E-03
Pu	243	94	0.00E+00	3.06E+03	2.60E+06
Pu	244	94	0.00E+00	5.31E-07	1.83E-05
Am	241	95	1.88E-02 ^②	1.14E-01	3.43E+00
Am	242	95	0.00E+00	9.54E+02	8.07E+05
Am	242m	95	4.09E+01	4.65E-03	1.05E+01
Am	243	95	6.92E-03 ^②	6.44E-03	2.00E-01
Am	245	95	0.00E+00	1.16E+04	6.24E+06
Cm	240	96	0.00E+00	7.61E+02	2.01E+04
Cm	242	96	7.50E-02 ^②	1.22E+02	3.31E+03
Cm	243	96	5.00E+00	1.81E+00	4.90E+01
Cm	244	96	4.09E-02 ^②	2.85E+00	8.14E+01
Cm	245	96	1.96E+01	5.72E-03	1.72E-01
Cm	246	96	2.81E-02 ^②	1.01E-02	3.07E-01
Cm	247	96	9.00E-01	2.87E-06	9.04E-05
Cm	248	96	0.00E+00	5.12E-04	4.24E-03
Cm	250	96	0.00E+00	1.42E-01	1.79E-01
Bk	247	97	0.00E+00	3.65E-02	1.05E+00
Bk	249	97	0.00E+00	3.22E-01	1.64E+03
Bk	250	97	0.00E+00	2.78E+04	3.90E+06
Cf	249	98	4.50E+01 ^②	1.52E-01	4.09E+00
Cf	250	98	0.00E+00	4.07E+00	1.09E+02
Cf	251	98	9.00E+01	5.81E-02	1.59E+00
Cf	252	98	0.00E+00	3.82E+01	5.36E+02
Cf	254	98	0.00E+00	9.22E-01	8.49E+03
Es	253	99	0.00E+00	1.01E+03	2.52E+04
Es	254	99	0.00E+00	7.27E+01	1.86E+03
Es	254m	99	0.00E+00	1.45E+03	3.14E+05

Notes:

- ① ANSI/ANS-8.1-2014³.
 ② ANSI/ANS-8.15-2014⁴.
 ③ Radcalc⁶.

- ④ These factors are only valid if ^{242m}Am content is ≤ 0.11 g (1% of subcritical mass); otherwise, a Pu-239 conversion factor of $4.09\text{E}+01$ must be used.⁵
- ⑤ These factors are only valid if ^{243}Cm content is ≤ 0.9 g, ^{245}Cm content is ≤ 0.23 g, and ^{247}Cm content is ≤ 5 g (1% of subcritical mass); otherwise, a Pu-239 conversion factor of $1.96\text{E}+01$ must be used.⁵
- ⑥ This factor is only valid if ^{251}Cf content is ≤ 0.05 g (1% of subcritical mass); otherwise, a Pu-239 conversion factor of $9.00\text{E}+01$ must be used.⁵

3.2 Radiation Dose Rates

3.2.1 Requirements

The external radiation surface dose rate of an individual payload container shall be less than or equal to the limits specified in [Table 3.2-1](#).

Table 3.2-1 – Maximum Surface Dose Rate Limits

Payload Container	Maximum Surface Dose (mrem/hr)
55-Gallon Drum	200
Standard Pipe Overpack	200
S100 Pipe Overpack	179
S200 Pipe Overpack	200
S300 Pipe Overpack	155
85-Gallon Drum	200
100-Gallon Drum	200
SWB	200
TDOP	200
SC-30G1 Shielded Container	200
SC-30G2 Shielded Container	200
SC-30G3 Shielded Container	200
SC-55G1 Shielded Container	200
SC-55G2 Shielded Container	200
CCO	200

The external radiation dose rates of the TRUPACT-II and the HalfPACT shall be less than or equal to 200 millirem per hour (mrem/hr) at the surface and less than or equal to 10 mrem/hr at 2 meters.

3.2.2 Methods of Compliance and Verification

The payload container surface dose rate shall be measured and compliance with the applicable limit recorded on the PCTCD or OPCTCD (see [Section 6.2.1](#)). Measurements shall be made with instruments traceable to a national standard. The dose rate for the TRUPACT-II and HalfPACT at the surface and at 2 meters shall be measured and compliance determined and documented in accordance with site procedures.

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3.3 Activity Limits

3.3.1 Requirements

3.3.1.1 Generic, Standard Pipe Overpack, Shielded Containers, and Criticality Control Overpack

The TRUPACT-II and HalfPACT are limited to a specific maximum total activity when packaging payloads of either Generic (55-gallon drums, 85-gallon drums, 100-gallon drums, SWBs, and TDOP), 6-in. standard pipe overpacks, 12-in standard pipe overpacks, SC-30G1 shielded containers, SC-30G2 shielded containers, SC-30G3 shielded container, SC-55G1 shielded containers, SC-55G2 shielded container, or CCOs. Each payload shall be acceptable for shipment only if the determined activity plus the error (i.e., one standard deviation) meets the specific limit as determined by the procedure provided in [Section 5.5.10](#) of the [TRUPACT-II SAR](#)¹.

3.3.1.2 S100 and S300 Pipe Overpack

The TRUPACT-II and HalfPACT are limited to a maximum total activity of 406 curies (Ci) when packaging payloads of either S100 or S300 pipe overpacks. The S100 and S300 pipe overpack payloads are limited to sealed neutron sources in the forms specified in [Table 4.2-1](#) of [Appendix 4.2](#) and [Table 4.4-1](#) of [Appendix 4.4](#), respectively, of the [CH-TRU Payload Appendices](#)⁷. Each payload shall be acceptable for shipment only if the determined activity plus the error (i.e., one standard deviation) meets this limit.

3.3.1.3 S200 Pipe Overpack

The contents of each S200 pipe overpack are limited to the radionuclides and associated activities listed in [Table 4.3-2](#) of [Appendix 4.3](#) of the [CH-TRU Payload Appendices](#) for both the S200-A and S200-B shield insert configurations. Each S200 pipe overpack shall be acceptable for shipment only if the determined activity plus the error (i.e., one standard deviation) meets the applicable limit.

3.3.1.4 $10^5 A_2$

As described in [Section 2.7.7](#), *Deep Water Immersion*, of the TRUPACT-II and HalfPACT SARs, a payload shall be acceptable for transport only if the activity plus error (i.e., one standard deviation) is less than or equal to $10^5 A_2$ curies. A_2 values are defined in 10 CFR §71.

3.3.2 Methods of Compliance and Verification

Compliance with the activity requirements is similar to the compliance methodology described in [Section 3.1.2](#). The activity of the payload shall be calculated from the isotopic composition and quantity of radionuclides comprising the payload.

3.3.2.1 Generic, Standard Pipe Overpack, Shielded Containers, and Criticality Control Overpack

The combined sum of “partial fractions” for the gamma and neutron source terms present in the package must be less than or equal to 0.9 (0.89 for the CCO, SC-30G2, SC-30G3, and SC-55G2 payload container configurations), or

$$\sum_{i=1}^m \frac{S_{G_i}}{S_{G_i}} + \sum_{j=1}^n \frac{S_{N_j}}{S_{N_j}} \leq 0.9 \text{ (or 0.89)}$$

where, for a particular payload container mix, s_{G_i} is the actual gamma source strength of the discrete gamma energy “i”, S_{G_i} is the allowable activity of the discrete gamma energy “i”, s_{N_j} is the actual neutron source strength of the neutron energy interval “j”, and S_{N_j} is the allowable activity of the neutron energy interval “j”. The limits for discrete gamma energies and neutron energy intervals are determined by the procedure provided in [Section 5.5.10](#) of the [TRUPACT-II SAR](#)¹.

3.3.2.2 S100 and S300 Pipe Overpack

The total payload activity plus error (i.e., one standard deviation) shall be used to determine compliance with the 406 Ci payload activity limit.

3.3.2.3 S200 Pipe Overpack

The sum of “partial fractions” for any combination of radionuclides present in each S200 pipe overpack must be less than or equal to one, or

$$\sum_{i=1}^n \frac{a_i}{A_{GN_i}} \leq 1$$

where, for a particular payload container mix, a_i is the actual curie content of isotope “i” and A_{GN_i} is the limiting curie content of radionuclide “i”. The limits for individual radionuclides for each S200 pipe overpack are specified in [Table 4.3-2](#) of [Appendix 4.3](#) of the [CH-TRU Payload Appendices](#).

3.3.2.4 $10^5 A_2$

The total payload activity plus error (i.e., one standard deviation) shall be used to determine compliance with the $10^5 A_2$ payload activity limit.

4.0 CHEMICAL PROPERTIES REQUIREMENTS

4.1 Pyrophoric Materials

As defined by 10 CFR §61.2:

A pyrophoric solid is any solid material, other than one classed as an explosive, which under normal conditions is liable to cause fires through friction, retained heat from manufacturing or processing, or which can be ignited readily and when ignited burns so vigorously and persistently as to create a serious transportation, handling, or disposal hazard. Included are spontaneously combustible and water-reactive materials.¹

Examples of pyrophoric radionuclides are metallic plutonium and americium. Examples of nonradioactive pyrophorics, or materials/wastes that may cause a pyrophoric-type event, are organic peroxides, sodium metal, and chlorates.

All waste generating sites administratively control the procurement, distribution, use, and disposal of nonradioactive pyrophoric materials. In general, pyrophoric materials are not permitted in TRU waste process areas. The quantity of pyrophoric materials that does enter any process is strictly limited and controlled by site safety considerations.

4.1.1 Requirements

Radioactive pyrophoric materials shall be present only in small residual amounts (≤ 1 percent [weight]) in payload containers. Radioactive pyrophorics in concentrations greater than 1 percent by weight and all nonradioactive pyrophorics shall be reacted (or oxidized) and/or otherwise rendered nonreactive prior to placement in the payload container.

4.1.2 Methods of Compliance and Verification

Compliance shall be by one, or a combination, of the following methods:

- Review of records and database information, which may include knowledge of process
- Administrative and procurement controls.

¹ U.S. Nuclear Regulatory Commission (NRC), *Pyrophoric definition*, Code of Federal Regulations, Title 10, Section 61.2, U.S. Nuclear Regulatory Commission, Washington, D.C.

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4.2 Explosives, Corrosives, and Compressed Gases

As defined by 49 CFR §173.50:

...an explosive means any substance or article, including a device, which is designed to function by explosion (i.e., an extremely rapid release of gas and heat) or which, by chemical reaction within itself, is able to function in a similar manner even if not designed to function by explosion, unless the substance or article is otherwise classed under the provisions of [49 CFR 173, Subpart C]. The term includes a pyrotechnic substance or article, unless the substance or article is otherwise classed under the provisions of [49 CFR 173, Subpart C].¹

Examples of explosives are ammunition, dynamite, black powder, detonators, nitroglycerine, urea nitrate, and picric acid.

As defined by 40 CFR §261.22:

- (a) A solid waste exhibits the characteristic of corrosivity if a representative sample of the waste has...the following property:
 - (1) It is aqueous and has a pH less than or equal to 2 or greater than or equal to 12.5.
- (b) A solid waste that exhibits the characteristic of corrosivity has the EPA Hazardous Waste Number of D002.²

The physical form of the waste and waste generating procedures at the sites ensure that the waste is in a nonreactive form. All waste generating sites control the procurement, distribution, use, and disposal of explosives. Most sites have lists of restricted materials that include explosives. Typically, the TRU waste generating and storage sites do not allow explosives in the same facility as TRU waste.

4.2.1 Requirements

Explosives, corrosives, and compressed gases (pressurized containers) are prohibited from the payload.

Used (i.e., empty) aerosol cans are allowed as they do not impact the package internal pressure or flammability. Verification that any aerosol cans present in retrievably stored waste are empty shall be by radiography and/or process knowledge and shall be documented in site-specific compliance documents. Any aerosol cans that are not empty are prohibited.

4.2.2 Methods of Compliance and Verification

Compliance shall be by one, or a combination, of the following methods:

- Visual examination of the waste

¹ U.S. Department of Transportation (DOT), *An Explosive; definition*, Code of Federal Regulations, Title 49, Section 173.50, U.S. Department of Transportation, Washington, D.C.

² U.S. Environmental Protection Agency (EPA), *Characteristic of corrosivity*, Code of Federal Regulations, Title 40, Section 261.22, U.S. Environmental Protection Agency, Washington, D.C.

- Administrative and procurement controls
- Radiography
- Sampling program
- Review of records and database information, which may include knowledge of process.

4.3 Chemical Composition

The chemical constituents allowed in a given waste material type (e.g., concreted inorganic particulate waste) are restricted so that a conservative bounding G value may be established for the gas generation potential in each waste material type.

Compliance with the lists of allowable materials in [Table 4.3-1](#) through [Table 4.3-8](#) has been demonstrated for each chemical list corresponding to each content code. The assignment of any content code to a waste material type will also be conservative with respect to G values. For example, if an inorganic solid waste material type (II.1) at a site contains materials that do not comply with the materials listed in [Table 4.3-4](#) (e.g., solid organics excluding packaging), it shall be classified as Waste Material Type III.1 (solid organics), which has twice the bounding G value, and the appropriate content code shall be applied. Similarly, Waste Material Type II.2 may only contain materials with no gas generation potential (G value of zero).

4.3.1 Requirements

Chemical constituents in a payload shall conform to the lists of allowable materials in [Table 4.3-1](#) through [Table 4.3-8](#). The total quantity of chemicals/materials not listed as allowed materials for a given waste material type in any payload container is restricted to less than 5 weight percent total. These materials, if present, are, in general, present as trace chemicals/materials (materials that occur individually in the waste in quantities less than 1 weight percent).

4.3.2 Methods of Compliance and Verification

Compliance shall be by one, or a combination, of the following methods:

- Review of records and database information, which may include knowledge of process
- Administrative and procurement controls
- Sampling program.

Content codes approved by the WIPP CH-TRU Payload Engineer comply with the chemical composition requirements. Any proposed change in process technology at a generator site for a given content code must be evaluated for compliance with the lists of allowable materials in [Table 4.3-1](#) through [Table 4.3-8](#). This change shall be evaluated and approved by the WIPP CH-TRU Payload Engineer for compliance with existing waste material type restrictions. All changes in the chemical characteristics of the waste shall be recorded, and the date of the new process, description of the process, and list of new chemicals submitted to the WIPP CH-TRU Payload Engineer. The WIPP CH-TRU Payload Engineer may allow transport of the waste under the approved content code if none of the restrictions are violated as a result of the change. If the WIPP CH-TRU Payload Engineer determines that the old content code and corresponding waste material type(s) are no longer valid, the waste may be assigned to a new content code for shipment under the appropriate approved waste material type. The NRC shall be notified of any change not covered by the authorized contents as defined by this document (e.g., addition of a new waste form with a new G value) through an amendment to the CH-TRAMPAC. All

changes exceeding currently authorized contents shall be submitted to the NRC for review and approval prior to incorporation into a chemical list or content code.

Table 4.3-1 – Allowable Materials for Waste Material Type I.1^①
Absorbed, Adsorbed, or Solidified Inorganic Liquid

Absorbents/adsorbents (e.g., Celite [®] , diatomaceous earth, diatomite, Florco [®] , Oil-Dri [®] , perlite, vermiculite)
Acids, inorganic
Alumina cement
Aquaset [®] products (for aqueous solutions)
Aqueous sludges
Aqueous solutions/water
Asbestos
Ash (e.g., ash bottoms, fly ash, soot)
Batteries, dry (e.g., flashlight)
Ceramics (e.g., molds and crucibles)
Clays (e.g., bentonite)
Concrete
Envirostone [®] (no organic emulsifiers allowed)
Fiberglass, inorganic
Filter media, inorganic
Firebrick
Glass (e.g., borosilicate glass, labware, leaded glass, Raschig rings)
Graphite (e.g., molds and crucibles)
Grit
Heel (e.g., ash heel; soot heel; firebrick heel; sand, slag, and crucible heel)
Insulation, inorganic
Magnesia cement (e.g., Ramcote [®] cement)
Metal hydroxides
Metal oxides (e.g., slag)
Metals (e.g., aluminum, cadmium, copper, steel, tantalum, tungsten, zinc)
Nitrates (e.g., ammonium nitrate, sodium nitrate)
Petroset [®] products (for aqueous solutions)
Portland cement
Sand/soil, inorganic
Salts (e.g., calcium chloride, calcium fluoride, sodium chloride)
Other inorganic materials

Note:

- ① Other chemicals or materials not identified in this table are allowed provided that they meet the requirements of [Section 4.3.1](#). All materials in the final waste form must be inert (nonreactive), be in a nonreactive form, or have been rendered nonreactive.

Table 4.3-2 – Allowable Materials for Waste Material Type I.2^①
Soils, Solidified Particulates, or Sludges Formed from Precipitation

Absorbents/adsorbents (e.g., Celite [®] , diatomaceous earth, diatomite, Florco [®] , Oil-Dri [®] , perlite, vermiculite)
Alumina cement
Aquaset [®] products (for aqueous solutions)
Aqueous sludges
Aqueous solutions/water
Asbestos
Ash (e.g., ash bottoms, fly ash, soot)
Batteries, dry (e.g., flashlight)
Ceramics (e.g., molds and crucibles)
Clays (e.g., bentonite)
Concrete
Fiberglass, inorganic
Filter media, inorganic
Firebrick
Glass (e.g., borosilicate glass, labware, leaded glass, Raschig rings)
Graphite (e.g., molds and crucibles)
Grit
Heel (e.g., ash heel; soot heel; firebrick heel; sand, slag, and crucible heel)
Insulation, inorganic
Magnesia cement (e.g., Ramcote [®] cement)
Metal hydroxides
Metal oxides (e.g., slag)
Metals (e.g., aluminum, cadmium, copper, steel, tantalum, tungsten, zinc)
Nitrates (e.g., ammonium nitrate, sodium nitrate)
Petroset [®] products (for aqueous solutions)
Portland cement
Sand/soil, inorganic
Salts (e.g., calcium chloride, calcium fluoride, sodium chloride)
Other inorganic materials

Note:

- ① Other chemicals or materials not identified in this table are allowed provided that they meet the requirements of [Section 4.3.1](#). All materials in the final waste form must be inert (nonreactive), be in a nonreactive form, or have been rendered nonreactive.

Table 4.3-3 – Allowable Materials for Waste Material Type I.3^①
Concreted Inorganic Particulate Waste

Absorbents/adsorbents (e.g., Celite [®] , diatomaceous earth, diatomite, Florco [®] , Oil-Dri [®] , perlite, vermiculite)
Asbestos
Ash (e.g., ash bottoms, fly ash, soot)
Batteries, dry (e.g., flashlight)
Ceramics (e.g., molds and crucibles)
Clays (e.g., bentonite)
Concrete
Fiberglass, inorganic
Filter media, inorganic
Firebrick
Glass (e.g., borosilicate glass, labware, leaded glass, Raschig rings)
Graphite (e.g., molds and crucibles)
Grit
Heel (e.g., ash heel; soot heel; firebrick heel; sand, slag, and crucible heel)
Insulation, inorganic
Metal hydroxides
Metal oxides (e.g., slag)
Metals (e.g., aluminum, cadmium, copper, steel, tantalum, tungsten, zinc)
Nitrates (e.g., ammonium nitrate, sodium nitrate)
Portland cement
Sand/soil, inorganic
Salts (e.g., calcium chloride, calcium fluoride, sodium chloride)
Water (maximum of 30 weight percent unbound water)
Other inorganic materials

Note:

- ① Other chemicals or materials not identified in this table are allowed provided that they meet the requirements of [Section 4.3.1](#). All materials in the final waste form must be inert (nonreactive), be in a nonreactive form, or have been rendered nonreactive.

Table 4.3-4 – Allowable Materials for Waste Material Types II.1 and II.2^①
Solid Inorganic Materials

Absorbents/adsorbents (e.g., Celite[®], Florco[®], Oil-Dri[®], diatomite, perlite, vermiculite)^②

Asbestos

Ash (e.g., ash bottoms, fly ash, soot)

Batteries, dry (e.g., flashlight)

Ceramics (e.g., molds and crucibles)

Clays (e.g., bentonite)

Concrete/Portland cement (surface contaminated only)

Fiberglass, inorganic

Filter media, inorganic

Firebrick

Glass (e.g., borosilicate glass, labware, leaded glass, Raschig rings)

Graphite (e.g., molds and crucibles)

Grit

Heel (e.g., ash heel; soot heel; firebrick heel; sand, slag, and crucible heel)

Insulation, inorganic

Magnesium alloy

Metal oxides (e.g., slag)

Metals (e.g., aluminum, cadmium, copper, steel, tantalum, tungsten, zinc)

Nitrates (e.g., ammonium nitrate, sodium nitrate)

Salts (e.g., calcium chloride, calcium fluoride, sodium chloride)

Sand/soil, inorganic

Other inorganic materials

Notes:

- ① Other chemicals or materials not identified in this table are allowed provided that they meet the requirements of [Section 4.3.1](#). All materials in the final waste form must be inert (nonreactive), be in a nonreactive form, or have been rendered nonreactive.
- ② Dry absorbents/adsorbents and other dry desiccants are allowed if they contain no absorbed or adsorbed liquids.

Table 4.3-5 – Allowable Materials for Waste Material Type II.3^①
Homogeneous Solid Inorganic Materials with Unbound Absorbed Ambient
Moisture ($\leq 6\%$ by weight)

Any material in Waste Material Types II.1 and II.2 (Table 4.3-4) and water as unbound absorbed ambient moisture ($\leq 6\%$ by weight).
--

Note:

- ① Other chemicals or materials not identified in this table are allowed provided that they meet the requirements of [Section 4.3.1](#). All materials in the final waste form must be inert (nonreactive), be in a nonreactive form, or have been rendered nonreactive.

Table 4.3-6 – Allowable Materials for Waste Material Type III.1^①
Solid Organic Materials

Any material in Waste Types I or II (Table 4.3-1 through Table 4.3-5)
Absorbent polymers, organic
Acids, solid, organic
Asphalt
Bakelite ^{® ②}
Cellulose (e.g., Benelex [®] , cotton Conwed [®] , paper, rags, rayon, wood)
Cellulose acetate butyrate
Cellulose propionate
Chlorinated polyether
Detergent, solid (e.g., emulsifiers, surfactants)
Fiberglass, organic
Filter media, organic
Greases, commercial brands
Insulation, organic
Leaded rubber (e.g., gloves, aprons, sheet material)
Leather
Oil (e.g., petroleum, mineral)
Organophosphates (e.g., tributyl phosphate, dibutyl phosphate, monobutyl phosphite)
Paint, dry (e.g., floor/wall paint, ALARA)
Plastics [e.g., polycarbonate, polyethylene, polymethyl methacrylate (Plexiglas [®] , Lucite [®]), polysulfone, polytetrafluoroethylene (Teflon [®]), polyvinyl acetate, polyvinyl chloride, polyvinylidene chloride (saran)]
Polyamides (nylon)
Polychlorotrifluoroethylene (e.g., Kel-F [®])
Polyesters (e.g., Dacron [®] , Mylar [®])
Polyethylene glycol (e.g., Carbowax [®])
Polyimides
Polyphenyl methacrylate
Polypropylene (e.g., Ful-Flo [®] filters)
Polyurethane
Polyvinyl alcohol
Resins (e.g., aniline-formaldehyde, melamine-formaldehyde, organic resins, phenol-formaldehyde, phenolic resins, urea-formaldehyde)
Rubber, natural or synthetic [e.g., chlorosulfonated polyethylene (Hypalon [®]), ethylene-propylene rubber, EPDM, polybutadiene, polychloroprene (neoprene), polyisobutylene, polyisoprene, polystyrene, rubber hydrochloride (pliofilm [®])]
Sand/Soil
Waxes, commercial brands

Notes:

- ① Other chemicals or materials not identified in this table are allowed provided that they meet the requirements of [Section 4.3.1](#). All materials in the final waste form must be inert (nonreactive), be in a nonreactive form, or have been rendered nonreactive.
- ② Bakelite is a trademark for materials that can be composed of several different polymers, including polyethylene, polypropylene, epoxy, phenolic, polystyrene, phenoxy, perylene, polysulfone, ethylene copolymers, ABS, acrylics, and vinyl resins and compounds.

Table 4.3-7 – Allowable Materials for Waste Material Types III.2 and III.3^①
Homogeneous Mixed Organic (10% by weight) and Inorganic (90% by weight) Materials

Any material in Waste Material Types I.1, I.2, I.3, II.1, II.2, II.3, or III.1 (Table 4.3-1 through Table 4.3-6), provided that the total amount of solid organic material and/or absorbed or adsorbed water is less than or equal to 10 weight percent of the total waste.

Note:

- ① Other chemicals or materials not identified in this table are allowed provided that they meet the requirements of [Section 4.3.1](#). All materials in the final waste form must be inert (nonreactive), be in a nonreactive form, or have been rendered nonreactive.

Table 4.3-8 – Allowable Materials for Waste Material Type IV.1^①
Solidified Organics

Any material in Waste Types I, II, or III (Table 4.3-1 through Table 4.3-7)
Acids, organic
Alcohols (e.g., butanol, ethanol, isopropanol, methanol)
Esters (e.g., ethyl acetate, polyethylene glycol ester)
Ethers (e.g., ethyl ether)
Halogenated organics (e.g., bromoform; carbon tetrachloride; chlorobenzene; chloroform; 1,1-dichloroethane; 1,2-dichloroethane; 1,1-dichloroethylene; cis-1,2-dichloroethylene; methylene chloride; 1,1,2,2-tetrachloroethane; tetrachloroethylene; 1,1,1-trichloroethane; 1,1,2-trichloroethane; trichloroethylene; 1,1,2-trichloro-1,2,2-trifluoroethane)
Hydrocarbons, aliphatic (e.g., cyclohexane, n-paraffin hydrocarbons)
Hydrocarbons, aromatic (e.g., benzene; ethyl benzene; toluene; 1,2,4-trimethylbenzene; 1,3,5-trimethylbenzene; xylene)
Ketones (e.g., acetone, methyl ethyl ketone, methyl isobutyl ketone)
Trioctyl phosphine oxide

Note:

- ① Other chemicals or materials not identified in this table are allowed provided that they meet the requirements of [Section 4.3.1](#). All materials in the final waste form must be inert (nonreactive), be in a nonreactive form, or have been rendered nonreactive.

4.4 Chemical Compatibility

The lists of allowable materials in [Table 4.3-1](#) through [Table 4.3-8](#) restrict the chemical composition of the payload. The basis for evaluating chemical compatibility is the U.S. Environmental Protection Agency (EPA) document “A Method for Determining the Compatibility of Hazardous Wastes” (EPA-600/2-80-076).¹ This method provides a systematic means of analyzing the chemical compatibility for specific combinations of chemical compounds and materials. Any incompatibilities between the payload and the packaging shall be evaluated separately if not covered by the EPA method.¹ As described in [Appendix 6.1](#) of the [CH-TRU Payload Appendices](#)², the EPA method classifies individual chemical compounds into chemical groups and identifies the potential adverse reactions resulting from incompatible combinations of the groups.

4.4.1 Requirements

Chemical compatibility shall be ensured for the following four conditions:

- Chemical compatibility of the waste form within each individual payload container
- Chemical compatibility between contents of payload containers during hypothetical accident conditions
- Chemical compatibility of the waste forms with the TRUPACT-II and HalfPACT ICV
- Chemical compatibility of the waste forms with the TRUPACT-II and HalfPACT O-ring seals.

4.4.2 Methods of Compliance and Verification

Compatibility of all waste material types has been demonstrated for transport in the TRUPACT-II and HalfPACT using the chemicals listed in [Table 4.3-1](#) through [Table 4.3-8](#). The restrictions imposed on the chemical constituents of the content codes approved by the WIPP CH-TRU Payload Engineer ensure compliance with the compatibility requirements (see also [Appendices 6.1, 6.2, 6.3, and 6.4](#) of the [CH-TRU Payload Appendices](#)²). The chemical list for each content code is formally documented by the site.

¹ Hatayama, H.K., Chen, J.J., de Vera, E.R., Stephens, R.D., and Storm, D.L., 1980, *A Method for Determining the Compatibility of Hazardous Wastes*, EPA-600/2-80-076, U.S. Environmental Protection Agency, Cincinnati, Ohio.

² U.S. Department of Energy (DOE), *CH-TRU Payload Appendices*, U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico.

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5.0 GAS GENERATION REQUIREMENTS

Gas generation, concentrations, and pressures during transport of CH-TRU wastes in a TRUPACT-II or HalfPACT payload are restricted as follows:

- For any package containing water and/or organic substances that could radiolytically generate combustible gases, determination must be made by tests and measurements or by analysis of a representative package such that the following criterion is met over a period of time that is twice the expected shipment time (defined in [Appendices 3.4, 3.5, and 3.6](#) of the [CH-TRU Payload Appendices](#)¹): While the primary flammable gas of concern is hydrogen generated by radiolysis, methane and flammable volatile organic compounds (VOCs) are also limited to ensure the absence of a flammable (gas/VOC) mixture in any layer of confinement during shipment. In the absence of other flammable gases, the hydrogen generated must be limited to a molar quantity that would be no more than 5 percent by volume of the innermost layer of confinement (or equivalent limits for other inflammable gases) if present at standard temperature and pressure (i.e., no more than 0.063 gram-moles/cubic foot at 14.7 pounds per square inch absolute and 32°F).
- The gases generated in the payload and released into the ICV cavity shall be controlled to maintain the pressure within the TRUPACT-II or HalfPACT ICV cavity below the acceptable design pressure of 50 pounds per square inch gauge.

The analyses presented in the [TRUPACT-II](#)² and [HalfPACT](#)³ SARs show that all payloads authorized for transport in the TRUPACT-II or the HalfPACT will comply with the design pressure limit.

The design decay heat limit for the TRUPACT-II package is 40 watts. The design decay heat limit for the HalfPACT package is 30 watts. The maximum decay heat in any single payload container is limited to the design decay heat limit for the package in which it is transported with the exception of any single CCO which is limited to 20 watts.

Specific requirements associated with the restrictions on gas generation during transport of a payload are described in detail below.

5.1 Payload Shipping Category

The CH-TRU waste at the DOE sites has been classified into “payload shipping categories” to evaluate and ensure compliance with the gas generation requirements. [Appendix 2.1](#) of the [CH-TRU Payload Appendices](#)¹ defines the methodology for determining a payload shipping category.

¹ U.S. Department of Energy (DOE), [CH-TRU Payload Appendices](#), U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico.

² U.S. Department of Energy (DOE), [TRUPACT-II Shipping Package Safety Analysis Report](#), USNRC Certificate of Compliance 71-9218, U.S. Department of Energy, Carlsbad Area Office, Carlsbad, New Mexico.

³ U.S. Department of Energy (DOE), [HalfPACT Shipping Package Safety Analysis Report](#), USNRC Certificate of Compliance 71-9279, U.S. Department of Energy, Carlsbad Area Office, Carlsbad, New Mexico.

The payload shipping category notation components represent the waste type, hydrogen gas generation potential of the waste, and the total resistance to hydrogen gas release of the waste packaging configuration. The rates of hydrogen gas generation and release are used in the evaluation of flammable gas generation compliance for a given payload container.

As shown in [Appendices 6.1, 6.5, and 6.6](#) of the [CH-TRU Payload Appendices](#)¹, gas generation due to chemical, biological, and thermal mechanisms is insignificant during transport, and radiolysis is the primary mechanism for potential flammable gas generation. Hydrogen gas is produced through radiolytic decomposition of waste resulting from the interaction of ionizing radiation (decay heat) with surrounding materials (waste material type). The radiolytic hydrogen gas generation potential of waste is quantified by the “G value” for hydrogen, which is the number of molecules of hydrogen generated per 100 electron volts (eV) of energy absorbed. Bounding G values for Waste Types I, II, and III are established based on a restricted composition of chemicals and materials defined for each waste material type in [Section 4.3](#). [Table 5.1-1](#) lists the G values associated with the various waste material types. These bounding G values are used in calculations of flammable gas generation rates (FGGR) and total gas generation rates (TGGR). A bounding G value has not been established for the chemical composition of Waste Type IV. [Appendices 3.1, 3.2, and 3.3](#) of the [CH-TRU Payload Appendices](#)¹ provide a discussion of radiolytic G values for chemicals and materials.

The hydrogen gas release is determined by the type of payload container and the type and number of inner confinement layers used. [Appendices 2.2, 3.8, 3.11, and 6.13](#) of the [CH-TRU Payload Appendices](#)¹ describe allowable inner confinement layers for which hydrogen resistance values have been determined. [Appendix 6.8](#) of the [CH-TRU Payload Appendices](#)¹ describes experiments relating to the quantification of hydrogen gas release from payload containers. The packaging configuration parameters that affect the release rate of hydrogen are defined in [Appendix 6.7](#) of the [CH-TRU Payload Appendices](#)¹.

Using the payload shipping category notation, [Appendices 2.3 and 6.7](#) of the [CH-TRU Payload Appendices](#)¹ define the selection of conservative calculational parameters for hydrogen gas generation and release and provide the derivation of maximum possible decay heat values for payload containers based on payload shipping category assignment. The bounding decay heat value for a container is determined when the container is at steady state (i.e., the hydrogen gas generation rate of the waste is equal to the hydrogen gas generation release of the packaging configuration), has an insignificant flammable VOC source term, and is assembled in a full payload with only containers assigned to the same payload shipping category. [Appendix 6.9](#) of the [CH-TRU Payload Appendices](#)¹ explains the temperature dependence of hydrogen gas generation and hydrogen gas release parameters that results in the conservative determination of bounding decay heat values for Waste Types I, II, and III. This temperature effect is reflected in the hydrogen gas resistance factors for the determination of payload shipping category total resistance specified in [Appendix 2.2](#) of the [CH-TRU Payload Appendices](#)¹. [Appendix 6.10](#) of the [CH-TRU Payload Appendices](#)¹ describes the impact to flammable (gas/VOC) limits when payload containers are overpacked. Note: [Appendix 6.9](#) of the [CH-TRU Payload Appendices](#) uses the term “decay heat limit” to refer to a bounding decay heat value per payload container. While the maximum possible decay heat value determined for a bounding set of conditions informs allowable payload container decay heat loading, it is not used in the flammable (gas/VOC) compliance evaluation described in [Section 5.2](#). The flammable (gas/VOC)

Table 5.1-1 – CH-TRU Waste Material Types and G Values

Waste Material Type	Typical Material Description^①	G Value^{②③}	Shipping Category Notation (G Value x 10²)
I.1	Absorbed, adsorbed, or solidified inorganic liquid	1.6	0160
I.2	Soils, solidified particulates, or sludges formed from precipitation	1.3	0130
I.3	Concreted inorganic particulate waste	0.4	0040
II.1	Solid inorganic materials in plastic bags (watt*year ≤0.012)	1.7	0170
II.1	Solid inorganic materials in plastic bags (watt*year >0.012)	0.32	0032
II.2	Solid inorganic materials in metal cans	0	0000
II.3	Homogeneous solid inorganic materials with unbound absorbed ambient moisture (≤6% by weight) in metal cans	0.08	0008
III.1	Solid organic materials (watt*year ≤0.012)	3.4	0340
III.1	Solid organic materials (watt*year >0.012)	1.09	0109
III.2	Homogeneous mixed organic (10% by weight) and inorganic (90% by weight) materials in metal cans (watt*year ≤0.012)	0.34	0034
III.2	Homogeneous mixed organic (10% by weight) and inorganic (90% by weight) materials in metal cans (watt*year >0.012)	0.11	0011
III.3	Homogeneous mixed organic (10% by weight) and inorganic (90% by weight) materials in plastic bags (watt*year ≤0.012)	1.85	0185
III.3	Homogeneous mixed organic (10% by weight) and inorganic (90% by weight) materials in plastic bags (watt*year >0.012)	0.4	0040
IV.1	Solidified organics	Unknown (test)	9999

Notes:

- ① [Appendix 3.3](#) of the [CH-TRU Payload Appendices](#)¹ provides a complete discussion of watt*year criteria.
- ② Dose-dependent G values for waste meeting the watt*year criteria (watt*year >0.012) cannot be used if absorbed, adsorbed, or solidified aqueous materials are present in the waste (see [Appendix 3.3](#) of the [CH-TRU Payload Appendices](#)¹). [Appendices 3.1](#) and [3.2](#) of the [CH-TRU Payload Appendices](#)¹ provide a complete discussion of G values.
- ③ Dose-dependent G values for waste meeting the watt*year criteria (watt*year >0.012) cannot be used for waste packaged in shielded containers (see [Appendix 3.3](#) of the [CH-TRU Payload Appendices](#)¹).

compliance evaluation uses the payload shipping category to define hydrogen gas generation and release along with the collective properties of payload containers as assembled in a payload to evaluate contributions from all flammable gases that may be present (i.e., hydrogen, methane, and flammable VOCs).

5.1.1 Requirements

Each payload container shall be assigned to a payload shipping category, in a content code approved by the WIPP CH-TRU Payload Engineer, that has information on the following components:

- Waste type
- Waste material type, which defines the gas generation potential
- Total resistance to gas release by the packaging configuration, which in addition to inner confinement layers surrounding the waste materials consists of the filtered payload container itself, any filtered container in an overpacked configuration, any filtered component of a fixed payload container configuration design, and the punctured/vented rigid liner if present in a drum. Total resistance has specific requirements associated with the following packaging configuration elements:
 - Inner Confinement Layers: Inner confinement layers are any >4-liter volume bag (fully expanded) or rigid container used to package waste within a payload container. The inner layers of confinement shall be plastic bags and/or rigid containers (e.g., metal cans) with closures that meet the applicable specifications outlined in [Appendices 2.2, 3.8, 3.11, and 6.13](#) of the [CH-TRU Payload Appendices](#).¹ For each type of inner confinement layer used at the sites, a minimum hydrogen release rate shall be determined, by testing or analysis, as described in [Appendix 3.8](#) of the [CH-TRU Payload Appendices](#).¹
 - Rigid Liner: The rigid liner and lid, if present, in a payload container shall contain a ≥ 0.3 -in. minimum diameter hole, or a filter with a hydrogen release rate equivalent to or greater than a 0.3-in. minimum diameter hole. Other liner puncture/vent configurations must be bound by the available options for rigid liner confinement layers in [Appendix 2.2](#) of the [CH-TRU Payload Appendices](#).¹
 - Shipping Period: The conditions specified in [Appendix 3.6](#) of the [CH-TRU Payload Appendices](#)¹ must be met for use of the Controlled Shipment shipping period (10 days). For other shipments (not Controlled Shipments), the appropriate shipping period [60 days ([Appendix 3.4](#) of the [CH-TRU Payload Appendices](#)¹) or 20 days ([Appendix 3.5](#) of the [CH-TRU Payload Appendices](#)¹)] may be applied based on the transport distance.

For each packaging configuration, a unique total resistance factor is determined by totaling the individual resistance factors (defined by [Appendix 2.2](#) of the [CH-TRU Payload Appendices](#)) for the inner confinement layers, the payload container (including a rigid liner, as applicable), and the “load type” associated with the shipping period.

5.1.2 Methods of Compliance and Verification

5.1.2.1 Shipping Category Notation

Compliance and verification of the shipping category requirements shall be by comparison of the assigned shipping category with the allowable shipping categories for the appropriate content code in the CH-TRUCON. The shipping category and content code information is recorded in the PCTCD and OPCTCD, as applicable.

5.1.2.2 Compliance with Specific Requirements for Total Resistance

The following paragraphs outline the methods of compliance and verification for inner confinement layers, rigid liner, and the shipping period.

Inner Confinement Layers

Radiography, visual examination, administrative and procedural controls, or a combination of these methods may be used to demonstrate that the method of closure for each layer of confinement is in accordance with [Appendices 2.2, 3.8, 3.11, and 6.13](#) of the [CH-TRU Payload Appendices](#).¹ The waste generation procedures shall specify the type and maximum number of inner confinement layers for each payload container. The type and maximum number of layers may be determined from the waste management practices in use at the time the waste was packaged and available records and database information.

Rigid Liner

The requirements for the rigid liner shall be met by procurement controls and site QA procedures. Venting of the lid of a liner (along with the minimum diameter of the hole in the liner) may be controlled administratively (i.e., buying only punctured liners) or by visual examination of the liner prior to closure. Alternatively, radiography or sampling programs and existing records may be used to verify that the liner meets the requirements.

Shipping Period

The requirements for the use of the Controlled Shipment (10 days) shipping period shall be met by administrative and procedural controls as specified in [Appendix 3.6](#) of the [CH-TRU Payload Appendices](#)¹ and [Section 6.2.3](#). No specific conditions exist for the use of the shipping periods defined in [Appendices 3.4 and 3.5](#) of the [CH-TRU Payload Appendices](#).¹

A given payload container may be conservatively assigned to a payload shipping category of an approved CH-TRUCON code if the combination of the actual packaging configuration and the shipping conditions result in a total resistance to hydrogen release and a representation of flammable VOC transport across confinement layers that are bound by the assigned payload shipping category total resistance factor ([Appendix 2.2](#) of the [CH-TRU Payload Appendices](#)¹) and the associated values for drum age criteria, concentration multipliers, and prediction factors ([Appendix 3.9](#) of the [CH-TRU Payload Appendices](#)¹).

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5.2 Flammable (Gas/VOC) Concentration Limits

5.2.1 Requirements

As discussed in [Appendices 6.1, 6.5, and 6.6](#) of the [CH-TRU Payload Appendices](#)¹, the primary mechanism for potential flammable gas generation in TRU wastes is radiolysis. TRU wastes to be transported in the TRUPACT-II and HalfPACT are restricted so that no flammable mixtures can occur in any layer of confinement during shipment. While the predominant flammable gas of concern is hydrogen, the presence of methane and flammable volatile organic compounds (VOCs) is also limited along with hydrogen to ensure the absence of flammable (gas/VOC) mixtures in TRU waste payloads.

5.2.2 Methods of Compliance and Verification

The evaluation of compliance with flammable (gas/VOC) limits occurs for an aggregated group of payload containers that are proposed for assembly in a single payload. The evaluation uses payload container-specific FGGR values determined by the analytical, measurement, or test category method and associated payload-dependent innermost confinement layer total flammable gas concentration (TFGC) values. The flammable (gas/VOC) concentration limits are established as payload-container-specific flammability indexes (FI), which give an indication of the status of the container at the end of the shipping period with respect to overall flammability due to hydrogen, methane, and flammable VOCs.

[Figure 5.2-1](#) presents the logic for performing the compliance evaluation for flammable (gas/VOC) limits. The step numbers identified in this section correspond to those identified in [Figure 5.2-1](#).

5.2.2.1 Determine Proposed Payload Assembly

Step 1. Container Data Package(s)

The starting point for the flammable (gas/VOC) compliance evaluation is the collection of payload-container-specific data for a selection of payload containers proposed for assembly in a single TRUPACT-II or HalfPACT payload. These data are gathered from one or more of the methods of payload compliance listed and defined in [Section 1.4](#) (i.e., visual examination, visual inspection, radiography, records and database information, administrative and procurement controls, sampling programs, and measurement). Headspace measurement methods and associated QA requirements are discussed in the Quality Assurance Program Plan (QAPP) for the TRUPACT-II Gas Generation Test Program.²

¹ U.S. Department of Energy (DOE), [CH-TRU Payload Appendices](#), U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico.

² U.S. Department of Energy (DOE), *Quality Assurance Program Plan for TRUPACT-II Gas Generation Test Program*, DOE/WIPP 01-3187, current revision, U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico).

Step 2. Aggregate Payload Containers into Proposed Payload Assembly

Select loaded payload containers and dunnage containers, as necessary, to complete one of the payload configurations authorized in [Section 2.1](#). These selected payload containers will be collectively evaluated for compliance with flammable (gas/VOC) limits. This compliance evaluation process is iterated for varied aggregations of payload containers until a payload assembly is determined to meet flammable (gas/VOC) limits.

5.2.2.2 Determine FGGR for Each Container in Payload Assembly**Step 3. Waste Type**

The payload-container-specific FGGR must be determined by one of three methods as dictated by waste type assignment.

- [A] If the payload container is classified as Waste Type I, II, or III, FGGR may be determined as described in Step 4, Compliance via Decay Heat (Analytical), Step 5, Compliance via Headspace Sampling (Measurement), or Step 6, Compliance via Full Container Testing (Test), for the analytical category, measurement category, or test category, respectively.
- Under the analytical category, the FGGR is determined analytically from the theoretical worst-case bounding G value (gas generation potential) associated with the container waste material type and decay heat loading as described in Step 4.
 - Under the measurement category, the FGGR is determined from the measurement of the payload container headspace gas as described in Step 5. Determination of FGGR under the measurement category is an option for payload containers that fail the flammable (gas/VOC) compliance evaluation using the analytically determined FGGR.
 - Under the test category, the FGGR and total gas generation rate (TGGR) are determined by full-container testing of the payload container as described in Step 6. Determination of FGGR and TGGR under the test category is an option for payload containers that fail the flammable (gas/VOC) compliance evaluation using the FGGR determined analytically or by measurement. Shielded containers shall not be evaluated by testing.
- [B] If the payload container is classified as Waste Type IV, test category is the only allowable option for FGGR and TGGR determination as described in Step 6. Waste Type IV is not authorized in shielded containers.

Because it does not have a fully characterized bounding theoretical radiolytic G value for hydrogen, Waste Type IV cannot be evaluated under the analytical category, which uses the bounding G value to determine FGGR, and cannot be evaluated under the measurement category, which uses a theoretical relationship between flammable gas G values and total gas G values to show compliance with allowable total gas generation rate (ATGGR).

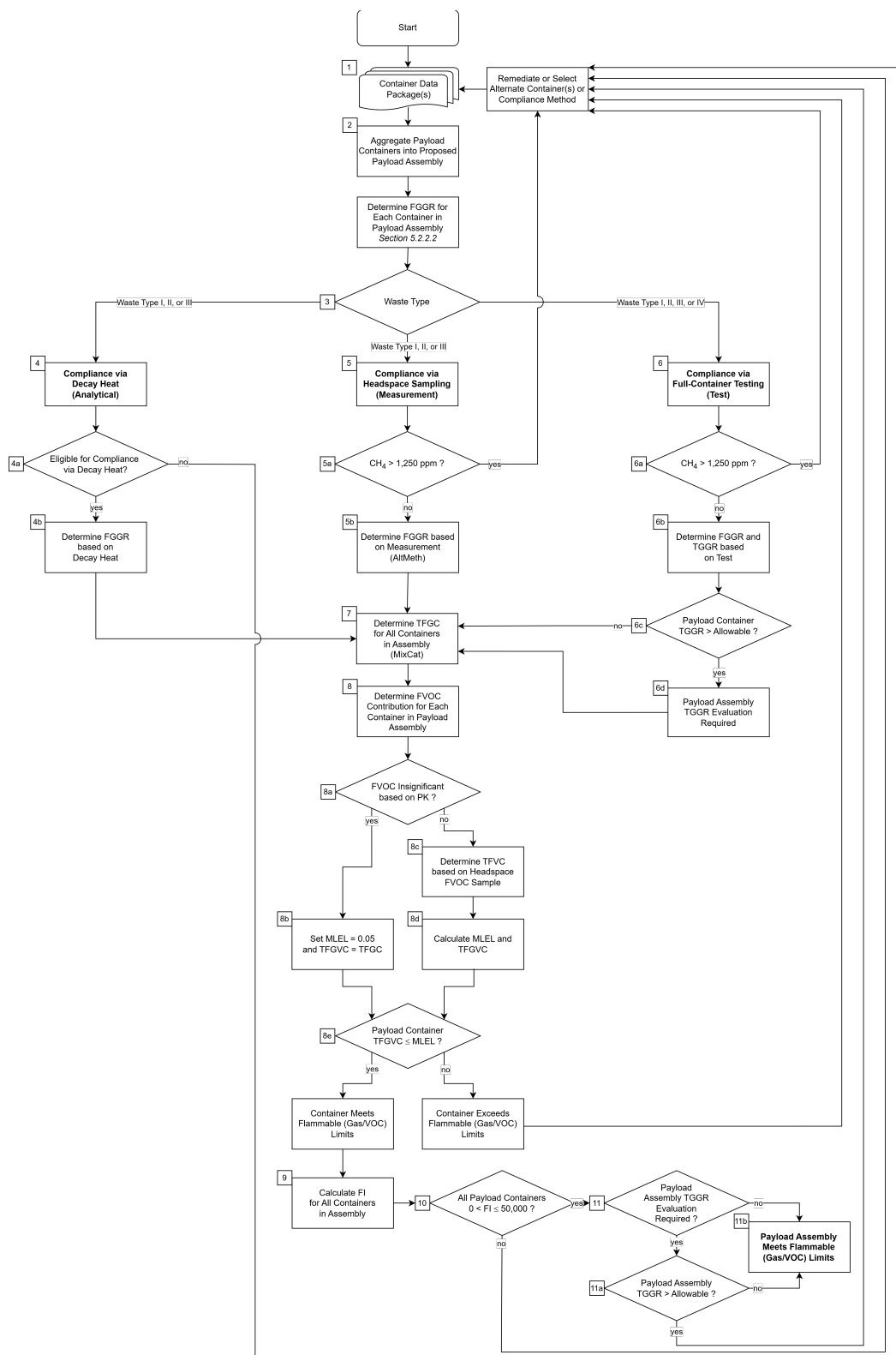


Figure 5.2-1 – Logic Diagram for Compliance with Flammable (Gas/VOC) Limits

Step 4. Compliance via Decay Heat (Analytical)

Compliance under the analytical category is an option only for payload containers with insignificant flammable VOC source terms as described in Step 4a[A] and that do not contain puck drums as described in Step 4a[B].

Step 4a. Eligible for Compliance via Decay Heat?

- [A] Determine if the payload container belongs to a waste population for which an insignificant flammable VOC source term has been documented based on the information contained in the applicable payload container data package. Data used to determine an insignificant flammable VOC source term may consist solely of process knowledge (e.g., knowledge of waste generation processes and chemical and material inputs to the process) as evaluated using the criteria defined in [Section 5.2.3](#).
- [B] Because headspace gas measurements of payload containers have shown elevated hydrogen concentrations upon loading with puck drums, the FGGR for payload containers with puck drum contents must be conservatively determined using the measurement or test category method. Payload containers containing one or more puck drums are not eligible for evaluation under the analytical category. Determine if the payload container contains puck drum(s).
- [C] If an insignificant flammable VOC source term cannot be determined per [Section 5.2.3](#) and/or if the payload container contains one or more puck drums, compliance cannot be demonstrated under the analytical category. Select an alternate compliance method for payload container evaluation.
- [D] If an insignificant flammable VOC source term can be determined per [Section 5.2.3](#) and the payload container does not contain puck drums, proceed to Step 4b, Determine FGGR based on Decay Heat.

Step 4b. Determine FGGR based on Decay Heat

- [A] Calculate the FGGR from the bounding G value and the measured decay heat plus error for the payload container using the following equation:

$$CG = Q \times G_{\text{eff}} \times C$$

where,

CG = FGGR due to radiolysis (mole/second)

Q = Decay heat plus error of payload container (watts)

G_{eff} = Effective G value for flammable gas [molecules of flammable gas formed per 100 eV of energy emitted (molecules/100 eV)]. The effective G values for each waste material type are provided in [Table 5.1-1](#).

C = Conversion constant for the units used as derived in [Appendix 3.1](#) of the [CH-TRU Payload Appendices](#) (1.03642E-5 (g-moles)(eV)/(molecule)(watt-sec))

- [B] Proceed to Step 7, Determine TFGC for All Containers in Assembly (MixCat)

Step 5. Compliance via Headspace Sampling (Measurement)

Step 5a. $\text{CH}_4 > 1,250$ ppm?

- [A] The payload container must be sampled to measure the headspace concentration of methane.
- [B] Compare the headspace methane concentration to the 1,250 ppm methane screening limit. The rationale for the methane screening limit is provided below:

Position: A methane screening limit or concentration of 1,250 ppm in the container headspace should be used in flammability evaluations. If the container headspace methane concentration is below this screening limit, the concentration of methane should be added to the hydrogen concentration and the FGGR should be determined. If the concentration is above the screening limit, the container is not eligible for shipment.

Rationale: Although the term “flammable gases” includes both hydrogen and methane, gas measurement data suggest methane is either not present or is present in very low concentrations.

The 1,250 ppm methane screening limit concentration is equivalent to:

- 2.5% of the methane lower explosive limit of 5 percent by volume
- Fraction of the methane contribution to the flammable gas G value for polyethylene (i.e., $[0.1 \text{ molecules methane}/100 \text{ eV}] / [4.1 \text{ molecules flammable gas}/100 \text{ eV}]$ or 2.4%).

- [C] If the methane concentration exceeds the 1,250 ppm screening limit, the payload container is not eligible for shipment and must be segregated for repackaging or other mitigation measures. Select alternate container for payload evaluation.
- [D] If the methane concentration is less than or equal to 1,250 ppm, proceed to Step 5b, Determine FGGR based on Measurement (AltMeth).

Step 5b. Determine FGGR based on Measurement (AltMeth)

- [A] The payload container must be sampled to measure the headspace concentration of hydrogen and methane.
- [B] Calculate the FGGR from the measured headspace hydrogen and methane concentrations based on the packaging configuration and the time history of the payload container using the AltMeth methodology as described in [Appendix 3.10](#) of the [CH-TRU Payload Appendices](#).
- [C] Proceed to Step 7, Determine TFGC for All Containers in Assembly (MixCat).

Step 6. Compliance via Full-Container Testing (Test)

Step 6a. $\text{CH}_4 > 1,250$ ppm?

- [A] The payload container must be sampled to measure the headspace concentration of methane.

- [B] Compare the headspace methane concentration to the 1,250 ppm methane screening limit. The rationale for the methane screening limit is provided in Step 5a[B].
- [C] If the methane concentration exceeds the 1,250 ppm screening limit, the payload container is not eligible for shipment and must be segregated for repacking or other mitigation measures. Select alternate container for payload evaluation.
- [D] If the methane concentration is less than or equal to 1,250 ppm, proceed to Step 6b, Determine FGGR and TGGR based on Test.

Step 6b. Determine FGGR and TGGR based on Test

- [A] The payload container must undergo full-container testing to determine the hydrogen/methane generation rate and TGGR over time as described in [Section 5.2.5.3](#).
- [B] Proceed to Step 6c, Payload Container TGGR > Allowable?

Step 6c. Payload Container TGGR > Allowable?

- [A] For Waste Types I, II, and III, TGGR compliance is through theoretical analysis as shown in [Section 5.2.4](#). Proceed to Step 7, Determine TFGC for All Containers in Assembly (MixCat).

For Waste Type IV, compare the TGGR as measured in Step 6b[A] to the ATGGR for the applicable payload container type as specified in [Table 5.2-1](#).
- [B] If the TGGR of the container is less than or equal to the applicable ATGGR in [Table 5.2-1](#), proceed to Step 7.
- [C] If the TGGR of the container exceeds the applicable ATGGR in [Table 5.2-1](#), compliance with total gas generation requirements must be assessed through the evaluation of compliance with the payload container total gas generation rate limit (PTGGRL). Proceed to Step 6d, Payload Assembly TGGR Evaluation Required.

Table 5.2-1 – Allowable Total Gas Generation Rates for Waste Type IV

Waste Type IV Payload Container	ATGGR (mol/s/container) ^①	
	TRUPACT-II	HalfPACT
55-Gallon Drum	3.951E-06	5.986E-06
85-Gallon Drum	5.953E-06	9.703E-06
55-Gallon Drum in 85-Gallon Drum Overpack	5.953E-06	9.703E-06
100-Gallon Drum	1.033E-05	1.538E-05
SWB	1.997E-05	3.489E-05
55-Gallon Drum in SWB/55 Overpack	9.955E-06	1.380E-05
85-Gallon Drum in SWB/85 Overpack	2.224E-05	2.997E-05
100-Gallon Drum in SWB/100 Overpack	2.110E-05	2.881E-05
TDOP	2.914E-05	NA ^②
CCO	3.979E-06	6.034E-06

Notes:

- ① The allowable total gas generation rates (in moles per second per payload container) were determined from analysis as described in [Section 3.4.4](#) of the [TRUPACT-II](#)³ and [HalfPACT](#)⁴ SARs.
- ② NA = Not applicable.

Step 6d. Payload Assembly TGGR Evaluation Required

Payload containers reaching this step require a payload assembly TGGR evaluation and must answer “yes” in Step 11, Payload Assembly TGGR Evaluation Required? Proceed to Step 7. Step 7 through Step 10 must be performed prior to the required payload assembly TGGR evaluation (Step 11).

5.2.2.3 Flammable (Gas/VOC) Compliance Evaluation for Payload Assembly**Step 7. Determine TFGC for All Containers in Assembly (MixCat)**

- [A] As described in [Section 6.2.4](#) and [Appendix 2.4](#) of the [CH-TRU Payload Appendices](#)¹, solve for the TFGC within the innermost confinement layer at the end of the shipping period inside a TRUPACT-II or HalfPACT for all payload containers in the assembly using the following equation:

$$\{TFGC\} = [R]\{CG\}$$

³ U.S. Department of Energy (DOE), [TRUPACT-II Shipping Package Safety Analysis Report](#), USNRC Certificate of Compliance 71-9218, U.S. Department of Energy, Carlsbad Area Office, Carlsbad, New Mexico.

⁴ U.S. Department of Energy (DOE), [HalfPACT Shipping Package Safety Analysis Report](#), USNRC Certificate of Compliance 71-9279, U.S. Department of Energy, Carlsbad Area Office, Carlsbad, New Mexico.

where,

CG = Calculated FGGR due to radiolysis per Step 4b[A] (analytical category), from measured hydrogen and methane per Step 5b[B] (measurement category), or from full-container testing per Step 6b[B] (test category) (mole/second).

R = Matrix of gas generation rate coefficients [i.e., resistance to hydrogen release (second/mole) (see [Appendix 2.2](#) of the [CH-TRU Payload Appendices](#)), time, number of generators].

- [B] Proceed to Step 8, Determine FVOC Contribution for Each Container in Payload Assembly.

Step 8. Determine FVOC Contribution for Each Container in Payload Assembly

The FVOC contribution to flammability must be determined for use in the evaluation of the mixture lower explosive limit (MLEL) and total flammable gas/VOC concentration (TFGVC) for the total flammable (gas/VOC) mixture.

Step 8a. FVOC Insignificant based on PK?

- [A] If the payload container has reached this step using the FGGR determined by Step 4b[A], proceed to Step 8b, Set MLEL = 0.05 and TFGVC = TFGC.
- [B] If the payload container has reached this step using the FGGR determined by Step 5b[B] or Step 6b[B], determine if it belongs to a waste population for which an insignificant flammable VOC source term has been documented based on the information contained in the applicable payload container data package. Data used to determine an insignificant flammable VOC source term may consist solely of process knowledge (e.g., knowledge of waste generation processes and chemical and material inputs to the process) as evaluated using the criteria defined in [Section 5.2.3](#).
- [C] If an insignificant flammable VOC source term can be determined per [Section 5.2.3](#), proceed to Step 8b.
- [D] If an insignificant flammable VOC source term cannot be determined per [Section 5.2.3](#), flammable VOC concentrations must be determined by sampling the payload container headspace. Proceed to Step 8c, Determine TFVC based on Headspace FVOC Sample.

Step 8b. Set MLEL = 0.05 and TFGVC = TFGC

If an insignificant flammable VOC source term can be determined per [Section 5.2.3](#), the payload container shall be evaluated with no contribution from flammable VOCs. In this case, the flammable gas within the innermost confinement layer is limited to a hydrogen/methane total concentration of less than or equal to 5% by volume.

- [A] Set MLEL = 0.05.
- [B] Set TFGVC equal to the TFGC determined in Step 7[A].
- [C] Proceed to Step 8e, Payload Container TFGVC $\leq$ MLEL?

Step 8c. Determine TFVC based on Headspace FVOC Sample

- [A] The payload container headspace must be sampled to measure the concentrations of flammable VOCs.
- [B] Determine the total flammable VOC concentration (TFVC) at steady state within the innermost confinement layer based on measured headspace flammable VOC concentrations as described in [Appendix 3.9](#) of the [CH-TRU Payload Appendices](#).
- [C] Proceed to Step 8d, Calculate MLEL and TFGVC.

Step 8d. Calculate MLEL and TFGVC

- [A] Calculate the payload-container-specific MLEL values using the following equation:

$$MLEL = \frac{100\%}{\sum f_i GCF_i}$$

where,

- f_i = Fraction of flammable gas i in mixture on an air-free and nonflammable VOC-free basis (i.e., the concentration of flammable compound i divided by the sum of the concentrations of flammable VOCs and flammable gas [hydrogen and methane])
- GCF_i = Group contribution factor for compound i . The group contribution factor values for various compounds are listed in [Appendix 3.10](#) of the [CH-TRU Payload Appendices](#)¹.

- [B] Calculate payload-container-specific TFGVC values using the following equation:

$$TFGVC = TFGC + TFVC$$

where,

TFGC = Total flammable gas concentration as determined in Step 7[A]

TFVC = Total flammable VOC concentration as determined in Step 8c[B]

- [C] Proceed to Step 8e, Payload Container $TFGVC \leq MLEL$?

Step 8e. Payload Container $TFGVC \leq MLEL$?

- [A] Compare the TFGVC within the innermost confinement layer as determined in either Step 8b[B] or Step 8d[B] to the calculated MLEL as determined in either Step 8b[A] or Step 8d[A].
- [B] If the TFGVC is greater than the MLEL, the payload container exceeds flammable (gas/VOC) limits and is not eligible for shipment with this payload assembly. The payload assembly shall be reconfigured without this container.

- [C] If the TFGVC is less than or equal to the MLEL, the payload container meets the flammable (gas/VOC) limits. Proceed to Step 9, Calculate FI for All Containers in Assembly.

Step 9. Calculate FI for All Containers in Assembly

- [A] For each payload container, an FI must be calculated as the ratio of the TFGVC and MLEL values determined in Step 8b or Step 8d multiplied by 50,000.
- [B] Proceed to Step 10, All Payload Containers $0 < FI \leq 50,000$?

Step 10. All Payload Containers $0 < FI \leq 50,000$?

- [A] If the FI for any payload container is a negative number or exceeds 50,000, the payload shall be reconfigured until this requirement is satisfied.
- [B] If the FI for each payload container in the payload assembly is a non-negative number less than or equal to 50,000, proceed to Step 11, Payload Assembly TGGR Evaluation Required?

5.2.2.4 Total Gas Compliance Evaluation for Payload Assembly

Step 11. Payload Assembly TGGR Evaluation Required?

- [A] For Waste Type IV payload containers that exceed the ATGGR as determined in Step 6c[C] and Step 6d, Payload Assembly TGGR Evaluation Required, proceed to Step 11a, Payload Assembly TGGR > Allowable? Otherwise, proceed to Step 11b, Payload Assembly Meets Flammable (Gas/VOC) Limits.

Step 11a. Payload Assembly TGGR > Allowable?

- [A] For Waste Type IV payload containers that exceed the ATGGR as determined in Step 6c[C] and Step 6d, compliance with the packaging design pressure limit may be demonstrated by compliance with the PTGGRL determined for the payload in which it is assembled as described in [Section 6.2.5](#). Calculate the payload total gas generation rate (PTGGR).
- [B] If the PTGGR is greater than the PTGGRL calculated as described in [Section 6.2.5](#), the payload shall be reconfigured until this requirement is satisfied.
- [C] If the PTGGR is less than the PTGGRL calculated as described in [Section 6.2.5](#), proceed to Step 11b.

Step 11b. Payload Assembly Meets Flammable (Gas/VOC) Limits

Payload assembly reaching this step qualifies for shipment after verification of all other transportation parameters.

5.2.3 Determination of Insignificant Flammable VOC Source Term

A determination of an insignificant flammable VOC source term may be made for a payload container or a population of containers with consistent flammable VOC properties. An insignificant flammable VOC source term is defined as a total flammable VOC concentration of ≤ 500 ppm in the innermost layer of confinement of the associated packaging configuration(s).

If an insignificant flammable VOC source term can be established, headspace gas measurement for flammable VOCs is not required.

5.2.3.1 Required Criteria

Containers that are identified by process knowledge or other characterization to have similar properties with respect to flammable VOCs may be grouped into a population. The basis for grouping individual containers into a population shall be documented (e.g., containers generated from the same waste stream). The following information shall be considered and documented for the payload container or population of containers:

- Knowledge of the waste generation processes. This may include the following:
 - Similarities among the facilities generating the waste and consistency of processes over time (e.g., negligible process changes to date)
 - Sources and quantities of potential flammable VOC sources (e.g., liquids, solvents, cleaners, mixtures, or coatings) in waste feed material, the waste generation process, and the waste packaging process
 - Solidification processes, treatment operations or additives that may remove, contain, or dilute the presence of potential flammable VOCs (e.g., use of absorbents or solidifying agents)
 - Headspace measurement of flammable VOC concentrations
 - Evaluation of any trend affecting flammable VOC concentrations with time (e.g., dates of waste generation and packaging, age of the waste, storage conditions)
- Knowledge of payload container packaging configurations, including consistency of packaging configurations used (which affect the measured headspace flammable VOC concentrations and predicted innermost layer concentrations)
- Ongoing evaluation of documented flammable VOC properties information. Periodic re-evaluation of documented information shall be performed to ensure the determination of insignificant flammable VOCs continues to bound process changes, packaging revisions, and inventory additions

Note: VOC absorbing or adsorbing material (such as granular activated carbon to adsorb carbon tetrachloride) may be placed in a payload container provided that site personnel can verify or demonstrate the following through testing, analysis, or knowledge of the process:

1. The absorbent/adsorbent remains effective in retaining VOCs from the time of waste packaging through the end of the maximum shipping period in the TRUPACT-II or HalfPACT,
2. A flammable mixture of gases does not exist in the innermost layer of confinement, and
3. The total concentration of potentially flammable VOCs in a payload container is insignificant as determined by the criteria defined by this section.

5.2.3.2 Assignment of Insignificant Flammable VOC Source Term

A determination of an insignificant flammable VOC source term shall be documented for a payload container or a population of containers with consistent flammable VOC properties in a

site-specific program approved by DOE-CBFO. If flammable VOC concentrations are determined to be insignificant, the compliance evaluation with the flammable gas/VOC limits considers only the radiolytic flammable gas contributions of hydrogen and methane.

5.2.4 Compliance with the Packaging Design Pressure Limit by Theoretical Analysis

As described in [Section 3.4.4](#) of the [TRUPACT-II³](#) and [HalfPACT⁴](#) SARs, compliance with the TRUPACT-II and HalfPACT design pressure limits can be demonstrated by conservative theoretical analysis up to the wattage limits shown in [Table 5.2-2](#) for Waste Types I, II, and III.

Table 5.2-2 – Wattage Limits for Theoretical Analysis of Design Pressure

Container Type	Maximum Decay Heat per Payload Container and per Package (watts)					
	Waste Type I		Waste Type II		Waste Type III	
	TRUPACT-II	HalfPACT	TRUPACT-II	HalfPACT	TRUPACT-II	HalfPACT
55-Gallon Drum	40	30	40	30	37	27
85-Gallon Drum	40	30	40	30	26	20
100-Gallon Drum	40	30	40	30	32	23
SWB	40	30	40	30	23	19
SWB/55 Overpack	40	30	40	30	38	25
SWB/85 Overpack	40	30	40	30	40	27
SWB/100 Overpack	40	30	40	30	39	26
TDOP	40	NA ^①	40	NA ^①	18	NA ^①
CCO ^②	40	30	40	30	39	29
SC-30G1	NA ^①	30	NA ^①	30	NA ^①	28
SC-30G2	NA ^①	30	NA ^①	30	NA ^①	30
SC-30G3	NA ^①	30	NA ^①	30	NA ^①	28
SC-55G1	NA ^①	30	NA ^①	30	NA ^①	30
SC-55G2	NA ^①	30	NA ^①	30	NA ^①	28

Notes:

① NA = Not applicable.

② The maximum decay heat in any single CCO is also limited to the payload container design limit of 20 watts.

For cases where the wattage limits specified in [Table 5.2-2](#) are exceeded but the packaging design limit of 40 watts per TRUPACT-II or 30 watts per HalfPACT is met, compliance with the container flammable (gas/VOC) limits by the methodologies described in [Section 5.2.2](#) can be used to evaluate compliance with the ATGGR. Because the primary mechanism for gas

generation for both flammable and total gas for Waste Types I, II, and III is radiolysis, compliance with the flammable (gas/VOC) limits implies actual G values (both flammable and total) that are much lower than those used to derive the limits in [Table 5.2-2](#). Therefore, compliance with the flammable (gas/VOC) limits will ensure compliance with the ATGGR for these cases (e.g., Waste Type III SWBs greater than 23 watts in a TRUPACT-II or 18 watts in a HalfPACT). An example calculation is provided below:

Assuming an SWB of Waste Material Type III.1 with no layers of confinement and no resistance to gas release from SWB filters and a decay heat loading of 30 watts, compliance with the ATGGR is ensured if compliance with the flammable (gas/VOC) limits can be demonstrated under the analytical, measurement, or test category. A maximum FGGR corresponding to an allowable TFGC of 0.05 for the SWB can be calculated as $3.333\text{E-}07$ moles/second. Based on the 30-watt decay heat loading of the SWB and an SWB hydrogen gas generation rate of less than or equal to $3.333\text{E-}07$ moles/second in order to comply with a $\text{TFGC}=0.05$, the actual G value (flammable gas) for this container can be calculated as 0.11 (see Equation 14 of [Appendix 2.4](#) of the [CH-TRU Payload Appendices](#)). This compares to a theoretical G value (flammable gas) of 1.09. Because the mechanism for both hydrogen and total gas generation is radiolysis, a similar ratio exists between the actual and theoretical gas generation rates for hydrogen and total gas. This means that if the SWB FGGR is less than or equal to $3.333\text{E-}07$ moles/second, the actual G value (total gas) for this container will be lower than the theoretical G value used to derive the limits in [Table 5.2-2](#) by a factor of $1.09/0.11 = 10$. Therefore, the actual total gas generation from this container will be extremely low and will easily comply with the packaging design pressure limits.

In order to more clearly illustrate this fact and directly address the purpose of the above example, two alternate pressure increase calculations for Waste Material Type III.1 are provided below. These calculations are based on the maximum allowable flammable gas generation rate (AFGGR) values and, for comparison, on actual FGGR values calculated from measurement data.

Example Pressure Increase Calculations Based on Maximum AFGGR Values

Pressure increase calculations that correspond to maximum AFGGR values provide a bounding assessment of pressure increase values (as opposed to the extremely conservative theoretical maximum values provided in [Table 3.4-6](#) through [Table 3.4-14](#) of the [TRUPACT-II SARI³](#)). The maximum AFGGR for each payload container type was determined by assuming an insignificant flammable VOC source term, zero confinement layers and no resistance to gas release from container filters. These assumptions allow the calculation of maximum AFGGR using only the load type resistance value specified in [Table 2.2-4](#) of [Appendix 2.2](#) of the [CH-TRU Payload Appendices](#). Resulting 60-day

shipping period payload shipping categories and maximum AFGGR values for each payload container type are shown below.

Container Type	Payload Shipping Category	AFGGR (mol/sec)
55-Gallon Drum	30 0109 0072	6.944E-08
85-Gallon Drum	30 0109 0048	1.042E-07
100-Gallon Drum	30 0109 0028	1.786E-07
SWB	30 0109 0015	3.333E-07
SWB/55 Overpack	30 0109 0058	8.621E-08
SWB/85 Overpack	30 0109 0029	1.724E-07
SWB/100 Overpack	30 0109 0029	1.724E-07
TDOP	30 0109 0010	5.000E-07

Using the above container specific maximum AFGGR values, the maximum possible “radiolytic gas generation rates” (see eighth column from left in [Table 3.4-6](#) through [Table 3.4-14](#) of the [TRUPACT-II SAR³](#)) were calculated using the following formula:

$$CG_{total} = \frac{G_{eff(T)}}{G} \times FGGR \times n_{containers}$$

where,

CG_{total}	=	Rate of total radiolytic gas generation (moles/second)
$G_{eff(T)}$	=	Temperature-corrected effective total G value [the total number of molecules of gas generated per 100 eV of energy absorbed (molecules/100 ev)]
G	=	Flammable gas G value
FGGR	=	Flammable gas generation rate
$n_{containers}$	=	Number of payload containers in package.

Assuming the maximum TRUPACT-II design decay heat limit of 40 watts, the maximum possible radiolytic gas generation rates (as a function of the container maximum AFGGR values) were used to calculate the AFGGR-limited pressures at the end of the 60-day shipping period. In each case, the pressure increase (shown below) is well below the 50-psig pressure design limit.

Container Type	Maximum Total Radiolytic Gas Generation Rate (mol/sec)	Maximum Pressure Increase at 60 days (psig)
55-Gallon Drum	1.224E-05	15.75
85-Gallon Drum	1.457E-05	20.09
100-Gallon Drum	1.873E-05	19.91
SWB	1.165E-05	19.40
SWB/55 Overpack	1.205E-05	12.64
SWB/85 Overpack	1.205E-05	11.88
SWB/100 Overpack	1.205E-05	12.23
TDOP	8.739E-06	19.80

Note: At a HalfPACT design decay heat limit of 30 watts, following the same calculational methodology, the SC-30G1 similarly has a maximum radiolytic gas generation rate of 3.618E-06 mol/second corresponding to a HalfPACT maximum pressure increase at 60 days of 9.54 psig. This bounds all authorized shielded container designs.

Example Pressure Increase Calculations Based on Actual FGGR Values

Sample 60-day shipment data for 4,008 payload containers shipped in 354 payloads, each of which included at least one measurement category container, were evaluated to determine the highest possible pressure increase based on the actual payload FGGR and wattage values. Using actual FGGR values calculated from headspace measurement data for each container, the highest total radiolytic gas generation rates per package were calculated using the following formula:

$$CG = \sum_{i=1}^N \frac{G_{eff(T)}}{G} \times FGGR_i$$

The evaluation used the actual as-shipped payload configuration where total wattage was determined by summing the individual payload container wattages. The shipment data calculations (shown below) validate the bounding pressure values calculated above as the highest pressure increase for each is less than the maximum pressure increase for the applicable payload container type.

Container Type	Highest Total Radiolytic Gas Generation Rate (mol/sec)	Highest Pressure Increase at 60 days (psig)
55-Gallon Drum	2.165E-06	5.55
100-Gallon Drum	8.068E-07	4.08
SWB	3.072E-07	3.73
SWB/55 Overpack	4.192E-07	4.49

If compliance with the flammable (gas/VOC) limits is demonstrated as described in [Section 5.2.2.2](#), then the maximum ATGGR is met for Waste Types I, II, and III as described above.

5.2.5 Unified Flammable Gas Test Procedure

5.2.5.1 Purpose

The Unified Flammable Gas Test Procedure (UFGTP) details the methodology used to determine FGGR and TGGR for test category payload containers. Note: Shielded containers shall not be evaluated by testing. It also provides instructions for the methodology by which each test activity will be performed. The objective of the UFGTP is to facilitate shipment of waste in the test category by testing individual payload containers to evaluate compliance with flammable (gas/VOC) limits. As described in [Section 5.2.6](#), the UFGTP long-term objective (LTO) may facilitate waste shippability for specific populations or subpopulations of waste by arriving at more realistic FGGR and TGGR values based on the results of previous measurement and testing.

5.2.5.2 Summary of Procedure

The test method consists of performing container tests to measure FGGR and TGGR. If the test method is performed on the outermost (overpacking) payload container of an overpacked configuration with multiple containers (drums in an SWB), the packaging configuration must be described as a direct load configuration in order to apply the steps specified in this section.

5.2.5.3 Procedure for Testing

5.2.5.3.1 Criteria for Testing

Containers selected for testing must be shown to meet all other applicable transportation limits (weight, FGE, dose rate, etc.). Waste containers chosen for testing shall be subject to the decay heat criteria shown in [Table 5.2-3](#).

Table 5.2-3 – Decay Heat Criteria for Container Selection

Waste Type	Payload Container	Maximum Decay Heat per Payload Container and per Package (watts)		Test Temperature (°F)
		TRUPACT-II	HalfPACT	
I, II and III	55-Gallon drum	40	30	RT ^①
	85-Gallon Drum	40	30	RT ^①
	100-Gallon Drum	40	30	RT ^①
	SWB	40	30	RT ^①
	55-Gallon Drum in SWB/55 Overpack	40	30	RT ^①
	85-Gallon Drum in SWB/85 Overpack	40	30	RT ^①
	100-Gallon Drum in SWB/100 Overpack	40	30	RT ^①
	TDOP	40	NA ^②	RT ^①
	CCO	40	30	RT ^①
IV	55-Gallon Drum	9.0	11.0	135
	85-Gallon Drum	2.9	4.0	135
	100-Gallon Drum	2.9	4.0	135
	SWB	2.9	4.0	135
	55-Gallon Drum in SWB/55 Overpack	2.9	4.0	135
	85-Gallon Drum in SWB/85 Overpack	2.9	4.0	135
	100-Gallon Drum in SWB/100 Overpack	2.9	4.0	135
	TDOP	2.9	NA ^②	135
	CCO	19.6	13.2	135

Notes:

① RT = Room temperature. For gas generation testing, should be at a minimum of 64°F. Higher temperatures are allowed and will yield conservative results.

② NA = Not applicable.

Table 5.2-3 also specifies the temperatures at which gas generation testing must be performed. Testing temperatures are based on compliance with TRUPACT-II and HalfPACT design pressure limits, and elevated temperature gas generation testing is only necessary in cases where compliance with the packaging design pressure limits cannot be demonstrated through theoretical analysis or if a correlation cannot be established between FGGR and TGGR (i.e., for Waste Type IV). As described in Section 5.2.4 and as described in Section 3.4.4 of the TRUPACT-II and HalfPACT SARs, compliance with the TRUPACT-II and HalfPACT design pressure limits can be demonstrated by conservative theoretical analysis up to the wattage limits shown in Table 5.2-2 for Waste Types I, II, and III. For Waste Types I, II, and III above the values in Table 5.2-2 and below the packaging design limits, a correlation exists between FGGR and TGGR as described in Section 5.2.4. For Waste Type IV, testing shall be performed at or above the elevated temperatures listed in Table 5.2-3.

5.2.5.3.2 Test Apparatus

A test apparatus will be used that allows measurement of the gas generation rate from a container. This setup will also allow sampling of the offgas during the test in order to measure the concentrations of hydrogen, methane, and possibly other compounds of interest. The test apparatus will enclose a container of waste, be heated to the predetermined temperature, if applicable (see [Table 5.2-3](#)), and be monitored for the test temperature during testing. Any combination of hardware and software that collects data that meet the QA objectives of the QAPP² may be used to perform testing. It is likely that facilities participating in the TRUPACT-II Gas Generation Test Program will use identical or similar equipment for ease of performance validation. The steps in the procedure are centered around the following activities:

- Measurement of barometric pressure
- Temperature control and measurement (for applicable containers)
- Gas generation measurement or determination (for applicable containers)
- Gas sampling
- Gas analyses.

In order to test the waste for gas generation, the temperature of the container will either be at room temperature or will be raised for Waste Type IV, as shown in [Table 5.2-3](#). Where elevated temperature testing is required, the selected waste container is placed in an insulated overpacking container enclosure or a temperature-controlled environment. Insulation, heat tape, thermocouples, and controllers may be used to monitor and control the temperature. The test temperature for Waste Type IV has been calculated based upon the maximum allowed wattage for shipped waste and is listed in [Table 5.2-3](#).

Barometric pressure will be determined in a location that will give sufficient information regarding the pressure of the sampled gas(es). Each sample analysis must have an associated barometric pressure measurement.

Data will be gathered from the test apparatus using a flow measurement or gas sampling device. Measurements or determinations of the total gas flow rate and/or associated hydrogen/methane gas concentrations will be used to calculate the hydrogen/methane gas generation rate. A mass flow meter, or other device, if used, may be placed outside of the waste container but within the overpacking container enclosure to avoid condensation of any gases within the flow device.

Sampling of gases may be accomplished using either of two methods. A sampling manifold will be a part of the waste container test unit. This manifold will be used to either collect discrete samples in containers or to divert a stream of gas to an analytical instrument. If discrete samples are collected, they will be sent to an analytical laboratory for analyses. If a stream is diverted to an analytical instrument, such as a mass spectrometer, this is called on-line sampling and analysis. Once the test begins, gas sampled during the test will be analyzed for hydrogen and methane, and possibly for other gases of interest (e.g., oxygen, nitrogen, carbon dioxide, hydrogen sulfide, sulfur dioxide, and sulfur trioxide).

The headspace of the containers will be sampled and analyzed for flammable VOCs, if necessary, before the container is qualified for shipment.

5.2.5.3.3 Data Quality Objectives

Data quality objectives (DQOs) are the qualitative and quantitative statements developed by data users to specify the quality of data required from a particular data collection activity. The DQOs for the testing are specified in the QAPP.²

The QAPP defines the QA objectives associated with these data measurements. Specific QA measures will also be followed for sample custody, calibration of equipment, data reporting, and data reduction.

5.2.5.3.4 Test Startup and Test Completion

The test startup consists of placing the waste container in the test apparatus and, if specified in [Table 5.2-3](#), heating the test unit to the required test temperature. Site-specific operating procedures describe the sampling and analysis required on a routine basis. The test will be terminated after sufficient data are obtained to calculate the hydrogen gas generation rate.

The term “sufficient data” is defined as data on the parameters needed to quantify a bounding and applicable gas generation rate for the container under the test conditions prescribed in the UFGTP. Sufficient data is measurement of FGGR and TGGR, temperature, and pressure during a testing period that is extended until the rates are shown to remain constant or decrease. In all cases, the collection of data as described herein ensures that the measured rates determined through testing are representative of the gas generation properties of the container over the allowed shipping period. The measured rates are then used to demonstrate compliance with the flammable (gas/VOC) limits.

5.2.6 UFGTP Long-Term Objective

The process for implementation of the UFGTP LTO is defined in the “UFGTP Long-Term Objective Implementation Methodology”.⁵

The UFGTP LTO is applied to a population of containers with consistent gas generation properties. The UFGTP LTO may be implemented once the required data have been collected through measurement and/or testing for a subpopulation of these containers. If a bounding FGGR value for compliance determination can be established and shown to be below the maximum AFGGR for the population (FGGR per innermost confinement layer that will yield a maximum flammable gas concentration equivalent to the TFGC for a bounding packaging configuration), no further need exists to perform measurement (headspace gas for hydrogen and/or flammable VOCs) or to test every container in the population. If the bounding FGGR value exceeds the maximum AFGGR, the population will continue to be processed under the measurement and testing methodology of the UFGTP. This methodology is consistent with that used in the determination of dose-dependent G values in [Appendix 3.3](#) of the [CH-TRU Payload Appendices](#).¹

Site implementation of the UFGTP LTO must be documented in site-specific programs approved by DOE-CBFO.

⁵ Shaw Environmental & Infrastructure, Inc., current revision, *UFGTP Long-Term Objective Implementation Methodology*, Shaw Environmental & Infrastructure, Inc., Albuquerque, New Mexico.

5.2.6.1 Required Data and Criteria for Subpopulation

Containers that are identified by process knowledge or other characterization to have similar properties with respect to gas generation are grouped into populations. The basis for grouping individual containers into a population shall be documented (e.g., by item description codes showing all containers are generated from the same waste stream). Within a given population, a subpopulation size may be determined and subjected to gas generation testing (i.e., measurement or container gas generation testing). The results of testing the subpopulation may be used to calculate flammable gas generation properties for the remainder of the population. The following data for each container of the subpopulation (i.e., that has undergone gas generation testing or measurement) shall be available:

- Content code and shipping category, from which the maximum AFGGR is determined
- Date of container closure, date of venting, and date of measurement or testing
- Methane concentration ([Section 5.2.2.2](#), Step 5a or [Step 6a](#))
- FGGR ([Section 5.2.2](#))
- TGGR for Waste Type IV ([Section 5.2.2](#)).

5.2.6.2 Statistics for Required Subpopulation Size

Containers selected for evaluation (i.e., the subpopulation) must be representative of the population with random or stratified sampling techniques used to avoid any bias in container selection. The required subpopulation size to implement the UFGTP LTO will be calculated through the following equation:

$$n = \frac{n_0}{1 + \frac{(n_0-1)}{N}}$$

where,

n_0 = A conservative subpopulation size for an infinite population for a confidence level of 95% and a $\pm 5\%$ precision level. In this case, the upper tail of the distribution curve is of interest, and n_0 has a value of 271.⁵

n = The required subpopulation size

N = The true population size.

As shown by the formula, the minimum number of required containers depends on the true population size.

5.2.6.3 Assignment of Flammable Gas Generation Properties for Population

A 95% upper tolerance limit (UTL) (the 95% upper confidence limit of the 95th percentile) of the methane concentration, FGGR, and TGGR (if required) shall be calculated for each subpopulation using a bootstrapping or comparable non-parametric technique.⁵ Non-parametric statistics are used to ensure the validity of this procedure regardless of the distribution shape of the subpopulation gas generation values. The 95% UTL of the methane concentration, FGGR, and TGGR values will then be assigned to the remainder of the containers in the

population in place of actual test values in evaluating compliance with the flammable gas/VOC limits using the methodology documented in [Section 5.2.2](#).

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5.3 Venting and Aspiration

5.3.1 Requirements

Payload containers that have been stored in an unvented condition (i.e., no filter and/or unpunctured liner) shall be aspirated for the specific length of time to ensure equilibration of any gases that may have accumulated in the closed container. For payload containers with Waste Material Type II.2 packaged in metal cans (which does not generate any flammable gas), aspiration is not required. Note: The aspiration requirement is applicable only to containers that have been stored in an unvented condition. The derivation of aspiration times is outlined in [Appendix 3.7](#) of the [CH-TRU Payload Appendices](#).¹

5.3.2 Methods of Compliance and Verification

The following sections describe the three options that are available for determining the required aspiration time. Tables in this section present aspiration times for common packaging configurations. Aspiration times for authorized configurations not covered by the tables in this section shall be determined as outlined in [Appendix 3.7](#) of the [CH-TRU Payload Appendices](#).¹

5.3.2.1 Option 1—Aspiration Time Based on Date of Payload Container Closure

Compliance shall be by records and database information. For sites using this option, the steps outlined below shall be implemented to determine the aspiration time for a payload container.

1. The potential shipping category of the payload container shall be known.
2. The duration for which the container has been in a closed condition should be computed from a knowledge of the date of closure of the drum. This closure time should be rounded up to the next highest time period listed in [Table 5.3-1](#), [Table 5.3-2](#), or [Table 5.3-3](#).
3. The required aspiration time corresponding to this closure time shall then be read from the same table under the column for the number of layers of confinement associated with the appropriate container shipping category.
4. The container shall be qualified for shipment only after being aspirated for at least this period of time.

5.3.2.2 Option 2—Headspace Gas Sampling at the Time of Venting

Compliance shall be by measurement of headspace gas concentrations and records. For sites using this option, the steps outlined below shall be implemented to determine the aspiration time for a payload container.

1. The potential shipping category of the payload container shall be known.

¹ U.S. Department of Energy (DOE), [CH-TRU Payload Appendices](#), U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico.

2. At the time of venting, a headspace sample shall be taken from the container headspace (Option 2A) or from the rigid liner headspace (Option 2B) and analyzed for the hydrogen concentration.
3. The measured headspace concentration of all hydrogen shall be rounded up to the next highest mole percentage listed in [Table 5.3-4](#), [Table 5.3-5](#), [Table 5.3-6](#), [Table 5.3-7](#), [Table 5.3-8](#), or [Table 5.3-9](#), determined by Option 2A or 2B, as appropriate.
4. The required aspiration time corresponding to this molar percentage shall be read from the same table under the column for the number of layers of confinement associated with the appropriate container shipping category.
5. The container shall be qualified for shipment only after being aspirated for at least this period of time.

5.3.2.3 Option 3—Headspace Gas Sampling During Aspiration

Compliance shall be by direct measurement of the headspace gas concentration. For sites using this option, the steps outlined below shall be implemented to determine the aspiration time for a payload container.

1. The potential shipping category of the payload container shall be known.
2. Prior to measuring the hydrogen concentration in the payload container headspace, the drum shall be vented for at least two weeks.
3. The measured headspace concentration of hydrogen shall be rounded up to the next highest mole percentage listed in [Table 5.3-10](#), [Table 5.3-11](#), or [Table 5.3-12](#).
4. The required aspiration time corresponding to this molar percentage shall be read from the same table under the column for the number of layers of confinement associated with the appropriate container shipping category.
5. The container shall be qualified for shipment only after being aspirated for this period of time.

**Table 5.3-1 – Aspiration Times Based on Date of Drum Closure
(Waste Type I[Ⓟ])**

Age of Waste		By Number of Layers of Confinement (Days)				
(Months)	(Years)	0	1	2	3	4
1	0.083	0	0	0	0	0
2	0.167	92	92	91	0	0
3	0.250	115	117	120	11	2
4	0.333	128	131	135	37	19
5	0.417	137	141	146	101	35
6	0.500	144	149	154	192	56
7	0.583	150	155	160	239	111
8	0.667	154	160	166	270	219
9	0.750	159	164	171	294	281
10	0.833	162	168	175	312	323
11	0.917	166	172	178	328	355
12	1.000	169	175	182	341	380
14	1.167	174	180	188	363	419
16	1.333	178	185	193	380	449
18	1.500	182	189	197	395	473
20	1.667	186	193	201	407	493
22	1.833	189	196	204	418	510
24	2.000	192	199	207	428	526
26	2.167	194	202	210	437	539
28	2.333	197	205	213	445	551
30	2.500	199	207	215	452	562
32	2.667	201	209	218	459	572
34	2.833	201	211	220	465	582
36	3.000	201	211	222	471	590
38	3.167	201	211	222	477	598
40	3.333	201	211	222	482	606
42	3.500	201	211	222	487	613
44	3.667	201	211	222	491	620
46	3.833	201	211	222	496	626
48	4.000	201	211	222	500	633

Age of Waste		By Number of Layers of Confinement (Days)				
(Months)	(Years)	0	1	2	3	4
50	4.167	201	211	222	504	638
52	4.333	201	211	222	508	644
54	4.500	201	211	222	512	649
56	4.667	201	211	222	515	654
58	4.833	201	211	222	519	659
60	5.000	201	211	222	522	664
66	5.500	201	211	222	531	677
72	6.000	201	211	222	540	690
78	6.500	201	211	222	548	701
84	7.000	201	211	222	555	711
90	7.500	201	211	222	561	720
96	8.000	201	211	222	568	729
102	8.500	201	211	222	568	737
108	9.000	201	211	222	568	745
114	9.500	201	211	222	568	753
120	10.000	201	211	222	568	760
132	11.000	201	211	222	568	760
144	12.000	201	211	222	568	760
156	13.000	201	211	222	568	760
168	14.000	201	211	222	568	760
180	15.000	201	211	222	568	760
192	16.000	201	211	222	568	760
204	17.000	201	211	222	568	760
216	18.000	201	211	222	568	760
228	19.000	201	211	222	568	760
240 ^②	20.000	201	211	222	568	760

Notes:

- ① Aspiration times for Waste Material Types I.1, I.2, and I.3 are identical.
- ② For wastes older than 240 months (20 years), the aspiration times for 240 months shall be used.

**Table 5.3-2 – Aspiration Times Based on Date of Drum Closure
(Waste Material Type II.1)**

Age of Waste		By Number of Layers of Confinement (Days)						
(Months)	(Years)	0	1	2	3	4	5	6
1	0.083	0	0	0	0	0	0	0
2	0.167	91	44	0	0	0	0	0
3	0.250	134	141	0	0	0	0	0
4	0.333	155	174	2	0	0	0	0
5	0.417	170	193	12	4	2	0	0
6	0.500	180	208	72	9	7	5	3
7	0.583	189	219	259	14	11	8	6
8	0.667	196	229	341	19	13	12	9
9	0.750	202	237	394	27	16	13	12
10	0.833	208	244	434	49	21	17	15
11	0.917	212	250	464	290	24	19	17
12	1.000	217	256	491	427	29	22	19
14	1.167	224	265	533	582	48	27	23
16	1.333	230	273	565	677	454	35	27
18	1.500	236	280	592	749	678	49	34
20	1.667	241	286	615	804	815	423	39
22	1.833	245	292	634	848	913	727	51
24	2.000	249	297	651	887	991	910	345
26	2.167	253	301	666	919	1055	1041	739
28	2.333	256	306	680	948	1110	1144	971
30	2.500	260	310	693	975	1157	1228	1136
32	2.667	262	313	706	997	1199	1298	1264
34	2.833	265	317	717	1021	1236	1359	1369
36	3.000	268	320	727	1041	1270	1414	1457
38	3.167	270	323	737	1059	1298	1463	1531
40	3.333	272	326	745	1074	1329	1507	1600
42	3.500	275	328	754	1091	1355	1547	1660
44	3.667	277	331	762	1107	1379	1582	1715
46	3.833	279	333	769	1122	1402	1618	1765
48	4.000	279	336	777	1135	1424	1650	1810

Age of Waste		By Number of Layers of Confinement (Days)						
(Months)	(Years)	0	1	2	3	4	5	6
50	4.167	279	338	784	1147	1444	1680	1851
52	4.333	279	340	791	1159	1463	1708	1892
54	4.500	279	340	797	1171	1479	1734	1929
56	4.667	279	340	803	1182	1498	1759	1963
58	4.833	279	340	809	1192	1515	1783	1996
60	5.000	279	340	815	1203	1530	1805	2027
66	5.500	279	340	831	1232	1572	1865	2110
72	6.000	279	340	845	1258	1613	1919	2183
78	6.500	279	340	858	1282	1646	1970	2247
84	7.000	279	340	870	1303	1678	2014	2306
90	7.500	279	340	881	1323	1709	2055	2358
96	8.000	279	340	892	1341	1737	2092	2407
102	8.500	279	340	892	1359	1762	2126	2451
108	9.000	279	340	892	1375	1785	2157	2493
114	9.500	279	340	892	1390	1808	2188	2531
120	10.000	279	340	892	1404	1829	2216	2567
132	11.000	279	340	892	1430	1868	2268	2633
144	12.000	279	340	892	1430	1903	2315	2692
156	13.000	279	340	892	1430	1935	2358	2746
168	14.000	279	340	892	1430	1963	2397	2795
180	15.000	279	340	892	1430	1963	2433	2842
192	16.000	279	340	892	1430	1963	2467	2885
204	17.000	279	340	892	1430	1963	2499	2924
216	18.000	279	340	892	1430	1963	2528	2960
228	19.000	279	340	892	1430	1963	2528	2995
240 ^①	20.000	279	340	892	1430	1963	2528	3028

Note:

① For wastes older than 240 months (20 years), the aspiration times for 240 months shall be used.

Table 5.3-3 – Aspiration Times Based on Date of Drum Closure (Waste Type III[®])

Age of Waste		By Number of Layers of Confinement (Days)						
(Months)	(Years)	0	1	2	3	4	5	6
1	0.083	0	0	0	0	0	0	0
2	0.167	94	73	0	0	0	0	0
3	0.250	130	138	0	0	0	0	0
4	0.333	149	165	9	3	0	0	0
5	0.417	162	182	28	9	6	4	2
6	0.500	172	195	193	16	11	8	6
7	0.583	179	205	285	23	15	12	10
8	0.667	186	213	338	32	20	16	13
9	0.750	191	220	376	60	24	19	17
10	0.833	196	226	406	274	30	23	20
11	0.917	201	232	430	392	36	26	23
12	1.000	205	237	450	468	46	30	25
14	1.167	212	245	483	570	338	39	31
16	1.333	218	252	509	639	571	56	38
18	1.500	223	259	531	691	703	382	46
20	1.667	227	264	550	733	795	653	63
22	1.833	231	269	566	769	865	813	419
24	2.000	235	274	581	799	923	927	721
26	2.167	238	278	594	825	971	1016	907
28	2.333	242	281	605	849	1013	1088	1042
30	2.500	244	285	616	870	1049	1149	1147
32	2.667	247	288	626	890	1082	1202	1234
34	2.833	250	291	635	907	1111	1249	1307
36	3.000	252	294	644	924	1138	1290	1371
38	3.167	254	297	652	939	1163	1328	1427
40	3.333	256	299	660	953	1186	1362	1477
42	3.500	258	302	667	966	1207	1394	1523
44	3.667	258	304	673	979	1227	1423	1565
46	3.833	258	306	680	991	1245	1450	1603
48	4.000	258	309	686	1002	1263	1476	1638

Age of Waste		By Number of Layers of Confinement (Days)						
(Months)	(Years)	0	1	2	3	4	5	6
50	4.167	258	309	692	1013	1279	1499	1672
52	4.333	258	309	698	1023	1295	1522	1702
54	4.500	258	309	703	1033	1310	1543	1732
56	4.667	258	309	708	1042	1324	1563	1759
58	4.833	258	309	713	1051	1338	1583	1785
60	5.000	258	309	718	1060	1351	1601	1810
66	5.500	258	309	731	1083	1387	1651	1877
72	6.000	258	309	743	1105	1419	1696	1936
78	6.500	258	309	755	1125	1449	1736	1989
84	7.000	258	309	765	1143	1475	1773	2036
90	7.500	258	309	765	1159	1500	1806	2079
96	8.000	258	309	765	1175	1523	1837	2119
102	8.500	258	309	765	1189	1544	1866	2156
108	9.000	258	309	765	1203	1564	1893	2190
114	9.500	258	309	765	1216	1583	1918	2222
120	10.000	258	309	765	1228	1600	1942	2253
132	11.000	258	309	765	1228	1633	1985	2308
144	12.000	258	309	765	1228	1662	2024	2358
156	13.000	258	309	765	1228	1689	2060	2403
168	14.000	258	309	765	1228	1689	2093	2444
180	15.000	258	309	765	1228	1689	2124	2483
192	16.000	258	309	765	1228	1689	2152	2518
204	17.000	258	309	765	1228	1689	2152	2552
216	18.000	258	309	765	1228	1689	2152	2583
228	19.000	258	309	765	1228	1689	2152	2612
240 ^②	20.000	258	309	765	1228	1689	2152	2612

Notes:

- ① Aspiration times for Waste Material Types III.1, III.2, and III.3 are identical.
- ② For wastes older than 240 months (20 years), the aspiration times for 240 months shall be used.

Table 5.3-4 – Aspiration Times Based on Container Headspace Gas Sampling at Time of Venting (Waste Type I[Ⓢ])

Mol % H₂ in headspace	By Number of Layers of Confinement (Days)				
	0	1	2	3	4
0.1	33	40	45	22	23
0.2	79	86	91	65	49
0.3	94	101	107	156	90
0.4	103	109	116	207	192
0.5	109	116	123	236	254
0.6	114	121	128	257	292
0.7	117	124	132	273	320
0.8	121	128	135	285	342
0.9	123	131	138	296	359
1	126	133	141	305	374
1.1	128	135	143	313	387
1.2	130	137	145	320	398
1.3	131	139	147	326	408
1.4	133	141	149	332	417
1.5	134	142	150	337	425
1.6	136	144	152	342	432
1.7	137	145	153	347	439
1.8	138	146	154	351	445
1.9	139	147	155	355	451
2	140	148	157	358	457
2.1	141	149	158	362	462
2.2	142	150	159	365	467
2.3	143	151	160	368	472
2.4	144	152	161	371	476
2.5	145	153	162	374	480
2.6	146	154	162	377	484
2.7	146	155	163	379	488
2.8	147	155	164	382	492
2.9	148	156	165	384	495

Mol % H ₂ in headspace	By Number of Layers of Confinement (Days)				
	0	1	2	3	4
3	149	157	166	386	499
3.1	149	158	166	389	502
3.2	150	158	167	391	505
3.3	150	159	168	393	508
3.4	151	159	168	395	511
3.5	152	160	169	397	514
3.6	152	161	169	399	517
3.7	153	161	170	400	519
3.8	153	162	171	402	522
3.9	154	162	171	404	524
4	154	163	172	405	527
4.1	155	163	172	407	529
4.2	155	164	173	409	532
4.3	156	164	173	410	534
4.4	156	165	174	412	536
4.5	156	165	174	413	538
4.6	157	166	175	415	540
4.7	157	166	175	416	542
4.8	158	166	176	417	544
4.9	158	167	176	419	546
5	158	167	176	420	548
6	162	171	180	432	565
7	165	174	184	442	580
8	168	177	186	450	593
9	170	179	189	458	604
10	172	181	191	465	614
11	174	183	193	471	623
12	175	185	195	477	631
13	177	187	197	482	639
14	178	188	198	487	646
15	180	189	200	491	653
16	181	191	201	496	660

Mol % H₂ in headspace	By Number of Layers of Confinement (Days)				
	0	1	2	3	4
17	182	192	203	500	666
18	183	193	204	504	671
19	184	194	205	507	677
20	185	195	206	511	682
21	186	196	207	514	687
22	187	197	208	518	692
23	188	198	209	521	697
24	189	199	210	524	701
25	190	200	211	527	706
26	191	201	212	530	710
27	191	202	213	532	714
28	192	203	214	535	718
29	193	203	214	538	722
30	193	204	215	540	725
35	197	207	219	552	742
40	199	210	222	562	758

Note:

① Aspiration times for Waste Material Types I.1, I.2 and I.3 are identical.

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Table 5.3-5 – Aspiration Times Based on Container Headspace Gas Sampling at Time of Venting (Waste Material Type II.1)

Mol % H₂ in headspace	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
0.1	52	0	1	1	2	4	5
0.2	112	119	4	7	9	10	12
0.3	132	149	14	13	14	16	17
0.4	143	165	57	18	19	19	20
0.5	152	177	215	24	22	22	22
0.6	158	185	287	34	26	25	26
0.7	164	192	333	106	31	29	28
0.8	168	199	366	287	38	32	31
0.9	171	203	392	389	51	36	35
1	174	207	413	459	221	42	37
1.1	177	210	431	511	393	51	41
1.2	179	214	446	554	506	140	47
1.3	182	216	460	589	590	384	55
1.4	184	219	471	619	657	541	126
1.5	186	223	483	646	711	655	410
1.6	187	224	493	668	758	745	596
1.7	189	226	502	688	798	820	734
1.8	191	228	509	707	834	882	844
1.9	192	230	516	725	865	937	934
2	194	232	525	742	894	985	1011
2.1	195	235	532	756	918	1027	1078
2.2	196	237	538	769	942	1066	1137
2.3	198	237	544	782	964	1101	1189
2.4	199	240	550	794	985	1131	1237
2.5	200	240	555	806	1006	1163	1281
2.6	201	243	560	816	1023	1189	1321
2.7	202	244	565	826	1040	1215	1358
2.8	203	244	570	835	1056	1239	1392
2.9	204	245	574	844	1071	1262	1422
3	205	248	579	853	1086	1285	1454

Mol % H ₂ in headspace	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
3.1	206	249	583	861	1099	1305	1481
3.2	207	249	587	869	1112	1324	1508
3.3	208	250	590	877	1125	1342	1533
3.4	209	251	594	885	1137	1360	1558
3.5	209	252	598	892	1148	1377	1582
3.6	210	253	601	899	1159	1393	1603
3.7	211	254	604	905	1170	1407	1624
3.8	212	255	608	911	1180	1423	1644
3.9	212	256	611	917	1190	1437	1663
4	213	258	614	923	1199	1451	1682
4.1	214	258	617	929	1208	1464	1700
4.2	214	258	620	934	1217	1477	1717
4.3	215	259	622	940	1226	1489	1733
4.4	216	260	625	945	1234	1501	1750
4.5	216	261	628	950	1242	1513	1766
4.6	217	261	630	955	1250	1524	1780
4.7	218	262	633	960	1258	1535	1795
4.8	218	263	635	965	1265	1546	1809
4.9	219	264	638	969	1273	1556	1823
5	219	264	640	974	1280	1566	1837
6	224	271	661	1013	1340	1655	1953
7	229	276	679	1046	1394	1727	2047
8	233	281	694	1075	1436	1788	2127
9	236	285	705	1100	1475	1838	2195
10	239	289	718	1123	1509	1887	2256
11	241	293	729	1143	1540	1929	2310
12	244	296	740	1161	1568	1967	2359
13	246	298	750	1178	1594	2002	2404
14	248	301	756	1194	1617	2035	2446
15	250	302	764	1208	1639	2065	2484
16	252	306	772	1222	1660	2094	2521
17	254	308	779	1235	1680	2120	2555

Mol % H₂ in headspace	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
18	255	310	785	1247	1698	2145	2587
19	257	312	791	1258	1715	2169	2618
20	258	312	797	1269	1732	2192	2647
21	260	314	803	1280	1748	2214	2674
22	261	317	808	1290	1763	2234	2701
23	262	317	813	1300	1778	2254	2726
24	263	320	818	1309	1792	2273	2750
25	264	320	823	1318	1805	2291	2773
26	266	323	828	1327	1818	2309	2795
27	267	323	832	1335	1830	2326	2816
28	268	324	837	1343	1842	2342	2837

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Table 5.3-6 – Aspiration Times Based on Container Headspace Gas Sampling at Time of Venting (Waste Type III[®])

Mol % H₂ in headspace	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
0.1	56	23	0	4	6	8	9
0.2	107	116	12	13	15	16	17
0.3	125	141	32	21	21	21	22
0.4	135	155	169	29	26	26	26
0.5	143	164	246	41	32	30	30
0.6	149	172	291	137	39	35	34
0.7	153	178	323	284	49	40	38
0.8	157	183	346	367	99	46	42
0.9	161	187	366	425	302	56	47
1.0	164	191	382	469	422	99	54
1.1	166	194	395	504	505	333	65
1.2	169	197	407	533	568	479	161
1.3	171	200	418	558	619	583	403
1.4	173	203	427	580	662	664	561
1.5	175	205	436	599	698	729	678
1.6	176	207	444	616	730	784	771
1.7	178	209	451	632	758	831	847
1.8	180	211	458	646	784	873	912
1.9	181	213	464	659	807	909	968
2.0	182	214	470	672	828	943	1018
2.1	184	216	475	683	847	973	1062
2.2	185	217	480	693	865	1000	1102
2.3	186	219	485	703	881	1026	1139
2.4	187	220	490	713	897	1049	1172
2.5	188	221	494	721	912	1071	1203
2.6	189	223	498	730	925	1092	1232
2.7	190	224	502	738	938	1111	1259
2.8	191	225	506	745	951	1129	1284
2.9	192	226	510	752	962	1146	1307
3	193	227	513	759	793	1163	1330

Mol % H ₂ in headspace	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
3.1	194	228	516	766	984	1178	1351
3.2	194	229	520	772	994	1193	1371
3.3	195	230	523	778	1004	1207	1390
3.4	196	231	526	784	1013	1220	1409
3.5	197	232	529	790	1022	1233	1426
3.6	197	233	532	795	1031	1246	1443
3.7	198	233	534	801	1039	1258	1459
3.8	199	234	537	806	1047	1269	1474
3.9	199	235	539	811	1055	1280	1489
4	200	236	542	815	1063	1291	1503
4.1	201	237	544	820	1070	1302	1517
4.2	201	237	547	825	1077	1312	1531
4.3	202	238	549	829	1084	1321	1544
4.4	202	239	551	833	1091	1331	1556
4.5	203	239	554	837	1097	1340	1568
4.6	203	240	556	841	1103	1349	1580
4.7	204	241	558	845	1109	1358	1592
4.8	204	241	560	849	1115	1366	1603
4.9	205	242	562	853	1121	1374	1614
5	205	243	564	857	1127	1382	1624
6	210	248	581	889	1178	1453	1717
7	214	253	596	917	1219	1511	1793
8	217	257	608	940	1255	1561	1857
9	220	260	619	961	1286	1604	1913
10	223	264	629	979	1314	1642	1962
11	225	266	638	995	1339	1677	2006
12	227	269	647	1010	1362	1708	2047
13	229	272	654	1024	1383	1737	2084
14	231	274	661	1037	1403	1764	2119
15	233	276	668	1049	1421	1789	2151
16	235	278	674	1061	1438	1812	2181
17	236	280	680	1071	1454	1834	2209

Mol % H₂ in headspace	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
18	238	282	685	1082	1470	1855	2236
19	239	283	690	1091	1484	1875	2261
20	240	285	695	1100	1498	1894	2285
21	241	286	700	1109	1511	1912	2308
22	243	288	705	1118	1524	1929	2330
23	244	289	709	1126	1536	1945	2351
24	245	291	713	1133	1548	1961	2371
25	246	292	717	1141	1559	1977	2390
26	247	293	721	1148	1570	1991	2409
27	248	294	725	1155	1580	2005	2427
28	249	296	729	1162	1591	2019	2444
29	250	297	732	1168	1600	2032	2460
30	251	298	736	1175	1610	2045	2477
35	255	303	752	1204	1653	2104	2550
40	259	307	766	1230	1692	2155	2615

Note:

① Aspiration times for Waste Material Types III.1, III.2, and III.3 are identical.

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Table 5.3-7 – Aspiration Times Based on Rigid Liner Headspace Gas Sampling at Time of Venting (Waste Type I[Ⓟ])

Mol % H ₂ in rigid liner	By Number of Layers of Confinement (Days)				
	0	1	2	3	4
1	0	0	0	0	6
2	0	0	0	28	40
3	0	0	0	113	141
4	21	47	62	218	291
5	63	77	88	267	360
6	81	93	103	300	406
7	92	103	113	324	439
8	101	111	121	344	467
9	107	118	128	360	489
10	113	123	133	374	508
11	117	128	138	386	525
12	121	132	142	397	540
13	125	136	146	407	554
14	128	139	149	416	566
15	131	142	153	424	577
16	134	145	155	432	588
17	136	147	158	439	597
18	139	150	161	445	606
19	141	152	163	451	615
20	143	154	165	457	623
21	145	156	167	463	630
22	147	158	169	468	637
23	149	160	171	473	644
24	150	161	173	477	651
25	152	163	175	482	657
26	153	165	176	486	663
27	155	166	178	490	668
28	156	168	179	494	674
29	158	169	181	498	679
30	159	170	182	502	684

Mol % H ₂ in rigid liner	By Number of Layers of Confinement (Days)				
	0	1	2	3	4
31	160	172	183	505	689
32	161	173	185	509	693
33	162	174	186	512	698
34	164	175	187	515	702
35	165	176	188	518	706
36	166	177	189	521	711
37	167	178	191	524	715
38	168	180	192	527	718
39	169	181	193	530	722
40	170	182	194	532	726
41	171	182	195	535	729
42	171	183	196	537	733
43	172	184	197	540	736
44	173	185	198	542	739
45	174	186	199	544	743
46	175	187	199	547	746
47	176	188	200	549	749
48	176	189	201	551	752
49	177	189	202	553	755
50	178	190	203	555	758
51	179	191	204	557	761
52	179	192	204	559	763
53	180	192	205	561	766
54	181	193	206	563	769
55	181	194	207	565	771
56	182	195	207	567	774
57	183	195	208	569	776
58	183	196	209	571	779
59	184	197	209	572	781
60	185	197	210	574	784
61	185	198	211	576	786
62	186	198	211	577	788

Mol % H ₂ in rigid liner	By Number of Layers of Confinement (Days)				
	0	1	2	3	4
63	186	199	212	579	790
64	187	200	213	581	793
65	188	200	213	582	795
66	188	201	214	584	797
67	189	201	215	585	799
68	189	202	215	587	801
69	190	203	216	588	803
70	190	203	216	590	805
71	191	204	217	591	807
72	191	204	217	593	809
73	192	205	218	594	811
74	192	205	218	595	813
75	193	206	219	597	815
76	193	206	220	598	817
77	194	207	220	599	819
78	194	207	221	601	820
79	195	208	221	602	822
80	195	208	222	603	824
81	196	209	222	605	826
82	196	209	223	606	827
83	197	210	223	607	829
84	197	210	224	608	831
85	198	210	224	609	832
86	198	211	224	611	834
87	198	211	225	612	836
88	199	212	225	613	837
89	199	212	226	614	839
90	200	213	226	615	840
91	200	213	227	616	842
92	200	214	227	617	843
93	201	214	228	619	845
94	201	214	228	620	846

Mol % H₂ in rigid liner	By Number of Layers of Confinement (Days)				
	0	1	2	3	4
95	202	215	228	621	848
96	202	215	229	622	849
97	202	216	229	623	851
98	203	216	230	624	852
99	203	216	230	625	853
100	203	217	230	626	855

Note:

① Aspiration times for Waste Material Types I.1, I.2, and I.3 are identical.

Table 5.3-8 – Aspiration Times Based on Rigid Liner Headspace Gas Sampling at Time of Venting (Waste Material Type II.1)

Mol % H ₂ in rigid liner	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
1	0	0	0	0	0	3	6
2	0	0	0	6	12	16	19
3	54	44	10	19	24	28	31
4	105	123	173	121	67	64	65
5	128	154	337	499	662	829	1001
6	143	173	415	647	881	1117	1356
7	154	188	466	741	1017	1295	1573
8	163	199	505	810	1116	1423	1730
9	171	208	536	865	1194	1523	1853
10	177	216	562	910	1258	1606	1955
11	182	223	584	948	1312	1676	2041
12	187	229	604	981	1359	1737	2115
13	192	235	621	1011	1401	1791	2181
14	196	240	636	1038	1439	1840	2241
15	200	245	650	1062	1473	1883	2294
16	203	249	663	1084	1504	1924	2344
17	206	253	675	1104	1533	1961	2389
18	209	257	687	1123	1559	1995	2431
19	212	260	697	1140	1584	2027	2470
20	215	263	707	1157	1607	2057	2507
21	217	266	716	1172	1629	2085	2541
22	219	269	724	1187	1649	2111	2574
23	222	272	733	1201	1669	2137	2605
24	224	275	740	1214	1687	2160	2634
25	226	277	748	1226	1705	2183	2662
26	228	280	755	1238	1722	2205	2689
27	230	282	762	1250	1738	2226	2714
28	231	284	768	1261	1753	2245	2738
29	233	286	774	1271	1768	2265	2762
30	235	289	780	1281	1782	2283	2784

Mol % H ₂ in rigid liner	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
31	236	291	786	1291	1796	2301	2806
32	238	292	792	1300	1809	2318	2827
33	240	294	797	1309	1822	2334	2847
34	241	296	802	1318	1834	2350	2867
35	242	298	807	1327	1846	2365	2885
36	244	300	812	1335	1858	2380	2904
37	245	301	817	1343	1869	2395	2921
38	246	303	822	1350	1880	2409	2939
39	248	304	826	1358	1890	2422	2955
40	249	306	830	1365	1901	2436	2972
41	250	307	835	1372	1911	2448	2987
42	251	309	839	1379	1920	2461	3003
43	252	310	843	1386	1930	2473	3018
44	253	312	847	1392	1939	2485	3032
45	255	313	850	1399	1948	2497	3047
46	256	314	854	1405	1957	2508	3061
47	257	315	858	1411	1965	2519	3074
48	258	317	861	1417	1974	2530	3087
49	259	318	865	1423	1982	2541	3100
50	260	319	868	1429	1990	2551	3113
51	261	320	871	1434	1998	2561	3126
52	262	321	875	1440	2006	2571	3138
53	262	323	878	1445	2013	2581	3150
54	263	324	881	1450	2020	2590	3161
55	264	325	884	1455	2028	2600	3173
56	265	326	887	1460	2035	2609	3184
57	266	327	890	1465	2042	2618	3195
58	267	328	893	1470	2049	2627	3206
59	268	329	896	1475	2055	2635	3217
60	268	330	899	1480	2062	2644	3227
61	269	331	901	1484	2068	2652	3237
62	270	332	904	1489	2075	2660	3247

Mol % H ₂ in rigid liner	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
63	271	333	907	1493	2081	2669	3257
64	271	334	909	1498	2087	2676	3267
65	272	334	912	1502	2093	2684	3277
66	273	335	914	1506	2099	2692	3286
67	274	336	917	1510	2105	2700	3295
68	274	337	919	1514	2111	2707	3304
69	275	338	922	1518	2116	2714	3313
70	276	339	924	1522	2122	2722	3322
71	276	340	926	1526	2128	2729	3331
72	277	340	929	1530	2133	2736	3340
73	278	341	931	1534	2138	2743	3348
74	278	342	933	1538	2144	2749	3356
75	279	343	935	1541	2149	2756	3365
76	280	344	937	1545	2154	2763	3373
77	280	344	939	1549	2159	2769	3381
78	281	345	942	1552	2164	2776	3389
79	281	346	944	1556	2169	2782	3396
80	282	346	946	1559	2174	2788	3404
81	283	347	948	1563	2179	2794	3412
82	283	348	950	1566	2183	2801	3419
83	284	349	952	1569	2188	2807	3427
84	284	349	954	1572	2193	2813	3434
85	285	350	956	1576	2197	2818	3441
86	285	351	957	1579	2202	2824	3448
87	286	351	959	1582	2206	2830	3455
88	286	352	961	1585	2210	2836	3462
89	287	353	963	1588	2215	2841	3469
90	287	353	965	1591	2219	2847	3476
91	288	354	967	1594	2223	2852	3483
92	288	354	968	1597	2228	2858	3489
93	289	355	970	1600	2232	2863	3496
94	289	356	972	1603	2236	2868	3502

Mol % H₂ in rigid liner	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
95	290	356	974	1606	2240	2874	3509
96	290	357	975	1609	2244	2879	3515
97	291	358	977	1612	2248	2884	3521
98	291	358	979	1614	2252	2889	3528
99	292	359	980	1617	2256	2894	3534
100	292	359	982	1620	2260	2899	3540

Table 5.3-9 – Aspiration Times Based on Rigid Liner Headspace Gas Sampling at Time of Venting (Waste Type III[®])

Mol % H ₂ in rigid liner	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
1	0	0	0	0	2	6	10
2	0	0	2	11	17	21	25
3	46	46	19	27	33	37	40
4	96	113	196	231	259	296	343
5	118	140	310	468	626	788	953
6	132	158	371	580	791	1004	1218
7	143	171	413	655	898	1143	1389
8	151	181	445	711	978	1246	1515
9	158	189	470	755	1041	1328	1615
10	164	196	492	793	1094	1396	1698
11	169	203	510	824	1139	1454	1769
12	174	208	527	852	1179	1505	1831
13	178	213	541	877	1213	1550	1886
14	181	218	554	899	1245	1590	1936
15	185	222	566	920	1273	1627	1981
16	188	226	577	938	1299	1661	2022
17	191	229	588	955	1324	1692	2060
18	194	233	597	971	1346	1720	2095
19	196	236	606	986	1367	1747	2128
20	199	239	614	1000	1386	1773	2159
21	201	242	622	1013	1405	1796	2188
22	203	244	629	1025	1422	1819	2216
23	205	247	636	1037	1439	1840	2242
24	207	249	643	1048	1454	1860	2266
25	209	251	649	1059	1469	1879	2290
26	211	254	655	1069	1484	1898	2313
27	213	256	661	1079	1497	1915	2334
28	215	258	667	1088	1510	1932	2355
29	216	260	672	1097	1523	1948	2375
30	218	262	677	1106	1535	1964	2394

Mol % H ₂ in rigid liner	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
31	219	263	682	1114	1546	1979	2412
32	221	265	687	1122	1558	1993	2430
33	222	267	692	1130	1568	2007	2447
34	224	268	696	1137	1579	2021	2463
35	225	270	700	1144	1589	2034	2479
36	226	272	704	1151	1599	2046	2495
37	227	273	709	1158	1608	2059	2510
38	229	275	712	1164	1618	2071	2525
39	230	276	716	1171	1627	2082	2539
40	231	277	720	1177	1635	2093	2552
41	232	279	724	1183	1644	2104	2566
42	233	280	727	1189	1652	2115	2579
43	234	281	731	1195	1660	2125	2592
44	235	282	734	1200	1668	2136	2604
45	236	284	737	1206	1676	2145	2616
46	237	285	740	1211	1683	2155	2628
47	238	286	743	1216	1690	2165	2640
48	239	287	746	1221	1698	2174	2651
49	240	288	749	1226	1705	2183	2662
50	241	289	752	1231	1711	2192	2673
51	242	290	755	1236	1718	2200	2683
52	243	291	758	1241	1725	2209	2694
53	244	292	761	1245	1731	2217	2704
54	244	293	763	1250	1737	2225	2714
55	245	294	766	1254	1744	2233	2724
56	246	295	769	1258	1750	2241	2733
57	247	296	771	1263	1755	2249	2743
58	248	297	774	1267	1761	2256	2752
59	248	298	776	1271	1767	2263	2761
60	249	299	778	1275	1773	2271	2770
61	250	300	781	1279	1778	2278	2778
62	251	301	783	1283	1784	2285	2787

Mol % H ₂ in rigid liner	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
63	251	302	785	1286	1789	2292	2796
64	252	302	788	1290	1794	2298	2804
65	253	303	790	1294	1799	2305	2812
66	253	304	792	1297	1804	2312	2820
67	254	305	794	1301	1809	2318	2828
68	255	306	796	1304	1814	2324	2836
69	255	306	798	1308	1819	2331	2843
70	256	307	800	1311	1824	2337	2851
71	257	308	802	1315	1829	2343	2858
72	257	309	804	1318	1833	2349	2866
73	258	309	806	1321	1838	2355	2873
74	258	310	808	1324	1842	2361	2880
75	259	311	810	1327	1847	2366	2887
76	260	311	812	1331	1851	2372	2894
77	260	312	814	1334	1855	2378	2901
78	261	313	815	1337	1860	2383	2907
79	261	314	817	1340	1864	2388	2914
80	262	314	819	1343	1868	2394	2921
81	262	315	821	1346	1872	2399	2927
82	263	315	822	1348	1876	2404	2934
83	263	316	824	1351	1880	2409	2940
84	264	317	826	1354	1884	2414	2946
85	264	317	828	1357	1888	2420	2952
86	265	318	829	1360	1892	2424	2958
87	265	319	831	1362	1896	2429	2964
88	266	319	832	1365	1899	2434	2970
89	266	320	834	1368	1903	2439	2976
90	267	320	835	1370	1907	2444	2982
91	267	321	837	1373	1910	2448	2988
92	268	321	839	1375	1914	2453	2993
93	268	322	840	1378	1918	2458	2999
94	269	323	842	1380	1921	2462	3004

Mol % H₂ in rigid liner	By Number of Layers of Confinement (Days)						
	0	1	2	3	4	5	6
95	269	323	843	1383	1925	2467	3010
96	270	324	844	1385	1928	2471	3015
97	270	324	846	1388	1931	2475	3021
98	271	325	847	1390	1935	2480	3026
99	271	325	849	1392	1938	2484	3031
100	272	326	850	1395	1941	2488	3036

Note:

① Aspiration times for Waste Material Types III.1, III.2, and III.3 are identical.

Table 5.3-10 – Aspiration Times (in Days) Based on Headspace Gas Sampling During Aspiration (Waste Type I^①)

mol %	0 ^②	mol %	1 ^②	mol %	2 ^②	mol %	3 ^②	mol %	4 ^②
0.100	0	0.100	0	0.100	0	0.100	0	0.100	0
0.200	0	0.200	0	0.200	0	0.200	0	0.200	0
0.300	0	0.300	0	0.300	0	0.300	0	0.300	0
0.400	0	0.400	0	0.400	0	0.400	0	0.432	0
0.500	0	0.500	0	0.500	0	0.591	0	0.500	6
0.600	0	0.600	0	0.600	0	0.600	1	0.600	13
0.700	0	0.700	0	0.700	0	0.700	8	0.700	19
0.800	0	0.800	0	0.800	0	0.800	15	0.800	26
0.900	0	0.900	0	0.900	0	0.900	21	0.900	32
1.000	0	1.000	0	1.000	0	1.000	28	1.000	38
1.100	0	1.100	0	1.100	0	1.100	36	1.100	45
1.200	0	1.200	0	1.200	0	1.200	45	1.200	53
1.300	0	1.300	0	1.300	0	1.300	56	1.300	64
1.400	0	1.400	0	1.400	0	1.400	74	1.400	78
1.500	0	1.500	0	1.500	0	1.500	100	1.500	104
1.600	0	1.600	0	1.600	0	1.600	130	1.600	146
1.700	0	1.700	0	1.700	0	1.700	155	1.700	185
1.800	0	1.800	0	1.800	0	1.800	175	1.800	215
1.900	0	1.900	0	1.900	0	1.900	191	1.900	240
2.000	0	2.000	0	2.024	0	2.000	205	2.000	261
2.100	0	2.100	0	2.100	36	2.100	218	2.100	278
2.200	0	2.282	0	2.200	50	2.200	228	2.200	294
2.300	0	2.300	16	2.300	60	2.300	238	2.300	308
2.400	0	2.400	34	2.400	68	2.400	246	2.400	320
2.500	0	2.500	45	2.500	74	2.500	254	2.500	331
2.616	0	2.600	54	2.600	79	2.600	261	2.600	342
2.700	17	2.700	60	2.700	83	2.700	268	2.700	351
2.800	30	2.800	66	2.800	87	2.800	274	2.800	360
2.900	39	2.900	71	2.900	91	2.900	280	2.900	368
3.000	46	3.000	75	3.000	94	3.000	286	3.000	376
3.100	51	3.100	78	3.100	97	3.100	291	3.100	383

mol %	0 ^o	mol %	1 ^o	mol %	2 ^o	mol %	3 ^o	mol %	4 ^o
3.200	56	3.200	81	3.200	100	3.200	296	3.200	390
3.300	60	3.300	84	3.300	102	3.300	300	3.300	396
3.400	64	3.400	87	3.400	104	3.400	305	3.400	402
3.500	67	3.500	90	3.500	106	3.500	309	3.500	408
3.600	70	3.600	92	3.600	108	3.600	313	3.600	414
3.700	73	3.700	94	3.700	110	3.700	316	3.700	419
3.800	75	3.800	96	3.800	112	3.800	320	3.800	424
3.900	78	3.900	98	3.900	114	3.900	323	3.900	429
4.000	80	4.000	100	4.000	115	4.000	327	4.000	434
4.100	82	4.100	101	4.100	117	4.100	330	4.100	438
4.200	84	4.200	103	4.200	118	4.200	333	4.200	442
4.300	85	4.300	105	4.300	120	4.300	336	4.300	446
4.400	87	4.400	106	4.400	121	4.400	339	4.400	450
4.500	89	4.500	107	4.500	122	4.500	342	4.500	454
4.600	90	4.600	109	4.600	124	4.600	344	4.600	458
4.700	92	4.700	110	4.700	125	4.700	347	4.700	462
4.800	93	4.800	111	4.800	126	4.800	350	4.800	465
4.900	94	4.900	112	4.900	127	4.900	352	4.900	469
5.000	96	5.000	114	5.000	128	5.000	354	5.000	472
6.000	106	6.000	123	6.000	137	6.000	375	6.000	500
7.000	114	7.000	131	7.000	145	7.000	392	7.000	523
8.000	120	8.000	137	8.000	151	8.000	405	8.000	542
9.000	126	9.000	142	9.000	156	9.000	417	9.000	558
10.000	130	10.000	146	10.000	160	10.000	427	10.000	572
11.000	134	11.000	150	11.000	164	11.000	436	11.000	585
12.000	137	12.000	154	12.000	167	12.000	444	12.000	596
13.000	141	13.000	157	13.000	170	13.000	452	13.000	606
14.000	143	14.000	160	14.000	173	14.000	458	14.000	616
15.000	146	15.000	162	15.000	176	15.000	465	15.000	624
16.000	148	16.000	165	16.000	178	16.000	470	16.000	632
17.000	151	17.000	167	17.000	180	17.000	476	17.000	640
18.000	153	18.000	169	18.000	182	18.000	481	18.000	647
19.000	155	19.000	171	19.000	184	19.000	486	19.000	654

mol %	0^②	mol %	1^②	mol %	2^②	mol %	3^②	mol %	4^②
20.000	157	20.000	173	20.000	186	20.000	490	20.000	660
21.000	158	21.000	174	21.000	188	21.000	495	21.000	666
22.000	160	22.000	176	22.000	189	22.000	499	22.000	672
23.000	162	23.000	178	23.000	191	23.000	503	23.000	678
24.000	163	24.000	179	24.000	192	24.000	506	24.000	683
25.000	164	25.000	181	25.000	194	25.000	510	25.000	688
26.000	166	26.000	182	26.000	195	26.000	513	26.000	693
27.000	167	27.000	183	27.000	197	27.000	517	27.000	698
28.000	168	28.000	185	28.000	198	28.000	520	28.000	702
29.000	170	29.000	186	29.000	199	29.000	523	29.000	707
30.000	171	30.000	187	30.000	200	30.000	526	30.000	711
35.000	176	35.000	192	35.000	205	35.000	540	35.000	730
40.000	181	40.000	197	40.000	210	40.000	551	40.000	747
45.000	184	45.000	200	45.000	213	45.000	562	45.000	762
50.000	188	50.000	204	50.000	217	50.000	571	50.000	776
55.000	191	55.000	207	55.000	220	55.000	580	55.000	788
60.000	194	60.000	210	60.000	223	60.000	588	60.000	799
65.000	197	65.000	212	65.000	225	65.000	595	65.000	809
70.000	199	70.000	215	70.000	228	70.000	602	70.000	819

Notes:

- ① Aspiration times for Waste Material Types I.1, I.2, and I.3 are identical.
- ② Numbers represent the number of layers of confinement.

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Table 5.3-11 – Aspiration Times (in Days) Based on Headspace Gas Sampling During Aspiration (Waste Material Type II.1)

mol %	0 ^o	mol %	1 ^o	mol %	2 ^o	mol %	3 ^o	mol %	4 ^o	mol %	5 ^o	mol %	6 ^o
0.100	0	0.100	0	0.100	0	0.100	0	0.100	0	0.100	0	0.100	0
0.200	0	0.200	0	0.200	0	0.200	0	0.200	0	0.200	0	0.265	0
0.300	0	0.300	0	0.300	0	0.300	0	0.300	0	0.320	0	0.300	2
0.400	0	0.400	0	0.400	0	0.400	0	0.403	0	0.400	4	0.400	6
0.500	0	0.500	0	0.500	0	0.546	0	0.500	4	0.500	8	0.500	10
0.600	0	0.600	0	0.600	0	0.600	2	0.600	8	0.600	11	0.600	13
0.700	0	0.700	0	0.700	0	0.700	5	0.700	9	0.700	12	0.700	16
0.800	0	0.800	0	0.846	0	0.800	7	0.800	13	0.800	16	0.800	18
0.900	0	0.900	0	0.900	2	0.900	10	0.900	14	0.900	18	0.900	20
1.000	0	1.000	0	1.000	5	1.000	13	1.000	16	1.000	20	1.000	21
1.100	0	1.100	0	1.100	9	1.100	15	1.100	20	1.100	22	1.100	24
1.200	0	1.200	0	1.200	13	1.200	18	1.200	21	1.200	23	1.200	26
1.300	0	1.300	0	1.300	19	1.300	22	1.300	24	1.300	26	1.300	27
1.400	0	1.400	0	1.400	31	1.400	25	1.400	26	1.400	28	1.400	31
1.500	0	1.500	0	1.500	109	1.500	30	1.500	30	1.500	31	1.500	33
1.600	0	1.600	0	1.600	183	1.600	38	1.600	33	1.600	33	1.600	35
1.700	0	1.700	0	1.700	232	1.700	71	1.700	39	1.700	37	1.700	37
1.800	0	1.874	0	1.800	269	1.800	212	1.800	47	1.800	42	1.800	40
1.900	0	1.900	30	1.900	299	1.900	304	1.900	97	1.900	47	1.900	44
2.000	0	2.000	62	2.000	323	2.000	371	2.000	275	2.000	59	2.000	49
2.192	0	2.100	82	2.100	344	2.100	424	2.100	395	2.100	200	2.100	57
2.200	4	2.200	96	2.200	362	2.200	467	2.200	483	2.200	385	2.200	122
2.300	33	2.300	107	2.300	378	2.300	503	2.300	554	2.300	517	2.300	374
2.400	49	2.400	116	2.400	393	2.400	535	2.400	612	2.400	619	2.400	547
2.500	61	2.500	124	2.500	406	2.500	563	2.500	662	2.500	701	2.500	680
2.600	71	2.600	131	2.600	417	2.600	588	2.600	706	2.600	771	2.600	786
2.700	78	2.700	137	2.700	430	2.700	611	2.700	744	2.700	831	2.700	875
2.800	85	2.800	142	2.800	439	2.800	631	2.800	778	2.800	884	2.800	952
2.900	90	2.900	147	2.900	448	2.900	648	2.900	810	2.900	931	2.900	1019
3.000	95	3.000	151	3.000	457	3.000	667	3.000	838	3.000	971	3.000	1078
3.100	100	3.100	155	3.100	465	3.100	684	3.100	864	3.100	1010	3.100	1132
3.200	104	3.200	159	3.200	473	3.200	700	3.200	890	3.200	1047	3.200	1180
3.300	108	3.300	163	3.300	480	3.300	715	3.300	910	3.300	1080	3.300	1225
3.400	111	3.400	166	3.400	487	3.400	726	3.400	930	3.400	1110	3.400	1267
3.500	114	3.500	169	3.500	494	3.500	739	3.500	950	3.500	1139	3.500	1303

mol %	0 ^o	mol %	1 ^o	mol %	2 ^o	mol %	3 ^o	mol %	4 ^o	mol %	5 ^o	mol %	6 ^o
3.600	117	3.600	172	3.600	501	3.600	751	3.600	968	3.600	1163	3.600	1340
3.700	120	3.700	175	3.700	506	3.700	762	3.700	986	3.700	1188	3.700	1373
3.800	123	3.800	178	3.800	511	3.800	773	3.800	1003	3.800	1213	3.800	1404
3.900	125	3.900	180	3.900	516	3.900	783	3.900	1019	3.900	1236	3.900	1431
4.000	127	4.000	183	4.000	523	4.000	793	4.000	1035	4.000	1257	4.000	1459
4.100	130	4.100	185	4.100	528	4.100	802	4.100	1049	4.100	1278	4.100	1486
4.200	132	4.200	187	4.200	532	4.200	811	4.200	1063	4.200	1298	4.200	1512
4.300	134	4.300	189	4.300	537	4.300	822	4.300	1077	4.300	1316	4.300	1536
4.400	136	4.400	191	4.400	542	4.400	830	4.400	1090	4.400	1334	4.400	1559
4.500	137	4.500	193	4.500	546	4.500	838	4.500	1102	4.500	1351	4.500	1582
4.600	139	4.600	195	4.600	550	4.600	845	4.600	1114	4.600	1367	4.600	1603
4.700	141	4.700	197	4.700	554	4.700	852	4.700	1125	4.700	1380	4.700	1624
4.800	142	4.800	199	4.800	558	4.800	859	4.800	1136	4.800	1395	4.800	1643
4.900	144	4.900	200	4.900	562	4.900	865	4.900	1147	4.900	1412	4.900	1662
5.000	145	5.000	202	5.000	566	5.000	872	5.000	1157	5.000	1426	5.000	1680
6.000	158	6.000	216	6.000	598	6.000	930	6.000	1242	6.000	1540	6.000	1830
7.000	168	7.000	226	7.000	623	7.000	975	7.000	1307	7.000	1633	7.000	1947
8.000	176	8.000	237	8.000	644	8.000	1010	8.000	1365	8.000	1708	8.000	2042
9.000	182	9.000	243	9.000	662	9.000	1043	9.000	1411	9.000	1771	9.000	2122
10.000	188	10.000	250	10.000	678	10.000	1073	10.000	1452	10.000	1826	10.000	2192
11.000	193	11.000	256	11.000	692	11.000	1096	11.000	1488	11.000	1875	11.000	2254
12.000	197	12.000	261	12.000	706	12.000	1119	12.000	1521	12.000	1918	12.000	2309
13.000	202	13.000	267	13.000	716	13.000	1139	13.000	1549	13.000	1956	13.000	2359
14.000	205	14.000	271	14.000	726	14.000	1157	14.000	1577	14.000	1995	14.000	2404
15.000	209	15.000	276	15.000	736	15.000	1174	15.000	1604	15.000	2028	15.000	2448
16.000	212	16.000	280	16.000	745	16.000	1190	16.000	1626	16.000	2059	16.000	2487
17.000	215	17.000	284	17.000	753	17.000	1205	17.000	1648	17.000	2088	17.000	2524
18.000	218	18.000	287	18.000	761	18.000	1219	18.000	1668	18.000	2116	18.000	2558
19.000	221	19.000	290	19.000	768	19.000	1232	19.000	1687	19.000	2142	19.000	2591
20.000	224	20.000	293	20.000	775	20.000	1244	20.000	1706	20.000	2166	20.000	2622
21.000	226	21.000	296	21.000	782	21.000	1256	21.000	1723	21.000	2189	21.000	2651
22.000	229	22.000	299	22.000	788	22.000	1267	22.000	1740	22.000	2211	22.000	2679
23.000	231	23.000	300	23.000	794	23.000	1278	23.000	1756	23.000	2233	23.000	2705
24.000	233	24.000	302	24.000	800	24.000	1288	24.000	1771	24.000	2253	24.000	2731
25.000	235	25.000	306	25.000	806	25.000	1298	25.000	1785	25.000	2272	25.000	2755
26.000	237	26.000	308	26.000	811	26.000	1308	26.000	1799	26.000	2291	26.000	2778
27.000	239	27.000	309	27.000	816	27.000	1317	27.000	1813	27.000	2309	27.000	2801
28.000	241	28.000	312	28.000	821	28.000	1325	28.000	1825	28.000	2326	28.000	2822

mol %	0 ^o	mol %	1 ^o	mol %	2 ^o	mol %	3 ^o	mol %	4 ^o	mol %	5 ^o	mol %	6 ^o
29.000	242	29.000	312	29.000	826	29.000	1334	29.000	1838	29.000	2342	29.000	2843
30.000	244	30.000	314	30.000	831	30.000	1342	30.000	1850	30.000	2359	30.000	2863
35.000	250	35.000	324	35.000	852	35.000	1380	35.000	1904	35.000	2431	35.000	2953
40.000	256	40.000	330	40.000	870	40.000	1412	40.000	1953	40.000	2495	40.000	3033
45.000	263	45.000	335	45.000	886	45.000	1441	45.000	1995	45.000	2550	45.000	3102
50.000	266	50.000	339	50.000	901	50.000	1467	50.000	2033	50.000	2600	50.000	3164
55.000	271	55.000	343	55.000	914	55.000	1491	55.000	2067	55.000	2645	55.000	3219
60.000	275	60.000	346	60.000	927	60.000	1513	60.000	2098	60.000	2686	60.000	3270
65.000	278	65.000	351	65.000	938	65.000	1533	65.000	2127	65.000	2724	65.000	3317
70.000	281	70.000	354	70.000	948	70.000	1551	70.000	2154	70.000	2759	70.000	3360

Note:

① Numbers represent the number of layers of confinement.

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Table 5.3-12 – Aspiration Times (in Days) Based on Headspace Gas Sampling During Aspiration (Waste Type III^①)

mol %	0 ^②	mol %	1 ^②	mol %	2 ^②	mol %	3 ^②	mol %	4 ^②	mol %	5 ^②	mol %	6 ^②
0.100	0	0.100	0	0.100	0	0.100	0	0.100	0	0.100	0	0.100	0
0.200	0	0.200	0	0.200	0	0.200	0	0.200	0	0.200	0	0.265	0
0.300	0	0.300	0	0.300	0	0.300	0	0.300	0	0.320	0	0.300	3
0.400	0	0.400	0	0.400	0	0.400	0	0.403	0	0.400	4	0.400	7
0.500	0	0.500	0	0.500	0	0.546	0	0.500	4	0.500	8	0.500	11
0.600	0	0.600	0	0.600	0	0.600	2	0.600	8	0.600	11	0.600	14
0.700	0	0.700	0	0.700	0	0.700	5	0.700	11	0.700	14	0.700	17
0.800	0	0.800	0	0.846	0	0.800	8	0.800	13	0.800	17	0.800	20
0.900	0	0.900	0	0.900	2	0.900	11	0.900	16	0.900	20	0.900	22
1.000	0	1.000	0	1.000	5	1.000	14	1.000	19	1.000	22	1.000	25
1.100	0	1.100	0	1.100	8	1.100	17	1.100	21	1.100	25	1.100	27
1.200	0	1.200	0	1.200	12	1.200	20	1.200	24	1.200	27	1.200	29
1.300	0	1.300	0	1.300	16	1.300	23	1.300	27	1.300	30	1.300	32
1.400	0	1.400	0	1.400	21	1.400	26	1.400	30	1.400	32	1.400	34
1.500	0	1.500	0	1.500	29	1.500	30	1.500	33	1.500	35	1.500	37
1.600	0	1.600	0	1.600	53	1.600	36	1.600	37	1.600	38	1.600	40
1.700	0	1.700	0	1.700	124	1.700	44	1.700	41	1.700	42	1.700	43
1.800	0	1.874	0	1.800	174	1.800	65	1.800	48	1.800	46	1.800	46
1.900	0	1.900	29	1.900	210	1.900	166	1.900	59	1.900	52	1.900	50
2.000	0	2.000	57	2.000	238	2.000	248	2.000	132	2.000	61	2.000	56
2.192	0	2.100	75	2.100	261	2.100	307	2.100	266	2.100	103	2.100	66
2.200	5	2.200	88	2.200	280	2.200	353	2.200	359	2.200	282	2.200	107
2.300	30	2.300	98	2.300	297	2.300	391	2.300	430	2.300	408	2.300	319
2.400	46	2.400	106	2.400	312	2.400	423	2.400	487	2.400	503	2.400	470
2.500	57	2.500	113	2.500	325	2.500	451	2.500	535	2.500	579	2.500	585
2.600	66	2.600	119	2.600	337	2.600	475	2.600	577	2.600	643	2.600	677
2.700	73	2.700	124	2.700	348	2.700	497	2.700	613	2.700	697	2.700	754
2.800	79	2.800	129	2.800	358	2.800	517	2.800	645	2.800	745	2.800	821
2.900	84	2.900	133	2.900	367	2.900	535	2.900	674	2.900	787	2.900	878
3.000	89	3.000	137	3.000	376	3.000	552	3.000	700	3.000	825	3.000	930
3.100	93	3.100	141	3.100	384	3.100	567	3.100	724	3.100	859	3.100	976
3.200	97	3.200	145	3.200	391	3.200	581	3.200	746	3.200	890	3.200	1018
3.300	100	3.300	148	3.300	398	3.300	594	3.300	767	3.300	919	3.300	1056
3.400	103	3.400	151	3.400	405	3.400	607	3.400	786	3.400	946	3.400	1092
3.500	106	3.500	154	3.500	411	3.500	618	3.500	804	3.500	971	3.500	1124

mol %	0°	mol %	1°	mol %	2°	mol %	3°	mol %	4°	mol %	5°	mol %	6°
3.600	109	3.600	156	3.600	417	3.600	629	3.600	820	3.600	995	3.600	1155
3.700	112	3.700	159	3.700	423	3.700	640	3.700	836	3.700	1017	3.700	1184
3.800	114	3.800	161	3.800	428	3.800	650	3.800	851	3.800	1037	3.800	1210
3.900	116	3.900	163	3.900	434	3.900	659	3.900	865	3.900	1057	3.900	1236
4.000	118	4.000	166	4.000	439	4.000	668	4.000	879	4.000	1075	4.000	1260
4.100	121	4.100	168	4.100	443	4.100	677	4.100	892	4.100	1093	4.100	1282
4.200	122	4.200	170	4.200	448	4.200	685	4.200	904	4.200	1110	4.200	1304
4.300	124	4.300	172	4.300	452	4.300	693	4.300	916	4.300	1126	4.300	1325
4.400	126	4.400	173	4.400	456	4.400	701	4.400	927	4.400	1141	4.400	1344
4.500	128	4.500	175	4.500	461	4.500	708	4.500	938	4.500	1156	4.500	1363
4.600	129	4.600	177	4.600	464	4.600	715	4.600	948	4.600	1170	4.600	1381
4.700	131	4.700	179	4.700	468	4.700	722	4.700	958	4.700	1183	4.700	1398
4.800	132	4.800	180	4.800	472	4.800	728	4.800	968	4.800	1197	4.800	1415
4.900	134	4.900	182	4.900	476	4.900	734	4.900	977	4.900	1209	4.900	1431
5.000	135	5.000	183	5.000	479	5.000	741	5.000	986	5.000	1221	5.000	1447
6.000	147	6.000	196	6.000	509	6.000	793	6.000	1063	6.000	1324	6.000	1577
7.000	156	7.000	206	7.000	532	7.000	833	7.000	1122	7.000	1404	7.000	1677
8.000	164	8.000	214	8.000	551	8.000	867	8.000	1171	8.000	1468	8.000	1759
9.000	170	9.000	221	9.000	568	9.000	895	9.000	1212	9.000	1523	9.000	1828
10.000	175	10.000	227	10.000	582	10.000	920	10.000	1247	10.000	1570	10.000	1887
11.000	180	11.000	233	11.000	595	11.000	941	11.000	1279	11.000	1612	11.000	1940
12.000	185	12.000	238	12.000	606	12.000	961	12.000	1307	12.000	1650	12.000	1987
13.000	188	13.000	242	13.000	616	13.000	978	13.000	1333	13.000	1684	13.000	2030
14.000	192	14.000	246	14.000	626	14.000	995	14.000	1356	14.000	1715	14.000	2070
15.000	195	15.000	250	15.000	634	15.000	1010	15.000	1378	15.000	1744	15.000	2106
16.000	198	16.000	253	16.000	642	16.000	1023	16.000	1398	16.000	1771	16.000	2139
17.000	201	17.000	256	17.000	650	17.000	1036	17.000	1417	17.000	1796	17.000	2171
18.000	204	18.000	259	18.000	657	18.000	1048	18.000	1435	18.000	1820	18.000	2200
19.000	206	19.000	262	19.000	663	19.000	1060	19.000	1451	19.000	1842	19.000	2228
20.000	208	20.000	264	20.000	669	20.000	1071	20.000	1467	20.000	1863	20.000	2254
21.000	210	21.000	267	21.000	675	21.000	1081	21.000	1482	21.000	1883	21.000	2279
22.000	213	22.000	269	22.000	681	22.000	1091	22.000	1496	22.000	1902	22.000	2303
23.000	215	23.000	271	23.000	686	23.000	1100	23.000	1510	23.000	1920	23.000	2326
24.000	216	24.000	273	24.000	691	24.000	1109	24.000	1523	24.000	1937	24.000	2347
25.000	218	25.000	275	25.000	696	25.000	1118	25.000	1535	25.000	1953	25.000	2368
26.000	220	26.000	277	26.000	701	26.000	1126	26.000	1547	26.000	1969	26.000	2388
27.000	222	27.000	279	27.000	706	27.000	1134	27.000	1559	27.000	1985	27.000	2407
28.000	223	28.000	281	28.000	710	28.000	1141	28.000	1570	28.000	1999	28.000	2425

mol %	0 ^o	mol %	1 ^o	mol %	2 ^o	mol %	3 ^o	mol %	4 ^o	mol %	5 ^o	mol %	6 ^o
29.000	225	29.000	282	29.000	714	29.000	1149	29.000	1581	29.000	2014	29.000	2443
30.000	226	30.000	284	30.000	718	30.000	1156	30.000	1591	30.000	2027	30.000	2460
35.000	233	35.000	291	35.000	737	35.000	1188	35.000	1638	35.000	2090	35.000	2538
40.000	238	40.000	297	40.000	753	40.000	1216	40.000	1679	40.000	2144	40.000	2605
45.000	243	45.000	302	45.000	767	45.000	1241	45.000	1715	45.000	2191	45.000	2664
50.000	248	50.000	306	50.000	780	50.000	1264	50.000	1748	50.000	2234	50.000	2717
55.000	252	55.000	310	55.000	791	55.000	1284	55.000	1777	55.000	2272	55.000	2764
60.000	255	60.000	314	60.000	802	60.000	1303	60.000	1804	60.000	2307	60.000	2808
65.000	259	65.000	318	65.000	812	65.000	1320	65.000	1829	65.000	2340	65.000	2848
70.000	262	70.000	321	70.000	821	70.000	1336	70.000	1851	70.000	2370	70.000	2885

Notes:

- ① Aspiration times for Waste Material Types III.1, III.2, and III.3 are identical.
- ② Numbers represent the number of layers of confinement.

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6.0 PAYLOAD ASSEMBLY REQUIREMENTS

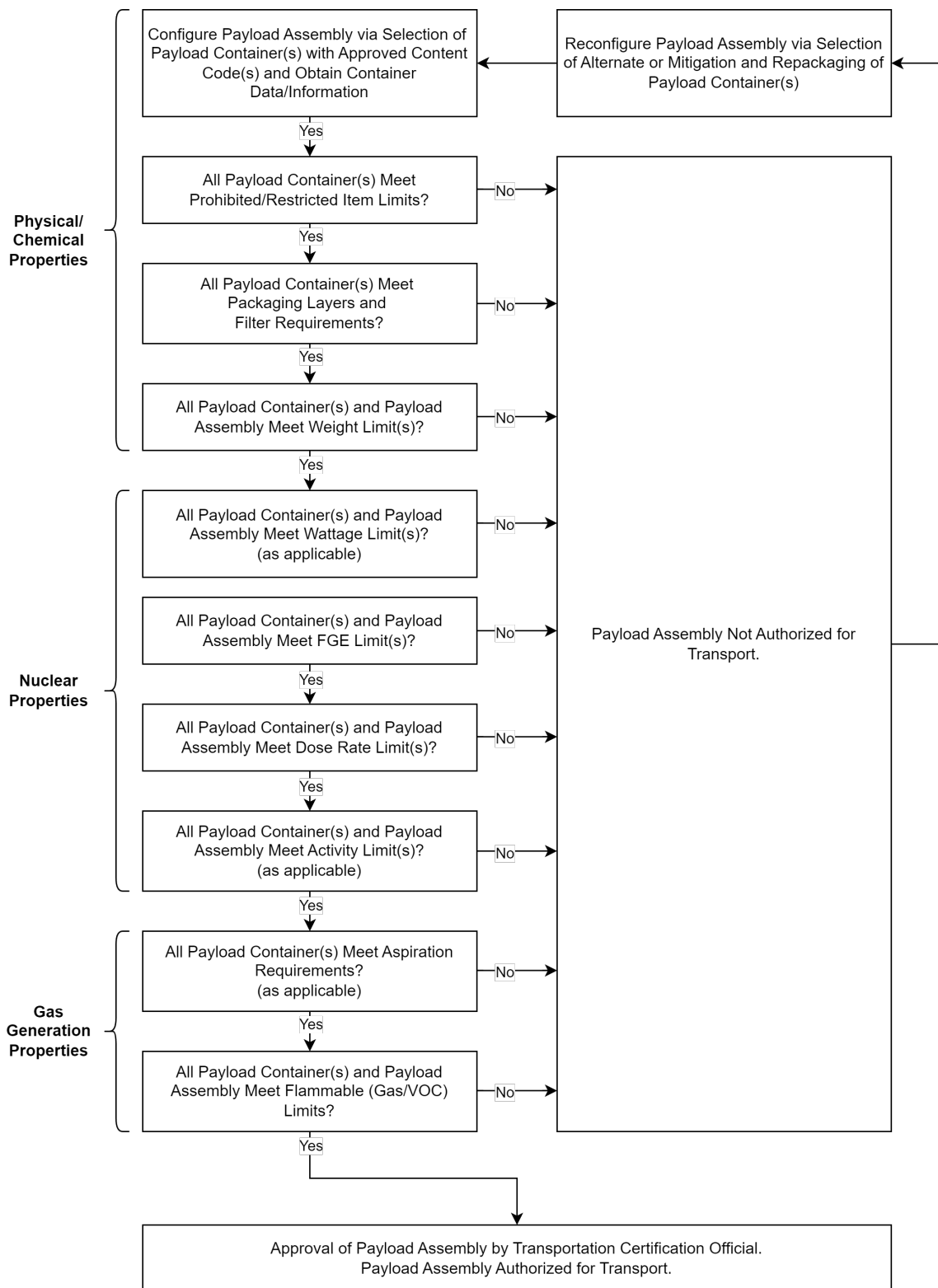
This chapter presents an overview of the control procedures that shall be used by the sites in order to assemble a payload qualified for transport in the TRUPACT-II or the HalfPACT. The parameters described in previous chapters shall be evaluated for selection of a payload. The evaluation of transportation compliance for individual payload container(s) comprising a payload is shown in [Figure 6.1-1](#). The container identification (ID) number(s) shall uniquely identify each payload container. Each payload container shall have an assigned shipping category and content code. Wherever applicable, the physical, nuclear, and gas generation parameter values (weight, fissile material, and wattage) shall be checked against the limits after addition of the associated error, as detailed in previous chapters. If any of the physical, chemical, nuclear, or gas generation property limits are not met, the container and payload assembly shall be rejected from transport (subject to payload container mitigation or repackaging or payload assembly reconfiguration).

Assembled payloads shall comply with the authorized payload configurations specified in [Section 2.1](#). Payload selection shall be made from only those payload containers that have been approved for payload assembly.

6.1 Requirements

Once compliance with all transportation parameter requirements is verified and the PCTCD(s) ([Table 6.2-1](#)) and/or OPCTCD(s) ([Table 6.2-2](#)) is completed, the TRUPACT-II or HalfPACT package shall be authorized for shipment by the site Transportation Certification Official by completing and signing the PATCD ([Table 6.2-3](#)).

The shipping records shall be maintained by the shipper for a minimum period of 3 years.



Note: Compliance determinations illustrated by this figure may occur in any sequence.

Figure 6.1-1 – Evaluation of Transportation Compliance for a Payload Container

6.2 Methods of Compliance and Verification

The following sections detail the procedures for completing the PCTCD (Table 6.2-1), the OPCTCD (Table 6.2-2), and the PATCD (Table 6.2-3). These tables may be reformatted for site use. All parameters noted on the forms shall be included in any modified version. Data on the parameters for specific payload containers and payloads shall be obtained by the methods outlined in this document. As an alternative to manually completing the PCTCD, OPCTCD, and/or PATCD, a verified/validated software package may be used to determine compliance and prepare equivalent forms.

6.2.1 Procedure for Certification of Individual Payload Containers

Generating and storage sites shall qualify an individual payload container for transport in a TRUPACT-II or HalfPACT by verifying that the container meets the parameter requirements/limits listed in the PCTCD (Table 6.2-1) and, if applicable, the OPCTCD (Table 6.2-2). Table 6.2-1 shall be completed for each payload container to be directly loaded into the ICV. If payload containers will be overpacked, Table 6.2-1 shall be completed for each payload container to be overpacked, and Table 6.2-2 shall be completed for the payload container overpacking other payload containers. For example, if four 55-gallon drums will be overpacked in one SWB, Table 6.2-1 must be completed separately for each 55-gallon drum, and Table 6.2-2 must be completed once for the SWB. The required completion of the PCTCD (Table 6.2-1) and as applicable the OPCTCD (Table 6.2-2) for direct-load, fixed, or overpacked configurations is as follows:

Direct Load Configurations:

- 55-Gallon Drum: Select Direct Load “55-Gallon Drum” for waste directly loaded into one 55-gallon drum to be loaded directly into the ICV. Complete Table 6.2-1 for the 55-gallon drum using the appropriate shipping category for the 55-gallon drum.
- 85-Gallon Drum: Select Direct Load “85-Gallon Drum” for waste directly loaded into one 85-gallon drum to be loaded directly into the ICV. Complete Table 6.2-1 for the 85-gallon drum using the appropriate shipping category for the 85-gallon drum.
- 100-Gallon Drum: Select Direct Load “100-Gallon Drum” for waste directly loaded into one 100-gallon drum to be loaded directly into the ICV. Complete Table 6.2-1 for the 100-gallon drum using the appropriate shipping category for the 100-gallon drum.
- SWB: Select Direct Load “SWB” for waste directly loaded into one SWB to be loaded directly into the ICV. Complete Table 6.2-1 for the SWB using the appropriate shipping category for the SWB.
- TDOP: Select Direct Load “TDOP” for waste directly loaded into one TDOP to be loaded directly into the ICV. Complete Table 6.2-1 for the TDOP using the appropriate shipping category for the TDOP.
- Shielded Container: Select Direct Load SC-30G1, SC-30G2, SC-30G3, SC-55G1, or SC-55G2 for waste directly loaded into one shielded container to be loaded directly into the ICV. Complete Table 6.2-1 for the SC-30G1, SC-30G2, SC-30G3, SC-55G1, or SC-55G2 using the appropriate shipping category for the shielded container.

Fixed Configurations:

- Six-Inch Standard Pipe Overpack: Select Fixed Configuration “6-in. Standard Pipe Overpack” for waste packaged in one 6-in. pipe component assembled in one 55-gallon drum, as described in [Section 2.9](#). Complete [Table 6.2-1](#) for the 6-in. standard pipe overpack using the appropriate shipping category for the standard pipe overpack. A 6-in. standard pipe overpack may only be combined in a seven-pack with other 6-in. standard pipe overpacks.
- Twelve-Inch Standard Pipe Overpack: Select Fixed Configuration “12-in. Standard Pipe Overpack” for waste packaged in one 12-in. pipe component assembled in one 55-gallon drum, as described in [Section 2.9](#). Complete [Table 6.2-1](#) for the 12-in. standard pipe overpack using the appropriate shipping category for the standard pipe overpack. A 12-in. standard pipe overpack may only be combined in a seven-pack with other 12-in. standard pipe overpacks.
- S100 Pipe Overpack: Select Fixed Configuration “S100 Pipe Overpack” for waste packaged in one S100 pipe component assembled in one 55-gallon drum, as described in [Section 2.9](#). Complete [Table 6.2-1](#) for the S100 pipe overpack using the appropriate shipping category for the S100 pipe overpack.
- S200 Pipe Overpack: Select Fixed Configuration “S200 Pipe Overpack” for waste packaged in one S200A or S200B pipe component assembled in one 55-gallon drum, as described in [Section 2.9](#). Complete [Table 6.2-1](#) for the S200 pipe overpack using the appropriate shipping category for the S200 pipe overpack.
- S300 Pipe Overpack: Select Fixed Configuration “S300 Pipe Overpack” for waste packaged in one S300 pipe component assembled in one 55-gallon drum, as described in [Section 2.9](#). Complete [Table 6.2-1](#) for the S300 pipe overpack using the appropriate shipping category for the S300 pipe overpack.
- Criticality Control Overpack: Select Fixed Configuration “Criticality Control Overpack” for waste packaged in one criticality control container assembled in one 55-gallon drum, as described in [Section 2.9](#). Complete [Table 6.2-1](#) for the CCO using the appropriate shipping category for the CCO.

Overpacked Configurations:

- 85-Gallon Drum with 55-Gallon Drum (85-Gallon Drum Overpack): Select Overpacked Configuration “85-Gallon Drum with 55-Gallon Drum (85-Gallon Drum Overpack)” for waste packaged in one 55-gallon drum overpacked in one 85-gallon drum. Complete [Table 6.2-1](#) for the 55-gallon drum using the appropriate shipping category for the 85-gallon drum overpack configuration and [Table 6.2-2](#) for the 85-gallon drum (no associated shipping category recorded for the 85-gallon drum).
- SWB with 55-Gallon Drum(s) (SWB /55 Overpack): Select Overpacked Configuration “SWB with 55-Gallon Drum(s) (SWB /55 Overpack)” for waste packaged in one 55-gallon drum to be overpacked in one SWB (up to four 55-gallon drums per SWB). Complete [Table 6.2-1](#) for each of the 55-gallon drums using the appropriate shipping category for each 55-gallon drum and [Table 6.2-2](#) for the SWB (no associated shipping category recorded for the SWB).

- SWB with 85-Gallon Drum(s) (SWB/85 Overpack): Select Overpacked Configuration “SWB with 85-Gallon Drum(s) (SWB/85 Overpack)” for waste packaged in one 85-gallon drum to be overpacked in one SWB (up to two 85-gallon drums per SWB). Complete [Table 6.2-1](#) for each of the 85-gallon drums using the appropriate shipping category for each 85-gallon drum and [Table 6.2-2](#) for the SWB (no associated shipping category recorded for the SWB).
- SWB with 100-Gallon Drum(s) (SWB/100 Overpack): Select Overpacked Configuration “SWB with 100-Gallon Drum(s) (SWB/100 Overpack)” for waste packaged in one 100-gallon drum to be overpacked in one SWB (up to two 100-gallon drums per SWB). Complete [Table 6.2-1](#) for each of the 100-gallon drums using the appropriate shipping category for each 100-gallon drum and [Table 6.2-2](#) for the SWB (no associated shipping category recorded for the SWB).
- TDOP with 55-Gallon Drum(s) (TDOP/55 Overpack): Select Overpacked Configuration “TDOP with 55-Gallon Drum(s) (TDOP/55 Overpack)” for waste packaged in one 55-gallon drum to be overpacked in one TDOP (up to ten 55-gallon drums per TDOP). Complete [Table 6.2-1](#) for each of the 55-gallon drums using the appropriate shipping category for each 55-gallon drum and [Table 6.2-2](#) for the TDOP (no associated shipping category recorded for the TDOP).
- TDOP with 85-Gallon Drum(s) (TDOP/85 Overpack): Select Overpacked Configuration “TDOP with 85-Gallon Drum(s) (TDOP/85 Overpack)” for waste packaged in one 85-gallon drum to be overpacked in one TDOP (up to six 85-gallon drums per TDOP). Complete [Table 6.2-1](#) for each of the 85-gallon drums using the appropriate shipping category for each 85-gallon drum and [Table 6.2-2](#) for the TDOP (no associated shipping category recorded for the TDOP).
- TDOP with 100-Gallon Drum(s) (TDOP/100 Overpack): Select Overpacked Configuration “TDOP with 100-Gallon Drum(s) (TDOP/100 Overpack)” for waste packaged in one 100-gallon drum to be overpacked in one TDOP (up to four 100-gallon drums per TDOP). Complete [Table 6.2-1](#) for each of the 100-gallon drums using the appropriate shipping category for each 100-gallon drum and [Table 6.2-2](#) for the TDOP (no associated shipping category recorded for the TDOP).
- TDOP with SWB (TDOP/SWB Overpack): Select Overpacked Configuration “TDOP with SWB (TDOP/SWB Overpack)” for waste packaged in one SWB to be overpacked in one TDOP (one SWB per TDOP). Complete [Table 6.2-1](#) for the SWB using the appropriate shipping category for the SWB and [Table 6.2-2](#) for the TDOP (no associated shipping category recorded for the TDOP).

6.2.1.1 Payload Container Transportation Certification

[Table 6.2-1](#) shall be completed as follows (section numbers in parentheses refer to sections in the CH-TRAMPAC that provide requirement and compliance and verification information for the transportation parameter described):

Identification Parameters

- Container ID#/Container Bar Code # ([Section 2.4](#)): The site-specific ID number(s) is unique to each container of waste and provides a means for tracking process data

records and package history. These records on the properties of the waste container are referred to as the data package. The container ID number is assigned to the container prior to waste packaging. The ID number appears on a label affixed to the payload container and can be read for visual verification or for electronic retrieval (i.e., bar codes). Information necessary for transporting payload containers is entered into the data package under this ID number.

- Container Designated for Controlled Shipment? ([Section 5.1](#)): Indicate if the container is designated for controlled shipment in accordance with the controls specified in [Appendix 3.6](#) of the [CH-TRU Payload Appendices](#).¹ If the payload container is designated for controlled shipment, the payload container may only be combined in a payload assembly that is designated for controlled shipment as described in [Section 6.2.2](#).
- Shipping Category ([Section 5.1](#)): The shipping category shall be assigned to the payload container by looking up the shipping category from the appropriate content code in the CH-TRUCON. The methodology by which a shipping category is assigned to a content code is shown in [Figure 6.2-1](#).
- Content Code ([Section 1.5](#)): The content code from the data package for the payload container shall be approved as described in [Section 1.5.2](#). The content code of the waste describes its physical and chemical form.
- Container Configuration ([Sections 2.1](#) and [2.9](#)): Select the appropriate authorized direct-load, fixed, or overpacked payload container configuration as summarized in [Section 6.2.1](#).
- Certification Site: The certification site shall be recorded as the location at which transportation certification occurs.

¹ U.S. Department of Energy (DOE), [CH-TRU Payload Appendices](#), U.S. Department of Energy, Carlsbad Field Office, Carlsbad, New Mexico.

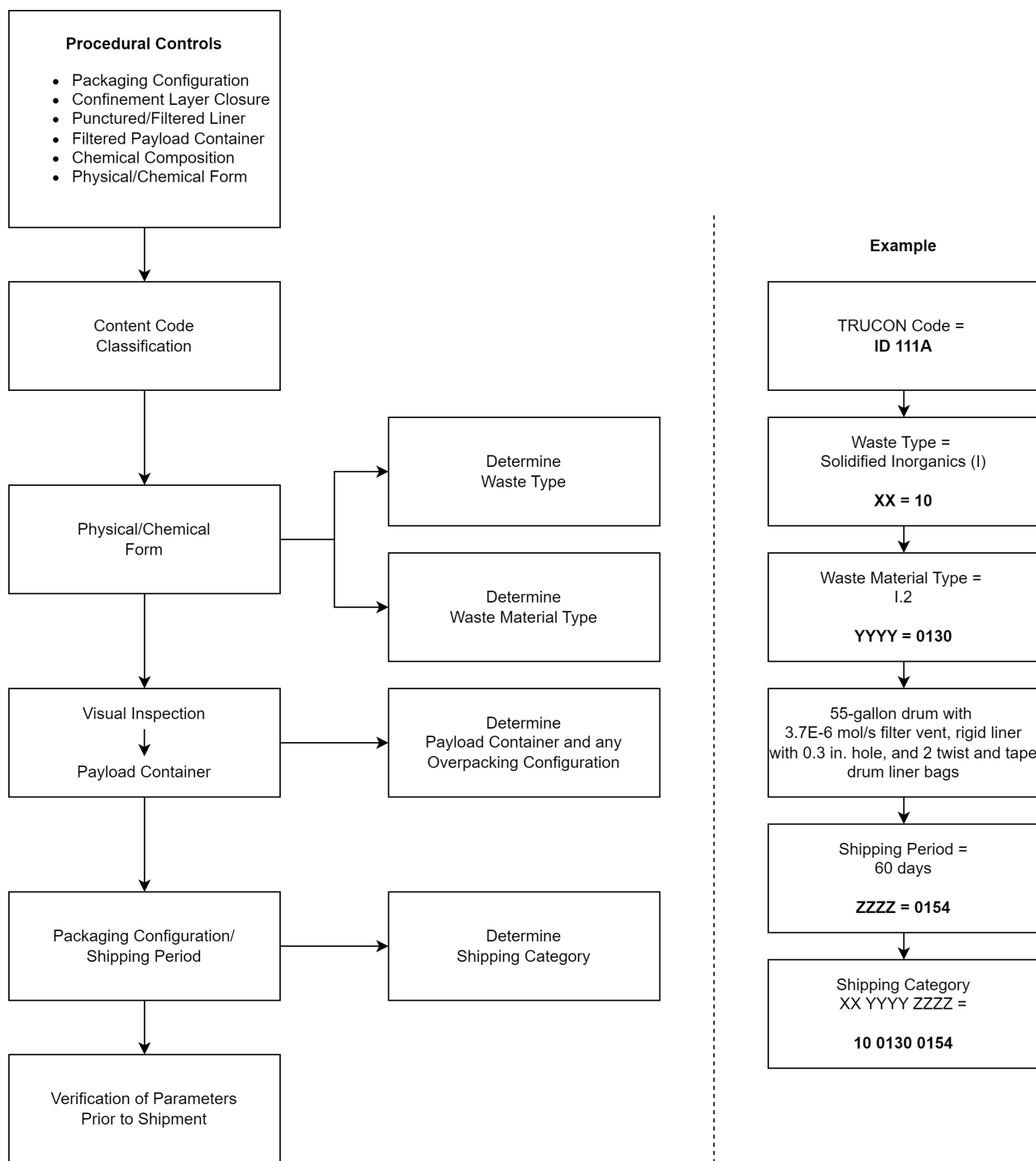


Figure 6.2-1 – Assignment of Shipping Categories

CH-TRAMPAC Transportation Parameters

Compliance information for the CH-TRAMPAC transportation parameters shall be obtained from the data package for the payload container.

- Criteria: The following criteria shall be met:
 - Residual liquids are <1% of payload container volume. ([Section 2.6](#))
 - Sharp/heavy objects are blocked/braced/suitably packaged. ([Section 2.7](#))
 - Nonradioactive pyrophorics are not present or have been reacted. ([Section 4.1](#))
 - Radioactive pyrophorics are ≤1% (weight). ([Section 4.1](#))
 - Explosives are not present. ([Section 4.2](#))
 - Corrosives are not present. ([Section 4.2](#))
 - Pressurized containers and compressed gases are not present. ([Section 4.2](#))
 - Sealed containers that are greater than 4 liters (nominal) are not present, except for Waste Material Type II.2 packaged in metal cans. ([Section 2.8](#))
 - Rigid liner (if present) is punctured/vented. ([Section 2.9](#))
 - Radiation dose rate at the surface of the payload container is ≤ the payload container limit provided in [Table 3.2-1](#) ([Section 3.2](#)). If the payload container will be overpacked, the limit applies only to the outermost payload container.
- Filter(s) Identification ([Section 2.5](#)): Identification shall be listed for the type and number of filters used in a payload container for compliance with the required total minimum hydrogen diffusivity requirement (see [Table 2.5-1](#)). If [Table 6.2-1](#) is being completed for a fixed configuration, filter information shall be listed for both payload containers. If inner filtered confinement layers (filtered metal cans, filtered inner lid, etc.) or equivalent venting mechanisms are used, documented compliance with the specifications in [Section 2.5](#) shall be verified.

Measured Parameters

- Weight ([Section 2.3](#)): The maximum allowable weight for the appropriate payload container type shall be recorded. If the payload container will be overpacked, the limit for the maximum allowable weight applies only to the outermost payload container of the overpacked configuration.

The weight of the payload container may be obtained from its data package. The measured payload container gross weight plus the measurement error (determined from the scale calibration tolerance) or the sum of measured waste weight and measured or reference weights of individually established payload container sub-components plus the root-sum-of-squares (RSS) of the individual errors shall be compared to the maximum allowable weight limit for the appropriate payload container type. If [Table 6.2-1](#) is being completed for a payload container that will be overpacked and the weight of the entire overpack configuration will be determined,

the recorded weight of the individual payload container to be overpacked is not required. Alternatively, the weights of the individual payload containers and the weight of the empty overpack container may be summed to determine the total weight of the overpacked configuration in the completion of [Table 6.2-2](#). The weight of the individual payload container may be recorded in [Table 6.2-1](#) for later use in completing [Table 6.2-2](#) for the overpacked configuration.

- Decay Heat (Section 5.0): The measured decay heat plus the measurement error (one standard deviation) of the payload container may be obtained from its data package. Measured decay heats are determined from the isotopic composition and quantity of radionuclides, as described in [Section 3.1](#). The measured decay heat plus the measurement error shall be recorded for each payload container and used to evaluate compliance of the payload assembly with the applicable design decay heat limits.
- Fissile Mass (FGE) (Section 3.1): The maximum allowable fissile mass limit for the appropriate payload container type shall be recorded. If the payload container will be overpacked, the limit for maximum allowable fissile mass applies only to the outermost payload container of the overpacked configuration.

The measured fissile mass of the payload container may be obtained from its data package. The measured payload container fissile mass value plus two times the measurement error (two standard deviations) shall be compared to the maximum allowable fissile mass limit for the appropriate payload container type. If [Table 6.2-1](#) is being completed for a payload container that will be overpacked, the measured fissile mass of the individual payload container may be recorded in [Table 6.2-1](#) for later use in completing [Table 6.2-2](#) for the overpack configuration.

- S200 Pipe Overpack Activity-Based Dose Sum of Partial Fractions (Section 3.3 and CH-TRU Payload Appendix 4.3): The sum of partial fractions for any combination of radionuclides present in the S200-A or S200-B shield insert configurations shall be recorded and shall be less than or equal to 1.0.

Unvented Waste Parameters (Unvented Waste Only) (Section 1.1)

- Aspiration Method: Option 1, 2A, 2B, or 3 shall be indicated.
- Required Aspiration Period: The required aspiration time for the selected option shall be noted from [Table 5.3-1](#) through [Table 5.3-12](#) of [Section 5.3](#) or determined in accordance with [Appendix 3.7](#) of the [CH-TRU Payload Appendices](#).¹ If the hydrogen concentration indicates that aspiration is not needed, a zero shall be entered.
- Container Closed Time (Option 1 Only): The period of time that a payload container has been unvented in storage shall be recorded.
- Headspace H₂ Concentration (Option 2A, 2B, or 3): The concentration of hydrogen measured in the headspace shall be recorded.
- Time Container Vented: Indicate the number of days the payload container has vented. The time that the container was vented must be greater than or equal to the required aspiration period recorded under “Required Aspiration Period” ([Section 1.1](#)).

Transportation Certification Official ([Section 1.6](#))

The site Transportation Certification Official shall verify that all of the requirements for the above transportation parameters are met as stated in this document. The site Transportation Certification Official shall sign and date the PCTCD upon completion, thereby authorizing the payload container for payload assembly or to be overpacked in another payload container (in which case [Table 6.2-2](#) must also be completed). If the requirements are not met, the payload container is rejected (nonconformance disposition) and is not qualified.

6.2.1.2 Overpack Payload Container Transportation Certification

[Table 6.2-2](#) (the OPCTCD) shall be completed for each payload container overpacking other payload containers. The container type shall be the same for all containers overpacked within the overpack payload container.

[Table 6.2-2](#) shall be completed as follows (section numbers in parentheses refer to the sections in the CH-TRAMPAC that provide requirements and compliance and verification information for the transportation parameters described):

Identification Parameters

- Overpack Container ID#/Overpack Container Bar Code # ([Section 2.4](#)): The site-specific ID number(s) is unique to each container of waste and provides a means for tracking process data records and package history. These records on the properties of the waste container are referred to as the data package. The container ID number is assigned to the container prior to waste packaging. The ID number appears on a label affixed to the payload container and can be read for visual verification or for electronic retrieval (i.e., bar codes). Information necessary for transporting payload containers is entered into the data package under this ID number.
- Container Designated for Controlled Shipment? ([Section 5.1](#)): Using information from [Table 6.2-1](#), which shall be completed separately for each overpacked container, determine if any overpacked container has been designated for controlled shipment. If any overpacked container is designated for controlled shipment, the overpack payload container must be qualified for controlled shipment. Indicate if the overpack payload container is designated for controlled shipment in accordance with the conditions specified in [Appendix 3.6](#) of the [CH-TRU Payload Appendices](#).¹
- Overpack/Overpacked Container Type ([Section 2.1](#)): Select the appropriate authorized overpacked payload container configuration (Overpack Container Type and Overpacked Container Type).
- Weight Limit ([Section 2.3](#)): The maximum allowable weight limit for the appropriate overpack container type (i.e., 85-gallon drum, SWB, or TDOP) shall be recorded. The limit applies only to the outermost payload container. For example, if an SWB will be overpacked in a TDOP, the limit for maximum allowable weight applies only to the TDOP of the overpacked configuration.
- Fissile Mass Limit ([Section 3.1](#)): The maximum allowable fissile mass limit for the appropriate overpack container type (i.e., 85-gallon drum, SWB, or TDOP) shall be recorded. The limit applies only to the outermost payload container.

- Certification Site: The certification site shall be recorded as the location at which transportation certification occurs.

CH-TRAMPAC Transportation Parameters

Compliance information for the CH-TRAMPAC transportation parameters shall be obtained from the data package for the payload container. The following criteria shall be met:

- Surface radiation dose rate $\leq$ payload container limit (Section 3.2): The radiation dose rate limit at the surface of the overpack container shall be less than or equal to the payload container limit provided in Table 3.2-1 of Section 3.2. The limit applies only to the outermost payload container. For example, if an SWB will be overpacked in a TDOP, the limit for radiation dose rate applies only to the TDOP.
- Filter(s) Identification (Section 2.5): Identification shall be listed for the type and number of filters used in an overpack container (i.e., 85-gallon drum, SWB, or TDOP) for compliance with the total minimum hydrogen diffusivity requirement (from Table 2.5-1 of Section 2.5). For overpacked container(s), documented compliance with the specifications in Section 2.5 shall be verified [e.g., as recorded on associated PCTCD(s) or OPCTCD].

Overpacked Container Measured Parameters

- Overpacked Container ID Number (Section 2.4): List the site-specific identification number for each of the overpacked payload containers.
- Weight and Associated Error (Section 2.3): The weight and associated error for each overpacked container and dunnage container (if applicable) or the entire overpack configuration shall be recorded. The weight and associated error may be obtained from the PCTCD(s) (Table 6.2-1) for each overpacked container. An established reference weight with an associated error of zero may be used for any dunnage container included in the overpack configuration. The weight values for the individual overpacked containers are not required if the entire overpack configuration will be weighed. If the weight of each individual overpacked container is recorded, enter the measured weight and measurement error (determined from the scale calibration tolerance) of the empty overpack container or the applicable reference weight with an associated error of zero on the last row. Calculate the total weight as the sum of each of the individual weights (including the weight of the empty overpack container) and record this value in the appropriate box. Calculate the RSS error for the overpack configuration as the square root of the sum of the squares of the individually listed errors and record this value in the appropriate box. If the entire overpack configuration is weighed, record the total measured weight of the overpack configuration and the measurement error (determined from the scale calibration tolerance) in the appropriate boxes (in this case, the measurement error is the same as the RSS error).

The total weight plus the total RSS error of the entire overpack configuration shall be recorded in the space provided and compared to the maximum allowable weight limit for the outermost payload container of the overpack configuration (i.e., 1,000 lbs. per 85-gallon drum, 4,000 lbs. per SWB, or 6,700 lbs. per TDOP).

- Measured Decay Heat and Measurement Error (Section 5.0): The measured decay heat value and measurement error (one standard deviation) for each overpacked container shall be recorded. The measured decay heat value and measurement error may be obtained from the PCTCD(s) (Table 6.2-1) for each overpacked payload container.

The spaces provided for “Total” and “Total RSS Error” are for use in completing Table 6.2-3.

- Measured Fissile Mass (FGE) and Measurement Error (Section 3.1): The measured fissile mass and two times the measurement error (two standard deviations) for each overpacked container shall be recorded as obtained from the PCTCD(s) (Table 6.2-1) for each overpacked payload container.

The subtotal fissile mass plus the total RSS error (RSS of twice each individual measurement error) of the entire overpack configuration shall be compared to the maximum allowable fissile mass limit for the outermost payload container of the overpack configuration.

- Flammability Index (Section 5.2 and Section 6.2.4): Payload container gas generation parameter values (decay heat; headspace hydrogen, methane, and flammable VOC concentrations; and/or UFGTP FGGR) are used to calculate FI values. Calculate the FI for each overpacked container as described in Section 5.2 and Section 6.2.4 and Appendix 2.4 of the CH-TRU Payload Appendices¹ after selecting the overpacking containers comprising the payload assembly. Calculation is typically performed by a verified/validated software package.

Certification of Compliance

The site Transportation Certification Official shall verify compliance with the requirements for weight, fissile mass, and FI, as applicable to the overpack configuration.

Transportation Certification Official (Section 1.6)

The site Transportation Certification Official shall verify that all of the requirements for the above transportation parameters are met as stated in this document. The site Transportation Certification Official shall sign and date the OPCTCD upon completion, thereby authorizing the payload container for payload assembly. If the requirements are not met, the payload container is rejected (nonconformance disposition) and is not qualified.

Table 6.2-1 – Payload Container Transportation Certification Document (PCTCD)

IDENTIFICATION PARAMETERS					
Container ID # _____		Container Bar Code # _____			
Container Designated for Controlled Shipment?		☐ YES ☐ NO			
Shipping Category _____		Content Code _____			
Container Configuration _____					
Certification Site _____					
CH-TRAMPAC TRANSPORTATION PARAMETERS					
Criteria			Filter(s) Identification		
Residual liquids are <1% of payload container volume Sharp/heavy objects are blocked/braced/suitably packaged Nonradioactive pyrophorics are not present or have been reacted Radioactive pyrophorics are ≤1% (weight) Explosives are not present Corrosives are not present Pressurized containers and compressed gases are not present Sealed containers >4 liters are not present (except in Waste Material Type II.2) Rigid liner (if present) is punctured/vented Surface radiation dose rate is ≤ payload container limit ^①			1. _____ 2. _____		
			3. _____ 4. _____		
			5. _____ 6. _____		
			7. _____ 8. _____		
			9. _____ 10. _____		
MEASURED PARAMETERS					
Container Parameter	Value	Error	Value + 1x Error	Value + 2x Error	Limit
Weight (lbs.)					①
Decay Heat (watts)					20 ^②
Fissile Mass (FGE)					①
S200 Pipe Overpack Activity-Based Dose Sum of Partial Fractions: _____ ≤1.0					
UNVENTED WASTE PARAMETERS					
Aspiration Method	☐ Option 1 ☐ Option 2A ☐ Option 2B ☐ Option 3		Required Aspiration Period _____ days		
Option 1					
Container closed time _____ months					
Options 2A, 2B, or 3					
Headspace H ₂ Concentration _____ mol%		Time Container Vented _____ days			

I certify that the above container meets all the requirements for shipment as stated in the CH-TRAMPAC, current revision. The container is approved for ☐ payload assembly or ☐ overpacking.

_____/_____
 TRANSPORTATION CERTIFICATION OFFICIAL / DATE

Notes:

① Limits are not applicable if container is to be overpacked.

② Applies only to CCO payload container ([Section 5.0](#)).

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Table 6.2-2 – Overpack Payload Container Transportation Certification Document (OPCTCD)

IDENTIFICATION PARAMETERS			
Overpack Container ID # _____	Overpack Container Bar Code # _____		
Container Designated for Controlled Shipment?	☐ YES ☐ NO		
Overpack Container Type	☐ 85-Gallon Drum	☐ SWB	☐ TDOP
Overpacked Container Type	☐ 55-Gallon Drum(s)	☐ 85-Gallon Drum(s)	☐ 100-Gallon Drum(s)
	☐ SWB		
Weight Limit _____ (lbs.)	Fissile Mass Limit _____ FGE		
Certification Site _____			

CH-TRAMPAC TRANSPORTATION PARAMETERS					
Surface Radiation Dose Rate $\leq$ payload container limit					
Filter(s) Identification	1. _____	2. _____	3. _____	4. _____	5. _____
	6. _____	7. _____	8. _____	9. _____	10. _____

OVERPACKED CONTAINER MEASURED PARAMETERS							
Overpacked Container ID Number	Weight (lbs.)	Error (lbs.)	Decay Heat (watts)	Error (watts)	Fissile Mass (FGE)	2x Error (FGE)	Flammability Index
			←Weight and error of empty overpack container (if applicable)				
Total (A)							
Total RSS Error							
Total + RSS Error							

CERTIFICATION OF COMPLIANCE

Total weight + weight error $\leq$ limit

Total fissile mass + 2x fissile mass error $\leq$ limit

FI of each overpacked container meets the following condition: $0 \leq FI \leq 50,000$

I certify that the above overpack container meets all the requirements for shipment as an overpack container as stated in the CH-TRAMPAC, current revision. The overpack is approved for

☐ payload assembly or ☐ overpacking.

_____/_____
TRANSPORTATION CERTIFICATION OFFICIAL / DATE

6.2.2 Procedure for Assembly and Certification of a CH-TRU Payload

Generating and storage sites shall qualify a payload for transport in the TRUPACT-II or HalfPACT by verifying that the payload meets the parameter requirements/limits listed in the PATCD ([Table 6.2-3](#)).

[Table 6.2-3](#) shall be completed as follows (section numbers in parentheses refer to sections in the CH-TRAMPAC that provide requirement and compliance and verification information for the transportation parameter described):

Identification Parameters

- Shipment #: The shipment number of the trailer or railcar of TRUPACT-IIs or HalfPACTs shall be recorded. For shipments by railcar, each railcar shall be assigned a separate shipment number or shipment number suffix.
- Packaging OCA Body/Lid #: The identification numbers on the TRUPACT-II or HalfPACT OCA body and lid shall be recorded.
- Payload Assembly Designated for Controlled Shipment? ([Section 5.1](#)): Using information from the PCTCD ([Table 6.2-1](#)) and OPCTCD ([Table 6.2-2](#)), determine if any payload container comprising the payload assembly has been designated for controlled shipment. If any container is designated for controlled shipment, the payload assembly must be qualified for controlled shipment. Indicate if the payload assembly is designated for controlled shipment in accordance with the conditions specified in [Appendix 3.6](#) of the [CH-TRU Payload Appendices](#).¹ If the payload assembly is designated for controlled shipment, [Table 6.2-4](#) must also be completed for the shipment of the payload ([Section 6.2.3](#)).
- Type of Payload ([Section 2.1](#)): The payload configuration shall consist of an approved type of payload container. If the payload type is “Standard Pipe Overpacks,” each seven-pack in the payload must be comprised of only 6-in. standard pipe overpacks or 12-in. standard pipe overpacks.
- Transportation Packaging: The type of packaging used (TRUPACT-II or HalfPACT) shall be recorded.
- Date ICV Closed: The date that the ICV is closed shall be recorded.
- Time ICV Closed: For payloads designated for controlled shipment only, the time that the ICV is closed shall be recorded. For other payloads, the time that the ICV is closed need not be recorded.

Bottom Payload Assembly Composition and Top Payload Assembly Composition

The following data shall be recorded from each PCTCD ([Table 6.2-1](#)) or from the OPCTCD ([Table 6.2-2](#)), as applicable, for each payload container/overpack comprising the payload:

- Payload container/overpack ID number or “DUNNAGE” or “EMPTY” ([Section 2.4](#))
- Weight and associated error for each payload container or applicable reference weight with an associated error of zero for any dunnage container included in the payload assembly ([Section 2.3](#))

- Measured decay heat and measurement error (one standard deviation) ([Section 5.0](#))
- Measured fissile mass and two times the measurement error (two standard deviations or one times the RSS error if values are taken from an OPCTCD) ([Section 3.1](#))

If the data are obtained from the OPCTCD, errors should be the calculated RSS errors (the square root of the sum of the squares of the individually listed errors).

Pursuant to [Section 6.2.4](#), calculate and record the FI for each payload container. If the payload assembly consists of overpacked configurations, indicate that the FI is not applicable and record the FI for each overpacked container on the appropriate OPCTCD ([Table 6.2-2](#)). This calculation is typically performed using a verified/validated software package.

Payload Totals

- Weight of Payload Handling Materials ([Section 2.3](#)): The total measured weight and measurement error of the payload handling materials (e.g., dunnage containers, pallets, spacers, guide tubes, slip sheets, reinforcing plates, banding material, slings, axial and radial dunnage, etc.) for TRUPACT-II or HalfPACT payloads or the applicable reference weight with an associated error of zero is recorded. If the weight is determined through a single measurement of the payload assembly, indicate the weight of the payload handling materials is not applicable (e.g., “NA”).
- Total Weight/Total RSS Weight Error ([Section 2.3](#)): The sum of the subtotal weights including the payload handling materials and the total RSS weight error shall be recorded. If the weight is determined through a single measurement of the payload assembly, the associated error is simply the error for that one measurement.
- Total Decay Heat/Total RSS Decay Heat Error ([Section 5.0](#)): The sum of the decay heats and the total RSS error shall be recorded.
- Total Fissile Mass/Total RSS Fissile Mass Error ([Section 3.1](#)): The sum of the fissile masses and the total RSS error shall be recorded.
- Bottom Assembly Weight plus Subtotal RSS Error (TRUPACT-II Only): For the TRUPACT-II, the subtotal weight plus the subtotal RSS error for the bottom layer of payload containers/overpacks shall be recorded.
- Top Assembly Weight plus Subtotal RSS Error (TRUPACT-II Only): For the TRUPACT-II, the subtotal weight plus the subtotal RSS error for the top layer of payload containers/overpacks shall be recorded.
- Total Weight plus Total RSS Error: The sum of the total weights including the payload handling materials plus the total RSS weight error shall be recorded.
- Total Decay Heat plus Total RSS Error: The sum of the total decay heat values plus the total RSS decay heat error shall be recorded.
- Total Fissile Mass plus Total RSS Error: The sum of the total fissile mass values plus the total RSS fissile mass error shall be recorded.

Payload Certification Parameters

The site Transportation Certification Official shall verify compliance with the following requirements.

- Flammability Index of Each Payload Container Greater Than Zero and Less Than or Equal to 50,000 (Section 5.2 and Section 6.2.4): Payload container gas generation parameter values (decay heat; headspace hydrogen, methane, and flammable VOC concentrations; and/or UFGTP FGGR) are used to calculate FI values. Calculate the FI for each container as described in Section 5.2, Section 6.2.4 and Appendix 2.4 of the CH-TRU Payload Appendices after selecting the containers comprising the payload assembly. Calculation is typically performed by a verified/validate software package.

Record the FI for each container. The FI for each container must be a non-negative number less than or equal to 50,000 for the payload to be eligible for shipment.

- Payload Total Gas Release Rate Less Than or Equal to Limit Calculated in Accordance With Section 6.2.5 (Section 6.2.5): Payload total gas release rate limits apply only to payloads containing one or more Waste Type IV payload containers that exceed the payload container total gas release rate limit specified in Table 5.2-1 of Section 5.2. The site Transportation Certification Official shall verify that the determined payload total gas release rate is less than or equal to the appropriate payload total gas release rate limit pursuant to the methodology defined in Section 6.2.5.
- Bottom Weight Greater Than or Equal to Top Weight (Section 2.3) (TRUPACT-II Only): For the TRUPACT-II, the subtotal weight plus subtotal RSS weight error of the bottom layer of payload containers/overpacks/drums overpacked in a TDOP shall be greater than or equal to that of the top layer.
- Total Weight plus RSS Error Less Than or Equal to Packaging Limit (Section 2.3): The total payload weight including the weight of the payload handling materials plus the total RSS weight error shall be less than or equal to 7,265 lbs. for TRUPACT-II payloads or 7,600 lbs. for HalfPACT payloads.
- Decay Heat plus RSS Error Less Than or Equal to Design Decay Heat Limits (Section 5.0): The measured decay heat value plus the RSS (as applicable) decay heat error shall be less than or equal to 40 W for TRUPACT-II payloads or 30 W for HalfPACT payloads.
- Fissile Mass (Pu-239 FGE) plus RSS Error Less Than or Equal to Payload Limit (Section 3.1): The total measured fissile mass (Pu-239 FGE) plus the total RSS fissile mass error shall be less than or equal to the maximum allowable fissile mass limit established for the payload configuration and waste contents.
- Payload Activity-Based Dose Sum of Partial Fractions (Section 3.3): The payload activity-based dose sum of partial fractions for a payload assembly plus the error (i.e., one standard deviation) shall be less than or equal to the applicable payload sum of partial fractions limit. Not applicable for payloads of S100, S200, or S300 pipe overpacks.
- S100 or S300 Pipe Overpack Payload Activity (Section 3.3): The total activity for a payload assembly of S100 or S300 pipe overpacks plus the error (i.e., one standard deviation) shall be less than or equal to 406 Ci.

- Payload A_2 Ci ([Section 3.3](#)): The total activity for a payload assembly plus the error (i.e., one standard deviation) shall be less than or equal to $10^5 A_2$ Ci.

Transportation Certification Official ([Section 1.6](#))

The site Transportation Certification Official shall verify that all of the requirements for the above transportation parameters are met as stated in this document. The site Transportation Certification Official shall sign and date the PATCD upon verifying that the CH-TRAMPAC transportation requirements are met and the payload is qualified for transport. If the requirements are not met, the payload is rejected (payload may be reconfigured and reevaluated against the payload requirements in this document).

Table 6.2-3— Payload Assembly Transportation Certification Document (PATCD)

IDENTIFICATION PARAMETERS	
Shipment #: _____	Packaging OCA Body/Lid #: _____
Payload Assembly Designated for Controlled Shipment? ^① ☐ YES ☐ NO	
Type of Payload: _____	
Transportation Packaging: ☐ TRUPACT-II ☐ HalfPACT	
Date ICV Closed: _____	Time ICV Closed: _____ ^②

BOTTOM PAYLOAD ASSEMBLY COMPOSITION							
Container/ Overpack ID	Weight (lbs.)	Error (lbs.)	Decay Heat (watts)	Error (watts)	Fissile Mass (FGE)	2x Error (FGE)	Flammability Index ^③
Subtotal (A)							
Subtotal RSS Error (C)							
TOP PAYLOAD ASSEMBLY COMPOSITION (TRUPACT-II Only)							
Container/ Overpack ID	Weight (lbs.)	Error (lbs.)	Decay Heat (watts)	Error (watts)	Fissile Mass (FGE)	2x Error (FGE)	Flammability Index ^③
Subtotal (B)							
Subtotal RSS Error (D)							

PAYLOAD TOTALS

Weight of Payload Handling Materials (E) _____ lbs. Error (F): _____ lbs.
 Total (A+E [+B (TRUPACT-II only)]) Weight: _____ lbs. Total RSS Weight Error (including F): _____ lbs.
 Total Decay Heat: _____ watts Total RSS Decay Heat Error: _____ watts
 Total Fissile Mass: _____ FGE Total RSS Fissile Mass Error: _____ FGE
 Bottom Assembly Weight plus Subtotal RSS Error (A+C) (TRUPACT-II Only) _____ lbs.
 Top Assembly Weight plus Subtotal RSS Error (B+D) (TRUPACT-II Only) _____ lbs.
 Total Weight plus Total RSS Error _____ lbs.
 Total Decay Heat plus Total RSS Error _____ watts
 Total Fissile Mass plus Total RSS Error _____ FGE

PAYLOAD CERTIFICATION PARAMETERS

Flammability Index of Each Payload Container Meets the Following Condition: $0 \leq FI \leq 50,000$

Payload Total Gas Release Rate _____ mol/s^⑤ Limit _____ mol/s^⑤

Bottom Weight $\geq$ Top Weight (for TRUPACT-II)

Total Weight plus RSS Error $\leq$ 7,265 lbs. for TRUPACT-II; $\leq$ 7,600 lbs. for HalfPACT

Payload Decay Heat Plus RSS Error _____ W Package Design Decay Heat Limit _____ W

Fissile Mass (Pu-239 FGE) plus RSS Error $\leq$ _____ FGE

Payload Activity-Based Dose Sum of Partial Fractions _____ $\leq$ _____^⑤

S100 or S300 pipe overpack payload activity _____ $\leq$ 406 Ci

_____ A₂ Ci $\leq$ 10⁵ A₂ Ci

I certify that the above payload assembly meets all the requirements for shipment as stated in the CH-TRAMPAC, current revision. The payload assembly is approved for shipment.

_____/_____
 TRANSPORTATION CERTIFICATION OFFICIAL / DATE

Notes:

- ① If the payload is designated for controlled shipment, [Table 6.2-4](#) must also be completed for the shipment of this payload as specified in [Section 6.2.3](#).
- ② Required only for payloads designated for Controlled Shipment (10 days).
- ③ If the payload assembly consists of overpacked configurations, the FI values for each overpacked container are recorded on the appropriate OPCTCD.
- ④ Applicable if shipping pursuant to [Section 6.2.5](#).
- ⑤ Not applicable for S100, S200, or S300 pipe overpack payloads.

6.2.3 Shipments Designated as Controlled Shipments

Compliance with the 10-day shipping period is administratively controlled in accordance with the conditions of [Appendix 3.6](#) of the [CH-TRU Payload Appendices](#)¹ and through the following steps. These steps must be completed by the site Transportation Certification Official, or designee, and the designated receiving site operations personnel, as applicable.

Loading Time

The loading time begins with the closure of the ICV and ends with the departure of the shipment from the site. The loading time is limited to a maximum of 24 hours. The following steps must be completed to ensure compliance with the 24-hour loading time:

- 6.2.3.1 Review PATCD to determine date and time that the ICV closure was completed. Record date and time on [Table 6.2-4 – Shipping Site Control Checklist for Controlled Shipments](#)^o may be reformatted for site use provided that the same information is recorded.
- 6.2.3.2 Note date and time that the shipment containing the loaded package is scheduled to depart the site. Record date and time on [Table 6.2-4](#).
- 6.2.3.3 Review dates and times recorded in Steps 6.2.3.1 and 6.2.3.2 to calculate total loading time. If total loading time is less than or equal to 24 hours, proceed to Step 6.2.3.4. If total loading time exceeds 24 hours, the package must be vented for a period at least as long as the period the TRUPACT-II or HalfPACT ICV was sealed, and the closure process must be repeated. Return to Step 6.2.3.1 above.
- 6.2.3.4 Indicate compliance with the 24-hour loading time by signature on [Table 6.2-4](#).

Transport and Unloading Time

The transport and unloading time begins with the departure of the shipment from the shipping site and ends with the venting of the package at the receiving site. The maximum transport and unloading time is 9 days. The following steps must be completed to document compliance:

- 6.2.3.5 Review [Table 6.2-4](#) to determine the date and time that the package was scheduled to depart from the shipping site. Record this date and time on [Table 6.2-5 – Receiving Site Control Checklist for Controlled Shipments](#)^o. [Table 6.2-5](#) may be reformatted for site use provided that the same information is recorded.
- 6.2.3.6 Using the date and time recorded in Step 6.2.3.5, ensure that the package is vented within 9 days of the departure of the shipment from the shipping site by implementing the site unloading procedures specific to controlled shipments. Record the date and time to show compliance. Operational procedures must be in place for unloading at the receiving site to ensure that the controlled shipments are not unattended beyond the maximum 9-day transport and unloading time. If total transport and unloading time exceeds 9 days, the shipment is noncompliant with the Certificate of Compliance for the applicable package and subject to the reporting requirements defined by 10 CFR §71.95.
- 6.2.3.7 Indicate compliance with the 9-day transport and unloading time by signature on [Table 6.2-5](#).

Table 6.2-4 – Shipping Site Control Checklist for Controlled Shipments^①

Shipment No. _____

Packaging No. _____

To be completed by Shipping Site Transportation Certification Official, or designee, for each package designated as a controlled shipment:

CH-TRAMPAC Section No.	Activity	Recorded Date	Recorded Time	Completion of Activity (Indicate by checkmark [✓])
6.2.3.1	Record date and time of ICV closure			
6.2.3.2	Record date and time the shipment containing the loaded package is scheduled to depart from the site			
6.2.3.3	Calculate and record total Loading Time [Limit = 24 hours]			
	<i>Total Loading Time ≤ 1 day, proceed to No. 6.2.3.4. Total Loading Time > 1 day, STOP. Vent package and repeat closure process.</i>			
6.2.3.4	I certify that the above data is accurate and compliant with the Loading Time limit of 24 hours, as specified in Section 6.2.3 of the CH-TRAMPAC. / TRANSPORTATION CERTIFICATION OFFICIAL (OR DESIGNEE) DATE			

Note:

- ① Controlled shipments (10 days) shall be made in accordance with the conditions specified in [Appendix 3.6](#) of the [CH-TRU Payload Appendices](#)¹ and [Section 6.2.3](#). This table may be reformatted for site use provided that the same information is recorded.

Table 6.2-5 – Receiving Site Control Checklist for Controlled Shipments^①

Shipment No. _____ Packaging No. _____

To be completed by designated Receiving Site Operations Personnel for each package designated as a controlled shipment:

CH-TRAMPAC Section No.	Activity	Recorded Date	Recorded Time	Completion of Activity (Indicate by checkmark [✓])
6.2.3.5	Record date and time that the package was scheduled to depart from the shipping site			
6.2.3.6	Vent package within 9 days of date and time recorded above and record vent date and time			
6.2.3.7	I certify that the above data is accurate and compliant with the Transport and Unloading Time limit of 9 days, as specified in Section 6.2.3 of the CH-TRAMPAC. _____ / _____ RECEIVING SITE OPERATIONS PERSONNEL DATE			

Note:

① Controlled shipments (10 days) shall be made in accordance with the conditions specified in [Appendix 3.6](#) of the [CH-TRU Payload Appendices](#)¹ and [Section 6.2.3](#). This table may be reformatted for site use provided that the same information is recorded.

6.2.4 Mixing of Shipping Categories

Appendix 2.4 of the [CH-TRU Payload Appendices](#)¹ describes the MixCat logic for assembling a payload of containers, which includes the determination of the FI.

The FIs are calculated based on each payload container in the direct-load payload or overpacked payload configurations to ensure that each payload container or overpacked container does not contain a flammable mixture of gases. Each payload container or overpacked container is assessed through the calculation of the FI for the container, which accounts for the properties of each of the other containers in the assembly or overpacked configuration, which may include dunnage containers.

A payload is qualified for shipment only if the FI of each payload container or overpacked container is a non-negative number less than or equal to 50,000. If one or more containers fail(s) the FI requirement, the payload or overpacked configuration shall be reconfigured until all containers satisfy this requirement. Otherwise, containers not meeting the FI requirement shall be rejected from transport and subject to mitigation or repackaging. The FI determination can be performed either manually or by the use of a verified/validated software package.

6.2.5 Payload Total Gas Generation Rate Compliance Evaluation

For containers of Waste Type IV that exceed the ATGGR in [Table 5.2-1](#), compliance with the packaging design pressure limit may be demonstrated by compliance with the PTGGRL determined for the payload in which it is assembled. These Waste Type IV payload assemblies may be comprised of other Waste Type IV containers, non-Waste Type IV containers, and/or dunnage containers. Each payload assembly is evaluated through the calculation of the PTGGR, which accounts for the TGGR of each container in the assembly into the ICV.

The TGGR values of containers of Waste Type IV ($TGGR_{i\ IV}$) shall be determined in accordance with [Section 5.2.2.4](#).

For payload assemblies of Waste Type IV containers mixed with containers of other waste types, the TGGR of the non-Waste Type IV containers ($TGGR_{i\ other}$) shall be calculated using the total decay heat of the container as follows:

$$TGGR = \frac{Q_i (G(\text{netgas})_i \text{ molecules}/100\text{eV})}{N_A 1.602(10)^{-19}} \quad (1)$$

where,

Q_i = Decay heat of container i (watts)

$G(\text{netgas})_i$ = Total gas temperature corrected effective G value (net gas) for container i (molecules of gas formed/100 eV emitted energy)^①

N_A = Avogadro's number = $6.0225(10)^{23}$ molecules/mole.

Note:

- ① Net gas G values in [Appendix 3.2](#) of the [CH-TRU Payload Appendices](#)¹ are corrected using the Arrhenius relation to Waste Type IV test temperatures corresponding to the maximum decay heat per packaging shown in [Table 5.2-3](#) of [Section 5.2.5](#).

The PTGGR is the sum of the TGGR values of the gas generating containers making up the payload assembly as follows:

$$PTGGR = \sum_{i=1}^{n_{IV\ containers}} TGGR_{i\ IV} + \sum_{i=1}^{n_{other\ containers}} TGGR_{i\ other} \quad (2)$$

A payload qualifies for shipment only if the PTGGR is less than the PTGGRL calculated below. If the PTGGR exceeds the PTGGRL, the payload shall be reconfigured until this requirement is satisfied. The determination of compliance with the PTGGRL can be performed either manually or by the use of a verified/validated software package. The PTGGRL is calculated as:

$$PTGGRL = 2.26(10)^{-8} \times V_{\text{void}} \quad (3)$$

where,

V_{void} = Available void volume inside the packaging ICV cavity, which includes the void volume of any dunnage containers.

From the values presented in Table 6.2-6, the void volume of a payload assembly is calculated as:

$$V_{\text{void}} = V_{\text{ICV}} + n_{\text{dunnage}} \times V_{\text{dunnage}} \quad (4)$$

where,

V_{ICV} = Void volume inside the ICV and outside the containers

n_{dunnage} = Number of dunnage containers

V_{dunnage} = Void volume of dunnage containers.

Table 6.2-6 – Void Volume Values for Payload Assembly Configurations

Type of Payload	TRUPACT-II Payload Assembly			HalfPACT Payload Assembly		
	Maximum Number of Containers	Void Volume Values (liters)		Maximum Number of Containers	Void Volume Values (liters)	
		ICV	Dunnage		ICV	Dunnage
55-Gallon Drums	14	2,450	208	7	1,846	208
Direct Load SWBs	2	1,750	1,750	1	1,496	NA ^②
One SWB/55 Overpack	2 ^①	2,620	1,750	1	2,366	NA ^②
One SWB/85 Overpack	2 ^①	2,824	1,750	1	2,570	NA ^②
One SWB/100 Overpack	2 ^①	2,724	1,750	1	2,470	NA ^②
Two SWB/55 Overpacks	2	3,490	NA ^②	NA ^②	NA ^②	NA ^②
Two SWB/85 Overpacks	2	3,898	NA ^②	NA ^②	NA ^②	NA ^②
Two SWB/100 Overpacks	2	3,698	NA ^②	NA ^②	NA ^②	NA ^②
85-Gallon Drums	8	2,087	321	4	1,664	321
100-Gallon Drums	6	2,715	378	3	1,978	378

Notes:

① Payload comprised of one SWB overpack of drums and one dunnage SWB.

② NA = Not applicable.

7.0 QUALITY ASSURANCE

This section describes the QA programs applicable to the CH-TRAMPAC. QA programs applicable to procurement, design, fabrication, assembly, testing, use, maintenance, and repair of the TRUPACT-II or HalfPACT packaging are found in [Section 9.0](#) of the [TRUPACT-II SAR](#)¹ or the [HalfPACT SAR](#)².

7.1 QA Requirements for Payload Compliance

Certification of authorized contents for shipment in the TRUPACT-II or HalfPACT shall be performed under a written QA program that provides confidence, for both the shipper and receiver, that the CH-TRAMPAC requirements have been met. All waste shall be described in an approved content code.

¹ U.S. Department of Energy (DOE), [TRUPACT-II Shipping Package Safety Analysis Report](#), USNRC Certificate of Compliance 71-9218, U.S. Department of Energy, Carlsbad Area Office, Carlsbad, New Mexico.

² U.S. Department of Energy (DOE), [HalfPACT Shipping Package Safety Analysis Report](#), USNRC Certificate of Compliance 71-9279, U.S. Department of Energy, Carlsbad Area Office, Carlsbad, New Mexico.

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7.2 QA Compliance and Verification

Compliance methods are documented in DOE-CBFO-approved programmatic TRAMPACs and/or waste-specific data TRAMPACs. The DOE-CBFO managing and operating contractor performs review of users' payload compliance procedures or data package to ensure the requirements of this CH-TRAMPAC are met. The DOE-CBFO will periodically audit users' payload compliance QA programs.

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